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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present specification provides the stage 3 definition of the Policy Authorization Service of the 5G System.

The 5G System Architecture is defined in 3GPP TS 23.501 [2]. The stage 2 definition and related procedures for the Npcf Policy Authorization Service are specified in 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

The 5G System stage 3 call flows are provided in 3GPP TS 29.513 [7].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in 3GPP TS 29.500 [5] and 3GPP TS 29.501 [6].

The Policy Authorization Service is provided by the Policy Control Function (PCF). This service creates policies as requested by the authorised AF for the PDU Session to which the AF session is bound.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 23.503: "Policy and Charging Control Framework for the 5G System; Stage 2".

[5] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[6] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[7] 3GPP TS 29.513: "5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3".

[8] 3GPP TS 29.512: "5G System; Session Management Policy Control Service; Stage 3".

[9] IETF RFC 9113: "HTTP/2".

[10] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[11] OpenAPI: "OpenAPI Specification Version 3.0.0", https://spec.openapis.org/oas/v3.0.0.

[12] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".

[13] 3GPP TS 29.508: "5G System; Session Management Event Exposure Service; Stage 3".

[14] 3GPP TS 29.554: "5G System; Background Data Transfer Policy Control Service; Stage 3".

[15] 3GPP TS 29.122: "T8 reference point for Northbound APIs".

[16] IEEE 802.3-2015: "IEEE Standard for Ethernet".

[17] IEEE 802.1Q-2014: "Bridges and Bridged Networks".

[18] IETF RFC 7042: "IANA Considerations and IETF Protocol and Documentation Usage for IEEE 802 Parameters".

[19] IETF RFC 3986: "Uniform Resource Identifier (URI): Generic Syntax".

[20] 3GPP TS 29.214: "Policy and Charging Control over Rx reference point".

[21] IETF RFC 7396: "JSON Merge Patch".

[22] 3GPP TS 32.291: "5G System; Charging service; Stage 3".

[23] 3GPP TS 22.153: "5G System; "Multimedia Priority Service".

[24] IETF RFC 9457: "Problem Details for HTTP APIs".

[25] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[26] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[27] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".

[28] 3GPP TR 21.900: "Technical Specification Group working methods".

[29] 3GPP TS 24.292: "IP Multimedia (IM) Core Network (CN) subsystem Centralized Services (ICS); Stage 3".

[30] 3GPP TS 26.114: "IP Multimedia Subsystem (IMS); Multimedia telephony; Media handling and interaction".

[31] IETF RFC 5761: "Multiplexing RTP Data and Control Packets on a Single Port".

[32] 3GPP TS 24.229: "IP Multimedia Call Control Protocol based on SIP and SDP; Stage 3".

[33] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS); Stage 2".

[34] IETF RFC 5031: "A Uniform Resource Name (URN) for Emergency and Other Well-Known Services".

[35] IETF RFC 5009: "Private Header (P-Header) Extension to the Session Initiation Protocol (SIP) for Authorization of Early Media".

[36] 3GPP TS 24.008: "Mobile radio interface Layer 3 specification; Core network protocols; Stage 3".

[37] IETF RFC 3556: "Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth".

[38] IETF RFC 3959 (December 2004): "The Early Session Disposition Type for the Session Initiation Protocol (SIP)".

[39] 3GPP TS 23.380: "IMS Restoration Procedures".

[40] 3GPP TS 23.167: "IP Multimedia Subsystem (IMS) emergency sessions".

[41] 3GPP TS 24.379: "Mission Critical Push To Talk (MCPTT) call control; Protocol specification".

[42] IETF RFC 8101: "IANA Registration of New Session Initiation Protocol (SIP), Resource-Priority Namespace for Mission Critical Push To Talk Service".

[43] 3GPP TS 24.281: "Mission Critical Video (MCVideo) signalling control; Protocol specification".

[44] 3GPP TS 23.316: "Wireless and wireline convergence access support for the 5G System (5GS)".

[45] 3GPP TS 22.179: "Mission Critical Push to Talk (MCPTT) over LTE; Stage 1".

[46] 3GPP TS 22.280: "Mission Critical (MC) services common requirements".

[47] 3GPP TS 22.281: "Mission Critical (MC) video over LTE".

[48] 3GPP TS 22.282: "Mission Critical (MC) data over LTE".

[49] Void.

[50] IETF RFC 4574: "The Session Description Protocol (SDP) Label Attribute".

[51] 3GPP TS 26.238: "Uplink Streaming".

[52] IETF RFC 6733: "Diameter Base Protocol".

[53] 3GPP TS 29.519: "5G System; Usage of the Unified Data Repository service for Policy Control Data, Application Data and Structured Data for Exposure; Stage 3".

[54] 3GPP TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3".

[55] Void.

[56] IETF RFC 8655: "Deterministic Networking Architecture".

[57] Void.

[58] IETF RFC 3550: "RTP: A Transport Protocol for Real-Time Applications".

[59] IETF RFC 9143: "Negotiating Media Multiplexing Using the Session Description Protocol (SDP)".

[60] IETF RFC 8285: "A General Mechanism for RTP Header Extensions".

[61] 3GPP TS 29.564: "5G System; User Plane Function Services; Stage 3".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

Application Function (AF): Element offering application(s) that use PDU session resources.

AF Application identifier: An identifier that refers to the particular service the NF service consumer session belongs to. In the context of application detection control, it refers to the application identifier used by the PCF in the PCC rule as specified in 3GPP TS 29.512 [8].

AF application session context: Application level session context established by an application level signalling protocol offered by the AF that requires a session context set-up with explicit session context description before the use of the service.

MCS session: A session for which priority treatment is applied for allocating and maintaining radio and network resources to support the Mission Critical Service (MCS). MCS is defined in 3GPP TS 22.179 [45], 3GPP TS 22.280 [46], 3GPP TS 22.281 [47], and 3GPP TS 22.282 [48].

MPS session: A session for which priority treatment is applied for allocating and maintaining radio and network resources to support the Multimedia Priority Service (MPS). MPS is defined in 3GPP TS 22.153 [23].

PCC rule: Set of information enabling the detection of a service data flow and providing parameters for policy control and/or charging control.

Service information: Set of information conveyed from the AF/NEF to the PCF by the Npcf_PolicyAuthorization service to be used as a basis for PCC decisions at the PCF, including information about the AF/NEF application session context (e.g. application identifier, type of media, bandwidth, IP address and port number).

Service data flow: An aggregate set of packet flows.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5G-RG	5G Residential Gateway
AF	Application Function
ARP	Allocation and Retention Priority
ATSSS	Access Traffic Steering, Switching and Splitting
BAT	Burst Arrival Time
BBF	Broadband Forum
BSSID	Basic Service Set IDentifier
CHEM	Coverage and Handoff Enhancements using Multimedia error robustness feature
CHF	Charging Function
DCCF	Data Collection Coordination Function
DEI	Drop Eligible Indicator
DetNet	Deterministic Networking
DNAI	DN Access Identifier
DNN	Data Network Name
DS-TT	Device-side TSN translator
DSL	Digital Subscriber Line
DTS	Data Transport Service
EAS	Edge Application Server
ECN	Explicit Congestion Notification
ePDG	evolved Packet Data Gateway
E-UTRA	Evolved Universal Terrestrial Radio Access
FLUS	Framework for Live Uplink Streaming
FN-RG	Fixed Network Residential Gateway
GEO	Geosynchronous Orbit
GPSI	Generic Public Subscription Identifier
HFC	Hybrid Fiber-Coaxial
H-PCF	PCF in the HPLMN
IMS	IP-Multimedia Subsystem
ISL	Intersatellite Link
JSON	JavaScript Object Notation
L4S	Low Latency Low Loss Scalable Throughput
LEO	Low Earth Orbit
MA	Multi-Access
MCPTT	Mission Critical Push to Talk Service
MCVideo	Mission Critical Video
MEO	Medium Earth Orbit
MoQ	Media over QUIC
MPS	Multimedia Priority Service
MTU	Maximum Transmission Unit
NEF	Network Exposure Function
NID	Network Identifier
NR	New Radio
NRF	Network Repository Function
NWDAF	Network Data Analytics Function
NW-TT	Network-side TSN translator
PCC	Policy and Charging Control
PCF	Policy Control Function
PCP	Priority Code Point
P-CSCF	Proxy Call Session Control Function
PDV	Packet Delay Variation
PEI	Permanent Equipment Identifier
PMIC	Port Management Information Container
PON	Passive Optical Network
PRA	Presence Reporting Area
PSA	PDU Session Anchor

PSDB	PDU Set Delay Budget
PSER	PDU Set Error Rate
PSIHI	PDU Set Integrated Handling Information
QoS	Quality of Service
QUIC	Quick UDP Internet Connections
RFSP	RAT Frequency Selection Priority
RSN	Redundancy Session Number
RTCP	Real Time Control Protocol
RTP	Real Time Protocol
SDF	Service Data Flow
SDP	Session Description Protocol
SFC	Service Function Chain
SIP	Session Initiation Protocol
SMF	Session Management Function
S-NSSAI	Single Network Slice Selection Assistance Information
SNPN	Stand-alone Non-Public Network
SPI	Security Parameter Index
SSC	Service and Session Continuity
SSID	Service Set IDentifier
SUPI	Subscription Permanent Identifier
TNAP	Trusted Non-3GPP Access Point
TSC	Time Sensitive Communication
TSCAI	Time Sensitive Communication Assistance Information
TSCTSF	Time Sensitive Communication and Time Synchronization Function
TSN	Time Sensitive Networking
TTNB	Time To Next Burst
UDP	User Datagram Protocol
UDR	Unified Data Repository
UMIC	User plane node Management Information Container
UPF	User Plane Function
URSP	UE Route Selection Policy
VID	VLAN Identifier
VLAN	Virtual Local Area Network
V-PCF	PCF in the VPLMN
W-5GAN	Wireline 5G Access Network
W-5GBAN	Wireline 5G BBF Access Network
W-5GCAN	Wireline 5G Cable Access Network
W-AGF	Wireline Access Gateway Function

4 Npcf_PolicyAuthorization Service

4.1 Service Description

4.1.1 Overview

The Npcf_PolicyAuthorization Service, as defined in 3GPP TS 23.502 [3] and in 3GPP TS 23.503 [4], is provided by the Policy Control Function (PCF).

The Npcf_PolicyAuthorization service authorises a NF service consumer request and creates policies as requested by the authorised NF service consumer for the PDU session to which the AF session is bound to. This service also allows the NF service consumer to subscribe/unsubscribe to notifications on events (e.g. access type change, PLMN change, usage report, access network information report).

4.1.2 Service Architecture

The 5G System Architecture is defined in 3GPP TS 23.501 [2]. The Policy and Charging control related 5G architecture is also described in 3GPP TS 23.503 [4] and 3GPP TS 29.513 [7].

The only known NF service consumers of the Npcf_PolicyAuthorization service are the Application Function (AF), the Network Exposure Function (NEF), the Time Sensitive Communication and Time Synchronization Function (TSCTSF) and the Policy Control Function for the UE (PCF for the UE).

The Npcf_PolicyAuthorization service is provided by the PCF and consumed by the AF, the NEF, the TSCTSF and, when the PCF for the PDU session and the PCF for the UE are different, the PCF for the UE, as shown in figure 4.1.2-1 for the SBI representation model and in figure 4.1.2-2 for the reference point representation model.

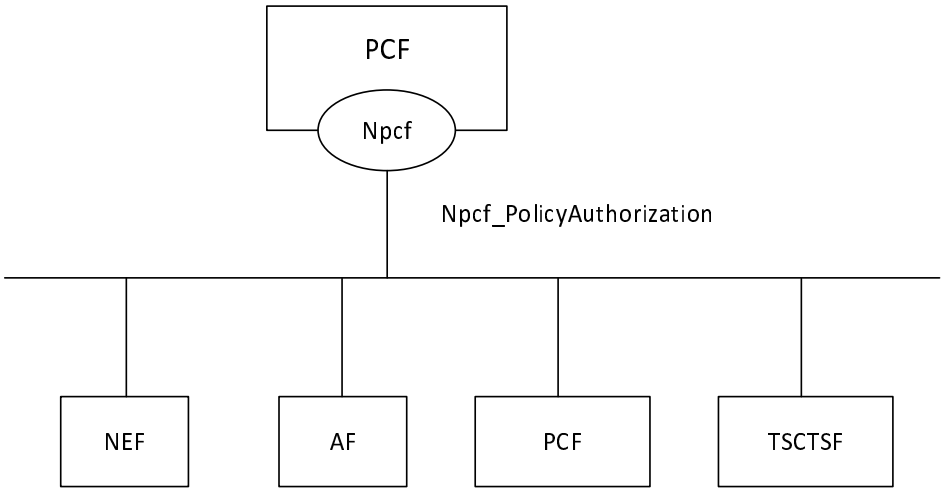


Figure 4.1.2-1: Npcf_PolicyAuthorization service Architecture, SBI representation

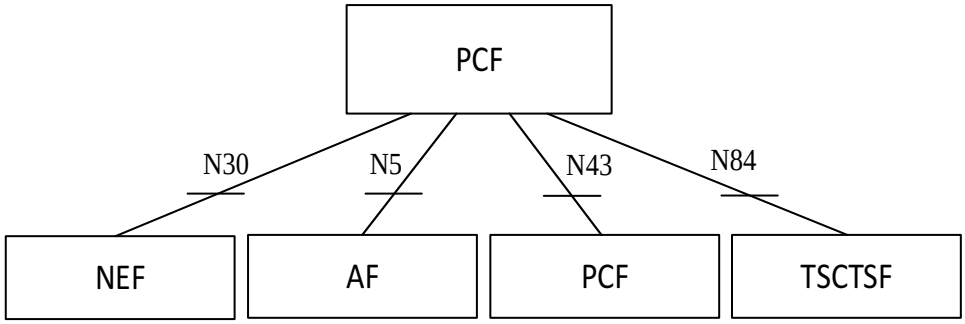


Figure 4.1.2-2: Npcf_PolicyAuthorization service Architecture, reference point representation

NOTE: When the N43 reference point exists, i.e. when the PCF is a NF service consumer of the Npcf_PolicyAuthorization service, the PCF for the UE interacts with the PCF for the PDU session.

The AF trusted by the operator interacts with the PCF via the N5 reference point. In the case of an untrusted AF, the AF interacts with the PCF via the NEF. The NEF interacts with the PCF via the N30 reference point in the same way that the AF interacts with the PCF via the N5 reference point.

4.1.3 Network Functions

4.1.3.1 Policy Control Function (PCF)

The PCF (Policy Control Function) performs policy and charging control for the PDU session and/or the flows indicated by the NF service consumer and according to the service requirements provided by the NF service consumer.

The policy and charging control for service data flows enable the PCF to provide network control regarding the service data flow detection, gating, QoS and flow based charging (except credit management) towards the SMF/UPF.

The PCF receives session and media related information from the Npcf_PolicyAuthorization service consumers and notifies them of subscribed traffic plane events.

The PCF may receive from the NF service consumers the request to monitor the requested service and media information and notifies them of the UL/DL/round-trip delay, and congestion information of the requested flows.

The PCF may receive service routing requirements and the indication of receiving notifications about user plane path changes from the Npcf_PolicyAuthorization service consumers.

The PCF may receive from the NF service consumers the specific required QoS and a prioritized list of alternative QoS profiles and notifies them about the QoS target the access network guarantees.

The PCF checks that the service information provided by the NF service consumer is consistent with the operator defined policy rules before storing the service information.

The PCF uses the received service information and the subscription information when it applies as basis for the policy and charging control decisions.

The PCF derives PCC rules and provisions them to the SMF via the Npcf_SMPolicyControl service and subscribes to traffic plane events via policy control request triggers as described in 3GPP TS 29.512 [8].

To enable Time Sensitive Communication, Time Synchronization and Deterministic Networking, the PCF:

- notifies the NF service consumer (i.e. TSN AF or TSCTSF) about the TSC user plane node and port number corresponding to the device side of a PDU session;
- enables the NF service consumer (i.e. TSN AF or TSCTSF) configures/reads information from the TSC user plane node and ports by forwarding TSC user plane node management containers and port management containers to the SMF as described in 3GPP TS 29.512 [8];
- notifies the NF service consumer (i.e. TSN AF or TSCTSF) about updated TSC user plane node configuration and port configuration information by forwarding TSC user plane node management containers and port management containers received from the SMF; and
- uses the received QoS and TSC assistance information to derive the policy information delivered in the PCC rule to the SMF as described in 3GPP TS 29.512 [8].

To enable PDU Set handling, the PCF may receive the PDU set related QoS information and/or Protocol Description related information from the NF service consumers to derive the policy information delivered in the PCC rule to the SMF as described in 3GPP TS 29.512 [8].

When the PCF for the UE is separated from the PCF for the PDU session, the PCF for the PDU session is responsible for notifying to the PCF for the UE about the URSP rule enforcement information received from the UE via the SMF as described in 3GPP TS 29.512 [8].

4.1.3.2 NF Service Consumers

The known NF service consumers are the AF, the NEF, the TSCTSF and the PCF (for a UE), as defined in 3GPP TS 23.502 [3].

The AF is an element offering control to applications that require the policy and charging control of traffic plane resources; specific user plane paths for the requested traffic, the monitoring of the required service QoS, and/or specific QoS and alternative QoS profiles. The AF uses the Npcf_PolicyAuthorization service to provide service information to the PCF.

In 5GS interworking with TSN networks, the TSN AF is an element offering to TSC control functions an interface to 5GS to forward TSC user plane node and port management configuration, and to set the QoS policy required to forward the TSC traffic making use of the 5GS traffic plane resources.

The AFs can be deployed by the same operator offering the access services or can be provided by external third-party service provider. If the AF is not allowed by the operator to access directly the PCF, the AF uses the external exposure framework via NEF to interact with the PCF, as described in clause 5.20 of 3GPP TS 23.501 [2].

The Network Exposure Function (NEF) supports external exposure of capabilities of network functions.

The AF trusted by the operator, the NEF or the DetNet controller can use the TSCTSF to interface with PCF to support time sensitive communication, time synchronization and deterministic networking. The TSCTSF is an element offering, to internal and/or external time sensitive AF (via NEF) and/or to the DetNet controller, control to handle from/towards the PCF the required TSC user plane node and port management configuration, and to set in the PCF the QoS policy required to forward TSC traffic.

The PCF providing session management policy control for a UE (i.e. PCF for a PDU session) and the PCF providing UE policy control and/or access and mobility control for this same UE (i.e. PCF for a UE) may be different PCFs. When access and mobility policies depend on traffic plane events (as e.g. application detection control), or the URSP may be adjusted based on the report of URSP rule enforcement information, the PCF for a UE may act as an NF service consumer of the PCF for the PDU session by subscribing to the corresponding events.

4.2 Service Operations

4.2.1 Introduction

Service operations defined for the Npcf_PolicyAuthorization Service are shown in table 4.2.1-1.

Table 4.2.1-1: Npcf_PolicyAuthorization Service Operations

Service Operation Name	Description	Initiated by
Npcf_PolicyAuthorization_Create	Determines and installs the policy according to the service information provided by an authorized NF service consumer. (NOTE)	NF service consumer (e.g. AF, NEF)
Npcf_PolicyAuthorization_Update	Determines and updates the policy according to the modified service information provided by an authorized NF service consumer. (NOTE)	NF service consumer (e.g. AF, NEF)
Npcf_PolicyAuthorization_Delete	Provides means to delete the application session context of the NF service consumer.	NF service consumer (e.g. AF, NEF)
Npcf_PolicyAuthorization_Notify	Notifies NF service consumers of the subscribed events.	PCF
Npcf_PolicyAuthorization_Subscribe	Allows NF service consumers to subscribe for an AF session to the notifications of events. (NOTE)	NF service consumer (e.g. AF, NEF, PCF for a UE)
Npcf_PolicyAuthorization_Unsubscribe	Allows NF service consumers to unsubscribe for an AF session from the notifications of events. (NOTE)	NF service consumer (e.g. AF, NEF, PCF for a UE)
NOTE: Subscription to notifications of events per flow level is only supported by the Npcf_PolicyAuthorization_Create and Npcf_PolicyAuthorization_Update as described in clauses 4.2.2.2 and 4.2.3.2. Unsubscription to notifications of events is only supported with the Npcf_PolicyAuthorization_Update service operation, as described in clause 4.2.3.		

NOTE 1: The NEF and the AF use the Npcf_PolicyAuthorization service in the same way.

NOTE 2: The PCF is the consumer when the PCF for the UE and the PCF for the PDU session are different in the Npcf_PolicyAuthorization_Notify/Subscribe/Unsubscribe operations.

NOTE 3: The NWDAF and the DCCF can be NF service consumers of the Npcf_PolicyAuthorization_Notify/Subscribe/Unsubscribe operations to perform data collection for UEs. However, there is no data collected from the PCF by the NWDAF or the DCCF defined in this Release of the specification.

4.2.2 Npcf_PolicyAuthorization_Create service operation

4.2.2.1 General

The Npcf_PolicyAuthorization_Create service operation authorizes the request from the NF service consumer, and optionally communicates with Npcf_SMPolicyControl service to determine and install the policy according to the information provided by the NF service consumer.

The Npcf_PolicyAuthorization_Create service operation creates an application session context in the PCF.

The following procedures using the Npcf_PolicyAuthorization_Create service operation are supported:

- Initial provisioning of service information.
- Gate control.
- Initial Background Data Transfer policy indication.
- Initial provisioning of sponsored connectivity information.
- Subscription to Service Data Flow QoS notification control.
- Subscription to Service Data Flow Deactivation.
- Initial provisioning of traffic routing, service function chaining information and handling of payload headers.
- Subscription to resources allocation outcome.
- Invocation of Multimedia Priority Services.
- Support of content versioning.
- Request of access network information.
- Initial provisioning of service information status.
- Provisioning of signalling flow information.
- Support of resource sharing.
- Indication of Emergency traffic.
- Invocation of MCPTT.
- Invocation of MCVideo.
- Priority sharing indication.
- Subscription to out of credit notification.
- Subscription to Service Data Flow QoS Monitoring information.
- Provisioning of TSCAI input information and TSC QoS related data.
- Provisioning of TSC user plane node management information and port management information.
- P-CSCF restoration enhancements.
- Support of CHEM feature.
- Support of FLUS feature.
- Subscription to EPS Fallback report.
- Subscription to TSC user plane node related events.
- Initial provisioning of required QoS information.

- Support of QoSHint feature.
- Subscription to reallocation of credit notification.
- Subscription to satellite backhaul category changes.
- Subscription to the report of extra UE addresses.
- Initial provisioning of Round-Trip latency requirements.
- Provisioning of multi-modal services.
- Provisioning of PDU Set handling related data.
- Subscription to BAT offset notification.
- Subscription to Packet Delay Variation monitoring.
- Provisioning of the indication of ECN marking for L4S support.
- Subscription of Round-Trip delay monitoring requirements over two QoS flows.
- Provisioning of the QoS timing information.
- Initial provisioning of traffic information for UE power saving management.
- Subscription to the report of network support for QoS Monitoring.
- Provisioning of dynamically changing traffic characteristics.
- Provisioning of deliver media related information for encrypted traffic.
- Request for 5GC network functions health monitoring information.
- Provisioning of the indication of RAN-Controlled UL Bitrate Recommendation Indication.
- Provisioning of the (S)RTP Multiplexed Media Identification Information.
- Subscription to Data Rate Limitation Information reporting.

4.2.2.2 Initial provisioning of service information

This procedure is used to set up an AF application session context for the service as defined in 3GPP TS 23.501 [2], 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

Figure 4.2.2.2-1 illustrates the initial provisioning of service information.

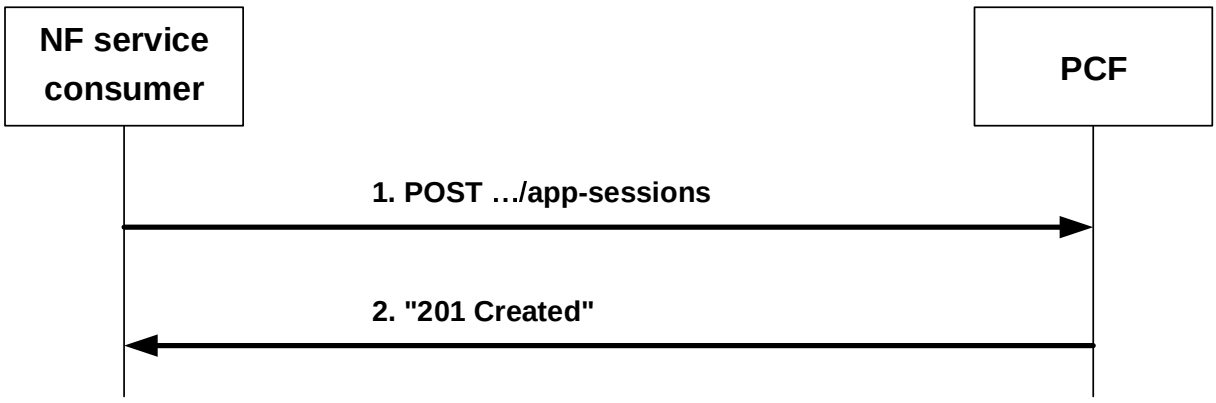


Figure 4.2.2.2-1: Initial provisioning of service information

When a new AF application session context is being established and media information for this application session context is available at the NF service consumer and the related media requires PCC control, the NF service consumer shall invoke the Npcf_PolicyAuthorization_Create service operation by sending the HTTP POST request to the

resource URI representing the "Application Sessions" collection resource of the PCF, as shown in figure 4.2.2.2-1, step 1.

The NF service consumer shall include in the "AppSessionContext" data type in the content of the HTTP POST request a partial representation of the "Individual Application Session Context" resource by providing the "AppSessionContextReqData" data type. The "Individual Application Session Context" resource and the "Events Subscription" sub-resource are created as described below.

The NF service consumer shall provide in the body of the HTTP POST request:

- for IP type PDU sessions, the IP address (IPv4 or IPv6) of the UE in the "ueIpv4" or "ueIpv6" attribute; and
- for Ethernet type PDU sessions, the MAC address of the UE in the "ueMac" attribute.

For Ethernet type PDU sessions, if the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported, the "ueMac" attribute containing the MAC address of the DS-TT port as received from the PCF during the reporting of TSC user plane node information as defined in clause 4.2.5.16.

NOTE 1: The determination of the DS-TT port MAC address is specified in clause 5.28.2 of 3GPP TS 23.501 [2]. The DS-TT port MAC address is used as identifier of the PDU session related to the reported TSC user plane node information.

For IP type PDU sessions, if the "TimeSensitiveCommunication" feature is supported, the "ueIpv4" or "ueIpv6" attribute containing the IPv4 or IPv6 address of the UE as received from the PCF during the reporting of user plane node information as defined in clause 4.2.5.16.

NOTE 2: The IP address of the PDU session is used as identifier of the PDU session related to the reported TSC user plane node information.

The NF service consumer shall provide the corresponding service information in the "medComponents" attribute, if available. The NF service consumer shall indicate to the PCF for each media component included within the "medComponents" attribute whether the media component service data flow(s) (IP or Ethernet) should be enabled or disabled with the "fStatus" attribute. The service data flow filters (IP or Ethernet) that identify the traffic of the media component, if available, shall be provided within the media subcomponent(s) elements included in the "medSubComps" attribute (one uplink and/or downlink service data flow filter per media subcomponent). If the "EnQoSMon" feature is supported, the NF service consumer may include the attribute "evSubsc" in the "MediaSubComponent" data type for QoS monitoring for each media component. Either the "evSubsc" in "MediaSubComponent" data type or attribute "evSubsc" in "AppSessionContextReqData" data type may be provided to subscribe to notifications for a specific event.

NOTE 3: The NF service consumer could provide more than one "MediaSubComponent" data type (within one or more media components) if the same or different events applies to different single-modal data flow.

An IP flow description is based on the definition of the packet filter for an IP flow (direction, IP source and destination address, protocol, and source and destination port) as defined by "FlowDescription" data type, the type of service or traffic class as defined in the "tosTrCl" attribute and, when the feature "DetNet" is supported, the flow label and the IPsec SPI as defined in the "flowLabel" and "spi" attributes respectively.

An Ethernet flow description is based on the definition of the packet filter for an Ethernet flow (direction, Ethertype, source and destination MAC address, vlan tags, IP flow description (when Ethertype is IP) and source and destination MAC address range) as specified by "EthFlowDescription" data type.

If the "AuthorizationWithRequiredQoS" feature as defined in clause 5.8 is supported, the AF may provide within the MediaComponent data structure required QoS information as specified in clause 4.2.2.32.

The AF may include the AF application identifier in the "afAppId" attribute into the body of the HTTP POST request in order to indicate the particular service that the AF session belongs to.

The AF application identifier may be provided at both "AppSessionContextReqData" data type level, and "MediaComponent" data type level. When provided at both levels, the AF application identifier provided at "MediaComponent" data type level shall have precedence.

The AF application identifier at the "AppSessionContextReqData" data type level may be used to trigger the PCF to indicate to the SMF/UPF to perform the application detection based on the operator's policy as defined in 3GPP TS 29.512 [8].

If the "IMS_SBI" feature is supported, the NF service consumer may include the AF charging identifier in the "afChargId" attribute for charging correlation purposes.

If the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported the NF service consumer may provide TSC information as specified in clauses 4.2.2.24 and 4.2.2.25.

If the "MultiMedia" feature is supported, the NF service consumer may provide the multi-modal service identifier in the "multiModalId" attribute for multi-modal communication purpose in clause 4.2.2.37.

If the "PDUSetHandling" feature is supported, the NF service consumer may provide PDU set handling related data as specified in clause 4.2.2.39.

If the "PowerSaving" feature is supported, the NF service consumer may provide UL and/or DL traffic periodicity and/or DL protocol description as described in clause 4.2.2.42.

The NF service consumer may also include the "evSubsc" attribute of "EventsSubscReqData" data type to request the notification of certain user plane events. The NF service consumer shall include the events to subscribe to in the "events" attribute, and the notification URI where to address the Npcf_PolicyAuthorization_Notify service operation in the "notifUri" attribute. The events subscription is provisioned in the "Events Subscription" sub-resource.

The AF shall also include the "notifUri" attribute in the "AppSessionContextReqData" data type to indicate the URI where the PCF can request to the AF the deletion of the "Individual Application Session Context" resource.

If the PCF cannot successfully fulfil the received HTTP POST request due to the internal PCF error or due to the error in the HTTP POST request, the PCF shall send the HTTP error response as specified in clause 5.7.

Otherwise, when the PCF receives the HTTP POST request from the NF service consumer, the PCF shall apply session binding as described in 3GPP TS 29.513 [7]. To allow the PCF to identify the PDU session for which the HTTP POST request applies, the NF service consumer shall provide in the body of the HTTP POST request:

- for IP type PDU session, either the "ueIpv4" attribute or "ueIpv6" attribute containing the IPv4 or the IPv6 address applicable to an IP flow or IP flows towards the UE; and
- for Ethernet type PDU session, the "ueMac" attribute containing the UE MAC address applicable to an Ethernet flow or Ethernet flows towards the UE.

The NF service consumer may provide DNN in the "dnn" attribute, SUPI in the "supi" attribute, GPSI in the "gpsi" attribute, the S-NSSAI in the "sliceInfo" attribute if available for session binding. The NF service consumer may also provide the domain identity in the "ipDomain" attribute.

NOTE 4: The "ipDomain" attribute is helpful in the following scenario: Within a network slice, there are several separate IP address domains, with SMF/UPF(s) that allocate Ipv4 IP addresses out of the same private address range to UE PDU sessions. The same IP address can thus be allocated to UE PDU sessions served by SMF/UPF(s) in different address domains. If one PCF controls several SMF/UPF(s) in different IP address domains, the UE IP address is thus not sufficient for the session binding. A NF service consumer can serve UEs in different IP address domains, either by having direct IP interfaces to those domains, or by having interconnections via NATs in the user plane between the UPF and the NF service consumer. If a NAT is used, the NF service consumer obtains the IP address allocated to the UE PDU session via application level signalling and supplies it for the session binding to the PCF in the "ueIpv4" attribute. The NF service consumer supplies an "ipDomain" attribute denoting the IP address domain behind the NAT in addition. The NF service consumer can derive the appropriate value from the source address (allocated by the NAT) of incoming user plane packets. The value provided in the "ipDomain" attribute is operator configurable.

NOTE 5: The "sliceInfo" attribute is helpful in the scenario where multiple network slices are deployed in the same DNN, and the same IPv4 address may be allocated to UE PDU sessions in different network slices. If one PCF controls several network slices, the UE IP address is not sufficient for the session binding. The NF service consumer supplies "sliceInfo" attribute denoting the network slice that allocated the IPv4 address of the UE PDU session. How the NF service consumer derives S-NSSAI is out of the scope of this specification.

NOTE 6: When the scenario described in NOTE 3 applies and the NF service consumer is a P-CSCF it is assumed that the P-CSCF has direct IP interfaces to the different IP address domains and that no NAT is located between the UPF and P-CSCF. How a non-IMS NF service consumer obtains the UE private IP address to be provided to the PCF is out of scope of the present release; it is unspecified how to support applications that use a protocol that does not retain the original UE's private IP address.

NOTE 7: As described in 3GPP TS 29.513 [7], in order to have a successful session binding, all attributes must match, if provided.

If the PCF fails in executing session binding, the PCF shall reject the Npcf_PolicyAuthorization_Create service operation with an HTTP "500 Internal Server Error" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "PDU_SESSION_NOT_AVAILABLE".

If the request contains the "medComponents" attribute the PCF shall store the received service information. The PCF shall process the received service information according to the operator policy and may decide whether the request is accepted or not. The PCF may take the priority information within the "resPrio" attribute into account when making this decision.

If the service information provided in the body of the HTTP POST request is rejected (e.g., the subscribed guaranteed bandwidth for a particular user is exceeded, the authorized data rate in that slice for a UE is exceeded), the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure that:

- shall contain the ProblemDetails data structure containing the "cause" attribute set to the "REQUESTED_SERVICE_NOT_AUTHORIZED" application error indicating the cause of the rejection; and

- may contain the acceptable QoS parameters within the "acceptableServInfo" attribute.

If the PCF detects that a temporary network failure has occurred (e.g. the SGW has failed as defined in clause B.3.3.3 or B.3.4.9 of 3GPP TS 29.512 [8]) or if the PCF supports "SmfFailureDetection" feature and the PCF due to operator configuration detects that the SMF has failed as defined in 3GPP TS 23.380 [39], and when the AF initiates an Npcf_PolicyAuthorization_Create service operation, the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "TEMPORARY_NETWORK_FAILURE".

If the service information provided in the HTTP POST request is rejected due to a temporary condition in the network (e.g. the NWDAF reported the network slice selected for the PDU session is congested), the PCF may reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUESTED_SERVICE_TEMPORARILY_NOT_AUTHORIZED". The PCF may also provide a retry interval within the "Retry-After" HTTP header field. When the NF service consumer receives the retry interval within the "Retry-After" HTTP header field, the NF service consumer shall not send the same service information to the PCF again (for the same application session context) until the retry interval has elapsed. The "Retry-After" HTTP header is described in 3GPP TS 29.500 [5] clause 5.2.2.2.

If the service information is invalid or in sufficient for the PCF to perform the requested action, e.g. invalid media type or invalid QoS reference, the PCF shall reject the request with an HTTP "400 Bad Request" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "INVALID_SERVICE_INFORMATION".

If the IP flow descriptions cannot be handled by the PCF because the restrictions defined in clause 5.3.8 of 3GPP TS 29.214 [20] are not observed, the PCF shall reject the request with an HTTP "400 Bad Request" status code with the response body including ProblemDetails data structure with the the "cause" attribute set to "FILTER_RESTRICTIONS".

If the AF provided the same AF charging identifier for a new Individual Application Session Context that is already in use for the other ongoing Individual Application Session, the PCF shall reject the request with an HTTP "400 Bad Request" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "DUPLICATED_AF_SESSION".

NOTE 8: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8] and the authorized data rate for any of those cases in a slice is exceeded due to the bandwidth demands of the new service information, it is also possible to accept the request based on operator policies. In this case the derived PCC rule(s) belonging to the authorized GBR service data flows can include a different MBR and/or have a different charging than the one applicable if the data rate is not exceeded as specified in 3GPP TS 29.512 [8].

If the "SignalingPathValidation" feature is supported, and the "User-Agent" HTTP header field indicates that the NF type of the NF that originated the request is "NEF" or "AF", and the PCF detects that the TSCTSF is the NF type required for the request (e.g., the PCF triggered a notification about TSC user plane node information towards the TSCTSF as described in clause 4.2.15.16), the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "INVALID_SIGNALING_PATH". When the NEF/AF receives this error from the PCF, the NEF/AF selects the TSCTSF for this request, as specified in 3GPP TS 29.522 [54].

If the "VPLMNErrorRep" feature is supported and the required QoS information provided by the AF as specified in clause 4.2.2.32 is not supported in the current serving PLMN where the UE is registered, the PCF may reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUEST_QOS_NOT_SUPPORTED_IN_PLMN" indicating the cause of the rejection.

If the "TrafficCharChange" feature is supported and the expedited data transfer with reflective QoS indication is provided by the AF as specified in clause 4.2.2.48, and the UE has not indicated support for reflective QoS as specified in 3GPP TS 29.512 [8], the PCF shall reject the request and indicate in an HTTP "403 Forbidden" response message the cause for the rejection including the "cause" attribute set to "REFLECTIVE_QOS_NOT_SUPPORTED_IN_UE".

If the "MpxMedia" feature is supported, the NF service consumer request includes media flows with an uplink direction and the SMF has not indicated UE support for (S)RTP Multiplexed Media Identification to the PCF as specified in 3GPP TS 29.512 [8], the PCF may, based on operator configuration, reject the request and indicate in an HTTP "403 Forbidden" response message the cause for the rejection including the "cause" attribute set to "MPX_MEDIA_NOT_SUPPORTED_IN_UE".

If the "ExtQoSR19" feature is supported, the NF service consumer may provide the RAN-Controlled UL Bitrate Recommendation Indication for the media flow as described in clause 4.2.2.52.

To allow the PCF and SMF/UPF to perform PCC rule authorization and QoS flow binding for the described service data flows, the NF service consumer shall supply:

- for IP type PDU session, both source and destination IP addresses and port numbers in the "fDescs" attribute within the "medSubComps" attribute, if such information is available; and

- for Ethernet type PDU session, the Ethernet Packet filters in the "ethfDescs" attribute within the "medSubComps" attribute, if such information is available.

The NF service consumer may specify the ToS traffic class (i.e. ToS (IPv4) or TC (IPv6) value) within the "tosTrCl" attribute for the described service data flows together with the "fDescs" attribute.

NOTE 9: A ToS/TC value can be useful when another packet filter attribute is needed to differentiate between packet flows. For example, packet flows encapsulated and encrypted by a tunnelling protocol can be differentiated by the ToS/TC value of the outer header if appropriately set by the application. To use ToS/TC for service data flow detection, network configuration needs to ensure there is no ToS/TC re-marking applied along the path from the application to the PSA UPF and the specific ToS/TC values are managed properly to avoid potential collision with other usage (e.g., paging policy differentiation).

The NF service consumer may include the "resPrio" attribute at the "AppSessionContextReqData" data type level to assign a priority to the AF Session as well as include the "resPrio" attribute at the "MediaComponent" data type level to assign a priority to the service data flow. The presence of the "resPrio" attribute in both levels does not constitute a conflict as they each represent different types of priority. The reservation priority at the "AppSessionContextReqData" data type level provides the relative priority for an AF session while the reservation priority at the "MediaComponent" data type level provides the relative priority for a service data flow within a session. If the "resPrio" attribute is not specified, the requested priority is PRIO_1.

The PCF shall check whether the received service information requires PCC rules to be created and provisioned as specified in 3GPP TS 29.513 [7]. Provisioning of PCC rules to the SMF shall be carried out as specified at 3GPP TS 29.512 [8].

Based on the received subscription information from the NF service consumer, the PCF may create a subscription to event notifications for a related PDU session from the SMF, as described in 3GPP TS 29.512 [8].

If the PCF created an "Individual Application Session Context" resource, the PCF shall send to the NF service consumer a "201 Created" response to the HTTP POST request, as shown in figure 4.2.2.2-1, step 2. The PCF shall include in the "201 Created" response:

- a Location header field; and
- an "AppSessionContext" data type in the content.

The Location header field shall contain the URI of the created individual application session context resource i.e. "{apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}".

When "Events Subscription" sub-resource is created in this procedure, the NF service consumer shall build the sub-resource URI by adding the path segment "/events-subscription" at the end of the URI path received in the Location header field.

The "AppSessionContext" data type the content shall contain the representation of the created "Individual Application Session Context" resource and may include the "Events Subscription" sub-resource.

The PCF shall include in the "evsNotif" attribute:

- if the NF service consumer subscribed to the event "PLMN_CHG" in the HTTP POST request, the "event" attribute set to "PLMN_CHG" and the "plmnId" attribute including the PLMN Identifier or the SNPN Identifier if the PCF has previously requested to be updated with this information in the SMF;

NOTE 10: The SNPN Identifier consists of the PLMN Identifier and the NID.

NOTE 11: Handover between non-equivalent SNPNs, and between SNPN and PLMN is not supported. When the UE is operating in SNPN access mode, the trigger reports changes of equivalent SNPNs.

- if the NF service consumer subscribed to the event "ACCESS_TYPE_CHANGE" in the HTTP POST request, the "event" attribute set to "ACCESS_TYPE_CHANGE" and:

  i. the "accessType" attribute including the access type, and the "ratType" attribute including the RAT type when applicable for the notified access type; and

  ii. if the "ATSSS" feature is supported, the "addAccessInfo" attribute with the additional access type information if available, where the access type is encoded in the "accessType" attribute, and the RAT type is encoded in the "ratType" attribute when applicable for the notified access type; and

NOTE 12: For a MA PDU session, if the "ATSSS" feature is not supported by the NF service consumer the PCF includes the "accessType" attribute and the "ratType" attribute with a currently active combination of access type and RAT type (if applicable for the notifed access type). When both 3GPP and non-3GPP accesses are available, the PCF includes the information corresponding to the 3GPP access.

  iii. the "anGwAddr" attribute including access network gateway address when available,

  if the PCF has previously requested to be updated with this information in the SMF;

- if the "IMS_SBI" feature is supported and if the NF service consumer subscribed to the "CHARGING_CORRELATION" event in the HTTP POST request, the "event" attribute set to "CHARGING_CORRELATION" and may include the "anChargIds" attribute containing the access network charging identifier(s) and the "anChargAddr" attribute containing the access network charging address; and

- if the "UEUnreachable" feature is supported and the NF service consumer subscribed to the "UE_REACH_STATUS_CH" event in the HTTP POST request, the "event" attribute set to "UE_REACH_STATUS_CH" together with the "ueReachStatus" attribute containing the corresponding UE status, and in case the "ueReachStatus" attribute is set to "UNREACHABLE", optionally the "retryAfter" attribute if available and the PCF has previously requested this information to the SMF.

The NF service consumer subscription to other specific events using the Npcf_PolicyAuthorization_Create request is described in the related clauses. Notification of events when the applicable information is not available in the PCF when receiving the Npcf_PolicyAuthorization_Create request is described in clause 4.2.5.

The acknowledgement towards the NF service consumer should take place before or in parallel with any required PCC rule provisioning towards the SMF.

NOTE 13: The behaviour when the NF service consumer does not receive the HTTP response message, or when it arrives after the internal timer waiting for it has expired, or when it arrives with an indication different than a success indication, are outside the scope of this specification and based on operator policy.

4.2.2.3 Gate control

This procedure is used by an NF service consumer to instruct the PCF about when the service data flow(s) are to be enabled or disabled for a PDU session.

The AF shall include in the HTTP POST request message described in subclause 4.2.2.2 the "fStatus" attribute for the flows to be enabled or disabled within the "medComponents" or "medSubComps" attributes.

If a "medSubComps" attribute contains a "flowUsage" attribute with the value "RTCP", then the IP Flows described by that media subcomponent shall be enabled in both directions irrespective of the value of the "fStatus" attribute of the corresponding media component.

As result of this action, the PCF shall set the appropriate gate status for the corresponding active PCC rule(s).

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.2.4 Initial Background Data Transfer policy indication

This procedure is used by a NF service consumer to indicate a transfer policy negotiated for background data transfer using the Npcf_BDTPolicyControl service as described in 3GPP TS 29.554 [14].

The NF service consumer may include in the HTTP POST request message described in clause 4.2.2.2 a reference identifier related to a transfer policy negotiated for background data transfer in the "bdtRefId" attribute.

NOTE 1: The PCF will retrieve the corresponding transfer policy from the UDR based on the reference identifier within the "bdtRefId" attribute. In case only one PCF is deployed in the network, transfer policies can be locally stored in the PCF and the interaction with the UDR is not required.

If the PCF cannot retrieve the transfer policy, the PCF shall set to TP_NOT_KNOWN the "servAuthInfo" attribute in the HTTP response message to the NF service consumer to indicate that the transfer policy is unknown.

If the time window of the received transfer policy has expired, the PCF shall set to TP_EXPIRED the "servAuthInfo" attribute in the HTTP response message to indicate to the NF service consumer that the transfer policy has expired. Otherwise, if the time window of the received transfer policy has not yet occurred, the PCF shall set to TP_NOT_YET_OCURRED the "servAuthInfo" attribute in the HTTP response message to the NF service consumer to indicate that the time window of the transfer policy has not yet occurred.

NOTE 2: In the case that the PCF cannot retrieve the transfer policy, the transfer policy time window has not yet occurred or the transfer policy expired, the PCF makes the decision without considering the transfer policy.

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.2.5 Initial provisioning of sponsored connectivity information

This procedure is used by a NF service consumer to indicate sponsored data connectivity when "SponsoredConnectivity" feature is supported.

The NF service consumer shall provide in the "AppSessionContext" data type of the HTTP POST request message described in clause 4.2.2.2 an application service provider identity and a sponsor identity within the "aspId" attribute and "sponId" attribute within the "ascReqData" attribute. Additionally, the NF service consumer may provide an indication to the PCF of sponsored data connectivity not enabled by including the "sponStatus" attribute set to "SPONSOR_DISABLED".

To support the usage monitoring of sponsored data connectivity, the NF service consumer may subscribe with the PCF to the notification of usage threshold reached. The NF service consumer shall include:

- an entry of the "AfEventSubscription" data type in the "events" attribute with the "event" attribute set to "USAGE_REPORT"; and
- the "usgThres" attribute of "UsageThreshold" data type in the "EventsSubscReqData" data type with:
  a) the total volume in the "totalVolume" attribute; or
  b) the uplink volume only in the "uplinkVolume" attribute; or
  c) the downlink volume only in the "downlinkVolume"; and/or
  d) the time in the "duration" attribute.

NOTE 1: If the NF service consumer is in the user plane, the AF can handle the usage monitoring and therefore it is not required to provide a usage threshold to the PCF as part of the sponsored connectivity functionality.

When the NF service consumer indicated to enable sponsored data connectivity, and the UE is roaming in a VPLMN, the following procedures apply:

- If the NF service consumer is located in the HPLMN, for home routed roaming case and when the operator policies do not allow accessing the sponsored data connectivity with this roaming case, the H-PCF shall reject the service request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "UNAUTHORIZED_SPONSORED_DATA_CONNECTIVITY".
- If the NF service consumer is located in the VPLMN, the V-PCF shall reject the service request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "UNAUTHORIZED_SPONSORED_DATA_CONNECTIVITY".

When the NF service consumer indicated to enable sponsored data connectivity, and the UE is non-roaming or roaming with the home routed case and the operator policies allow accessing the sponsored data connectivity with this roaming case, the following procedures apply:

- If the SMF does not support sponsored connectivity and the required reporting level for that service indicates a sponsored connectivity level according to 3GPP TS 29.512 [8], then the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUESTED_SERVICE_NOT_AUTHORIZED".
- If the SMF supports sponsored data connectivity feature or the required reporting level is different from sponsored connectivity level as described in 3GPP TS 29.512 [8], then the PCF, based on operator policies, shall check whether it is required to validate the sponsored connectivity data. If it is required, it shall perform the authorizations based on sponsored data connectivity profiles. If the authorization fails, the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "UNAUTHORIZED_SPONSORED_DATA_CONNECTIVITY".

NOTE 2: The PCF is not required to verify that a trust relationship exists between the operator and the sponsors.

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.2.6 Subscriptions to Service Data Flow QoS notification control

The subscription to Service Data Flow QoS notification control is used by a NF service consumer to subscribe to receive a notification when the GBR QoS targets for one or more service data flows can no longer (or can again) be guaranteed.

NOTE: It may happen that the GBR QoS targets for one or more PCC rules (i.e. Service Data Flows) cannot be guaranteed, either permanently or temporarily in the radio access network.

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "events" attribute with the "event" attribute set to "QOS_NOTIF".

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As result of this action, the PCF shall set the appropriate subscription to QoS notification control for the corresponding PCC rule(s) as described in in 3GPP TS 29.512 [8].

4.2.2.7 Subscription to Service Data Flow Deactivation

This procedure is used by NF service consumer to subscribe to the notification of deactivation of one or more Service Data Flows within the AF application session context.

NOTE: It may happen that one or more PCC rules (i.e. Service Data Flows) are deactivated at the SMF at certain time, either permanently or temporarily, due to e.g. release of resources or out of credit condition.

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "events" attribute with the "event" attribute set to "FAILED_RESOURCES_ALLOCATION".

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As result of this action, the PCF shall set the appropriate subscription to service data flow deactivation for the corresponding PCC rule(s) as described in in 3GPP TS 29.512 [8].

4.2.2.8 Initial provisioning of traffic routing, service function chaining information and handling of payload headers

This procedure is used by a NF service consumer to:

when the "InfluenceOnTrafficRouting" feature is supported:

- influence SMF traffic routing decisions to a local access to a Data Network identified by a DNAI; and/or
- request subscriptions to notifications about UP path management events related to the PDU session.

when the "UeSatUeComm" feature is supported:

- request subscriptions to notifications about UP path management events related to the PDU session.

when the "SFC" feature is supported:

- influence the steering of user traffic to service function chain(s) on N6-LAN.

when the "HeaderHandling" feature is supported:

influence the handling of payload headers on the existing or future PDU session(s).

NOTE 1: The NF service consumer uses the Npcf_PolicyAuthorization service for requests targeting specific on-going PDU sessions of individual UE(s). The NF service consumer requests that target existing or future PDU Sessions of multiple UE(s) or any UE are sent via the NEF and may target multiple PCF(s), as described in 3GPP TS 29.513 [7].

NOTE 2: In the non-roaming scenario, the AF can request simultaneously to influence on traffic routing, steering of user traffic to service function chain(s) on N6-LAN and header handling and payload header handling.

When the "CommonEASDNAI" feature is supported, the procedure is also used by a NF service consumer to request to select a common EAS or EAS(es) corresponding to a common DNAI for a set of UE associated with the same traffic correlation Id accessing the application identified by the provided service information.

NOTE 3: Common EAS selection means the common DNAI is selected.

In order to influence on traffic routing, the NF service consumer shall include in the HTTP POST request message described in clause 4.2.2.2 the "afRoutReq" attribute of "AfRoutingRequirement" data type with specific routing

requirements for the application traffic flows either within "AppSessionContextReqData" data type for the service indicated in the "afAppId" attribute, or within the "medComponents" attribute. When provided at both levels, the "afRoutReq" attribute value in the "medComponents" attribute shall have precedence over the "afRoutReq" attribute included in the "AppSessionContextReqData" data type.

In order to influence on N6-LAN traffic steering, the NF service consumer shall include in the HTTP POST request message described in clause 4.2.2.2 the "afSfcReq" attribute of "AfSfcRequirement" data type with specific N6-LAN traffic steering requirements for the application traffic flows either within "AppSessionContextReqData" data type for the service indicated in the "afAppId" attribute, or within the "medComponents" attribute. When provided at both levels, the "afSfcReq" attribute value in the "medComponents" attribute shall have precedence over the "afSfcReq" attribute included in the "AppSessionContextReqData" data type.

In order to influence on handling of payload headers, the NF service consumer shall include in the HTTP POST request message described in clause 4.2.2.2 the "afHdrReq" attribute of "AfHeaderHandlingControlInfo" data type with specific handling of payload headers requirements for the application traffic flows either within "AppSessionContextReqData" data type for the service indicated in the "afAppId" attribute, or within the "medComponents" attribute. When provided at both levels, the "afHdrReq" attribute value in the "medComponents" attribute shall have precedence over the "afHdrReq" attribute included in the "AppSessionContextReqData" data type. In order to retrieve header handling event report for the performed header handling action, the AF shall set the "notifFlag" attribute to true for the related "HeaderHandlingActionRequest" data type and shall also provide the "notifUri" attribute and the "notifId" attribute within the "AfHeaderHandlingControlInfo" data type.

The NF service consumer may include traffic routing, N6-LAN traffic steering requirements, and handling of payload headers together with service information.

The NF service consumer may request to influence on N6-LAN traffic steering, to influence SMF traffic routing decisions and/or to influence handling of payload headers to a DNAI.

If the "SFC" feature is supported, when the NF service consumer requests to influence N6-LAN traffic steering, it shall include in the "afSfcReq" attribute:

a) the pre-defined Service Function Chain identifier for downlink in "sfcIdDl" and/or for uplink in "sfcIdUl";

and may also include:

b) spatial validity which the NF service consumer request is valid shall be indicated in terms of validity areas encoded in the "spVal" attribute of "SpatialValidity" data type. The "SpatialValidity" data type consists of a list of presence areas included in the "presenceInfoList" attribute, where each element shall include the presence reporting area identifier in the "praId" attribute and may include the elements composing a presence area encoded in the attributes: "trackingAreaList", "ecgList", "ncgList", "globalRanNodeIdList".

c) The metadata information wich should be sent to the UPF via SMF transparently as defined in 3GPP TS 29.512 [8].

If the "InfluenceOnTrafficRouting" feature is supported, when the NF service consumer request to influence on traffic routing, the NF service consumer shall include in the "afRoutReq" attribute:

a) A list of routes to locations of applications in the "routeToLocs" attribute. Each element of the list shall contain:

- a DNAI in the "dnai" attribute to indicate the location of the application towards which the traffic routing is applied; and

- a routing profile identifier in the "routeProfId" attribute, and/or the explicit routing information in the "routeInfo" attribute.

and may also include:

a) Indication of application relocation possibility in the "appReloc" attribute.

b) Temporal validity during which the NF service consumer request is valid shall be indicated with the "startTime" and "stopTime" attributes.

c) Spatial validity during which the NF service consumer request is valid shall be indicated in terms of validity areas encoded in the "spVal" attribute of "SpatialValidity" data type. The "SpatialValidity" data type consists of a list of presence areas included in the "presenceInfoList" attribute, where each element shall include the

presence reporting area identifier in the "praId" attribute and may include the elements composing a presence area encoded in the attributes: "trackingAreaList", "ecgList", "ncgList", "globalRanNodeIdList".

d) Indication of UE IP address preservation in the "addrPreserInd" attribute if the URLLC feature is supported.

e) If the SimultConnectivity feature is supported:

- indication of simultaneous connectivity temporarily maintained in the source and target PSA during the edge re-location procedure in the "simConnInd" attribute; and
- if the "simConnInd" attribute is set to true, the minimum time interval to be considered for inactivity of the traffic routed via the source PSA in the "simConnTerm" attribute.

f) EAS IP replacement information in the "easIpReplaceInfos" attribute if the EASIPreplacement feature is supported.

g) Indication of EAS rediscovery in the "easRedisInd" attribute if the EASDiscovery feature is supported.

h) Maximum allowed user plane latency in the "maxAllowedUpLat" attribute if the AF_latency feature is supported.

NOTE 4: The EAS IP Replacement information and the information indicating the EAS rediscovery are not provided simultaneously.

i) If the CommonEASDNAI feature is supported, traffic correlation information in the "tfcCorreInfo" attribute.

j) Indication of whether to consider the N6 delay measurement in the "n6DelayInd" attribute if the N6DelayMeasurement feature is supported.

k) If the EnCommonDnai feature is supported, an indication about reporting candidate DNAI(s) in the "candDnaiInd" attribute.

If the "HeaderHandling" feature is supported, when the NF service consumer requests to influence handling of payload headers, it shall include in the "afHdrReq" attribute:

a) The pre-defined header handling detection reference in the "hDetectionReference" attribute.

and may also include:

a) Dynamic information not pre-configured in a UPF used to detect headers within "hDetectionSuppInfo" attribute.

b) The list of header handling action request parameters in the uplink or downlink direction within "hHndlgUl" or "hHndlgDl" attributes respectively.

c) The time interval(s) during which the NF service consumer request is valid shall be indicated within the "tempVals" attribute.

d) Spatial validity which the NF service consumer request is valid shall be indicated in terms of validity areas encoded in the "spVal" attribute of "SpatialValidity" data type. The "SpatialValidity" data type consists of a list of presence areas included in the "presenceInfoList" attribute, where each element shall include the presence reporting area identifier in the "praId" attribute and may include the elements composing a presence area encoded in the attributes: "trackingAreaList", "ecgList", "ncgList", "globalRanNodeIdList".

e) The notification URI where the NF service consumer is receiving the Nupf_EventExposure_Notify service operation in the "notifUri" attribute and the notification correlation identifier assigned by the NF service consumer in the "notifId" attribute.

When the "InfluenceOnTrafficRouting" feature and/or the "UeSatUeComm" feature is(are) supported, the NF service consumer may also subscribe to notifications about UP path management events. The NF service consumer shall include in the "upPathChgSub" attribute:

- notifications of early and/or late DNAI change, using the attribute "dnaiChgType" indicating whether the subscription is for "EARLY", "LATE" or "EARLY_LATE";
- the notification URI where the NF service consumer is receiving the Nsmf_EventExposure_Notify service operation in the "notificationUri" attribute; and

- the notification correlation identifier assigned by the NF service consumer in the "notifCorreId" attribute.

When the NF service consumer subscribes to notifications about UP path management events, it may include the "3gpp-Sbi-Consumer-Info" custom HTTP header as described in clause 6.6.2 of 3GPP TS 29.500 [5] to indicate the features supported by the NF service consumer over the Nsmf_EventExposure service related to UP path management event handling as described in 3GPP TS 29.508[13].

If the URLLC feature and/or the "UeSatUeComm" feature is(are) supported, the NF service consumer may include an indication of NF service consumer acknowledgement to be expected as an "afAckInd" attribute within the "upPathChgSub" attribute.

When the feature "RoutingReqOutcome" is supported:

- the PCF may set the "servAuthInfo" attribute in the HTTP response message to "ROUT_REQ_NOT_AUTHORIZED" when the PCF determines, e.g. based on subscription, the AF influence on traffic routing is not allowed for the PDU session;

- when the NF service consumer requests the steering of traffic to a DNAI and/or the subscription to notifications about UP path management events, the NF service consumer may subscribe to notifications of failures in the enforcement of UP path changes including within the "evSubsc" attribute the "event" attribute value "UP_PATH_CHG_FAILURE" in an entry of the "events" array.

NOTE 5: In the case that the PCF determines that the requested AF routing requirements cannot be applied and returns the "servAuthInfo" attribute in the HTTP response, the PCF makes the decision without considering the requested AF routing requirements.

When "TraffRouteReqOutcome" feature is supported, the NF service consumer may subscribe to the traffic routing requirement installation outcome event. The NF service consumer shall include for that in the "outcomeSub" attribute:

- the notification URI via which the NF service consumer desires to receive the notifications within the "notificationUri" attribute; and

- the notification correlation identifier assigned by the NF service consumer within the "notifCorreId" attribute.

NOTE 6: If the NF service consumer subscribes for traffic routing requirements installation outcome event notification using the "outcomeSub" attribute, it is not required to subscribe to the notifications of failures in the enforcement of UP path changes through the "UP_PATH_CHG_FAILURE" event within the "evSubsc" attribute.

When "SimConnFailure" feature is supported, the NF service consumer may subscribe to the simultaneous connectivity failure event. The NF service consumer shall include for that in the "simConnFailSub" attribute:

- the notification URI via which the NF service consumer desires to receive the notifications within the "notificationUri" attribute; and

- the notification correlation identifier assigned by the NF service consumer within the "notifCorreId" attribute.

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

The PCF shall store the routing requirements included in the "afRoutReq" attribute and/or if the SFC feature is supported, the N6-LAN traffic steering requirements within the "afSfcReq" attribute and/or if the "HeaderHandling" feature is supported, the header handling control information within the "afHdrReq".

The PCF shall check whether the received routing requirements and/or N6-LAN traffic steering requirements require PCC rules to be created or provisioned to include or modify traffic steering policies (for both routing requirements and/or N6-LAN traffic steering requirements) and the application relocation possibility (only for routing requirements) as specified in 3GPP TS 29.513 [7]. If the "SFC" feature is supported, the PCF shall check whether the received N6-LAN traffic steering requirements require PCC rules to be created or provisioned and/or if the"HeaderHandling" feature is supported, the PCF shall check whether the received header handling control information require PCC rules to be created or provisioned. Provisioning of PCC rules to the SMF shall be carried out as specified in 3GPP TS 29.512 [8].

NOTE 7: The NF service consumer receives the notification about UP path management events by the Nsmf_EventExposure_Notify service operation as defined in clause 4.2.2.2 of 3GPP TS 29.508 [13].

4.2.2.9 Void

4.2.2.10 Subscription to resources allocation outcome

This procedure is used by a NF service consumer to subscribe to notifications when the resources associated to the corresponding service information have been allocated and/or cannot be allocated.

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message:

- if the NF service consumer requests the PCF to provide a notification when the resources associated to the service information have been allocated, an event entry within the "events" attribute with the "event" attribute set to "SUCCESSFUL_RESOURCES_ALLOCATION";
- if the NF service consumer requests the PCF to provide a notification when the resources associated to the service information cannot be allocated, an event entry within the "events" attribute with the "event" attribute set to "FAILED_RESOURCES_ALLOCATION"; and/or
- if the feature "UEUnreachable" is supported and the NF service consumer request the PCF to provide a notification when the resources associated to the service information are not allocated because the UE is unreachable, an event entry within the "events" attribute with the "event" attribute set to "UE_TEMPORARILY_UNAVAILABLE".

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As a result of this action, the PCF shall set the appropriate subscription to notification of resources allocation outcome for the corresponding PCC Rule(s) as described in 3GPP TS 29.512 [8].

4.2.2.11 Void

4.2.2.12 Invocation of Multimedia Priority Services

4.2.2.12.1 General

This procedure allows a NF service consumer, as per 3GPP TS 22.153 [23], to request prioritized access to system resources in situations such as during congestion.

The NF service consumer may include the "mpsId" attribute to indicate that the new AF session relates to an MPS session.

The "mpsId" attribute shall contain the national variant for the MPS service name indicating an MPS session. The "resPrio" attribute shall include the priority value of the related priority service.

If the NF service consumer supports the SBI Message Priority mechanism for an MPS session, it shall include the "3gpp-Sbi-Message-Priority" custom HTTP header towards the PCF as described in clause 6.8.2 of 3GPP TS 29.500 [5].

NOTE 1: If the NF service consumer supports the SBI Message Priority mechanism for an MPS session, the NF service consumer will include the "3gpp-Sbi-Message-Priority" custom HTTP header with a priority value equivalent to the value of the "resPrio" attribute. Highest user priority value is mapped in the corresponding lowest value of the "3gpp-Sbi-Message-Priority" custom HTTP header.

When the PCF receives the "mpsId" attribute indicating an MPS session, the PCF shall take specific actions on the corresponding PDU session to ensure that the MPS session is prioritized as specified in 3GPP TS 29.512 [8].

NOTE 2: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8], it is possible that, subject to operator policy and national/regional regulations, prioritised services are exempted from the limitation of data rate per network slice. In that case the PCF will handle the request from the NF service consumer even if the authorized data rate per network slice is exceeded.

4.2.2.12.2 MPS for DTS

MPS for DTS is the means for an NF service consumer to invoke/revoke the Priority PDU connectivity service for the default QoS flow only, i.e. without designating a particular service data flow for priority service. MPS for DTS applies only to non-IMS DNNs.

When the "MPSforDTS" feature is supported, to invoke MPS for DTS, the NF service consumer includes the "mpsAction" attribute, set to "ENABLE_MPS_FOR_DTS" or "AUTHORIZE_AND_ENABLE_MPS_FOR_DTS". These "mpsAction" attribute values signal a QoS change to the default QoS flow and service data flows mapped to the default QoS flow without the creation of a new QoS flow.

When the "ENABLE_MPS_FOR_DTS" value is received, and allowed by local policy, the PCF does not check the user's MPS subscription details. When the "AUTHORIZE_AND_ENABLE_MPS_FOR_DTS" value is received, and allowed by local policy, the PCF shall check the user's MPS subscription to authorize the request. When the request is to authorize and enable, and the request is not authorized (e.g. not allowed by MPS subscription), the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with "cause" attribute set to "REQUESTED_SERVICE_NOT_AUTHORIZED" indicating the cause of the rejection.

NOTE: How the NF service consumer checks the MPS for DTS authorization is out of scope of the present document.

When creating an Individual Application Session Context resource, due to the invocation or revocation of MPS for DTS, the NF service consumer may subscribe to the outcome of the default QoS updates by setting within the "evSubsc" attribute an event in the "events" array with:

- the "event" attribute set to the value "SUCCESSFUL_QOS_UPDATE" to report that the invocation/revocation requested by the NF service consumer was successful; and/or
- the "event" attribute set to the value "FAILED_QOS_UPDATE" to report that the invocation/revocation requested by the NF service consumer has failed to be successful.

The NF service consumer may use the procedure specified in clause 4.2.2.12.3 to open a new priority PDU session related to the AF signalling IP flow between the UE and NF service consumer.

4.2.2.12.3 Provisioning of MPS for DTS signalling flow information

This clause is applicable to provisioning of signalling flow information for MPS for DTS if the "MPSforDTS" feature is supported as described in clause 5.8.

This procedure allows the NF service consumer to provision information about the AF signalling IP flows between the UE and the NF service consumer.

The NF service consumer shall provide:

- the IP address (IPv4 or IPv6) of the UE in the "ueIpv4" or "ueIpv6" attribute;
- when the "AuthorizationForMpsSignalling" feature is supported, the "mpsAction" attribute set to "AUTHORIZE_AND_ENABLE_MPS_FOR_AF_SIGNALLING";
- the "mpsId" attribute; and
- a media component within the "medComponents" attribute including:
  - the "medCompN" attribute set to "0"; and
  - the media subcomponent within the "medSubComps" attribute representing the AF signalling IP flow, where the media subcomponent shall contain:
    - the "flowUsage" attribute set to the value "AF_SIGNALLING";
    - the "fNum" attribute set according to the rules described in Annex C;
    - the "fDesc" attribute containing the IP flows of the AF signalling flow; and
    - the "fStatus" set to the value "ENABLED".

The PCF shall determine whether the request is accepted or not. If the "AuthorizationForMpsSignalling" feature is supported and the "mpsAction" attribute is set to "AUTHORIZE_AND_ENABLE_MPS_FOR_AF_SIGNALLING" the PCF shall determine if the request is authorized according to the UE MPS Subscription information. If the request is accepted, the PCF shall perform session binding and shall reply to the NF service consumer as described in clause 4.2.2.2. If rejected, the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUESTED_SERVICE_NOT_AUTHORIZED" indicating the cause of the rejection.

The PCF shall set appropriate QoS values for the AF signalling IP flow and shall install the corresponding dynamic PCC rule for the AF signalling IP flows.

The NF service consumer may de-provision the information about the AF signalling IP flows at any time. To do that, if the "Individual Application Session Context" resource is only used to provide information about the AF Signalling IP flows, the NF service consumer shall remove the resource by sending an Npcf_PolicyAuthorization_Delete service operation towards the PCF as defined in clause 4.2.4.2. Otherwise, the NF service consumer shall remove the IP flow within the media component invoking the Npcf_PolicyAuthorization_Update service operation as defined in clause 4.2.3.17.

NOTE: Combining the request for the AF signalling flow with an MPS for DTS invocation/revocation request is not supported in this release.

4.2.2.12.4 Priority for AF Signalling flow for MPS for Messaging

The support of the MPSforMessaging feature is optional. When the MPSforMessaging feature is supported as described in clause 5.8, the AF and the PCF shall comply with the procedures specified in this clause.

This procedure allows the NF service consumer to provision information about the AF signalling IP flows between the UE and the NF service consumer.

The NF service consumer shall provide:

- the IP address (IPv4 or IPv6) of the UE in the "ueIpv4" or "ueIpv6" attribute;
- the "mpsAction" attribute set to "ENABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING" or "DISABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING"; and
- a media component within the "medComponents" attribute including:
  - the "medCompN" attribute set to "0"; and
  - the media subcomponent within the "medSubComps" attribute representing the AF signalling IP flow, where the media subcomponent shall contain:
    - the "flowUsage" attribute set to the value "AF_SIGNALLING";
    - the "fNum" attribute set according to the rules described in Annex C;
    - the "fDesc" attribute containing the IP flows of the AF signalling flow; and
    - the "fStatus" set to the value "ENABLED".

When the "mpsAction" attribute is set to "ENABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING", the PCF shall check if the AF Signalling IP flows are already prioritized. The PCF may upgrade the QoS of the related PCC Rule(s) based on the PCF decision.

When the "mpsAction" attribute is set to "DISABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING", the PCF shall check if there are other services that require priority treatment and may update the QoS of the related PCC Rule(s) based on the PCF decision.

The PCF shall install the corresponding dynamic PCC rule(s) for the AF signalling IP flows.

The NF service consumer may de-provision the information about the AF signalling IP flows at any time. To do that, if the "Individual Application Session Context" resource is only used to provide information about the AF Signalling IP flows, the NF service consumer shall remove the resource by sending an Npcf_PolicyAuthorization_Delete service operation towards the PCF as defined in clause 4.2.4.2. Otherwise, the NF service consumer shall remove the IP flow

within the media component invoking the Npcf_PolicyAuthorization_Update service operation as defined in clause 4.2.3.17.

4.2.2.13 Support of content versioning

The support of the media component versioning is optional. When the "MediaComponentVersioning" feature is supported, the NF service consumer and the PCF shall comply with the procedures specified in this clause.

If required by operator policies, the NF service consumer shall assign a content version to the media component related to certain service and shall provide assigned content version to the PCF in the "contVer" attribute included in the corresponding media component entry of the "medComponents" attribute.

If the PCF receives the "contVer" attribute for a certain media component, the PCF shall follow the procedures described in 3GPP TS 29.512 [8], clause 4.2.6.2.14.

4.2.2.14 Request of access network information

This procedure is used by a NF service consumer to request the PCF to report the access network information (i.e. user location and/or user timezone information) at the creation of the "Individual Application Session Context" resource, when the "NetLoc" feature is supported and to report serving satellite identifier within the access network information, if available and when the "UeSatUeComm" feature is supported.

The NF service consumer shall include in the HTTP POST request message described in clause 4.2.2.2:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:

  a) the "event" attribute set to the value "ANI_REPORT"; and

  b) the "notifMethod" attribute set to the value "ONE_TIME"; and

- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information and/or if the "UeSatUeComm" feature is supported, identifier of the serving satellite of the UE.

When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature, the PCF shall respond to the NF service consumer including in the "EventsNotification" data type the "noNetLocSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2.

4.2.2.15 Initial provisioning of service information status

When the "IMS_SBI" feature is supported, the NF service consumer may provide the status of the service information.

If the NF service consumer provides service information that has been fully negotiated (e.g. based on the SDP answer), the NF service consumer may include the "servInfStatus" attribute set to "FINAL". In this case the PCF shall authorize the session and provision the corresponding PCC rules to the SMF.

The NF service consumer may additionally provide preliminary service information not fully negotiated yet (e.g. based on the SDP offer) at an earlier stage. To do so, the NF service consumer shall include the "servInfStatus" attribute set to "PRELIMINARY". Upon receipt of such preliminary service information, the PCF shall perform an early authorization check of the service information. If the NF service consumer requests the PCF to report the access network information together with preliminary service information, the PCF shall immediately configure the SMF to provide the access network information.

If the "SmfFailureDetection" feature is suppported and the PCF is not aware whether the SMF has failed, the PCF may install PCC rules related to the preliminary service information to check whether the CN NF is available or not based on operator configuration. If PCF detects that the SMF has failed, it may behave as defined in clause 4.2.2.2, clause 4.2.3.2 and clause 4.2.5.3.

4.2.2.16 Provisioning of signalling flow information

This clause is applicable when IMS restoration is supported according to the supported feature "ProvAFsignalFlow" as described in clause 5.8.

This procedure allows NF service consumer to provision information about the AF signalling IP flows between the UE and the NF service consumer.

The NF service consumer shall provide:

- the IP address (IPv4 or IPv6) of the UE in the "ueIPv4" or "ueIPv6" attribute; and
- a media component within the "medComponents" attribute including:
  - the "medCompN" attribute set to "0"; and
  - one or more media subcomponents within the "medSubComps" attribute representing the AF signalling IP flows, where each media subcomponent shall contain:
    - the "flowUsage" attribute set to the value "AF_SIGNALLING";
    - the "fNum" attribute set according to the rules described in Annex C;
    - the "fDesc" attribute containing the IP flows of the AF signalling flow;
    - the "fStatus" set to the value "ENABLED"; and
    - the "afSigProtocol" set to the value corresponding to the signalling protocol used between the UE and the NF service consumer.

The PCF shall perform session binding and shall reply to the NF service consumer as described in clause 4.2.2.2.

PCC rules related to the AF signalling IP flows could have been provisioned to SMF using the corresponding procedures specified in 3GPP TS 29.512 [8] at an earlier stage (e.g. typically at the establishment of the QoS flow for AF Signalling IP Flows). The PCF shall install the corresponding dynamic PCC rule for the AF signalling IP flows.

The NF service consumer may de-provision the information about the AF signalling IP flows at any time. To do that, if the "Individual Application Session Context" resource is only used to provide information about the AF Signalling IP flows, the NF service consumer shall remove the resource by sending an Npcf_PolicyAuthorization_Delete service operation as service operation towards the PCF as defined in clause 4.2.4.2. Otherwise, the NF service consumer shall remove the IP flows within the media component invoking the Npcf_PolicyAuthorization_Update service operation as defined in clause 4.2.3.17.

4.2.2.17 Support of resource sharing

This procedure is used by a NF service consumer to indicate that a media component of an Individual Application Session Context resource may share resources with other media components in the related direction in other Individual Application Session Context resources when the "ResourceSharing" feature is supported.

The NF service consumer may include the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute within a media component of the "medComponents" attribute to indicate that the related media of the created new Individual Application Session Context resource may share resources with other media components in the related direction that include the same value for the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute.

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2.

If the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute are provided within a media component of the "medComponents" attribute, the PCF may apply the mechanisms for resource sharing as described in 3GPP TS 29.512 [8], clause 4.2.6.2.8.

4.2.2.18 Indication of Emergency traffic

When the "IMS_SBI" feature is supported, this procedure allows a NF service consumer to indicate that the new AF session context relates to emergency traffic.

The NF service consumer may include the "servUrn" attribute to indicate that the new AF session context relates to emergency traffic. Additionally, the NF service consumer may include the "afReqData" attribute to indicate the additional information requested for the AF session context.

When the PCF receives the "servUrn" attribute indicating an emergency session, the PCF may apply special policies, for instance prioritising service flows relating to the new AF session context, allowing these service flows free of charge or exempting the service flows from the limitation of data rate per network slice when the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8]).

If the "servUrn" attribute indicates that the new NF service consumer session context relates to emergency traffic and the "afReqData" attribute is received, the PCF shall reply to the NF service consumer as described in clause 4.2.2.2 and shall provide the requested available user information in the "ueIds" attribute included within the "ascRespData" attribute in the HTTP "201 Created" response.

NOTE 1: The "supi" attribute within the "ueIds" attribute contains an IMSI, if available, when the UE accesses a PLMN and contains either an IMSI or a network-specific identifier that takes the form of an NAI, if available, when the UE accesses a SNPN. For both, PLMN access and SNPN access, the "gpsi" attribute within the "ueIds" attribute contains an MSISDN, if available, and the "pei" attribute contains an IMEI(SV).

If the NF service consumer supports the SBI Message Priority mechanism for an emergency session, it shall include the "3gpp-Sbi-Message-Priority" custom HTTP header towards the PCF as described in clause 6.8.2 of 3GPP TS 29.500 [5].

NOTE 2: If the NF service consumer supports the SBI Message Priority mechanism for an emergency session, the NF service consumer includes the "3gpp-Sbi-Message-Priority" custom HTTP header based on NF service consumer policies in relation to valid values of the "servUrn" attribute. The highest user priority value is mapped to the corresponding lowest value of the "3gpp-Sbi-Message-Priority" custom HTTP header.

When the new AF session context does not indicate emergency traffic and the session binding function detects that the binding is to a PDU session established to the Emergency DNN, the PCF shall reject the HTTP POST request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure including the "cause" attribute set to "UNAUTHORIZED_NON_EMERGENCY_SESSION" indicating the cause of the rejection.

4.2.2.19 Invocation of MCPTT

When the feature "MCPTT" is supported by the NF service consumer and the PCF, this procedure allows a NF service consumer to request prioritized access to system resources in situations such as an MCPTT session with priority call.

The NF service consumer may include the "mcpttId" attribute to indicate that the new "Individual Application Session Context" resource relates to an MCPTT session with priority call.

When the PCF receives the "mcpttId" attribute indicating an MCPTT session and the "resPrio" attribute, the PCF shall take specific actions on the corresponding PDU session to ensure that the MCPTT session is prioritized as specified in 3GPP TS 29.512 [8].

NOTE: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8], it is possible that, subject to operator policy and national/regional regulations, prioritised services are exempted from the limitation of data rate per network slice. In that case the PCF will handle the request from the NF service consumer even if the authorized data rate per network slice is exceeded.

Additionally, when the "PrioritySharing" feature is supported, the PCF may receive the "prioSharingInd" attribute within the media component received in the "medComponents" attribute as described in clause 4.2.2.21. In this case, and if "MCPTT-Preemption" feature is supported, the PCF may receive pre-emption information as also described in clause 4.2.2.21.

For the handling of MCPTT session with priority call, see Annex B.13

4.2.2.20 Invocation of MCVideo

When the feature "MCVideo" is supported by the NF service consumer and the PCF, this procedure allows a NF service consumer to request prioritized access to system resources in situations such as an MCVideo session with priority call.

The NF service consumer may include the "mcVideoId" attribute to indicate that the new "Individual Application Session Context" resource relates to an MCVideo session with priority call.

When the PCF receives the "mcVideoId" attribute indicating an MCVideo session and the "resPrio" attribute, the PCF shall take specific actions on the corresponding PDU session to ensure that the MCVideo session is prioritized as specified in 3GPP TS 29.512 [8].

NOTE: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8], it is possible that, subject to operator policy and national/regional regulations, prioritised services are exempted from the limitation of data rate per network slice. In that case the PCF will handle the request from the NF service consumer even if the authorized data rate per network slice is exceeded.

For the handling of MCVideo session with priority call, see Annex B.15.

4.2.2.21 Priority sharing indication

When the "PrioritySharing" feature is supported, the NF service consumer may indicate to the PCF that the related media flow is allowed to use the same Allocation and Retention Priority (ARP) as media flows belonging to other "Individual Application Session Context" resources.

The NF service consumer may include the "prioSharingInd" attribute set to "ENABLED" within a media component of the "medComponents" attribute to indicate to the PCF that the related media flow is allowed to use the same Allocation and Retention Priority as media flows which:

- are assigned the same 5QI in the PCF; and
- belong to other "Individual Application Session Context" resources bound to the same PDU session that also contain the "prioSharingInd" attribute set to "ENABLED".

If the "MCPTT-Preemption" feature is supported, the NF service consumer may also include:

- within a media component of the "medComponents" attribute, the "preemptCap" attribute containing the suggested pre-emption capability value and the "preemptVuln" attribute containing the suggested pre-emption vulnerability value, for the PCF to determine ARP values;
- within the "ascReqData" attribute in the request body, the "preemptControlInfo" attribute containing the pre-emption control information for the PCF to perform pre-emption control as described in 3GPP TS 29.512 [8], clause 4.2.6.2.9; and
- within the "evSubsc" attribute, the "event" attribute set to "FAILED_RESOURCES_ALLOCATION" to request the notification for resource allocation failure.

Upon reception of this information, the PCF shall behave as described in 3GPP TS 29.512 [8], clause 4.2.6.2.9. For the handling of MCPTT sessions, see Annex B.10.

NOTE 1: Service data flow deactivation procedures will apply according to clauses 4.2.2.7, 4.2.3.7, 4.2.5.5.

NOTE 2: This enhancement avoids the risk that a QoS flow establishment request is rejected if the maximum number of active QoS flows is exceeded.

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2.

4.2.2.22 Subscription to Out of Credit notification

This procedure is used by the NF service consumer if the "IMS_SBI" feature is supported to subscribe to notifications of credit not available for the Service Data Flows within the AF application session context.

NOTE: It can happen that there are one or more PCC rules (i.e. Service Data Flows) with credit not available, each one with their corresponding termination action (terminate, redirect, access restricted).

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "evSubsc" attribute with the "event" attribute set to the value "OUT_OF_CREDIT".

As result of this action, the PCF shall set the appropriate subscription to out of credit notification for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2.

4.2.2.23 Subscriptions to Service Data Flow QoS Monitoring Information

4.2.2.23.1 Subscriptions to QoS Monitoring at AF session level

The subscription at AF session level to Service Data Flow QoS monitoring information is used by a NF service consumer to receive a notification about the real-time measurements of QoS monitoring parameters for a QoS Flow (i.e. when the AF request involves only flows with the same QoS requirements), e.g. packet delay between UPF and UE, when the "QoSMonitoring" feature is supported.

NOTE 1: When the subscription for QoS Monitoring is for multiple QoS flows, i.e, when the AF-session can contain multiple media components, the subscription for QoS monitoring can only be indicated within the corresponding "medSubComps" entry.

When the requested measurement(s) are for the packet delay QoS monitoring parameter, the NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include:

- the requested QoS monitoring parameter(s) to be measured (i.e. DL, UL and/or round trip packet delay) within the "reqQosMonParams" attribute; and
- an entry of the "AfEventSubscription" data type per requested notification method in the "events" attribute with:
  a) the "event" attribute set to the value "QOS_MONITORING"; and
  b) the "notifMethod" attribute set to the value "EVENT_DETECTION" or "PERIODIC"; and
  c) when the "notifMethod" attribute is set to the value "PERIODIC", the periodic time for reporting and, if the feature "PacketDelayFailureReport" is supported, the maximum period with no QoS measurement results reported within the "repPeriod" attribute; and
  d) when the "notifMethod" attribute is set to the value "EVENT_DETECTION", the minimum waiting time between subsequent reports within the "waitTime" attribute and, if the feature "PacketDelayFailureReport" is supported, the maximum period with no QoS measurement results reported within the "repPeriod" attribute;
- when the "notifMethod" attribute set to the value "EVENT_DETECTION":
  a) the delay threshold for downlink with the "repThreshDl" attribute;
  b) the delay threshold for uplink with the "repThreshUl" attribute; and/or
  c) the delay threshold for round trip with the "repThreshRp" attribute.
- The NF service consumer may include in "EventsSubscReqData" data type the notification correlation identifier assigned by the AF within the "notifCorreId" attribute and, if the feature "ExposureToEAS" is supported, the "directNotifInd" attribute set to true to indicate direct event notification of QoS Monitoring data from the UPF.

When the "EnQoSMon" feature is supported, the NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2, as follows:

- the requested QoS monitoring parameter(s) to be measured (i.e. DL/UL/round trip packet delay, UL/DL data rate information, UL/DL congestion information, and/or, if the "EnQoSMon_v2" feature is supported, UL/DL available bitrate) within the "reqQosMonParams" attribute; and
- within the "events" attribute, the NF service consumer shall include an entry of the "AfEventSubscription" data type per QoS monitoring parameter, with:
  1. the "event" attribute set to the value "QOS_MONITORING";

2. the requested QoS monitoring parameter type within the "qosMonParamType" attribute;

3. the notification method within the "notifMethod" ("EVENT_DETECTION" or "PERIODIC), as applicable for the indicated QoS monitoring within the "qosMonParamType" attribute;

NOTE 2: When the "reqQosMonParams" attribute indicates UL/DL congestion information, and/or UL/DL available bitrate, the "notifMethod" attribute can only indicate "EVENT_DETECTION".

4. when the "notifMethod" attribute is set to the value "PERIODIC", the periodic time for reporting and the maximum period with no QoS measurement results reported within the "repPeriod" attribute; and

5. when the "notifMethod" attribute is set to the value "EVENT_DETECTION", the minimum waiting time between subsequent reports within the "waitTime" attribute and the maximum period with no QoS measurement results reported within the "repPeriod" attribute; and

- when the "notifMethod" attribute is set to the value "EVENT_DETECTION":

1. when the "qosMonParamType" attribute is omitted or indicates packet delay, the "qosMon" attribute with the required QoS Monitoring information as described above;

2. when the "qosMonParamType" attribute indicates congestion information, the "congestMon" attribute with:

   a) the congestion threshold for downlink with the "conThreshDl" attribute; and/or

   b) the congestion threshold for uplink with the "conThreshUl" attribute; or

3. when the "qosMonParamType" attribute indicates QoS monitoring for data rate, the "qosMonDatRate" attribute with:

   a) the data rate threshold for the downlink within the "repThreshDatRateDl" attribute; and/or

   b) the data rate threshold for the uplink within the "repThreshDatRateUl" attribute;

4. when the "qosMonParamType" attribute indicates QoS monitoring for available bitrate, the "avlBitRateMon" attribute with:

   a) a list of thresholds for uplink available bitrate reporting within the "avlBitrateUlThrs" attribute; and/or

   b) a list of thresholds for downlink available bitrate reporting within the "avlBitrateDlThrs" attribute;

and may include:

1. the notification correlation identifier assigned by the AF within the "notifCorreId" attribute;

2. the "directNotifInd" attribute set to true to indicate that the one or more QoS monitoring parameter(s) in the request require direct event notification data from the UPF; and

3. when the "qosMonParamType" attribute indicates QoS monitoring for data rate, an averaging window within the "avrgWndw" attribute.

The NF service consumer may include the "3gpp-Sbi-Consumer-Info" custom HTTP header as described in clause 6.6.2 of 3GPP TS 29.500 [5] to indicate the support of one or more QoS monitoring features (e.g. "QoSMonitoring" feature and "EnQoSMon" feature) by the NF service consumer over the Nsmf_EventExposure service as described in 3GPP TS 29.508[13].

The NF service consumer shall include more than one "AfEventSubscription" data type within the "EventsSubscReqData" data type if more than one notification method is required.

The PCF shall reply to the AF as described in clause 4.2.2.2.

If the AF provided an indication of direct event notification in the request and PCF determines that the direct notification of QoS Monitoring reports applies (i.e. the AF request does not include QoS parameter measurements that are derived by PCF), the PCF behaves as specified in 3GPP TS 29.512 [8].

If the AF provided an indication of direct event notification and PCF determines that the direct notification of QoS Monitoring reports does not apply (i.e. the AF request includes QoS parameter measurements that are derived by PCF as specified in clause 4.2.2.41 (AF request for monitoring packet delay variation), and clause 4.2.2.44 (AF request for

monitoring round trip packet delay when UL and DL are on different service data flows) and the information cannot be consolidated in the QoS monitoring policy in the PCC rule), the PCF generates a successful response to the AF and indicates that direct event notification is not possible by including within the "servAuthInfo" attribute the value "DIRECT_NOTIF_NOT_POSSIBLE". In this case, the PCF shall not indicate direct notification in the QoS Monitoring policy provided to the SMF and instead subscribe to receive QoS Monitoring reports from SMF as specified in 3GPP TS 29.512 [8].

As result of this action, the PCF shall set the appropriate subscription to QoS Monitoring information for the corresponding PCC rule as described in 3GPP TS 29.512 [8].

4.2.2.23.2 Subscriptions to QoS Monitoring for one or more service data flow(s)

When the "EnQoSMon" feature is supported, the subscription to QoS monitoring for one or more service data flow(s) is used by the NF service consumer to receive notification(s) about the real-time measurements of the indicated QoS parameter(s) for the concerned service data flow(s).

To indicate the subscription to one or more QoS monitoring parameter(s) for one or more service data flow(s), the NF service consumer shall include the "medComponents" attribute as described in clause 4.2.2.2. For each media component that requires QoS monitoring, the NF service consumer shall include the service information within the "medSubComps" attribute and the QoS monitoring subscription information for the indicated flows within the "evSubsc" attribute.

For the requested QoS parameter(s), if the AF provided an indication of direct event notification and the PCF determines that the direct notification of QoS Monitoring reports applies (i.e., the AF request does not include QoS parameter measurements that are derived by PCF), the PCF behaves as specified in 3GPP TS 29.512 [8].

If the AF provided an indication of direct event notification and the PCF determines that the direct notification of QoS Monitoring report does not apply (i.e. the AF request includes QoS parameter measurements that are derived by PCF as specified in clause 4.2.2.41 (AF request for monitoring packet delay variation), and clause 4.2.2.44 (AF request for monitoring round trip packet delay when UL and DL are on different service data flows) and the information cannot be consolidated in the QoS monitoring policy in the PCC rule), the PCF generates a successful response to the AF and indicates that direct event notification is not possible by including within the "servAuthInfo" attribute the value "DIRECT_NOTIF_NOT_POSSIBLE" and within the "directNotifReports" attribute, the QoS monitoring parameter that is not notified directly within the "qosMonParamType" attribute and the affected flows within the "flows" attribute. In this case, the PCF shall not indicate direct notification in the QoS Monitoring policy provided to the SMF for the indicated flows and QoS monitoring parameter, and instead shall subscribe to receive QoS Monitoring reports from SMF as specified in 3GPP TS 29.512 [8].

As result of this action, the PCF shall set the appropriate subscription to QoS Monitoring information for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

4.2.2.24 Provisioning of TSCAI input Information and QoS related data

If the "TimeSensitiveNetworking" feature is supported the NF service consumer (i.e. TSN AF or TSCTSF) may provide TSCAI input information within the TSC assistance container and QoS related data to the PCF by the Npcf_PolicyAuthorization_Create service operation to describe the TSC traffic pattern and QoS characteristics for use in the 5G System.

The NF service consumer (i.e. TSN AF or TSCTSF) shall derive the TSCAI input information and the QoS related data for a given TSC stream or flow of aggregated TSC streams. The TSCTSF may determine the TSCAI input information and the related QoS data based on information provided by an AF/NEF, and may provide it for IP type and Ethernet type of PDU sessions as specified in clauses 4.15.6.6 and 4.15.6.6a of TS 23.502 [3]. In case of integration with IEEE TSN network, the TSN AF determines the TSCAI input information as defined in clause 5.27.2.2 of 3GPP TS 23.501 [2] and the QoS related data as defined in clause 5.28.4 of 3GPP TS 23.501 [2].

To indicate the TSCAI input information of a TSC stream or aggregated set of TSC streams, the NF service consumer (i.e. TSN AF or TSCTSF) may include for the uplink flow direction (ingress interface of the DS-TT/UE) in the "tscaiInputUl" attribute and/or for the downlink flow direction (ingress interface of the NW-TT) the "tscaiInputDl" attribute included in a media component entry of the "medComponents" attribute:

- the time period between the start of two bursts of a TSC stream or aggregated TSC streams in reference to the external GM encoded in the "periodicity" attribute, and the acceptable lower bound and upper bound of the periodicity of the start two bursts of a TSC stream or aggregated TSC streams in reference to the external GM or

acceptable list of the periodicity value(s) encoded in the "periodicityRange" attribute if the "EnTSCAC" feature is supported;

- the arrival time of the first data burst of a TSC stream or aggregated TSC streams in reference to the external GM encoded in the "burstArrivalTime" attribute, and the acceptable earliest and latest arrival time of the first data burst of the TSC stream or aggregated TSC streams in reference to the external GM encoded in the "burstArrivalTimeWnd" attribute if the "EnTSCAC" feature is supported; and

- if the "TimeSensitiveCommunication" feature is supported, the time period an application can survive without any burst, i.e., the survival time, in terms of maximum number of messages encoded in the "surTimeInNumMsg" attribute or in time units encoded in the "surTimeInTime" attribute.

NOTE: A single burst (message is equivalent to burst) is expected within a single periodicity. The survival time in terms of maximum number of messages represents the time period result of multiplying the periodicity by the indicated number of messages.

The uplink and/or downlink flow of the TSC stream or aggregated set of TSC streams shall be encoded within the corresponding "MediaSubComponent" entries of the "medSubComps" attribute, for PDU sessions of Ethernet type in the "ethfDescs" attribute and for PDU sessions of IP type in the "fDescs" attribute.

When the feature "TimeSensitiveCommunication" is supported, to indicate the time domain the NF service consumer is located in (i.e. the (g)PTP domain), the NF service consumer may include the "tscaiTimeDom" attribute in the corresponding media component entry of the "medComponents" attribute.

When the feature "EnTSCAC" is supported, the NF service consumer may include within the "capBatAdaptation" attribute in the corresponding media component entry of the "medComponents" attribute the capability of AF to adjust the burst sending time according to the network provided Burst Arrival Time offset if burst arrival time window is not provided.

When the feature "EnTSCAC" is supported, an NF service consumer may request to be notified about the network-determined BAT offset and the optionally adjusted periodicity by using the "EventsSubscReqData" data type and shall include in the HTTP POST request message an event within the "events" attribute with the "event" attribute set to "BAT_OFFSET_INFO".

To indicate the TSC QoS related data of a TSC stream or aggregated set of TSC streams, the NF service consumer (i.e. TSN AF or TSCTSF) may include in the "tsnQos" attribute included in a media component entry of the "medComponents" attribute;

- the maximum burst size encoded either:

  - in the "maxTscBurstSize" attribute, if it is between 4096 and 2000000 bytes; or

  - in the "maxTscBurstSizeSmall" attribute, if it is between 1 and 4095 bytes and the "SmallBurstSizes" feature is supported;

- the maximum time a packet may be delayed encoded in the "tscPackDelay" attribute;

- the maximum packet error rate encoded in the "maxPer" attribute, if the "ExtQoS" feature is supported;

- the TSC traffic priority in scheduling resources among other TSC streams encoded in the "tscPrioLevel" attribute.

The NF service consumer (i.e. TSN AF or TSCTSF) may also include the max bitrates in uplink and downlink within the "marBwUl" attribute and the "marBwDl" attribute of the "medComponents" attribute respectively. In case of integration with IEEE TSN network, the TSN AF determines the maximum flow bit rate as defined in Annex I of 3GPP TS 23.501 [2]. In case of integration with a TSC network other than IEEE TSN network, the TSCTSF may additionally include the "mirBwUl" attribute and the "mirBwDl" attribute of the "medComponents" attribute to indicate the requested guaranteed bit rates in uplink and downlink respectively.

When the feature "TimeSensitiveCommunication" is supported, and the feature "AuthorizationWithRequiredQoS" is supported as specified in clause 4.2.2.32, the NF service consumer (i.e. TSCTSF or TSN AF) may provide within an entry of the "medComponents" attribute a reference to pre-defined QoS information within the "qosReference" attribute instead of providing the attributes "tsnQos", "marBwUl", "marBwDl", "mirBwUl", and/or "mirBwDl". Additionally, if the NF service consumer supports adjustments to different QoS parameter combinations, the NF service consumer may provide a prioritized list of one or more QoS references within the "altSerReqs" attribute as specified in clause 4.2.2.32.

When the feature "TimeSensitiveCommunication" is supported, the feature "AltSerReqsWithIndQoS" is supported as specified in clause 4.2.2.32, and the NF service consumer (i.e. TSCTSF or TSN AF) provides within an entry of the "medComponents" attribute individual QoS information (e.g. within the "tsnQos", "marBwUl" and/or "marBwDl" attributes as described in this clause, then the NF service consumer may provide adjustments to different QoS parameter combinations within a prioritized list of one or more Requested Alternative QoS Parameter set(s) within the "altSerReqsData"attribute as specified in clause 4.2.2.32.

The PCF shall reply to the NF service consumer (i.e. TSN AF or TSCTSF) as described in clause 4.2.2.2.

The PCF shall check whether the received TSCAI input container and TSC QoS related data require to create PCC rules to provide the SMF with derived QoS characteristics and the received TSCAI input container. Provisioning of PCC rule(s) to the SMF shall be carried out as specified in 3GPP TS 29.512 [8].

4.2.2.25 Provisioning of TSC user plane node management information and port management information

During the lifetime of a PDU session enabling Time Sensitive Communications, Time Synchronization and Deterministic Networking, the PCF may receive from the NF service consumer TSC user plane node management information and/or, when the DS-TT or the NW-TT functions are used, port management information for a port located in DS-TT and/or NW-TT and/or, the direct event notification of TSC management information from the UPF if the feature "ExposureToTSC" is supported.

NOTE: The 5GS Architecture to support IETF Deterministic Networking (IETF RFC 8655 [56]) does not require the DS-TT functionality to be supported in the device nor require the user plane NW-TT functionality to be supported in the UPF. However, it can co-exist with such functions.

If the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported, the NF service consumer (i.e., the TSN AF or the TSCTSF) may provide a UMIC with TSC user plane node management information for the UPF/NW-TT and PMIC(s) with port management information for the DS-TT port and/or the NW-TT ports, to configure the 5G system as a TSC user plane node by invoking the Npcf_PolicyAuthorization_Create service operation to the PCF.

The NF service consumer may include in the "AppSessionContextReqData" data type:

- the DS-TT PMIC encoded in the attribute "tsnPortManContDstt" and/or the one or more NW-TT PMIC(s) encoded in the "tsnPortManContNwtts" attribute, if available, for the DS-TT port and NW-TT ports allocated for a PDU session. The PMIC(s) are encoded in the "PortManagementContainer" data type, which includes the port management information in the "portManCont" attribute and the related port number in the "portNum" attribute;

- the UMIC encoded in the "tsnBridgeManCont", if available, for the TSC user plane node functionality of the UPF/NW-TT allocated for a PDU session. The UMIC is encoded in the "BridgeManagementContainer" data type;and/or

- if the feature "ExposureToTSC" is supported, the notification URI within the "tscNotifUri" attribute and the notification correlation identifier assigned by the TSCTSF or TSN AF within the "tscNotifCorreId" attribute, which, if available, indicates that the direct event notification of TSC management information from the UPF is requested.

As result of this action, the PCF shall provide the received DS-TT and/or NW-TT PMIC(s) and/or UMIC and/or the direct event notification of TSC management information for the corresponding PDU session as described in 3GPP TS 29.512 [8].

4.2.2.26 Invocation of Mission Critical Services

This procedure allows a NF service consumer, as per 3GPP TS 22.179 [45], to request prioritized access to system resources in situations such as during congestion.

The NF service consumer may include the "mcsId" attribute to indicate that the new AF session relates to an MCS session.

The "mcsId" attribute shall contain the national variant for the MCS service name indicating an MCS session. The "resPrio" attribute shall include the priority value of the related priority service.

If the NF service consumer supports the SBI Message Priority mechanism for an MCS session, it shall include the "3gpp-Sbi-Message-Priority" custom HTTP header towards the PCF as described in clause 6.8.2 of 3GPP TS 29.500 [5].

NOTE: If the NF service consumer supports the SBI Message Priority mechanism for an MCS session, the NF service consumer will include the "3gpp-Sbi-Message-Priority" custom HTTP header with a priority value equivalent to the value of the "resPrio" attribute. Highest user priority value is mapped in the corresponding lowest value of the "3gpp-Sbi-Message-Priority" custom HTTP header.

When the PCF receives the "mcsId" attribute indicating an MCS session, the PCF shall take specific actions on the corresponding PDU session to ensure that the MCS session is prioritised as specified in 3GPP TS 29.512 [8].

4.2.2.27 P-CSCF restoration enhancements

The P-CSCF restoration custom operation is applicable when the PCF based Restoration Enhancement, as defined in 3GPP TS 23.380 [39], represented by the supported feature "PCSCF-Restoration-Enhancement" is supported by both P-CSCF and PCF.

Figure 4.2.2.27-1 illustrates the P-CSCF restoration enhancements.

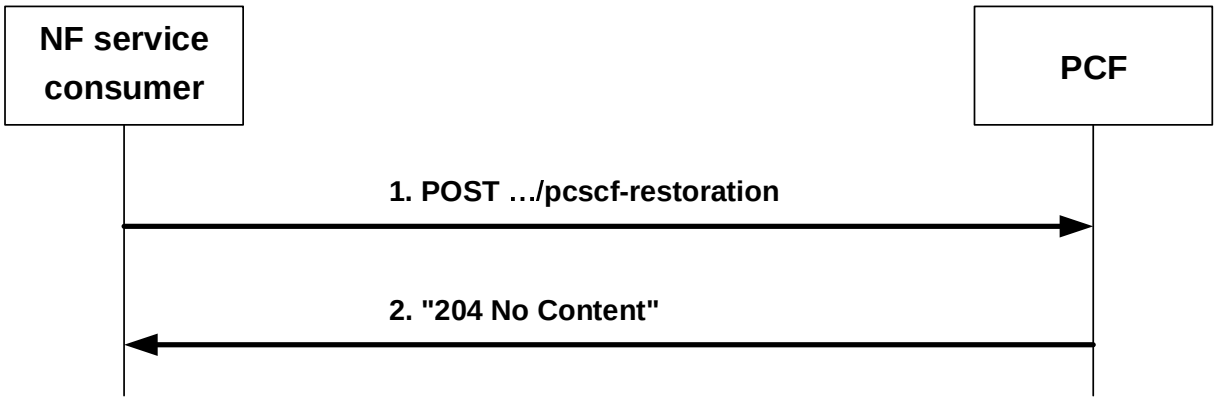


Figure 4.2.2.27-1: P-CSCF restoration enhancements

The P-CSCF acting as a NF service consumer shall invoke the "P-CSCF restoration" custom operation sending an HTTP POST request to the resource URI representing the custom operation (POST …/pcscf-restoration) as shown in figure 4.2.2.27-1, step 1, in case P-CSCF restoration needs to be performed.

The P-CSCF shall include in the "PcscfRestorationRequestData" data type in the content of the HTTP POST request:

- the IP address (IPv4 or IPv6) of the UE in the "ueIpv4" or "ueIpv6" attribute, and if the IP address is not unique (e.g. private IPv4 case), the "ipDomain" attribute or the "sliceInfo" attribute if available; or
- if the IP address is not available or if the IP address is not unique and the "ipDomain" attribute and the "sliceInfo" attribute are not available, the SUPI in the "supi" attribute and the DNN in the "dnn" attribute.

The PCF shall identify the PDU session for which the HTTP POST request applies. If the PCF fails in identifying the PDU session, the PCF shall reject the "P-CSCF restoration" custom operation with an HTTP "500 Internal Server Error" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "PDU_SESSION_NOT_AVAILABLE".

Otherwise, the PCF shall acknowledge the request and shall send to the NF service consumer a "204 No content" response to the HTTP POST request, as shown in figure 4.2.2.27-1, step 2.

The PCF shall send a request for P-CSCF restoration to the SMF for the corresponding PDU session as described in 3GPP TS 29.512 [8], clause 4.2.3.18.

4.2.2.28 Support of CHEM feature

When CHEM feature is supported, the NF service consumer may include the value of Maximum Packet Loss Rate for UL within the "maxPacketLossRateUl" attribute and/or the value of Maximum Packet Loss Rate for DL within the "maxPacketLossRateDl" attribute in "medComponents" attribute. For CHEM feature, see Annex B.14.

4.2.2.29 Support of FLUS feature

When "FLUS" feature is supported by the NF service consumer, the NF service consumer may include the "flusId" attribute within a media component of the "medComponents" attribute to indicate that the related media of the created new Individual Application Session Context resource corresponds to a FLUS media stream. Additional QoS information for the treatment of FLUS media may be provided within "desMaxLatency" attribute and/or "desMaxLoss" attribute.

4.2.2.30 Subscription to EPS Fallback report

When the "EPSFallbackReport" feature is supported, the NF service consumer subscribes to EPS Fallback report to be notified of the rejection in 5GS of the requested resources associated to service information for voice media type and the subsequent fallback to EPS of the resources associated to the voice media and other medias requested by this NF service consumer.

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "events" attribute with the "event" attribute set to "EPS_FALLBACK". The NF service consumer shall request to the PCF to report EPS Fallback in conjuction with providing the PCF with NF service consumer service information for voice media type as described in clause 4.2.2.2.

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As result of this action, the PCF shall set the appropriate subscription to EPS Fallback report for the corresponding PCC rule(s) as described in in 3GPP TS 29.512 [8].

4.2.2.31 Subscription to TSC user plane node related events

This procedure is used by the NF service consumer (i.e. TSN AF or TSCTSF) if the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported to subscribe to notifications of updated TSC user plane node information, e.g., DS-TT PMIC and/or NW-TT PMIC(s) and/or UMIC availability within the Individual Application Session Context resource created to handle the TSC user plane node in the context of a PDU session.

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message within the "evSubsc" attribute an event within "events" attribute with the "event" attribute set to the value "TSN_BRIDGE_INFO" to subscribe to the reception of TSC user plane node information.

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2. If the PCF stores updated PMIC/UMIC updated information, the PCF shall include the "evsNotif" attribute with an entry in the "evNotifs" array with the "event" attribute set to "TSN_BRIDGE_INFO" and the "tsnBridgeManCont" attribute and/or the "tsnPortManContDstt" attribute and/or the "tsnPortManContNwtts" attribute as received from the SMF if not previously reported.

As result of this action, the PCF shall set the corresponding subscription to the report of TSC user plane node management information and port management information for the corresponding PDU session , if not previously done, as described in 3GPP TS 29.512 [8].

4.2.2.32 Initial provisioning of required QoS information

This procedure is used by a NF service consumer to request that a data session to a UE is set up with a specific QoS (e.g. low latency or Packet Delay Variation) and priority handling when the "AuthorizationWithRequiredQoS" feature is supported.

The NF service consumer may provide within one or more entries of the "medComponents" attribute included in the "ascReqData" attribute of the HTTP POST request message described in clause 4.2.2.2 a reference to pre-defined QoS information within the "qosReference" attribute.

Additionally, if the NF service consumer supports adjustment to different QoS parameter combinations, the NF service consumer may provide a prioritized list of one or more QoS references within the "altSerReqs" attribute, where the lower the index of the array for a given entry, the higher the priority.

If the "AltSerReqsWithIndQoS" feature is supported, and the NF service consumer requests that the data session to a UE is set up with individual QoS parameters (i.e., with QoS information within "medComponents" attribute, e.g. the "tsnQos", "marBwUl" and/or "marBwDl" attributes, instead of a QoS reference within the "qosReference" attribute), the NF service consumer may instead of the "altSerReqs" attribute provide a prioritized list of alternative service requirements that include Requested Alternative QoS Parameter set(s) within the "altSerReqsData" attribute, where the lower the index of the array for a given entry, the higher the priority.

If the "DisableUENotification" feature is supported, the AF may also indicate to the PCF that the UE does not need to be informed about changes related to Alternative QoS Profiles by including the "disUeNotif" attribute set to true.

When the NF service consumer provides the "altSerReqs" attribute or the "altSerReqsData" attribute, the NF service consumer shall also subscribe to receive notifications from the PCF when the resources associated to the corresponding service information have been allocated as described in clause 4.2.2.10 and when the GBR QoS targets for one or more service data flows can no longer (or can again) be guaranteed, as described in clause 4.2.2.6.

Due to the received QoS information, the PCF may need to provision or modify the related PCC rules as specified in 3GPP TS 29.513 [7] and provide the related information towards the SMF following the corresponding procedures specified in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.2.33 Support of QoSHint feature

If the QoSHint feature is supported by the NF service consumer, the NF service consumer may include the "desMaxLatency" attribute and/or "desMaxLoss" attribute within a media component of the "medComponents" attribute to indicate that the related media of the created Individual Application Session Context resource has specific latency and/or loss demands.

4.2.2.34 Subscription to Reallocation of Credit notification

This procedure is used by the NF service consumer if the "IMS_SBI" and the "ReallocationOfCredit" features are supported to subscribe to notifications of reallocation of credit for the Service Data Flows within the AF application session context.

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "evSubsc" attribute with the "event" attribute set to the value "REALLOCATION_OF_CREDIT".

As result of this action, the PCF shall set the appropriate subscription to reallocation of credit notification for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2.

4.2.2.35 Subscription to satellite backhaul category changes

When the feature "SatelliteBackhaul"is supported, the subscription to satellite backhaul category changes is used by a NF service consumer to subscribe to receive a notification when the satellite backhaul category changes and when the backhaul category changes between satellite backhaul and non-satellite backhaul. When the feature "EnSatBackhaulCatChg" is supported, the subscription is also used to receive a notification when the satellite backhaul category change comprises a dynamic satellite backhaul category.

The NF service consumer shall use the "evSubsc" attribute as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "events" attribute with the "event" attribute set to "SAT_CATEGORY_CHG".

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2. The PCF shall include the "evsNotif" attribute with an entry in the "evNotifs" array with the "event" attribute set to "SAT_CATEGORY_CHG" and the "satBackhaulCategory" attribute including the satellite backhaul category or the indication of non-satellite backhaul if the PCF has previously requested to the SMF to be updated with this information.

As result of this action, the PCF shall set the appropriate subscription to satellite backhaul changes for the PDU session, if not previously subscribed, as described in in 3GPP TS 29.512 [8].

4.2.2.36 Subscription to the report of extra UE addresses

When the feature "ExtraUEaddrReport" is supported, the subscription to the report of extra UE addresses is used to report information about the extra IP addresses or address ranges allocated to the PDU session due to framed routes or IPv6 prefix delegation. The report shall include the actual list of IPv4 addresses or list of IPv6 prefixes as currently allocated.

NOTE: In case of Deterministic Networking, the 5GS DetNet Node, as described in 3GPP TS 23.501 [2], can forward via its device side interface IP packets destined not only to the UE's IP address or prefix but also to a range of IPv4 addresses or IPv6 IP prefixes according to one or more Framed Routes or prefixes delegated to the UE by IPv6 prefix delegation.

The NF service consumer shall use the "evSubsc" attribute as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "events" array with the "event" attribute set to "EXTRA_UE_ADDR".

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2.

If the PCF received from the SMF the framed routes as described in 3GPP TS 29.512 [8], clause 4.2.2.2, or the PCF receives updated information of the extra one or more IPv6 prefixes allocated to the UE as described in 3GPP TS 29.512 [8], clauses 4.2.4.2, 5.6.2.19, and C.3.4.1, the PCF shall include in the response the "evsNotif" attribute with an entry in the "evNotifs" array with the "event" attribute set to "EXTRA_UE_ADDR" and:

- the actual list of IPv4 addresses within the "ipv4AddrList" attribute, if one or more IPv4 framed routes are associated to the PDU session and are available in the PCF; or
- the actual list of IPv6 prefixes allocated to the UE within the "ipv6PrefixList" attribute, if one or more IPv6 framed routes are associated to the PDU session and are available in the PCF, or if the PCF keeps updated IPv6 prefix(es) information.

As result of this action, the PCF shall set the appropriate subscription to the report of UE IP addresses, if not previously subscribed, as described in in 3GPP TS 29.512 [8].

4.2.2.37 Provisioning of multi-modal services

This procedure is used by a NF service consumer to:

- provide service requirements for each media flow that comprise the multi-modal service; and/or
- provide QoS monitoring requirements for each media flow that comprise the multi-modal service;

when "MultiMedia" feature is supported.

The NF service consumer may include the multi-modal Service Identifier within the "multiModalId" attribute to indicate that the new AF session relates to a multi-modal service.

NOTE: For the delivery of multi-modal services, the NF service consumer may request to the PCF multiple media flows for a single UE or for multiple UEs via multiple PDU Sessions and separate request(s) per PDU Session with the same "multiModalId" attribute.

To provide service requirements for a multi-modal service, the NF service consumer shall follow the procedures described in clause 4.2.2.2 for initial provisioning of service information, with the following additional considerations:

- When the multi-modal service combines several media, the NF service consumer shall provide the service information of each media within the "medComponents" attribute. The media subcomponent(s), when provided for a media component, only contain the description of the service data flow(s).
- To provide distinct QoS monitoring requirements for multiple service data flows, the NF service consumer shall follow the procedures described in clause 4.2.2.23.2 for subscriptions to QoS Monitoring Information.

4.2.2.38 Initial provisioning of Round-Trip latency requirements

When the "RTLatency" feature is supported, this procedure is used by a NF service consumer to provision Round-Trip (RT) latency indication and RT latency requirement for an XR or other interactive media services via the AF session with required QoS procedure. The RT latency indication and RT Latency requirement may be provided for the UL and DL flow directions of a service data flow or for the UL and DL flow directions of two service data flows.

When the RT latency indication and requirement is provided for the UL and DL flow directions of a service data flow, the AF may provide the "rTLatencyInd" attribute contained in MediaComponent data type to indicate that RT delay of the UL and DL flow directions of the service data flow needs to meet the RT latency requirement of the service, which is twice of the single direction delay requirement between the UE and the PSA UPF derived from the "qosReference" attribute or included in the "pdb" attribute.

When the RT latency indication and requirement is provided for the UL and DL flow directions of two service data flows, the AF may provide:

- the DL flow description and the UL flow description, each one in a MediaSubComponent entry of separate MediaComponent entries;
- the RT latency requirement for the UL or DL flow directions included in the MediaSubComponent entry within the "qosReference" attribute or within the "pdb" attribute included in each MediaComponent entry. The sum of the UL and DL direction delay between the UE and PSA needs to meet the RT latency requirement;
- the "rTLatencyIndCorreId" attribute within each MediaComponent entry, to indicate that RT Latency requirement needs to be fulfilled by the two MediaComponent entries that share the same "sharedKey" attribute value.

Due to the received Round-Trip latency requirements, the PCF may need to provision or modify the related PCC rules as specified in 3GPP TS 29.513 [7] and provide the related information towards the SMF following the corresponding procedures specified in clause 4.2.6.2.22 of 3GPP TS 29.512 [8].

NOTE: When more than one media subcomponents are provided in the media component with RT latency requirement, the PCF determines the monitored flows (the related PCC rules) based on implementation specific means.

4.2.2.39 Provisioning of PDU Set handling related data

If the "PDUSetHandling" feature is supported, the NF service consumer may provide PDU Set handling related data, i.e., PDU Set QoS related information and/or Protocol Description related information.

To indicate the PDU Set QoS related information for the PDU Set of a service data flow, the NF service consumer may include the "pduSetQosDl" and/or "pduSetQosUl" attribute(s) within a media component entry of the "medComponents" attribute. Within the "PduSetQosPara" data type, the NF service consumer may include:

- both:
  a. the PDU Set Delay Budget (PSDB), i.e., the upper bound for the delay that a PDU Set may experience for the transfer between the UE and the N6 termination point at the UPF (see 3GPP TS 23.501 [2], clause 5.7.7.2) encoded in the "pduSetDelayBudget" attribute as described in 3GPP TS 29.571 [12]; and
  b. the PDU Set Error Rate (PSER), i.e., the upper bound for the non-congestion related PDU Set loss rate (see 3GPP TS 23.501 [2], clause 5.7.7.3) encoded in the "pduSetErrRate" attribute as described in 3GPP TS 29.571 [12]; and/or
- the PDU Set Integrated Handling information (PSIHI), that indicates whether all PDUs of the PDU Set are needed for the usage of the PDU Set by the application layer in the receiver side (see 3GPP TS 23.501 [2], clause 5.7.7.4) in the "pduSetHandlingInfo" attribute as described in 3GPP TS 29.571 [12].

If the "ExtQoSR19" feature is supported, and the PSDB (UL and/or DL) and PSER (UL and/or DL) in "pduSetQosDl" and/or "pduSetQosUl" attribute(s) within "medComponents" attribute are provided, the NF service consumer shall also provide the PSDB (UL and/or DL) and PSER (UL and/or DL) in Alternative QoS Parameter set(s) by including the "pduSetQosDl" and/or "pduSetQosUl" attribute(s) within "altSerReqsData"; otherwise, the corresponding Alternative QoS Parameter set(s) shall not contain the PSDB and PSER in the respective direction(s).

The NF service consumer may also include Protocol Description information for the downlink direction within the "protoDescDl" attribute (to indicate the transport protocol (e.g. RTP, SRTP) and additional information used for identifying the PDU Set information by the PSA UPF) and/or for the uplink direction within the "protoDescUl" attribute (to indicate the transport protocol and additional information used for identifying the PDU Set information by the UE).

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As result of this action, the PCF shall set the appropriate PDU Set handling parameters for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

4.2.2.40 Subscription to BAT offset notification

When the "EnTSCAC" feature is supported, the subscription to BAT offset is used by a NF service consumer to subscribe to receive a notification when NF service consumer provides the Capability for BAT adaptation or BAT Window as defined in clause 4.2.2.24. When the NF service consumer provides the periodicity range, the BAT offset subscription is also used to receive the adjusted periodicity.

The NF service consumer shall use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "evSubsc" attribute with the "event" attribute set to the value "BAT_OFFSET_INFO".

As result of this action, the PCF shall set the appropriate subscription to BAT offset notification for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

4.2.2.41 Subscription to Packet Delay Variation monitoring

The subscription to Packet Delay Variation is used by an NF service consumer to receive a notification about the variation of packet delay between UE and PSA UPF when the "EnQoSMon" feature is supported.

When the subscription to measure Packet Delay Variation is at AF session level and for a QoS flow, the NF service consumer may use the "EventsSubscReqData" data type as described in clause 4.2.2.2 and shall include:

- an entry of the "AfEventSubscription" data type per requested notification method in the "events" attribute with:
  a) the "event" attribute set to the value "PACK_DEL_VAR"; and
  b) the "notifMethod" attribute set to the value "EVENT_DETECTION" or "PERIODIC"; and
  c) when the "notifMethod" attribute is set to the value "PERIODIC", the periodic time for reporting within the "repPeriod" attribute; and
  d) when the "notifMethod" attribute is set to the value "EVENT_DETECTION", the minimum waiting time between subsequent reports within the "waitTime" attribute;
- the requested Packet Delay Variation parameter(s) to be measured (i.e. DL, UL and/or round trip packet delay variation) within the "pdvReqMonParams" attribute;
- when the "notifMethod" attribute set to the value "EVENT_DETECTION", the "pdvMon" attribute, with the required Packet Delay Variation monitoring information:
  a) the delay threshold for downlink with the "repThreshDl" attribute;
  b) the delay threshold for uplink with the "repThreshUl" attribute; and/or
  c) the delay threshold for round trip with the "repThreshRp" attribute.

The NF service consumer shall include more than one "AfEventSubscription" data types within the "EventsSubscReqData" data type if more than one notification methods are required.

NOTE 1: When the subscription to Packet Delay Variation is for one or more media components of the AF session, the subscription to Packet Delay Variation can only be indicated within the corresponding "medSubComps" entry.

When the NF service consumer subscribes to the monitoring of Packet Delay Variation for one or more service data flow(s), the NF service consumer shall include the "medComponents" attribute as described in clause 4.2.2.2. For each

media component that requires Packet Delay Variation measurements, the NF service consumer shall include the service data flow filter information within the "medSubComps" attribute and Packet Delay Variation subscription information for the indicated flows within the "evSubsc" attribute.

If the AF also subscribed to packet delay measurements and provided "directNotifInd" attribute in the request as described in clause 4.2.2.23, and the PCF determines that to calculate the Packet Delay Variations, based on the packet delay measurements, the packet delay measurements cannot be notified directly, the PCF shall set the "servAuthInfo" attribute to "DIRECT_NOTIF_NOT_POSSIBLE" and shall include the "directNotifReports" attribute with the QoS parameter that cannot be notified directly within the "qosMonParamType" attribute and the affected flows within the "flows" attribute in the HTTP response message. The PCF shall not provide the notification addresses and direct notification indication in the QoS monitoring policy of the PCC rule and for the indicated QoS monitoring parameter. The PCF shall subscribe to receive the QoS Monitoring reports from SMF by setting the QoS Monitoring Policy Control Request Trigger.

The PCF shall reply to the AF as described in clause 4.2.2.2.

As result of this action, the PCF shall determine the QoS Monitoring information to derive packet delay variation measurements requested by the AF and shall set the appropriate subscription for QoS Monitoring with the SMF to receive packet delay monitoring reports for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

NOTE 2: When packet delay measurement failure happens, how the PCF calculates the variation of packet delay is implementation specific.

4.2.2.42 Initial provisioning of traffic information for UE power saving management

If the "PowerSaving" feature is supported, the NF service consumer may provide traffic information to enable 5GS configures UE power saving management schemes for a service data flow as described in 3GPP TS 23.501 [2]. The AF may include within the MediaComponent entry of the "medComponents" attribute:

- the Uplink and/or Downlink Periodicity information which indicates the time period between the start of the two data bursts in Uplink and/or Downlink direction within "periodUl" and "periodDl" attributes respectively contained in MediaComponent data structure; and/or
- the DL protocol description within the "protoDescDl" attribute.

As a result of this action, when the UL and/or DL periodicity is received, the PCF determines the policies for periodicity and N6 jitter information to provide in the PCC rule the PCF shall send the Periodicity information to the SMF as described in 3GPP TS 29.512 [8].

When the DL protocol description is received, the PCF determines the policies for End of Data Burst handling to provide in the PCC rule as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.2.43 Provisioning of the indication of ECN marking for L4S support

When the "L4S" feature is supported, this procedure is used by a NF service consumer to explicitly indicate that the UL and/or DL service data flow of a media component supports ECN marking for L4S support.

The NF service consumer may include in the HTTP POST request message described in clause 4.2.2.2, within the corresponding media component(s) entries of the "medComponents" attribute, the "l4sInd" attribute set to "UL", "DL" or "UL_DL" to indicate respectively whether the UL, the DL, or both, UL and DL, service data flow(s) supports ECN marking for L4S support.

The NF service consumer shall also subscribe to receive notifications from the PCF when the ECN marking for L4S support is not available or available again in 5GS by including within the "evSubsc" attribute the "events" attribute with the "event" attribute set to "L4S_SUPP".

The PCF may indicate to the SMF to enable for ECN marking for L4S support following the procedures specified in clause 4.2.6.2.21 of 3GPP TS 29.512 [8].

4.2.2.44 Subscription to Round-Trip delay over two QoS flows

When the "EnQoSMon" feature is supported, this procedure is used by an NF service consumer to receive a notification about the Round-Trip delay measurements over two Service Data Flows.

The Round-Trip delay measurement over two QoS flows includes two aspects,

- Round-Trip delay measurement for one service data flow, the UL traffic and DL traffic are mapped two QoS flows;
- Round-Trip delay for two service data flows, one UL direction service data flow and one DL direction service data flow, which are over two QoS flows respectively.

The NF service consumer shall include the "medComponents" attribute as described in clause 4.2.2.2. For each media component that requires Round-Trip delay monitoring, the NF service consumer shall include the Round-Trip delay monitoring subscription information for the indicated flows within the "evSubsc" attribute as follows as described in clause 4.2.2.23.2:

- an entry of the "AfEventSubscription" data type per requested notification method in the "events" attribute with:
  a) the "event" attribute set to the value "RT_DELAY_TWO_QOS_FLOWS"; and
  b) the "notifMethod" attribute set to the value "EVENT_DETECTION", or "PERIODIC"; and
  c) when the "notifMethod" attribute is set to the value "PERIODIC", the periodic time for reporting within the "repPeriod" attribute; and
  d) when the "notifMethod" attribute is set to the value "EVENT_DETECTION", the minimum waiting time between subsequent reports within the "waitTime" attribute;
- when the "notifMethod" attribute set to the value "EVENT_DETECTION", the "rttMon" attribute, with the delay threshold for round trip with the "repThreshRp" attribute; and
- when the UL and DL flows to be measured are defined in different media subcomponents, the "rttFlowRef" attribute with the shared key between the measured UL and DL flows and, optionally, an indication of whether the media subcomponent represents the UL or DL direction for the round trip time measurement.

If the QoS parameters requested for both UL and DL flows are same and the "RT_DELAY_TWO_QOS_FLOWS" is subscribed, the NF service consumer shall use one entry of the "MediaComponent" data type as described in clause 4.2.2.2 for the two Service Data Flows of which the Round-Trip delay will be measured and shall include:

- an entry of the "MediaSubComponent" data type with the "fDescs" attribute contains the flow description for the monitored Uplink and Downlink IP flows, and with the "evSubsc" attribute with the subscription information to Round-Trip delay measurements over two SDFs as described above.

If the QoS parameters requested for both UL and DL flows are same and different events are subscribed for UL and DL flows, the NF service consumer shall use an entry of the "MediaComponent" data type as described in clause 4.2.2.2 and shall include:

- for the uplink flow, an entry of the "MediaSubComponent" data type with the "fDescs" attribute with the monitored flow description for the Uplink IP flow, and with the "evSubsc" attribute including the subscription to Round-Trip delay measurements over two SDFs as described above, where the "rttFlowRef" attribute may be included to indicate the shared key with the media subcomponent where the corresponding DL subscription is defined
- for the downlink flow, an entry of the "MediaSubComponent" data type with the "fDescs" attribute with the monitored flow description for the Downlink IP flow, and with the "evSubsc" attribute with the subscription to Round-Trip delay measurements over two SDF as described above, where the "rttFlowRef" attribute may be included to indicate the shared key with the media subcomponent where the corresponding UL subscription is defined.

If the QoS parameters requested for both UL and DL flows are different and the "RT_DELAY_TWO_QOS_FLOWS" is subscribed, the NF service consumer shall use two "MediaComponent" data types as described in clause 4.2.2.2 for the two Service Data Flows of which the Round-Trip delay will be measured and shall include:

- for the uplink flow, an entry of the "MediaSubComponent" data type with the "fDescs" attribute with the monitored flow description for the Uplink IP flow, and optionally, the flow description for the Downlink IP flow. The "MediaSubComponent" data type may also include the "evSubsc" attribute with the subscription to Round-Trip delay measurements over two SDFs as described above, where the "rttFlowRef" attribute, if provided, includes the shared key with the media subcomponent where the corresponding DL subscription is defined and, when the "fDescs" attribute includes the UL and DL flows, the that the UL direction of the provided flows needs to be monitored; and

- for the downlink flow, an entry of the "MediaSubComponent" data type with the "fDescs" attribute with the monitored flow description for the Downlink IP flow, and optionally, the flow description for the Uplink IP flow. The "MediaSubComponent" data type may also include the "evSubsc" attribute with the subscription to Round-Trip delay measurements over two SDF as described above, where the "rttFlowRef" attribute, if provided, includes the shared key with the media subcomponent where the corresponding UL subscription is defined and, when the "fDescs" attribute includes the UL and DL flows, an indication that the DL direction of the provided flows need to be monitored.

The NF service consumer shall include more than one "AfEventSubscription" data type within the "EventsSubscReqData" data type if more than one notification method is required.

If the NF service consumer also subscribed to packet delay measurements and provided "directNotifInd" attribute in the request as described in clause 4.2.2.23, and the PCF determines that to calculate the Round-Trip delay over two QoS flows, based on the packet delay measurements, the packet delay measurements cannot be notified directly, the PCF shall set the "servAuthInfo" attribute to "DIRECT_NOTIF_NOT_POSSIBLE" and shall include the "directNotifReports" attribute with the QoS parameter that cannot be notified directly within the "qosMonParamType" attribute and the affected flows within the "flows" attribute in the HTTP response message. The PCF shall not provide the notification addresses and direct notification indication in the QoS monitoring policy of the PCC rule and for the indicated QoS monitoring parameter. The PCF shall subscribe to receive the QoS Monitoring reports from SMF by setting the QoS Monitoring Policy Control Request Trigger.

Based on the NF service consumer request and local policy, the PCF derives the separate QoS monitoring policies of each direction as defined in clause 4.2.2.23.

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As result of this action, the PCF shall determine the QoS Monitoring information to derive round-trip delay measurements requested by the NF service consumer and shall set the appropriate subscription for QoS Monitoring information for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

NOTE: When packet delay measurement failure happens, how the PCF calculates the Round-Trip delay over two QoS flows is implementation specific.

4.2.2.45 Provisioning of the QoS timing information

When the "QoSTiming_5G" feature is supported, this procedure is used by the NF service consumer to indicate the time period when the requested QoS information should be applied. The NF service consumer may include in the HTTP POST request message described in clause 4.2.2.2:

- the "qosDuration" attribute to indicate the time period when the PCF shall allocate and release the resources for the corresponding PCC rult(s) that should be applied. The PCF may allocate resources at the beginning of each QoS duration interval and release the resources at the end of the corresponding QoS duration interval. This process is repeated until the AF session is revoked.

- the "qosInactInt" attribute to indicate the inactivity time interval of the requested QoS information.

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As result of this action, the PCF shall allocate and release the resources for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

4.2.2.46 Subscription to the report of network support for QoS Monitoring

The QoS monitoring capabilities in the network may change, e.g. when the serving NG-RAN is changed (e.g., QoS monitoring is not supported in the target NG-RAN while it was supported in the source NG-RAN). When the feature

"QoSMonCapRepo" is supported, the NF service consumer may subscribe to receive notifications from the PCF of whether the requested QoS Monitoring Capability is supported or not by including within the "evSubsc" attribute the "events" attribute with the "event" attribute set to "QOS_MON_CAP_REPO". If the "QOS_MON_CAP_REPO" event is subscribed, the NF service consumer shall also include the monitoring type(s) within the "capTypes" attribute as described in clause 4.2.2.2.

NOTE: When the AF subscribes to QoS monitoring parameters for a functionality not directly related to the capability type within "capTypes" attribute and still is interested on the report of the network support for QoS monitoring, the AF will subscribe to receive notifications of whether the requested QoS monitoring types is not supported or supported again by including within the "evSubsc" attribute the "events" attribute with the "event" attribute set to "QOS_MON_CAP_REPO" and the monitoring type within the "capTypes" attribute that is needed for the support of that functionality, e.g. packet delay variation and round-trip delay over two QoS flows monitoring requires the support of packet delay monitoring thus "capTypes" must contain "PACKET_DELAY".

4.2.2.47 Provisioning of the (S)RTP Multiplexed Media Identification Information

When the "MpxMedia" feature is supported, the NF service consumer may specify the (S)RTP Multiplexed Media Identification Information for the Uplink and/or Downlink IP flows within the "mpxMediaUlInfos" and/or "mpxMediaDlInfos" attributes, respectively to uniquely identify each media flow of multiplexed media.

NOTE: Data traffic of different media components with different QoS requirements could be multiplexed on the same end-to-end transport layer connection. (S)RTP Multiplexed Media Identification Information can be useful when another packet filter attribute is needed to differentiate between the multiplexed media flows.

If the PCF has not received UE support for (S)RTP Multiplexed Media Identification from SMF, the PCF may, based on local configuration, generate a successful response to the NF service consumer and set "servAuthInfo" attribute to MPX_MEDIA_NOT_SUPPORTED_IN_UE in the HTTP response message to the NF service consumer to indicate that the requested differentiated QoS for multiplexed media flows is not supported in the UE.

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

As result of this action, the PCF shall determine the PCC rules and provide to the SMF as described in 3GPP TS 29.512 [8].

4.2.2.48 Initial provisioning of dynamically changing traffic characteristics

If the "TrafficCharChange" feature is supported, this procedure is used by the NF service consumer to indicate that the Data Burst Size is marked in the header of the protocol identified by the DL Protocol Description.

The NF service consumer may include within the corresponding media component(s) entries of the "medComponents" attribute in the HTTP POST request message described in clause 4.2.2.2:

- the Data Burst Size Marking Indication in the "datBurstSizeInd" attribute; and
- the DL Protocol Description in the "protoDescDl" attribute.

The PCF determines the policies for the identification and marking of the Data Burst Size and provisions the PCC rule as described in 3GPP TS 29.512 [8].

If the "TrafficCharChange" feature is supported, this procedure is used by the NF service consumer to indicate that the TTNB is marked in the header of the protocol identified by the DL Protocol Description.

The NF service consumer may include within the corresponding media component(s) entries of the "medComponents" attribute in the HTTP POST request message described in clause 4.2.2.2:

- the TTNB Indication in the "timetoNextBurstInd" attribute; and
- the DL Protocol Description in the "protoDescDl" attribute.

The PCF determines the policies for the identification and marking of the TTNB and provisions the PCC rule as described in 3GPP TS 29.512 [8].

If the "TrafficCharChange" feature is supported, this procedure is used by the NF service consumer to request expedited data transfer with reflective QoS to expedite the data transfer of larger payload for Non-GBR service flow(s) of XR application.

The NF service consumer may include within the corresponding media component(s) entries of the "medComponents" attribute in the HTTP POST request message described in clause 4.2.2.2:

- the two media flows that contains the same SDF fileter associated with different QoS requirements. One flow with Expedited Data Transfer Indication within the "expTranInd" attribute set to true and the other flow with Expedited Data Transfer Indication within the "expTranInd" attribute set to false; and/or

- the DL Protocol Description in the "protoDescDl" attribute.

The PCF determines the policies for the expedited data transfer with reflective QoS and provisions the PCC rule as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.2.49 Initial provisioning of deliver media related information for encrypted traffic

If the "OnPathN6MediaInfo" feature is supported, this procedure is used by the NF service consumer to indicate that the support of deliver media related information for encrypted traffic.

The NF service consumer may include within the corresponding media component(s) entries of the "medComponents" attribute in the HTTP POST request message described in clause 4.2.2.2:

- the on-path N6 Signaling Information to indicate the supported method to deliver media related information and the AS proxy address in the "onPathN6SigInfo" attribute; or

- the UDP Option in the "protoDescDl" attribute when the "onPathN6Method" within "onPathN6SigInfo" indicates method "CONNECT_UDP"; or

- the MoQ indication in the "protoDescDl" attribute and the "onPathN6SigInfo" attribute shall not be present.

The PCF determines the policies and provisions the PCC rule as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.2.50 Subscription to Data Rate Limitation Information reporting

When the "RateLimitReport" feature is supported, the subscription is used by the NF service consumer to receive a notification about the report of uplink and downlink Data Rate Limitation information for Non-GBR service data flow.

The NF service consumer shall use the EventsSubscReqData data type as described in clause 4.2.2.2 and shall include in the HTTP POST request message an event within the "evSubsc" attribute with the "event" attribute set to the value "RATE_LIMIT_INFO_REPO".

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.2.2. If the PCF can determine the authorized maximum bitrate for uplink and/or downlink for each media component, the PCF shall include the EventsNotification data type in response with the "evNotifs" array including an entry with the "event" attribute set to the value "RATE_LIMIT_INFO_REPO" and the "rateLimitRepoUl" and/or "rateLimitRepoDl" attribute including the Data Rate Limitation information for uplink and/or downlink that is authorized for non-GBR service data flows. Alternatively, the PCF may not include the authorized rate limitation information in the response message and the PCF shall notify the NF service consumer using the Notification service as described in clause 4.2.5.23 when the Data Rate Limitation information is available.

4.2.2.51 Request of 5GC network functions health monitoring information

This procedure is used by a NF service consumer to request the PCF to check whether the PDU session is affected by 5GC Network Functions (i.e. SMF or UPF) failure and prioritize the recovery of this PDU session, as defined in 3GPP TS 23.380 [39], when the "CnHealthMonitor" feature is supported.

The NF service consumer shall include in the HTTP POST request message described in clause 4.2.2.2:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:

  a) the "event" attribute set to the value "CN_HEALTH_MONITOR"; and

  b) the "notifMethod" attribute set to the value "ONE_TIME"; and

The PCF shall reply to the NF service consumer with an HTTP POST response message following the procedure described in clause 4.2.2.2. After that, PCF shall send contact SMF to request 5GC network functions health monitoring information as it is specified in 3GPP TS 29.512 [8].

4.2.2.52 Initial provisioning of the RAN-Controlled UL Bitrate Recommendation Indication

If the "ExtQoSR19" feature is supported, this procedure is used by the NF service consumer to indicate the RAN-Controlled UL Bitrate Recommendation support for the media flow.

The NF service consumer may include within the corresponding media component(s) entries of the "medComponents" attribute in the HTTP POST request message described in clause 4.2.2.2:

- the RAN-Controlled UL Bitrate Recommendation indication in the "ulBrRecInd" attribute.

The PCF determines the policies for the RAN-Controlled UL Bitrate Recommendation indication and provisions the PCC rule as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.3 Npcf_PolicyAuthorization_Update service operation

4.2.3.1 General

The Npcf_PolicyAuthorization_Update service operation provides updated application level information from the NF service consumer and optionally communicates with the Npcf_SMPolicyControl service to determine and install the policy according to the information provided by the NF service consumer.

The Npcf_PolicyAuthorization_Update service operation updates an application session context in the PCF.

The following procedures using the Npcf_PolicyAuthorization_Update service operation are supported:

- Modification of service information.
- Gate control.
- Background Data Transfer policy indication at policy authorization update.
- Modification of sponsored connectivity information.
- Modification of Subscription to Service Data Flow QoS notification control.
- Modification of Subscription to Service Data Flow Deactivation.
- Update of traffic routing, service function chaining information and handling of payload headers.
- Modification of subscription to resources allocation outcome.
- Modification of Multimedia Priority Services.
- Support of content versioning.
- Request of access network information.
- Modification of service information status.
- Support of SIP forking.
- Provisioning of signalling flow information.

- Support of resource sharing.
- Modification of MCPTT.
- Modification of MCVideo.
- Priority sharing indication.
- Modification of subscription to out of credit notification.
- Modification of Subscription to Service Data Flow QoS Monitoring Information.
- Update of TSCAI Input Information and TSC QoS related data.
- Provisioning of TSC user plane node management information and port management information.
- Support of CHEM feature.
- Support of FLUS feature.
- Subscription to EPS Fallback report.
- Modification of required QoS information.
- Support of QoSHint feature.
- Modification of subscription to reallocation of credit notification.
- Modification of subscription to satellite backhaul category changes.
- Modification of the subscription to the report of extra UE addresses.
- Modification of multi-modal services
- Modification of Round-Trip latency requirements.
- Update of PDU Set handling related data.
- Modification of subscription to BAT offset notification.
- Modification of subscription to Packet Delay Variation monitoring.
- Provisioning of the indication of ECN marking for L4S support.
- Modification of Round-Trip delay monitoring requirements over two QoS flows.
- Provisioning of the QoS timing information.
- Modification of traffic information for UE power saving management.
- Modification of the subscription to the report of network support for QoS Monitoring.
- Modification of dynamically changing traffic characteristics.
- Provisioning of the indication of RAN-Controlled UL Bitrate Recommendation Indication.
- Modification of the (S)RTP Multiplexed Media Identification Information.
- Modification of the subscription to Data Rate Limitation Information reporting.

4.2.3.2 Modification of service information

This procedure is used to modify an existing application session context as defined in 3GPP TS 23.501 [2], 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4] when the feature "PatchCorrection" is supported.

Figure 4.2.3.2-1 illustrates the modification of service information using HTTP PATCH method.

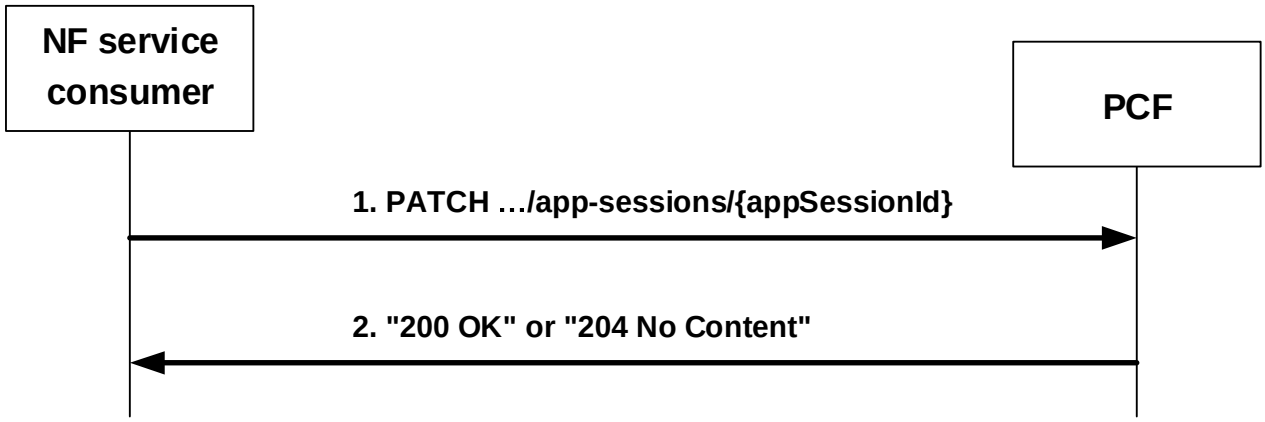


Figure 4.2.3.2-1: Modification of service information using HTTP PATCH

The NF service consumer may modify the application session context information at any time (e.g. due to an AF session modification or internal NF service consumer trigger) and invoke the Npcf_PolicyAuthorization_Update service operation by sending the HTTP PATCH request message to the resource URI representing the "Individual Application Session Context" resource, as shown in figure 4.2.3.2-1, step 1, with the modifications to apply.

The JSON body within the PATCH request shall include the "AppSessionContextUpdateDataPatch" data type and shall be encoded according to "JSON Merge Patch", as defined in IETF RFC 7396 [21]. The modifications to apply are encoded within the attributes of the "ascReqData" attribute, as described below and in subsequent clauses.

The NF service consumer may include the updated service information in the "medComponents" attribute of the "ascReqData" attribute. The NF service consumer may update the service data flow filter(s) (IP or Ethernet) that identify the traffic of the media component by replacing, within the concerned media subcomponent(s), the previously provided value(s) with the updated one(s).

If the "AuthorizationWithRequiredQoS" feature as defined in clause 5.8 is supported, the NF service consumer may provide within the MediaComponentRm data structure an update of the required QoS information as specified in clause 4.2.3.30.

The NF service consumer may include in the "ascReqData" attribute an AF application identifier in the "afAppId" attribute to trigger the PCF to indicate to the SMF/UPF to perform the application detection based on the operator's policy as defined in 3GPP TS 29.512 [8].

If the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported, the NF service consumer may provide TSC user plane node related information as specified in clauses 4.2.3.24 and 4.2.3.25.

If the "PDUSetHandling" feature is supported, the NF service consumer may update PDU set handling related data as specified in clause 4.2.3.36.

If the "PowerSaving" feature is supported, the NF service consumer may update the UL and/or DL traffic periodicity and/or DL protocol description as described in clause 4.2.3.41.

The NF service consumer may also create, modify or remove events subscription information by sending the HTTP PATCH request message to the resource URI representing the "Individual Application Session Context" resource.

The NF service consumer shall create event subscription information by including in the "ascReqData" attribute the "evSubsc" attribute of "EventsSubscReqDataRm" data type with the corresponding list of events to subscribe to; and the "notifUri" attribute with the notification URI where the PCF shall send the notifications.

The NF service consumer shall update existing event subscription information by including in the "ascReqData" attribute an updated value of the "evSubsc" attribute of the "EventsSubscReqDataRm" data type as follows:

- The "events" attribute shall include the new complete list of subscribed events.
- When the NF service consumer requests to update the additional information related to an event (e.g. the NF service consumer needs to provide new thresholds to the PCF in the "usgThres" attribute related to the "USAGE_REPORT" event) the NF service consumer shall include the additional information, which shall completely replace the previously provided one.

NOTE 1: Note that when the NF service consumer requests to remove an event, this event is not included in the "events" attribute.

NOTE 2: When an event is included in the "events" attribute and its related additional information is set to null, the PCF considers the subscription to this event is active, but the related procedures stop applying.

NOTE 3: When an event is removed from the "events" attribute but its related information is not set to null, the PCF considers the subscription to this event is terminated, the related additional information is removed, and the related procedures stop applying.

The NF service consumer shall remove existing event subscription information by setting to null the "evSubsc" attribute included in the "ascReqData" attribute.

If the "EnQoSMon" feature is supported, the NF service consumer may include attribute "evSubsc" in "MediaSubComponentRm" data type for QoS monitoring for each media component. Either the attribute "evSubsc" in "MediaSubComponentRm" data type or attribute "evSubsc" in "AppSessionContextReqDataRm" data type may be provided to subscribe to notifications for a specific event. An event subscription modification shall not create simultaneous subscriptions, for the provided event, within the media subcomponent and within the application session context.

The NF service consumer shall update the existing event subscription information of each media component by updating the value of the "evSubsc" attribute in "MediaSubComponentRm" data type.

The NF service consumer shall remove the existing event subscription information of each media component by setting to null the "evSubsc" attribute in "MediaSubComponentRm" data type.

Events with "notifMethod" set to "ONE_TIME" shall only apply at the time the NF service consumer requests their subscription. Once the event report is performed, the subscription to this event is automatically terminated in the PCF and the related information is removed. The presence of a one-time event, together with its related additional information when applicable, during an update procedure shall represent the recreation of the subscription to this event in the PCF.

NOTE 4: The "notifUri" attribute within the EventsSubscReqData data structure can be modified to request that subsequent notifications are sent to a new NF service consumer.

If the PCF cannot successfully fulfil the received HTTP PATCH request due to the internal PCF error or due to the error in the HTTP PATCH request, the PCF shall send the HTTP error response as specified in clause 5.7.

If the feature "ES3XX" is supported, and the PCF determines the received HTTP PATCH request needs to be redirected, the PCF shall send an HTTP redirect response as specified in clause 6.10.9 of 3GPP TS 29.500 [5].

Otherwise, the PCF shall process the received service information according the operator policy and may decide whether the HTTP request message is accepted or not.

If the updated service information is not acceptable (e.g., the subscribed guaranteed bandwidth for a particular user is exceeded, the authorized data rate in that slice for the UE is exceeded), the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure that:

- shall contain the ProblemDetails data structure containing the "cause" attribute set to the "REQUESTED_SERVICE_NOT_AUTHORIZED" application error indicating the cause of the rejection; and

- may contain the acceptable QoS parameters within the "acceptableServInfo" attribute.

If the PCF detects that a temporary network failure has occurred (e.g. the SGW has failed as defined in clause B.3.3.3 or B.3.4.9 of 3GPP TS 29.512 [8]), or if the PCF supports "SmfFailureDetection" feature and the PCF due to operator configuration detects that the SMF has failed as defined in 3GPP TS 23.380 [39], and when the AF initiates an Npcf_PolicyAuthorization_Update service operation, the PCF shall reject the request with an HTTP "403 Forbidden" response including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "TEMPORARY_NETWORK_FAILURE".

If the service information provided in the HTTP PATCH request is rejected due to a temporary condition in the network (e.g. the NWDAF reported the network slice selected for the PDU session is congested), the PCF may reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUESTED_SERVICE_TEMPORARILY_NOT_AUTHORIZED". The PCF may also provide a retry interval

within the "Retry-After" HTTP header field. When the NF service consumer receives the retry interval within the "Retry-After" HTTP header field, the NF service consumer shall not send the same service information to the PCF again (for the same application session context) until the retry interval has elapsed. The "Retry-After" HTTP header is described in 3GPP TS 29.500 [5] clause 5.2.2.2.

If the service information is invalid or in sufficient for the PCF to perform the requested action, e.g. invalid media type or invalid QoS reference, the PCF shall reject the request with an HTTP "400 Bad Request" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "INVALID_SERVICE_INFORMATION".

If the IP flow descriptions cannot be handled by the PCF because the restrictions defined in clause 5.3.8 of 3GPP TS 29.214 [20] are not observed, the PCF shall reject the request with an HTTP "400 Bad Request" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "FILTER_RESTRICTIONS".

If the AF provided the same AF charging identifier for a new Individual Application Session Context that is already in use for the other ongoing Individual Application Session, the PCF shall reject the request with an HTTP "400 Bad Request" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "DUPLICATED_AF_SESSION".

NOTE 5: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8] and the authorized data rate in a slice is exceeded due to the bandwidth demands of the modified service information, it is also possible to accept the request based on operator policies. In this case the derived PCC rule(s) belonging to the authorized GBR service data flows can include a different MBR and/or have a different charging than the one applicable if the data rate is not exceeded as specified in 3GPP TS 29.512 [8].

If the "VPLMNErrorRep" feature is supported and the required QoS, provided by the AF in the updated information, is not supported in the current serving PLMN where the UE is registered, the PCF may reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUEST_QOS_NOT_SUPPORTED_IN_PLMN" indicating the cause of the rejection.

If the "TrafficCharChange" feature is supported and the expedited data transfer with reflective QoS indication is provided by the AF in the updated information as specified in clause 4.2.3.47, and the UE has not indicated support for reflective QoS as specified in 3GPP TS29.512 [8], the PCF shall reject the request and indicate in an HTTP "403 Forbidden" response message the cause for the rejection including the "cause" attribute set to "REFLECTIVE_QOS_NOT_SUPPORTED_IN_UE".

If the "MpxMedia" feature is supported, the NF service consumer request includes media flows with an uplink direction in the updated information as specified in clause 4.2.3.46 and the SMF has not indicated UE support for (S)RTP Multiplexed Media Identification to the PCF as specified in 3GPP TS 29.512 [8], the PCF may, based on operator configuration, reject the request and indicate in an HTTP "403 Forbidden" response message the cause for the rejection including the "cause" attribute set to "MPX_MEDIA_NOT_SUPPORTED_IN_UE".

If the "ExtQoSR19" feature is supported, the NF service consumer may update RAN-Controlled UL Bitrate Recommendation Indication for the media flow as specified in clause 4.2.3.50.

If the request is accepted, the PCF shall update the service information with the new information received. Due to the updated service information, the PCF may need to create, modify or delete the related PCC rules as specified in 3GPP TS 29.513 [7] and provide the updated information towards the SMF following the corresponding procedures specified in 3GPP TS 29.512 [8].

Based on the received subscription information from the NF service consumer, the PCF may create a subscription to event notifications or may modify the existing subscription to event notifications, for a related PDU session from the SMF, as described in 3GPP TS 29.512 [8].

The PCF shall reply with the HTTP response message to the NF service consumer and may include the "AppSessionContext" data type content with the representation of the modified "Individual Application Session Context" resource and may include the "Events Subscription" sub-resource.

The PCF shall include in the "evsNotif" attribute:

- if the NF service consumer subscribed to the "PLMN_CHG" event in the HTTP PATCH request, the "event" attribute set to "PLMN_CHG" and the "plmnId" attribute including the PLMN Identifier or the SNPN Identifier if the PCF has previously requested to be updated with this information in the SMF;

NOTE 6: The SNPN Identifier consists of the PLMN Identifier and the NID.

NOTE 7: Handover between non-equivalent SNPNs, and between SNPN and PLMN is not supported. When the UE is operating in SNPN access mode, the trigger reports changes of equivalent SNPNs.

- if the NF service consumer subscribed to the event "ACCESS_TYPE_CHANGE" event in the HTTP PATCH request, the "event" attribute set to "ACCESS_TYPE_CHANGE" and:

  i. the "accessType" attribute including the access type, and the "ratType" attribute including the RAT type when applicable for the notified access type; and

  ii. if the "ATSSS" feature is supported, the "addAccessInfo" attribute with the additional access type information if available, where the access type is encoded in the "accessType" attribute, and the RAT type is encoded in the "ratType" attribute when applicable for the notified access type; and

NOTE 8: For a MA PDU session, if the "ATSSS" feature is not supported by the NF service consumer, the PCF includes the "accessType" attribute and the "ratType" attribute with a currently active combination of access type and RAT type (if applicable for the notifed access type). When both 3GPP and non-3GPP accesses are available, the PCF includes the information corresponding to the 3GPP access.

  iii. the "anGwAddr" attribute including access network gateway address when available,

  if the PCF has previously requested to be updated with this information in the SMF;

- if the "IMS_SBI" feature is supported and if the NF service consumer subscribed to the "CHARGING_CORRELATION" event in the HTTP PATCH request, the "event" attribute set to "CHARGING_CORRELATION" and may include the "anChargIds" attribute containing the access network charging identifier(s) and the "anChargAddr" attribute containing the access network charging address; and

- if the "UEUnreachable" feature is supported and the NF service consumer subscribed to the "UE_REACH_STATUS_CH" event in the HTTP PATCH request, the "event" attribute set to "UE_REACH_STATUS_CH" together with the "ueReachStatus" attribute containing the corresponding UE status, and in case the "ueReachStatus" attribute is set to "UNREACHABLE", optionally the "retryAfter" attribute if available and the PCF has previously requested this information to the SMF.

The NF service consumer subscription to other specific events using the Npcf_PolicyAuthorization_Update request is described in the related clauses. Notification of events when the applicable information is not available in the PCF when receiving the Npcf_PolicyAuthorization_Update request is described in clause 4.2.5.

The HTTP response message towards the NF service consumer should take place before or in parallel with any required PCC rule provisioning towards the SMF.

If the PCF does not have an existing application session context for the application session context being modified (such as after a PCF failure), the PCF shall reject the HTTP request message with the HTTP response message with the applicable rejection cause.

4.2.3.3 Gate control

This procedure is used by a NF service consumer to modify in the PCF the service data flow(s) that are to be enabled or disabled to pass through the PDU session.

The NF service consumer shall use the HTTP PATCH method to modify the gate control information.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, in the media component(s) included in the "medComponents" attribute at media and/or media subcomponent level, the "fStatus" attribute for the flows to be enabled or disabled with the appropriate value.

If a "medSubComps" attribute contains a "flowUsage" attribute with the value "RTCP", then the IP Flows described by that media subcomponent shall be enabled in both directions irrespective of the value of the "fStatus" attribute of the corresponding media component.

As result of this action, the PCF shall set the appropriate gate status for the corresponding active PCC rule(s).

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.4 Background Data Transfer policy indication at policy authorization update

This procedure is used by a NF service consumer to indicate at policy authorization update a transfer policy negotiated for background data transfer using the Npcf_BDTPolicyControl service as described in 3GPP TS 29.554 [14].

The NF service consumer may include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, a new reference id in the "bdtRefId" attribute.

NOTE 1: The PCF will retrieve the corresponding transfer policy from the UDR based on the reference identifier within the "bdtRefId" attribute. In case only one PCF is deployed in the network, transfer policies can be locally stored in the PCF and the interaction with the UDR is not required.

If the PCF cannot retrieve the transfer policy, the PCF shall set to TP_NOT_KNOWN the "servAuthInfo" attribute in the HTTP response message to the NF service consumer to indicate that the transfer policy is unknown.

If the time window of the received transfer policy has expired, the PCF shall set to TP_EXPIRED the "servAuthInfo" attribute in the HTTP response message to indicate to the NF service consumer that the transfer policy has expired. Otherwise, if the time window of the received transfer policy has not yet occurred, the PCF shall set to TP_NOT_YET_OCURRED the "servAuthInfo" attribute in the HTTP response message to the NF service consumer to indicate that the time window of the transfer policy has not yet occurred.

NOTE 2: In the case that the PCF cannot retrieve the transfer policy, the transfer policy time window has not yet occurred or the transfer policy expired, the PCF makes the decision without considering the transfer policy.

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.5 Modification of sponsored connectivity information

This procedure is used by a NF service consumer to modify sponsored data connectivity when "SponsoredConnectivity" feature is supported.

The NF service consumer shall use the HTTP PATCH method to modify the sponsored connectivity information.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, an application service provider identity and a sponsor identity within the "aspId" attribute and "sponId" attribute, and optionally an indication of whether to enable or disable sponsored data connectivity within the "sponStatus" attribute set to the applicable value to provide sponsored connectivity information or to update existing sponsored connectivity information.

If the NF service consumer requests to enable sponsored data connectivity the NF service consumer shall change the "sponStatus" attribute value to "SPONSOR_ENABLED".

If the NF service consumer requests to disable sponsored data connectivity the NF service consumer shall provide an indication to disable sponsored data connectivity to the PCF by setting the "sponStatus" attribute to "SPONSOR_DISABLED".

To support the usage monitoring of sponsored data connectivity, the NF service consumer may also include in the HTTP PATCH a new or modified "evSubsc" attribute of "EventsSubscReqDataRm" data type with:

- the usage thresholds to apply in the "usgThres" attribute; and
- the subscription to usage monitoring for sponsored data connectivity in an entry of the "events" attribute of the "AfEventSubscription" data type with the "event" attribute set to "USAGE_REPORT".

NOTE 1: If the NF service consumer is in the user plane, the NF service consumer can handle the usage monitoring and therefore it is not required to provide a usage threshold to the PCF as part of the sponsored data connectivity information.

When the NF service consumer indicated to enable sponsored data connectivity, and the UE is roaming with the visited access case, the following procedures apply:

- If the NF service consumer is located in the HPLMN, for home routed roaming case and when operator policies do not allow accessing the sponsored data connectivity with this roaming case, the H-PCF shall reject the service request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "UNAUTHORIZED_SPONSORED_DATA_CONNECTIVITY".

- If the NF service consumer is located in the VPLMN, the V-PCF shall reject the service request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "UNAUTHORIZED_SPONSORED_DATA_CONNECTIVITY".

When the NF service consumer indicated to enable sponsored data connectivity, and the UE is in the non-roaming case or roaming with the home routed case and the operator policies allow accessing the sponsored data connectivity with this roaming case, the following procedures apply:

- If the SMF does not support sponsored connectivity and the required reporting level for that service indicates a sponsored connectivity level according to 3GPP TS 29.512 [8], then the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUESTED_SERVICE_NOT_AUTHORIZED".

- If the SMF supports sponsored data connectivity feature or the required reporting level is different from sponsored connectivity level as described in 3GPP TS 29.512 [8], then the PCF, based on operator policies, shall check whether it is required to validate the sponsored connectivity data. If it is required, it shall perform the authorizations based on sponsored data connectivity profiles. If the authorization fails, the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "UNAUTHORIZED_SPONSORED_DATA_CONNECTIVITY".

NOTE 2: The PCF is not required to verify that a trust relationship exists between the operator and the sponsors.

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.6 Modification of Subscription to Service Data Flow QoS notification control

This procedure is used in the NF service consumer to modify in the PCF the subscription to notification about whether the GBR QoS targets can no longer (or can again) be guaranteed.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "EventsSubscReqDataRm" data type, which either shall include in the "events" attribute a new element with the "event" attribute set to "QOS_NOTIF" to indicate the subscription to QoS notification control, or shall not include in the "events" attribute an existing element with the "event" attribute set to "QOS_NOTIF" to indicate the termination of the subscription to QoS notification control.

As result of this action, the PCF shall set the appropriate subscription to QoS notification control for the corresponding active PCC rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.7 Modification of Subscription to Service Data Flow Deactivation

This procedure is used by a NF service consumer to modify in the PCF the subscription to the notification of deactivation of one or more Service Data Flows within the AF application session context.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "EventsSubscReqDataRm" data type, which either shall include in the "events" attribute a new element with the "event" attribute set to "FAILED_RESOURCES_ALLOCATION" to indicate

the subscription to service data flow deactivation, or shall not include in the "events" attribute an existing element with the "event" attribute set to "FAILED_RESOURCES_ALLOCATION".

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

As result of this action, the PCF shall set the appropriate subscription to service data flow deactivation for the corresponding PCC rule(s) as described in in 3GPP TS 29.512 [8].

4.2.3.8 Update of traffic routing, service function chaining information and handling of payload headers

When the "InfluenceOnTrafficRouting" feature is supported, this procedure is used by the NF service consumer to modify the traffic routing information to a local access to a DNN, and/or to modify the subscription to notifications about UP path management events. Additionally:

- when the "SimultConnectivity" feature is supported, this procedure may be used to modify (create, delete, update) the indication of simultaneous connectivity temporarily maintained for the source and target PSA and/or the indication of the minimum time interval to be considered for inactivity for the traffic routed via the source PSA;

- when the "URLLC" feature is supported, this procedure may be used to modify (create, delete, update) the indication of UE IP address preservation; and

- when the "EASIPreplacement" feature is supported, this procedure may be used to modify (initially provide, delete, update) the EAS IP replacement information to the PCF.

When the "UeSatUeComm" feature is supported, this procedure is used by the NF service consumer to modify the subscription to notifications about UP path management events.

When the "SFC" feature is supported, this procedure is used by the NF service consumer to modify service chaining information.

When the "HeaderHandling" feature is supported, this procedure is used by the NF service consumer to modify header handling control information.

The NF service consumer shall use the HTTP PATCH method.

To modify traffic routing information, the NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, an updated "afRoutReq" attribute(s) with the modified traffic routing information. To modify the indication of simultaneous connectivity and/or the termination of the simultaneous connectivity, the NF service consumer shall include an updated "simConnInd" attribute and/or an updated "simConnTerm" attribute and/or if feature "SimConnFailure" is supported, an updated "simConnFailSub" attribute, if applicable. To modify the indication of UE IP address preservation, the NF service consumer shall include the updated indication of UE IP address preservation in the "addrPreserInd" attribute, if applicable. To modify the EAS IP replacement information, the NF service consumer shall include the updated/new "easIpReplaceInfos" attribute, if applicable. To modify the maximum allowed user plane latency, the NF service consumer shall include the updated/new "maxAllowedUpLat" attribute, if applicable. To modify the traffic correlation information, the NF service consumer shall include an updated/new "tfcCorreInfo" attribute. To modify the indication about reporting candidate DNAI(s), the NF service consumer shall include an updated/new "candDnaiInd" attribute, if applicable. To send a new indication of EAS rediscovery, the NF service consumer shall include the indication in the "easRedisInd" attribute, if applicable. To modify the indication about considering the N6 delay measurement, the NF service consumer shall include an updated/new indication in the "n6DelayInd" attribute, if applicable.

To modify (create, delete or modify) the service function chaining information, the NF service consumer shall include the"afSfcReq" attribute including the modified service function chaining information within the AppSessionContextUpdateData.

To modify header handling control information, the NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, an updated "afHdrReq" attribute with the modified header handling control information. In order to retrieve header handling event report for the performed header handling action, the AF shall set the"notifFlag" attribute to true true for the related "HeaderHandlingActionRequest" data type and shall also provide the "notifUri" attribute and the "notifId" attribute within the "AfHeaderHandlingControlInfo" data type if not provided yet.

To modify the subscription to notifications about UP path management events (create, delete or modify), the NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "upPathChgSub" attribute with the modified subscription to UP path management events.

When the feature "RoutingReqOutcome" is supported:

- and the NF service consumer is creating or modifying AF routing information, the PCF may set the "servAuthInfo" attribute in the HTTP response message to "ROUT_REQ_NOT_AUTHORIZED" when the PCF determines, e.g. based on subscription, the AF influence on traffic routing is not allowed for the PDU session;

- when the NF service consumer requests the update of the steering of traffic to a DNAI and/or the subscription to notifications about UP path management events, the NF service consumer may subscribe to notifications of failures in the enforcement of UP path changes including within the "evSubsc" attribute the "event" attribute value "UP_PATH_CHG_FAILURE" in an entry of the "events" array, or may remove the subscription to notification of failures in the enforcement of UP path changes by not including the the "event" attribute value "UP_PATH_CHG_FAILURE" in an entry of the "events" array of the "evSubsc" attribute.

NOTE: In the case that the PCF determines that the requested AF routing requirements cannot be applied and returns the "servAuthInfo" attribute in the HTTP response, the PCF makes the decision without considering the requested AF routing requirements.

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

The PCF shall store the application routing requirements included in the "afRoutReq" attribute when the "InfluenceOnTrafficRouting" feature is supported, the N6-LAN traffic steering requirements within the "afSfcReq" attribute when the "SFC" feature is supported and/or the header handling control information included in the "afHdrReq" attribute when the "HeaderHandling" feature is supported.

The PCF shall check whether the updated application routing requirements, N6-LAN traffic steering requirements and/or the header handling control information requirements require PCC rules to be created or modified to include updated traffic steering policies (for routing requirements, N6-LAN traffic steering requirements and/or the header handling control information), or to update the application relocation possibility (only for routing requirements) as specified in 3GPP TS 29.513 [7]. Provisioning of PCC rules to the SMF shall be carried out as specified at 3GPP TS 29.512 [8].

4.2.3.9 Void

4.2.3.10 Modification of subscription to resources allocation outcome

This procedure is used in the NF service consumer to modify in the PCF the subscription to notification about resources allocation outcome.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "EventsSubscReqDataRm" data type, which either include in the "events" attribute a new element with the "event" attribute set to "SUCCESSFUL_RESOURCES_ALLOCATION","FAILED_RESOURCES_ALLOCATION" and/or "UE_TEMPORARILY_UNAVAILABLE" or remove in the "events" attribute an existing element with the "event" attribute set to "SUCCESSFUL_RESOURCES_ALLOCATION", "FAILED_RESOURCES_ALLOCATION" and/or "UE_TEMPORARILY_UNAVAILABLE".As a result of this action, the PCF shall set the appropriate subscription to notification of resources allocation outcome in the corresponding PCC Rule(s) as described in 3GPP TS 29.512 [8].

4.2.3.11 Void

4.2.3.12 Modification of Multimedia Priority Services

The NF service consumer may include, in the "ascReqData" attribute, the "mpsId" attribute if it was not previously provided in order to indicate that the modified AF session relates to an MPS session.

If the NF service consumer supports the SBI Message Priority mechanism for an MPS session, the NF service consumer shall include the "3gpp-Sbi-Message-Priority" custom HTTP header towards the PCF as described in clause 4.2.2.12.1.

If the PCF receives the "mpsId" attribute, the PCF shall take specific actions on the corresponding PDU session to ensure that the MPS session is prioritized as defined in 3GPP TS 29.512 [8].

NOTE: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8], it is possible that, subject to operator policy and national/regional regulations, prioritised services are exempted from the limitation of data rate per network slice. In that case the PCF will handle the request from the NF service consumer even if the authorized data rate per network slice is exceeded.

When the feature "MPSforMessaging" is supported, if the NF service consumer wants to change the priority treatment for the AF Signalling IP flows related to Messaging, the NF service consumer includes the "mpsAction" attribute set to "ENABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING" or "DISABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING" depending on whether the NF service consumer wants to enable priority to the AF Signalling IP flows or wants to disable it. The PCF shall behave as described in clause 4.2.2.12.4.

When the feature "MPSforDTS" is supported, the NF service consumer includes the "mpsAction" attribute to invoke or revoke MPS for DTS, as specified in clause 4.2.2.12.2. When invoking MPS for DTS, the NF service consumer shall include the "mpsAction" attribute set to "ENABLE_MPS_FOR_DTS" or "AUTHORIZE_AND_ENABLE_MPS_FOR_DTS". When the "ENABLE_MPS_FOR_DTS" value is received in the "mpsAction" attribute, and allowed by local policy, the PCF does not check the authorization of the request. When the "AUTHORIZE_AND_ENABLE_MPS_FOR_DTS" value is received in the "mpsAction" attribute, and allowed by local policy, the PCF shall check the user's MPS subscription to authorize the request. When the request is to authorize and enable, and the request is not authorized (e.g. not allowed by MPS subscription), the PCF shall reject the request with an HTTP "403 Forbidden" status code with the response body including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "REQUESTED_SERVICE_NOT_AUTHORIZED" indicating the cause of the rejection.

To revoke MPS for DTS, the NF service consumer shall include the "mpsAction" attribute set to "DISABLE_MPS_FOR_DTS". When the "DISABLE_MPS_FOR_DTS" value is received in the "mpsAction" attribute, and allowed by local policy, the PCF does not check the authorization of the request.

When modifying an Individual Application Session Context resource due to the invocation or revocation of the MPS for DTS service, the NF service consumer may subscribe to the outcome of the QoS updates by setting within the "evSubsc" attribute an event within the "events" array with:

- the "event" attribute set to the value "SUCCESSFUL_QOS_UPDATE" to report that the invocation/revocation requested by the NF service consumer was successful; and/or
- the "event" attribute set to the value "FAILED_QOS_UPDATE" to report that the invocation/revocation requested by the NF service consumer has failed.

4.2.3.13 Support of content versioning

The support of the media component versioning is optional. When the "MediaComponentVersioning" feature is supported, the NF service consumer and the PCF shall comply with the procedures specified in this clause.

Upon each media component modification encoded in the "medComponents" attribute included in the "ascReqData" attribute, if the content version was previously assigned to a media component, the NF service consumer shall assign a new content version. All the content related to that media component shall be included and the content version shall be unique for the lifetime of the media component.

NOTE: The NF service consumer will include all the content of the media component in each media component modification in order to ensure that the media component is installed with the proper information regardless of the outcome of the QoS flow procedure related to previous interactions that are not reported to the PCF yet.

If the PCF receives the "contVer" attribute for a certain media component, the PCF shall follow the procedures described in 3GPP TS 29.512 [8], clause 4.2.6.2.14.

4.2.3.14 Request of access network information

This procedure is used by a NF service consumer to request access network information for an existing "Individual Application Session Context" resource at service information modification when the "NetLoc" feature is supported and to request the serving satellite identifier within the access network information when the "UeSatUeComm" feature is supported.

NOTE 1: Clause 4.2.6.6 describes the NF service consumer request of access network information without providing service information.

The NF service consumer shall create event subscription information by including in the "AppSessionContextUpdateData" data type the "evSubsc" attribute of "EventsSubscReqData" data type with the corresponding list of events to subscribe to.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:

  a) the "event" attribute set to the value "ANI_REPORT"; and

  b) the "notifMethod" attribute set to the value "ONE_TIME"; and

- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information and/or if the "UeSatUeComm" feature is supported, identifier of the serving satellite of the UE.

When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature, the PCF shall respond to the NF service consumer including in the "EventsNotification" data type the "noNetLocSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.3.2.

NOTE 2: The NF service consumer does not invoke the Npcf_PolicyAuthorization_Update service operation to remove subscription to access network information report since the "Access Network Information Notification" is the one-time reported event. Once the access network information is reported to the NF service consumer the subscription to the access network information report is automatically terminated in the PCF and the related information is removed.

4.2.3.15 Modification of service information status

When the "IMS_SBI" feature is supported, the NF service consumer may update the status of the service information. If the NF service consumer provides service information that has been fully negotiated (e.g. based on the SDP answer), the NF service consumer may include in the "ascReqData" attribute the "servInfStatus" attribute set to "FINAL". In this case the PCF shall authorize the session and provision the corresponding PCC rules to the SMF.

The NF service consumer may additionally provide preliminary service information not fully negotiated yet (e.g. based on the SDP offer) at an earlier stage. To do so, the NF service consumer shall include the "servInfStatus" attribute set to "PRELIMINARY". Upon receipt of such preliminary service information, the PCF shall perform an early authorization check of the service information. If the NF service consumer requests the PCF to report the access network information together with preliminary service information, the PCF shall immediately configure the SMF to provide the access network information.

4.2.3.16 Support of SIP forking

When the "IMS_SBI" feature is supported, this procedure is used by a NF service consumer to indicate that an existing "Individual Application Session Context" resource comprises service information about several SIP dialogues.

The NF service consumer shall use the HTTP PATCH method to modify the service information.

The NF service consumer may include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the "sipForkInd" attribute and include the updated service information.

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

When the "sipForkInd" attribute gets the value:

- "SEVERAL_DIALOGUES", the PCF shall send additional PCC rules or individual data flow filters to already provided PCC rules as described in Annex B.3.1.
- "SINGLE_DIALOGUE", the PCF shall update installed PCC rules and Authorized-QoS information as described in Annex B.3.2.

4.2.3.17 Provisioning of signalling flow information

This procedure is used by a NF service consumer to provision or de-provision information about the AF signalling IP flows between the UE and the NF service consumer.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute:

- when the procedure is used to provision information about the AF signalling IP flows, a media component within the "medComponents" attribute including the attributes described in clause 4.2.2.16 in the case of IMS restoration or clause 4.2.2.12.3 otherwise;
- when the procedure is used to de-provision information about the AF signalling IP flows, for the media subcomponents containing the AF signalling IP flows, the "fStatus" attribute set to the value "REMOVED".

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.18 Support of resource sharing

When the "ResourceSharing" is supported by the NF service consumer and the PCF, the NF service consumer may include, in the "ascReqData" attribute, the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute within a media component of the "medComponents" attribute to indicate to the PCF that the related media of the modified Individual Application Session Context resource may share resources with other media components in the related direction that include the same value in the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute.

The NF service consumer may modify the conditions for resource sharing by including the media component within the "medComponents" attribute with a new value for the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute. The NF service consumer may indicate that the related media of the modified Individual Application Session resource is not sharing resources with other media components in the related direction setting the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute to "null".

The NF service consumer shall use the HTTP PATCH method to update the "Individual Application Session Context resource" as described in clause 4.2.3.2.

If the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute are provided within a media component of the "medComponents" attribute, the PCF may apply the mechanisms for resource sharing as described in 3GPP TS 29.512 [8], clause 4.2.6.2.8.

4.2.3.19 Modification of MCPTT

The NF service consumer may include, in the "ascReqData" attribute, the "mcpttId" attribute in order to indicate that the modified "Individual Application Session Context" resource relates to the priority adjustment of an MCPTT session. When the PCF receives the "mcpttId" attribute related to that MCPTT session, the PCF may take specific actions on the corresponding PDU session to ensure that the MCPTT session is prioritized. For the handling of MCPTT session with priority call, see Annex B.13.

NOTE: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8], it is possible that, subject to operator policy and national/regional regulations, prioritised services are exempted from the limitation of data rate per network slice. In that case the PCF will handle the request from the NF service consumer even if the authorized data rate per network slice is exceeded.

Additionally, when the "PrioritySharing" feature is supported, the PCF may receive the "prioSharingInd" attribute within the media component received in the "medComponents" attribute as described in clause 4.2.2.21. In this case,

and if "MCPTT-Preemption" feature is supported, the PCF may receive pre-emption information as also described in clause 4.2.3.21.

4.2.3.20 Modification of MCVideo

The NF service consumer may include, in the "ascReqData" attribute, the "mcVideoId" attribute in order to indicate that the modified "Individual Application Session Context" resource relates to the priority adjustment of an MCVideo session. When the PCF receives the "mcVideoId" attribute related to that MCVideo session, the PCF may take specific actions on the corresponding PDU session to ensure that the MCVideo session is prioritized. For the handling of MCVideo session with priority call, see Annex B.15.

NOTE: When the PCF supports data rate control per network slice and/or data rate control per network slice for a UE as specified in 3GPP TS 29.512 [8], it is possible that, subject to operator policy and national/regional regulations, prioritised services are exempted from the limitation of data rate per network slice. In that case the PCF will handle the request from the NF service consumer even if the authorized data rate per network slice is exceeded.

4.2.3.21 Priority sharing indication

When the "PrioritySharing" feature is supported, the NF service consumer may include the "prioSharingInd" attribute set to "ENABLED" within a media component of the "medComponents" attribute included in the "ascReqData" attribute to indicate to the PCF that the related media flow is allowed to use the same Allocation and Retention Priority (ARP) as media flows belonging to other "Individual Application Session Context" resources as described in clause 4.2.2.21. In this case, if the "MCPTT-Preemption" feature is supported, the NF service consumer may also include the "preemptCap", "preemptVuln" and "preemptControlInfo" attributes as described in clause 4.2.2.21.

When the "preemptControlInfo" attribute is modified, the latest provided value shall be applied to all potential media flow candidates.

If the NF service consumer earlier has indicated a media flow priority sharing to the PCF by setting the "prioSharingInd" attribute to "ENABLED", the NF service consumer may include the Priority-Sharing-Indicator AVP set to "DISABLED" within a media component of the "medComponents" attribute to indicate to the PCF that the related media flow shall not be part of the mechanism for sharing the Allocation and Retention Priority with other media flows any longer.

If this media flow was in priority sharing with other media flows the PCF should readjust the Allocation and Retention Priority for the remaining services sharing priority as described in 3GPP TS 29.512 [8], clause 4.2.6.2.9 and handle the media flow excluded from priority sharing according to normal PCC/QoS rule provisioning procedures described in 3GPP TS 29.512 [8], clause 4.2.6.2.

If the NF service consumer earlier has indicated a media flow priority sharing to the PCF by setting the "prioSharingInd" attribute to "ENABLED" for media flows and the NF service consumer indicates to remove one or more of the media flows in priority sharing with other media flows, the PCF should readjust the Allocation and Retention Priority for the remaining services sharing priority as described in 3GPP TS 29.512 [8], clause 4.2.6.2.9 and handle the media flow removed according to normal PCC/QoS rule provisioning procedures described in 3GPP TS 29.212 [8], clause 4.2.6.2.

4.2.3.22 Modification of Subscription to Out of Credit notification

This procedure is used by the NF service consumer if the "IMS_SBI" feature is supported to modify in the PCF the subscription to notification about credit unavailability for the Service Data Flows within the AF application session context.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "EventsSubscReqDataRm" data type, which either include in the "events" attribute a new element with the "event" attribute set to the value "OUT_OF_CREDIT" or remove from the "events" attribute the existing element with the "event" attribute set to the value "OUT_OF_CREDIT".

As a result of this action, the PCF shall set the appropriate subscription to out of credit notification for the corresponding PCC Rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.3.2.

4.2.3.23 Modification of Subscription to Service Data Flow QoS Monitoring Information

4.2.3.23.1 Modification of Subscriptions to QoS Monitoring at AF session level

This procedure is used by NF service consumer to modify the PCF subscription for notification about real-time measurements of QoS monitoring parameters for a QoS Flow (i.e. when the AF request involves only flows with the same QoS requirements), e.g. packet delay and/or congestion information between UPF and UE, when the "QoSMonitoring" feature is supported.

NOTE 1: When the subscription for QoS Monitoring is for multiple QoS flows, i.e, when the AF-session can contain multiple media components, the subscription for QoS monitoring can only be indicated within the corresponding "medSubComps" entry.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "evSubsc" attribute of "EventsSubscReqDataRm" data type, as follows:

- to create a subscription to notifications of QoS monitoring report for packet delay:

  a) shall include the "events" array with an array that contains a new entry per requested notification method with the "event" attribute set to "QOS_MONITORING", and notification related information as described in clause 4.2.2.23;

  b) when the "notifMethod" of the new entry is "EVENT_DETECTION", shall include a "qosMon" attribute with the QoS monitoring information for packet delay as described in clause 4.2.2.23.

NOTE 2: When the "congestMon" attribute is included, only the "EVENT_DETECTION" reporting frequency is applicable.

  c) shall include the new requested QoS monitoring parameter(s) to be measured (i.e. DL,/UL/round trip packet delay) within the "reqQosMonParams" attribute;

  d) may include the notification correlation identifier assigned by the AF within the "notifCorreId" attribute;

  e) if the feature "ExposureToEAS" is supported, may include the "directNotifInd" attribute set to true to indicate the direct event notification of QoS Monitoring data from the UPF; and

- when the "EnQoSMon" feature is supported, to create an event subscription to QoS monitoring, shall include:

  a) the "events" array with a new entry with the "event" attribute set to "QOS_MONITORING", the requested QoS monitoring parameter type within "qosMonParamType" attribute, and notification related information as described in clause 4.2.2.23.1; and

  b) the requested QoS monitoring parameter(s) to be measured within the "reqQosMonParams" attribute as described in clause 4.2.2.23.1; and

  c) when the "notifMethod" attribute is set to the value "EVENT_DETECTION", the corresponding threshold information, as described in clause 4.2.2.23.1; and

  may include the notification correlation identifier and direct notification indication, and/or averaging window, as specified in clause 4.2.2.23.1; and

- to remove a subscription to QoS monitoring information for packet delay:

  a) shall include the "events" array containing an array that shall omit the corresponding entry with the "event" attribute value "QOS_MONITORING";

  b) when the "notifMethod" attribute of the removed entry is "EVENT_DETECTION", it shall contain the "qosMon" attribute set to null;

c) if the "directNotifInd" attribute was previously provided, it shall contain the "directNotifInd" attribute set to null.

- when the "EnQoSMon" feature is supported, to remove a subscription to QoS monitoring:

a) shall include the "events" array and shall omit the corresponding entry(ies) with the "event" attribute value "QOS_MONITORING" and "qosMonParamType" attribute, if applicable;

b) if the notification method of the removed entry is "EVENT_DETECTION", it shall include the "qosMon" and/or "qosMonDatRate" and/or "congestMon" and/or "avlBitRateMon" attribute(s) set to null; and

c) if the "directNotifInd" attribute and/or the "avrgWndw" attribute was previously provided, it shall contain the "directNotifInd" attribute and/or the "avrgWndw" attribute set to null.

If the AF provided an indication of direct event notification, and the PCF determines that the QoS Monitoring reports cannot be notified directly (e.g. the AF requests for monitoring packet delay variation or round trip packet delay when UL and DL are on different service data flows and the information cannot be consolidated in the QoS monitoring policy in the PCC rule), the PCF generates a successful response to the AF and indicates that direct event notification is not possible by including within the "servAuthInfo" attribute the value "DIRECT_NOTIF_NOT_POSSIBLE", as described in clause 4.2.2.23.

As result of this action, the PCF shall set the appropriate subscription to QoS monitoring information for the corresponding active PCC rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.23.2 Modification of Subscriptions to QoS Monitoring for one or more service data flow(s)

When the "EnQoSMon" feature is supported, the NF service consumer may update the subscription to the monitoring of different QoS monitoring parameters provided per service data flow.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the provided or updated or deleted subscription values within the "evSubsc" attribute included within the "medSubComps" attribute of the "medComponents" array.

If the AF provided an indication of direct event notification, and the PCF determines that the QoS Monitoring reports cannot be notified directly (e.g. the AF requests for monitoring packet delay variation or round trip packet delay when UL traffic and DL traffic are mapped to twoQoS flows and the information cannot be consolidated in the QoS monitoring policy in the PCC rule), the PCF generates a successful response to the AF and indicates that direct event notification is not possible by including within the "servAuthInfo" attribute the value "DIRECT_NOTIF_NOT_POSSIBLE" and within the "directNotifReports" attribute, the QoS parameter that is not notified directly within the "qosMonParamType" attribute and the affected flows within the "flows" attribute, as described in clause 4.2.2.23.2.

As result of this action, the PCF shall set the appropriate subscription to QoS monitoring information for the corresponding active PCC rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.24 Update of TSCAI Input Information and TSC QoS related data

If the "TimeSensitiveNetworking" feature is supported, the NF service consumer may update the TSCAI Input container and the TSC QoS related data held in an "Individual Application Session Context" resource using the Npcf_PolicyAuthorization_Update service operation to modify the TSCAI input information and QoS characteristics delivered to the SMF for use in the 5G System.

The NF service consumer shall use the HTTP PATCH method as described in clause 4.2.3.2 to modify TSCAI input container and the TSC QoS related information.

The NF service consumer may indicate TSCAI input information and/or TSC QoS related information for new TSC streams by adding, in the "ascReqData" attribute, one or more media component entries within the "medComponents" attribute including the "tsnQos" attribute and including the "tscaiInputUl" attribute and/or the "tscaiInputDl" attribute and, when the feature "TimeSensitiveCommunication" is supported, the "tscaiTimeDom" attribute, if available, when

the feature "EnTSCAC" feature is supported, the "capBatAdaptation" attribute, if available, as described in clause 4.2.2.24.

The NF service consumer may update the TSCAI input information and/or the TSC QoS related information for existing TSC traffic by including the updated values in the "tscaiInputUl" attribute and/or "tscaiInputDl"attribute and/or updated values in the "tsnQos" attribute included in a media component entry of the "medComponents" attribute included in the "ascReqData" attribute.

The NF service consumer may delete the TSCAI input information and TSC QoS related information of removed TSC traffic by removing the corresponding media component entries within the "medComponents" attribute included in the "ascReqData" attribute.

Alternatively, when the "TimeSensitiveCommunication" and "AuthorizationWithRequiredQoS" features are supported, the NF service consumer (i.e., the TSCTSF or TSN AF) may update TSC QoS related information updating the "qosReference" attribute, and/or may indicate the update of the alternative service requirements updating the "altSerReqs" attribute as specified in clause 4.2.3.30.

When the "TimeSensitiveCommunication" and "AltSerReqsWithIndQoS" features are supported, the NF service consumer (i.e., the TSCTSF or TSN AF) may update TSC QoS related information updating the individual QoS requirement within the "tsnQos" attribute, and/or may indicate the update of the alternative service requirements updating the "altSerReqsData" attribute as specified in clause 4.2.3.30.

When the "EnTSCAC" feature is supported, the NF service consumer may update the request to notify the network-determined BAT offset and the optionally adjusted periodicity by using the "EventsSubscReqDataRm" data type and may update by including the event within the "events" attribute with the "event" attribute set to "BAT_OFFSET_INFO", if available, as described in clause 4.2.2.40.

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

The PCF shall check whether the received TSCAI input information and TSC QoS related information require to modify or to remove PCC rules in the SMF. Provisioning of PCC rule(s) to the SMF shall be carried out as specified in 3GPP TS 29.512 [8].

4.2.3.25 Provisioning of TSC user plane node management information and port management information

During the lifetime of a PDU session enabling Time Sensitive Communications, Time Synchronization and Deterministic Networking, the NF service consumer may provision or update, at any time, TSC user plane node management information and/or, when the DS-TT or the NW-TT functions are used, port management information for a port located in DS-TT and/or NW-TT and/or, the direct event notification of TSC management information from the UPF if the feature "ExposureToTSC" is supported.

If the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported the NF service consumer (i.e., the TSN AF or the TSCTSF) may provide a UMIC with TSC user plane management information for the UPF/NW-TT and/or a PMIC for the DS-TT port and/or PMIC(s) for the NW-TT ports with the respective port management information, to read and/or to update the configuration of the 5G system as a TSC user plane node by invoking the Npcf_PolicyAuthorization_Update service operation to the PCF.

The NF service consumer shall use the HTTP PATCH method as described in clause 4.2.3.2 to modify the "Individual Application Session Context" resource holding the UMIC and/or the DS-TT PMIC and/or NW-TT PMIC(s).

The NF service consumer may include in the "ascReqData" attribute:

- the DS-TT PMIC encoded in the "tsnPortManContDstt" and/or the one or more NW-TT PMIC(s)encoded in the "tsnPortManContNwtts", if available;
- the UMIC encoded in the "tsnBridgeManCont", if available; and/or
- if the feature "ExposureToTSC" is supported, the notification URI within the "tscNotifUri" attribute and the notification correlation identifier assigned by the TSCTSF or TSN AF within the "tscNotifCorreId" attribute, which, if available, indicates that the direct event notification of TSC management information from the UPF is requested.

As result of this action, the PCF shall provide the received DS-TT and/or NW-TT PMIC(s) and/or UMIC and/or the direct event notification of TSC management information for the corresponding PDU session as described in 3GPP TS 29.512 [8].

4.2.3.26 Modification of Mission Critical Services

The NF service consumer may include, in the "ascReqData" attribute, the "mcsId" attribute if it was not previously provided in order to indicate that the modified AF session relates to an MCS session.

If the NF service consumer supports the SBI message priority mechanism for an MCS session, the NF service consumer shall include the "3gpp-Sbi-Message-Priority" custom HTTP header towards the PCF as described in clause 4.2.2.12.

If the PCF receives the "mcsId" attribute, the PCF shall take specific actions on the corresponding PDU session to ensure that the MCS session is prioritised as defined in 3GPP TS 29.512 [8].

4.2.3.27 Support of CHEM feature

When CHEM feature is supported, the NF service consumer may include the value of Maximum Packet Loss Rate for UL within the "maxPacketLossRateUl" attribute and/or the value of Maximum Packet Loss Rate for DL within the "maxPacketLossRateDl" attribute in "medComponents" attribute. For CHEM feature, see Annex B.14.

4.2.3.28 Support of FLUS feature

If the "FLUS" feature is supported by the NF service consumer, the NF service consumer may include the "flusId" attribute within a media component of the "medComponents" attribute to indicate that the related media of the modified Individual Application Session Context resource corresponds to a FLUS media stream. Additional QoS information for the treatment of FLUS media may be provided within "desMaxLatency" attribute and/or "desMaxLoss" attribute.

4.2.3.29 Subscription to EPS Fallback report

When the "EPSFallbackReport" feature is supported, this procedure is used in the NF service consumer to subscribe to the notification of EPS Fallback events, if this event was not previously provisioned.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "evSubsc" attribute of the "EventsSubscReqDataRm" data type, which shall include in the "events" attribute a new element with the "event" attribute set to "EPS_FALLBACK". The NF service consumer shall request to the PCF to report EPS Fallback in conjunction with providing the PCF with NF service consumer service information for voice media type as described in clause 4.2.3.2, if the event was not previously provisioned as described in clause 4.2.2.30.

As result of this action, the PCF shall set the appropriate subscription to EPS Fallback for the corresponding active PCC rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.30 Modification of required QoS information

When the "AuthorizationWithRequiredQoS" feature is supported, this procedure is used by a NF service consumer to modify the required QoS by providing a different QoS reference(s) parameter while the AF session is ongoing. When the "AltSerReqsWithIndQoS" feature is supported, this procedure is used by a NF service consumer to modify the Requested Alternative QoS Parameter set(s).

The NF service consumer shall use the HTTP PATCH method to modify the required QoS information.

When the "AuthorizationWithRequiredQoS" feature is supported, the NF service consumer may include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, within one or more entries of the "medComponents" attribute included in the AppSessionContextUpdateData data type:

- a "qosReference" attribute, which may contain:

i. a QoS reference, that replaces an existing QoS reference value if the "qosReference" attribute was previously provisioned, or creates a new one if no "qosReference" attribute was previously provisioned;

ii. a "null" value, which removes a previously provisioned "qosReference" attribute value.

- an "altSerReqs" attribute, which may contain:

i. a prioritized list of alternative QoS references, which replaces an existing alternative QoS references list if the "altSerReqs" attribute was previously provisioned, or creates a new one if no "altSerReqs" attribute was previously provisioned;

ii. a "null" value, which removes a previously provisioned alternative QoS references list.

When the "AltSerReqsWithIndQoS" feature is supported, and the service QoS is provided, or was previously provided using individual QoS parameters (e.g. "marBwUl" and/or "marBwDl", attributes) instead of a QoS reference, the NF service consumer may include within one or more entries of the "medComponents" attribute:

- an "altSerReqsData" attribute, which may contain:

i. a prioritized list of alternative service requirements that include Requested Alternative QoS Parameter set(s), which replaces an existing list of alternative service requirements that include Requested Alternative QoS Parameter set(s) if the "altSerReqsData" attribute was previously provisioned, or creates a new one if no "altSerReqsData" attribute was previously provisioned;

ii. a "null" value, which removes a previously provisioned list of alternative service requirements that include individual QoS parameter sets.

NOTE: The modification of the individual QoS parameters is performed by provisioning within the "medComponents" attribute an update of the existing values or deleting the previously provided values, as described in clause 4.2.3.2.

When the "DisableUENotification" feature is supported, the NF service consumer may include a "disUeNotif" attribute, which may contain:

i. a "true" value if it was not provided or it was provided and set to "false";

ii. a "false" value if it was provided and set to "true".

When the NF service consumer provides the "altSerReqs" attribute containing a prioritized list of alternative QoS references or "altSerReqsData" attribute containing a prioritized list of alternative service requirements that include individual QoS parameter sets, the NF service consumer shall subscribe to receive notifications from the PCF when the resources associated to the corresponding service information have been allocated as described in clause 4.2.3.10 and when the GBR QoS targets for one or more service data flows can no longer (or can again) be guaranteed, as described in clause 4.2.3.6, if not previously subscribed.

Due to the updated required QoS information, the PCF may need to modify the related PCC rules as specified in 3GPP TS 29.513 [7] and provide the updated information towards the SMF following the corresponding procedures specified in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.31 Support of QoSHint feature

If the QoSHint feature is supported by the NF service consumer, the NF service consumer may include the "desMaxLatency" attribute and/or "desMaxLoss" attribute within a media component of the "medComponents" attribute to indicate that the related media of the modified Individual Application Session Context resource has specific latency and/or loss demands.

4.2.3.32 Modification of Subscription to Reallocation of Credit notification

This procedure is used by the NF service consumer if the "IMS_SBI" and the "ReallocationOfCredit" features are supported to modify in the PCF the subscription to notification about reallocation of credit for the Service Data Flows within the AF application session context.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "EventsSubscReqDataRm" data type, which either include in the "events" attribute a new element with the "event" attribute set to the value "REALLOCATION_OF_CREDIT" or remove from the "events" attribute the existing element with the "event" attribute set to the value "REALLOCATION_OF_CREDIT".

As a result of this action, the PCF shall set the appropriate subscription to reallocation of credit notification for the corresponding PCC Rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.3.2.

4.2.3.33 Modification of Subscription to satellite backhaul category changes

When the feature "SatelliteBackhaul"is supported, this procedure is used in the NF service consumer to modify in the PCF the subscription to notification about satellite backhaul category changes. When the feature "EnSatBackhaulCatChg" is supported, the procedure is also used to modify the subscription to notification about dynamic satellite backhaul category changes.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "EventsSubscReqDataRm" data type, which shall include in the "events" attribute a new element with the "event" attribute set to "SAT_CATEGORY_CHG" to indicate the subscription to changes of satellite backhaul category or changes between satellite backhaul and non-satellite backhaul.

As result of this action, the PCF shall set the appropriate subscription to satellite backhaul changes for the PDU session as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2. The PCF shall include the "evsNotif" attribute with an entry in the "evNotifs" array with the "event" attribute set to "SAT_CATEGORY_CHG" and the "satBackhaulCategory" attribute including the satellite backhaul category or the indication of non-satellite backhaul if the PCF has previously subscribed with the SMF to changes in this information.

4.2.3.34 Modification of Subscription to TSC user plane node related events

When the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported, this procedure is used to modify in the PCF the subscription to TSC user plane node related events.

The NF service consumer shall use the HTTP PATCH request message described in clause 4.2.3.2. To subscribe to this event, the NF service consumer shall include in the "evSubsc" attribute within the "ascReqData" attribute, the "events" attribute with a new element with the "event" attribute set to the value "TSN_BRIDGE_INFO". To unsubscribe to this event, the NF service consumer shall omit the event "TSN_BRIDGE_INFO" within the "events" attribute or, if all the subscribed events are being removed, shall set the "evSubsc" attribute to null.

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.3.2.

If the PCF stores updated PMIC/UMIC updated information, and the NF service consumer subscribed to the reception of TSC user plane information, the PCF shall include the "evsNotif" attribute with an entry in the "evNotifs" array with the "event" attribute set to "TSN_BRIDGE_INFO" and the "tsnBridgeManCont" attribute and/or the "tsnPortManContDstt" attribute and/or the "tsnPortManContNwtts" attribute as received from the SMF, if not previously reported.

As result of this action, the PCF shall modify the corresponding subscription to the report of TSC user plane node management information and port management information for the corresponding PDU session, if applicable, as described in 3GPP TS 29.512 [8].

4.2.3.35 Modification of the subscription to the report of extra UE addresses

When the feature "ExtraUEaddrReport" is supported, this procedure is used to modify in the PCF the subscription to the report of extra UE addresses.

The NF service consumer shall use the HTTP PATCH request message described in clause 4.2.3.2. To subscribe to this event, the NF service consumer shall include in the "evSubsc" attribute within the "ascReqData" attribute, the "events" array with a new element with the "event" attribute set to the value "EXTRA_UE_ADDR". To unsubscribe to this event, the NF service consumer shall omit the event "EXTRA_UE_ADDR" within the "events" attribute or, if all the subscribed events are being removed, shall set the "evSubsc" attribute to null.

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.3.2.

If the PCF received from the SMF the framed routes as described in 3GPP S 29.512 [8], clause 4.2.2.2, or the PCF received updates of the one or more IPv6 prefixes allocated to the UE as described in 3GPP TS 29.512 [8], clauses 4.2.4.2 and C.3.4.1, and the NF service consumer is subscribing to this event, the PCF shall include in the response the "evsNotif" attribute with an entry in the "evNotifs" array with the "event" attribute set to "EXTRA_UE_ADDR" and:

- the actual list of IPv4 addresses within the "ipv4AddrList" attribute, if one or more IPv4 framed routes are available in the PCF; or
- the actual list of IPv6 prefixes allocated to the UE within the "ipv6PrefixList" attribute, if one or more IPv6 framed routes are associated to the PDU session and are available in the PCF, or if the PCF keeps updated IPv6 prefix(es) information.

As result of this action, the PCF shall set the appropriate subscription to the report of UE IP addresses, if not previously subscribed, as described in in 3GPP TS 29.512 [8].

4.2.3.36 Modification of multi-modal services

This procedure is used by a NF service consumer to modify the provisioning of multi-modal services when "MultiMedia" feature is supported.

The NF service consumer shall use the HTTP PATCH method to modify the provisioning of multi-modal services.

The NF service consumer may include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, as follows:

- To modify one provisioned media flow of a multi-modal services, the NF service consumer shall provide the updated service information within the concerned entry of the "medComponents" map and the updated QoS monitoring requirements per media subcomponent within the "evSubsc" attribute included in the "medSubComps" map entry as specified in clause 4.2.3.23.2.
- To remove one of the provisioned media flow of a multi-modal service, the NF service consumer shall provide the media component number of the media flow that needs to be removed as the key of the "medComponents" map and set the value to NULL.

4.2.3.37 Modification of Round-Trip latency requirements

If the "RTLatency" feature is supported, the NF service consumer may update the RT latency requirement (i.e., provide RT latency requirement for a media component or for the UL and DL flows of two media components as described in clause 4.2.2.38 or update/delete RT latency requirements previously provided) using the HTTP PATCH method as described in clause 4.2.3.2 by invoking the Npcf_PolicyAuthorization_Update service operation.

Due to the change of Round-Trip latency requirements, the PCF may need to provision or modify the related PCC rules as specified in 3GPP TS 29.513 [7] and provide the related information towards the SMF following the corresponding procedures specified in clause 4.2.6.2.22 of 3GPP TS 29.512 [8].

4.2.3.38 Update of PDU Set handling related data

If the "PDUSetHandling" feature is supported, the NF service consumer may update the PDU Set handling related data held in an "Individual Application Session Context" resource using the Npcf_PolicyAuthorization_Update service

operation to modify the PDU Set QoS characteristics and/or Protocol Description delivered to the SMF for PDU Set handling in the 5G System.

The NF service consumer may indicate PDU Set handling related data for new service data flow(s) by adding, in the "ascReqData" attribute, one or more media component entries within the "medComponents" attribute including:

- the "pduSetQosDl" and/or "pduSetQosUl" attribute(s) to update PDU Set QoS related information;
- the "protoDescDl" and/or "protoDescUl" attribute(s) to update Protocol Description related information, as described in clause 4.2.2.39; and/or
- the "pduSetQosDl" and/or "pduSetQosUl" attribute(s) within the "altSerReqsData" attribute to update PDU Set QoS information in the Alternative QoS Parameter set(s).

The NF service consumer may modify the previously provided PDU Set handling related information for one or more service data flow(s) by providing the updated attribute(s) of the "pduSetQosDl", "pduSetQosUl", "protoDescDl" and/or "protoDescUl" attribute(s) within the concerned entry(ies) of the "medComponents" map.

The NF service consumer may delete the PDU Set handling related information of service data flow(s) by removing the previously provided "pduSetQosDl", "pduSetQosUl", "protoDescDl" and/or "protoDescUl" attribute(s) of the corresponding media component entry(ies) within the "medComponents" attribute included in the "ascReqData" attribute. The "pduSetErrRate" and the "pduSetDelayBudget" attribute shall be deleted simultaneously.

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

As result of this action, the PCF shall set the appropriate PDU Set handling parameters for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

4.2.3.39 Modification of Subscription to TSC user plane node related events

When the "EnTSCAC" feature is supported, this procedure is used to modify in the PCF the subscription to the BAT offset information notification.

The NF service consumer shall use the HTTP PATCH method to update the "Events Subscription" sub-resource together with the modifications to the "Individual Application Session Context" resource.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "EventsSubscReqDataRm" data type, which shall include in the "events" attribute a new element with the "event" attribute set to "BAT_OFFSET_INFO" to indicate the subscription to changes of the BAT offset and the optionally adjusted periodicity.

As result of this action, the PCF shall set the appropriate subscription to BAT offset notification for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

4.2.3.40 Modification of Packet Delay Variation monitoring requirement

This procedure is used by an NF service consumer to modify the subscription for notification about the variation of packet delay between UE and PSA UPF when the "EnQoSMon" feature is supported.

To modify the subscription at AF-session level to Packet Delay Variation, the NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the created or deleted or updated values of the "evSubsc" attribute of "EventsSubscReqDataRm" data type, as follows:

- to create a subscription to notifications of Packet Delay Variation reports, the NF service consumer shall behave as specified in clause 4.2.2.41;
- to remove a subscription to Packet Delay Variation reports, the NF service consumer shall include the "events" array without any entry with the "event" attribute value "PACK_DEL_VAR" and shall set to null the "pdvMon" attribute if previously provided.
- to modify a subscription to Packet Delay Variation reports, the NF service consumer shall replace the Packet Delay Variation monitoring parameter(s) and/or the Packet Delay Variation information within the "pdvMon" attribute and/or the event reporting information within the corresponding entry in the "events" attribute, as necessary.

NOTE: When the subscription to Packet Delay Variation is for one or more of the media components of the AF session, the subscription to Packet Delay Variation can only be indicated within the corresponding "medSubComps" entry.

To modify the subscription to Packet Delay Variation when the NF service consumer subscribes to the monitoring of Packet Delay Variation to one or more service data flow(s), the NF service consumer shall include the "medComponents" attribute as described in clause 4.2.3.2. For each media component that requires Packet Delay Variation measurements, the NF service consumer shall include the service data flow filter information within the "medSubComps" attribute and the Packet Delay Variation updated subscription information for the indicated flows within the "evSubsc" attribute.

As result of this action, the PCF shall determine the QoS Monitoring information to derive Packet Delay Variation measurements requested by the AF, if applicable, and shall modify the subscription for QoS Monitoring for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.41 Modification of traffic information for UE power saving management

If the "PowerSaving" feature is supported, the NF service consumer may provide the Uplink and/or Downlink Periodicity information and/or the DL Protocol Description.

The NF service consumer shall use the HTTP PATCH method as described in clause 4.2.3.2 to modify the "Individual Application Session Context" resource and shall provide the traffic information for a new or previously provided MediaComponentRm element within the "medComponents" attribute as defined in clause 4.2.2.42. The NF service consumer shall remove the periodicity and/or the DL protocol description by setting to null the "periodUl" and/or "periodDl" attributes and/or the "protoDescDl" attribute respectively contained in MediaComponentRm data structure.

As a result of this action, the PCF shall update the policies for periodicity and N6 jitter information and/or for End of Data Burst Handling to the SMF as described in 3GPP TS 29.512 [8].

4.2.3.42 Provisioning of the indication of ECN marking for L4S support

When the "L4S" feature is supported, this procedure is used by a NF service consumer to provide the explicit indication of whether the UL and/or DL service data flow of a new media component supports ECN marking for L4S.

The NF service consumer may include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, in the corresponding new media component(s) entries of the "medComponents" attribute, the "l4sInd" attribute with the indication of ECN marking for L4S support as described in clause 4.2.2.43. The NF service consumer shall also subscribe to receive notifications when the ECN marking for L4S support is not available or available again as described in clause 4.2.2.43, if not previously subscribed.

As result of this action, the PCF may indicate to the SMF to enable for ECN marking for L4S support for the service data flow of the new media component(s) following the procedures specified in clause 4.2.6.2.21 of 3GPP TS 29.512 [8].

4.2.3.43 Modification of Round-Trip delay over two QoS flows

This procedure is used by an NF service consumer to modify the subscription for notification about the Round-Trip delay measurements over two QoS flows when the "EnQoSMon" feature is supported.

The NF service consumer shall include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, the updated values of the "evSubsc" attribute of "EventsSubscReqDataRm" data type, as follows:

- to create a subscription to notifications of Round-Trip delay measurements over two QoS flows reports, the NF service consumer shall behave as specified in clause 4.2.2.44; and
- to remove a subscription to Round-Trip delay measurements over two QoS flows reports, the NF service consumer shall include the "events" array without any entry with the "event" value "RT_DELAY_TWO_QOS_FLOWS";

- to modify a subscription to Round-Trip delay measurements over two QoS flows reports, the NF service consumer shall replace the Round-Trip delay over two QoS flows and/or the event reporting information, as necessary.

As result of this action, the PCF shall determine the QoS Monitoring information to derive the Round-Trip delay measurements over two QoS flows requested by the AF, if applicable, and shall modify the subscription for QoS Monitoring for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8],

The PCF shall reply to the NF service consumer as described in clause 4.2.3.2.

4.2.3.44 Modification of the QoS timing information

When the "QoSTiming_5G" feature is supported, this procedure is used by the NF service consumer to indicate the time period when the requested QoS information should be applied. The NF service consumer may include in the HTTP PATCH request message described in clause 4.2.3.2.

Due to the change of QoS timing information, the PCF needs to adjust the timing information for allocating and releasing resources for the corresponding PCC rule(s).

As result of this action, the PCF shall allocate and release the resources for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

4.2.3.45 Modification of the subscription to the report of network support for QoS Monitoring

When the feature "QoSMonCapRepo" is supported, the NF service consumer may subscribe to receive notifications from the PCF of whether the requested QoS Monitoring Capability is not supported or is supported again as described in clauses 4.2.2.46 and 4.2.6.15, and may delete the subscription to QoS monitoring Capability report by providing within the "evSubsc" attribute the "events" attribute without event information for the "QOS_MON_CAP_REPO" event as described in clause 4.2.3.2.

4.2.3.46 Modification of the (S)RTP Multiplexed Media Identification Information

When the "MpxMedia" feature is supported, the NF service consumer may include in the HTTP PATCH request message described in clause 4.2.3.2, in the "ascReqData" attribute, in the corresponding "medSubComponent" entries of the "medComponents" attribute, the "mpxMediaUlInfos" and/or "mpxMediaDlInfos" attributes with the (S)RTP Multiplexed Media Identification Information.

If the PCF has not received UE support for (S)RTP Multiplexed Media Identification from SMF, the PCF may, based on local configuration, generate a successful response to the NF service consumer and set "servAuthInfo" attribute to MPX_MEDIA_NOT_SUPPORTED_IN_UE in the HTTP response message to the NF service consumer to indicate that the requested differentiated QoS for multiplexed media flows is not supported in the UE.

As a result of this action, the PCF shall update the policies for the multiplexed media flows to the SMF as described in 3GPP TS 29.512 [8].

4.2.3.47 Provisioning of dynamically changing traffic characteristics

If the "TrafficCharChange" feature is supported, this procedure is used by the NF service consumer to indicate that the Data Burst Size is marked in the header of the protocol identified by the DL Protocol Description.

The NF service consumer may modify the previously provided Data Burst Size Marking Indication by providing the updated attribute(s) in the "datBurstSizeInd" and "protoDescDl" within the "medComponents" map contained in the HTTP PATCH request message described in clause 4.2.3.2.

The PCF determines the policies for the identification and marking of the Data Burst Size and provisions the PCC rule as described in 3GPP TS 29.512 [8].

If the "TrafficCharChange" feature is supported, this procedure is used by the NF service consumer to indicate that TTNB is marked in the header of the protocol identified by the DL Protocol Description.

The NF service consumer may modify the previously provided TTNB Indication by providing the updated attribute(s) in the " timetoNextBurstInd" and "protoDescDl" within the "medComponents" map contained in the HTTP PATCH request message described in clause 4.2.3.2.

The PCF determines the policies for the identification of the TTNB and provisions the PCC rule as described in 3GPP TS 29.512 [8].

If the "TrafficCharChange" feature is supported, this procedure is used by the NF service consumer to request expedited data transfer with reflective QoS to expedite the data transfer of larger payload for Non-GBR service flow(s) of XR application.

The NF service consumer may provision the expedited data transfer with reflective QoS indication by providing the updated attribute(s) in the "expTranInd" and/or "protoDescDl" within the "medComponents" map contained in the HTTP PATCH request message described in clause 4.2.3.2.

The PCF determines the policies for the expedited data transfer with reflective QoS and provisions the PCC rule as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.3.48 Provisioning of deliver media related information for encrypted traffic

If the "OnPathN6MediaInfo" feature is supported, this procedure is used by the NF service consumer to indicate that the support of deliver media related information for encrypted traffic.

The NF service consumer may modify the previously provided on-path N6 Signaling Information by providing the updated attribute(s) in the "onPathN6SigInfo" and "protoDescDl" within the "medComponents" map contained in the HTTP PATCH request message described in clause 4.2.3.2.

The PCF determines the policies and provisions the PCC rule as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.3.49 Modification of the subscription to the report of Data Rate Limitation Information

When the feature "RateLimitReport" is supported, this procedure is used to modify in the PCF the subscription to the report of Data Rate Limitation information.

The NF service consumer shall use the HTTP PATCH request message described in clause 4.2.3.2. To subscribe to this event, the NF service consumer shall include in the "evSubsc" attribute within the "ascReqData" attribute, the "events" array with a new element with the "event" attribute set to the value "RATE_LIMIT_INFO_REPO". To unsubscribe from this event, the NF service consumer shall omit the event "RATE_LIMIT_INFO_REPO" within the "events" attribute or, if all the subscribed events are being removed, shall set the "evSubsc" attribute to null.

The PCF shall reply to the NF service consumer with an HTTP response message as described in clause 4.2.3.2. If the PCF determines the Data Rate Limitation Information for uplink and/or downlink for each media conponent, the PCF shall include the EventsNotification data type in response with the "evNotifs" array including an entry with the "event" attribute set to the value "RATE_LIMIT_INFO_REPO" and the "rateLimitRepoUl" and/or "rateLimitRepoDl" attribute including the maximum bitrate for uplink and/or downlink that is authorized for non-GBR service data flows.

4.2.3.50 Provisioning of RAN-Controlled UL Bitrate Recommendation Indication

If the "ExtQoSR19" feature is supported, this procedure is used by the NF service consumer to indicate the RAN-Controlled UL Bitrate Recommendation support for the media flow.

The NF service consumer may modify the previously provided the RAN-Controlled UL Bitrate Recommendation indication by providing the updated attribute in the "ulBrRecInd" within the "medComponents" map contained in the HTTP PATCH request message described in clause 4.2.3.2.

The PCF determines the policies for the RAN-Controlled UL Bitrate Recommendation indication and provisions the PCC rule as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.2.2.

4.2.4 Npcf_PolicyAuthorization_Delete service operation

4.2.4.1 General

The Npcf_PolicyAuthorization_Delete service operation provides means for the NF service consumer to delete the context of application session information.

The following procedures using the Npcf_PolicyAuthorization_Delete service operation are supported:

- AF application session context termination.
- Reporting usage for sponsored data connectivity.
- Termination of Multimedia Priority Services.
- Request and report of access network information.
- Termination of MCPTT.
- Termination of MCVideo.
- Priority sharing indication.
- Report of RAN-NAS release cause.
- Termination of Mission Critical Services.

4.2.4.2 AF application session context termination

This procedure is used to terminate an AF application session context for the service as defined in 3GPP TS 23.501 [2], 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

Figure 4.2.4.2-1 illustrates the application session context termination.

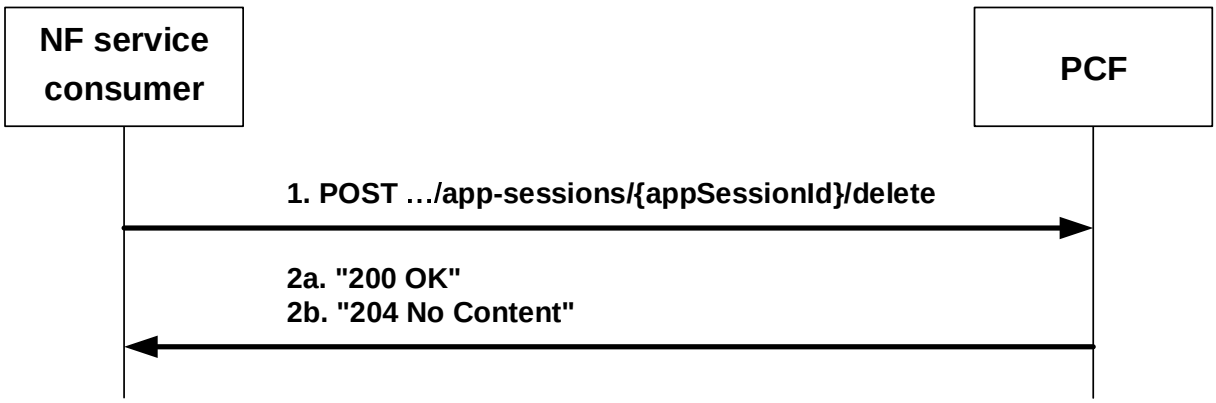


Figure 4.2.4.2-1: Application session context termination

When an AF session is terminated, and if the AF application session context was created as described in clause 4.2.2 or in clause 4.2.6.3, the NF service consumer shall invoke the Npcf_PolicyAuthorization_Delete service operation to the PCF using an HTTP POST request, as shown in figure 4.2.4.2-1, step 1.

The NF service consumer shall set the request URI to "{apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}/delete".

The NF service consumer may include in the body of the HTTP POST the "EventsSubscReqData" data type with the "events" attribute indicating the corresponding list of events to subscribe to.

When the PCF receives the HTTP POST request from the NF service consumer, indicating the termination of the AF application session context information, the PCF shall acknowledge that request by sending an HTTP response message with the corresponding status code.

If the HTTP POST request from the NF service consumer is accepted, the PCF shall send to the NF service consumer:

a) if event information is reported, a "200 OK" response to HTTP POST request, as shown in figure 4.2.4.2-1, step 2a, including in the "AppSessionContext" data type the "evsNotif" attribute, which encodes within "evNotifs" attribute the event to report to the NF service consumer, if available, as described in clause 4.2.5.2. If the event information is not available at the PCF the PCF shall defer sending the response to the NF service consumer and shall immediately configure the SMF to provide such information, as specified in 3GPP TS 29.512 [8];

b) otherwise, the PCF shall send to the NF service consumer a "204 No Content".

Afterwards, the PCF shall free the network resources allocated for the Service Data Flow(s) corresponding to the deleted AF application session context information. In order to do that, the PCF shall initiate the request for the removal of any related PCC rules from the SMF, if not previously done, following the corresponding procedures specified in 3GPP TS 29.512 [8].

If the HTTP POST request from the NF service consumer is rejected, the PCF shall indicate in the response to HTTP POST request the cause for the rejection as specified in clause 5.7.

4.2.4.3 Reporting usage for sponsored data connectivity

When "SponsoredConnectivity" is supported, and the NF service consumer indicated to enable sponsored data connectivity and the NF service consumer provided usage thresholds for such sponsor to the PCF, the PCF shall report accumulated usage to the NF service consumer using the response of the Npcf_PolicyAuthorization_Delete service operation.

This procedure is initiated when:

- the "Individual Application Session Context" is deleted by the NF service consumer; or

- the PCF requests the deletion of the "Individual Application Session Context" to the NF service consumer, as described in clause 4.2.5.3, due to PDU session termination, the termination of all the service data flows of the AF session or the home operator policy disallowing the UE accessing the sponsored data connectivity in the roaming case.

To report the accumulated usage, the PCF shall immediately configure the SMF to retrieve the accumulated usage as specified in 3GPP TS 29.512 [8]. When the PCF receives the usage information from the SMF, the PCF shall notify the NF service consumer by including the "EventsNotification" data type in the response of the HTTP POST request as described in clause 4.2.4.2.

The PCF shall include:

- an event of the "AfEventNotification" data type in the "evNotifs" attribute with the matched event "USAGE_REPORT" in the "event" attribute; and

- the usage encoded in the "usgRep" attribute.

4.2.4.4 Void

4.2.4.5 Termination of Multimedia Priority Services

If the AF session being terminated corresponds to an MPS session, the PCF shall delete the PCC rules corresponding to the MPS session and the PCF shall revoke the actions related to the prioritization of the MPS session in the corresponding PDU Session as defined in 3GPP TS 29.512 [8].

If the AF session being terminated corresponds to an MPS for DTS session, the PCF shall revoke MPS for DTS session in the corresponding PDU Session as defined in 3GPP TS 29.512 [8].

4.2.4.6 Request and report of access network information

This procedure is used by a NF service consumer to request the PCF to report the access network information (i.e. user location and/or user timezone information) at the deletion of the "Individual Application Session Context" resource when the "NetLoc" feature is supported and to report serving satellite identifier within the access network information, if available and when the "UeSatUeComm" feature is supported.

This procedure is initiated when:

- the "Individual Application Session Context" is deleted by the NF service consumer; or
- the PCF requests the deletion of the "Individual Application Session Context" from the NF service consumer, as described in clause 4.2.5.3, due to PDU session termination or the termination of all the service data flows of the AF session.

The NF service consumer shall include in the HTTP POST request message described in clause 4.2.4.2:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
  a) the "event" attribute set to the value "ANI_REPORT"; and
  b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information and/or if the "UeSatUeComm" feature is supported, identifier of the serving satellite of the UE.

When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature, the PCF shall respond to the NF service consumer including in the "EventsNotification" data type the "noNetLocSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

When the PCF receives the access network information from the SMF, the PCF shall provide the corresponding access network information to the NF service consumer by including the "EventsNotification" data type in the "200 OK" response to the HTTP POST request. The PCF shall include:

- in case of 3GPP access, the user location information in the "eutraLocation" or in the "nrLocation" attribute in the "ueLoc" attribute, if available and required;
- in case of untrusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:
  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available;
  b) the UDP source port or the TCP source port in the "portNumber" and "protocol" attributes, if available; and
  c) if the "WLAN_Location" feature is supported, the WLAN location information encoded in the "twapId" attribute, if available, that shall consist of:
     i. the SSID in the "ssId" attribute;
     ii. the BSSID the "bssId" attribute if available; and
     iii. the civic address in the "civicAddress" attribute if available;

NOTE 1: When the UE reaches the ePDG via a NAT, the combination of UE local IP address and the UE source port is needed for lawful interception purposes. The UE source port may be either a UDP or a TCP port, and it is indicated in the "protocol" attribute.

- in case of trusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:
  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available; and
  b) the UDP source port in the "portNumber" attribute if available; and

NOTE 2: The UDP protocol can be used between the UE and the TNGF to enable NAT traversal.

c) either the TNAP identifier encoded in the "tnapId" attribute or the TWAP identifier encoded in the "twapId" attribute. The TNAP identifier and the TWAP identifier shall consist of:

i. the SSID in the "ssId" attribute;

ii. the BSSID the "bssId" attribute if available; and

iii. the civic address in the "civicAddress" attribute if available;

- if user location was required, the time when it was last known in the "ueLocTime" attribute if available;

NOTE 3: The PCF derives the value of the "ueLocTime" attribute from the "userLocationInfoTime" attribute received from the SMF as specified in 3GPP TS 29.512 [8].

- the serving network identity i.e. the PLMN Identifier (the PLMN network code and the country code) or the SNPN Identifier (the PLMN Identifier and the NID) in the "plmnId" attribute, if user location information is required but not available in any access; and/or

- the UE timezone in the "ueTimeZone" attribute if required and available; and/or

- if the "UeSatUeComm" feature is supported, the serving satellite identifier in the "servSatId" attribute if available, when the UE is accessible via regenerative satellite access.

NOTE 4: The PCF forwards both 3GPP and non-3GPP access UE locations in the "ueLoc" attribute when both UE locations are provided by the SMF as defined in 3GPP TS 29.512 [8].

When the PCF receives from the SMF that the access network does not support access network information report, the PCF shall include the "noNetLocSupp" attribute set to "ANR_NOT_SUPPORTED", "TZR_NOT_SUPPORTED", "UE_SAT_NOT_SUPPORTED" or "LOC_NOT_SUPPORTED" value received from the SMF in the "EventsNotification" data type in the "200 OK" response to the HTTP POST request.

The PCF shall also include an event of the "AfEventNotification" data type in the "evNotifs" attribute with the "event" attribute set to the value "ANI_REPORT".

4.2.4.7 Termination of MCPTT

If the "Individual Application Session Context" resource being removed corresponds to an MCPTT session, the PCF shall delete the PCC rules corresponding to the MCPTT session and the PCF shall revoke the actions related to the prioritization of the MCPTT session in the corresponding PDU Session as defined in 3GPP TS 29.512 [8].

4.2.4.8 Termination of MCVideo

If the "Individual Application Session Context" resource being removed corresponds to an MCVideo session, the PCF shall delete the PCC rules corresponding to the MCVideo session and the PCF shall revoke the actions related to the prioritization of the MCVideo session in the corresponding PDU Session as defined in 3GPP TS 29.512 [8].

4.2.4.9 Priority sharing indication

If the "Individual Application Session Context" resource being removed included the "prioSharingInd" attribute set to "ENABLED" within a media component of the "medComponents" attribute, if the related media flow(s) was in priority sharing with other media flows the PCF should readjust the Allocation and Retention Priority for the remaining services sharing Allocation and Retention Priority as described in 3GPP TS 29.512 [8], clause 4.2.6.2.9 and handle the media flow removed according to normal PCC/QoS rule provisioning procedures described in 3GPP TS 29.512 [8], clause 4.2.6.2.

4.2.4.10 Report of RAN-NAS release cause

This procedure is used by a PCF to report about the RAN-NAS release cause together with access network information (i.e. user location and/or user timezone information) at the deletion of the "Individual Application Session Context" resource when the "RAN-NAS-Cause" feature is supported and also include serving satellite identifier within the access network information, if available and when the "UeSatUeComm" feature is supported.

This procedure is initiated when:

- the "Individual Application Session Context" is deleted by the NF service consumer; or
- the PCF requests the deletion of the "Individual Application Session Context" from the NF service consumer, as described in clause 4.2.5.3, due to PDU session termination or the termination of all the service data flows of the AF session.

The PCF shall immediately configure the SMF to provide such RAN-NAS release cause together with access information, as specified in 3GPP TS 29.512 [8].

NOTE 1: The PCF does not need to interact with the SMF during this delete operation if it has already received the RAN-NAS release cause from the SMF.

When the PCF receives the RAN-NAS release cause and access network information from the SMF, the PCF shall provide the corresponding access network information and RAN-NAS release cause to the NF service consumer by including the "EventsNotification" data type in the "200 OK" response to the HTTP POST request. The PCF shall include:

- in case of 3GPP access, the user location information in the "eutraLocation" or in the "nrLocation" attribute in the "ueLoc" attribute, if available;
- in case of untrusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if available, as follows:

  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute;

  b) the UDP source port or the TCP source port in the "portNumber" and "protocol" attributes if available; and

  c) if the "WLAN_Location" feature is supported, the WLAN location information encoded in the "twapId" attribute, if available, that shall consist of:

     i. the SSID in the "ssId" attribute;

     ii. the BSSID the "bssId" attribute if available; and

     iii. the civic address in the "civicAddress" attribute if available;

NOTE 2: When the UE reaches the ePDG via a NAT, the combination of UE local IP address and the UE source port is needed for lawful interception purposes. The UE source port may be either a UDP or a TCP port, and it is indicated in the "protocol" attribute.

- in case of trusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if available, as follows:

  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available; and

  b) the UDP source port in the "portNumber" attribute if available; and

NOTE 3: The UDP protocol can be used between the UE and the TNGF to enable NAT traversal.

  c) either the TNAP identifier encoded in the "tnapId" attribute or the TWAP identifier encoded in the "twapId" attribute. The TNAP identifier and the TWAP identifier shall consist of:

     i. the SSID in the "ssId" attribute;

     ii. the BSSID the "bssId" attribute if available; and

     iii. the civic address in the "civicAddress" attribute if available;

- the serving network identity i.e. the PLMN Identifier (the PLMN network code and the country code) or the SNPN Identifier (the PLMN Identifier and the NID) in the "plmnId" attribute, if user location information is not available in any access;
- the UE timezone in the "ueTimeZone" attribute if available;
- the RAN and/or NAS release cause in the "ranNasRelCauses" attribute, if available; and

- if the "UeSatUeComm" feature is supported, the serving satellite identifier within the "servSatId" attribute if available, when the UE is accessible via regenerative satellite access.

The PCF shall also include an event of the "AfEventNotification" data type in the "evNotifs" attribute with the "event" attribute set to the value "RAN_NAS_CAUSE".

4.2.4.11 Termination of Mission Critical Services

If the AF session being terminated corresponds to an MCS session, the PCF shall delete the PCC rules corresponding to the MCS session and the PCF shall revoke the actions related to the prioritisation of the MCS session in the corresponding PDU Session as defined in 3GPP TS 29.512 [8].

4.2.4.12 Void

4.2.5 Npcf_PolicyAuthorization_Notify service operation

4.2.5.1 General

The Npcf_PolicyAuthorization_Notify service operation enables notification to NF service consumers that the previously subscribed event for the existing application session context occurred or that the application session context is no longer valid.

The following procedures using the Npcf_PolicyAuthorization_Notify service operation are supported:

- Notification about application session context event.
- Notification about application session context termination.
- Notification about Service Data Flow QoS notification control.
- Notification about service data flow deactivation.
- Reporting usage for sponsored data connectivity.
- Notification of resources allocation outcome.
- Reporting access network information.
- Notification of signalling path status.
- Notification about out of credit.
- Notification about TSC user plane node management information and/or port management information, Individual Application Session Context exists.
- Notification about Service Data Flow QoS Monitoring control.
- Report of EPS Fallback.
- Notification about TSC user plane node Information, no Individual Application Session Context exists.
- Notification about reallocation of credit.
- Notification of MPS for DTS outcome.
- Notification about application detection information.
- Notification about satellite backhaul category changes.
- Notification about UP path change enforcement failure.
- Notification about PDU session established/terminated events.
- Notification about extra UE addresses.

- Notification about BAT offset.
- Notification about URSP rule enforcement information.
- Notification about Packet Delay Variation.
- Notification about 5GS support for Policy Control for L4S.
- Notification about Round-Trip delay monitoring measurements over two QoS flows.
- Event notification for AF requested QoS for a UE or group of UE(s) not identified by UE address(es).
- Notification about network support of QoS Monitoring.
- Notification about released the application session context.
- Notification about Data Rate limitation information.
- Notification about UP path management events.
- Reporting 5GC network functions health monitoring information.

4.2.5.2 Notification about application session context event

This procedure is invoked by the PCF to notify the NF service consumer when a certain, previously subscribed, application session context event occurs, as defined in 3GPP TS 23.501 [2], 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

Figure 4.2.5.2-1 illustrates the notification about application session context event.

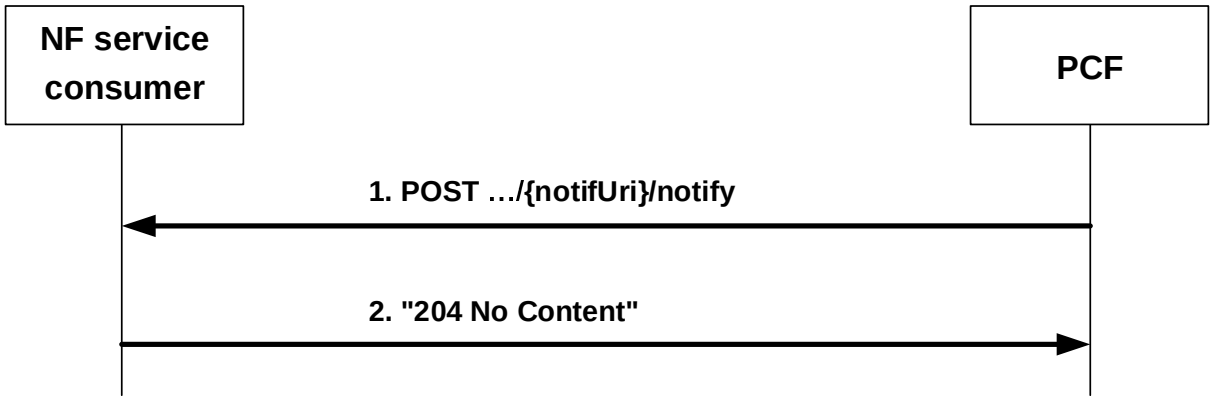


Figure 4.2.5.2-1: Notification about application session context event

When the PCF determines that the event for the existing AF application session context, to which the NF service consumer has subscribed to, occurred e.g. upon reception of an event notification for a PDU session from the SMF as described in 3GPP TS 29.512 [8], the PCF shall invoke the Npcf_PolicyAuthorization_Notify service operation by sending the HTTP POST request (as shown in figure 4.2.5.2-1, step 1) to the NF service consumer using the notification URI received in the subscription creation (or modification), as specified in clause 4.2.6, and appending the "notify" segment path at the end of the URI. The PCF shall provide in the body of the HTTP POST request the "EventsNotification" data type including:

- the Events Subscription resource identifier related with the notification in the "evSubsUri" attribute; and
- the list of the reported events in the "evNotifs" attribute. For each reported event, the "AfEventNotification" data type shall include the event identifier and may include additional event information.

The PCF shall include:

- if the NF service consumer subscribed to the "PLMN_CHG" event, the "event" attribute set to "PLMN_CHG" and the "plmnId" attribute including the PLMN Identifier or the SNPN Identifier if the PCF has requested to be updated with this information in the SMF;

NOTE 1: The SNPN Identifier consists of the PLMN Identifier and the NID.

NOTE 2: Handover between non-equivalent SNPNs, and between SNPN and PLMN is not supported. When the UE is operating in SNPN access mode, the trigger reports changes of equivalent SNPNs.

- if the NF service consumer subscribed to the event "ACCESS_TYPE_CHANGE", the "event" attribute set to "ACCESS_TYPE_CHANGE" and:

  i. the "accessType" attribute including the access type, and the "ratType" attribute including the RAT type when applicable for the notified access type; and/or

  ii. if the "ATSSS" feature is supported and the PDU session is a MA PDU session:

    a. if it is the first access type report, and both, 3GPP and non-3GPP access information is available, the "addAccessInfo" attribute. The "addAccessInfo" attribute contains the additional access type information, where the access type is encoded in the "accessType" attribute, and the RAT type is encoded in the "ratType" attribute when applicable for the notified access type;

    b. if it is a subsequent access type change report:

      - if a new access type is added to the MA PDU session, the"addAccessInfo" attribute with the added access type encoded in the "accessType" attribute, and the RAT type encoded in the "ratType" attribute when applicable for the notified access type;

      - if an access type is released to the MA PDU session, the "relAccessInfo" attribute with the released access type encoded in the "accessType" attribute, and the RAT type encoded in the "ratType" attribute when applicable for the notified access type; and

NOTE 3: For a MA PDU session, if the "ATSSS" feature is not supported by the NF service consumer the PCF shall include the "accessType" attribute and the "ratType" attribute with a currently active combination of access type and RAT type. When both 3GPP and non-3GPP accesses are available, the PCF includes the information corresponding to the 3GPP access and only changes on activation and deactivation of 3GPP access are reported.

  iii. the "anGwAddr" attribute including access network gateway address when available;

- if the "IMS_SBI" feature is supported and if the NF service consumer subscribed to the "CHARGING_CORRELATION" event, the "event" attribute set to "CHARGING_CORRELATION" and may include the "anChargIds" attribute containing the access network charging identifier(s) and the "anChargAddr" attribute containing the access network charging address; and

- if the "UEUnreachable" feature is supported and the NF service consumer subscribed to the "UE_REACH_STATUS_CH" event, the "event" attribute set to "UE_REACH_STATUS_CH" together with the "ueReachStatus" attribute containing the corresponding UE reachability status, and in case the "ueReachStatus" attribute is set to "UNREACHABLE", optionally the "retryAfter" attribute if received from the SMF.

The NF service consumer notification of other specific events using the Npcf_PolicyAuthorization_Notify request is described in the related clauses.

Upon the reception of the HTTP POST request from the PCF indicating that the PDU session and/or service related event occurred, the NF service consumer shall acknowledge that request by sending an HTTP response message with the corresponding status code.

If the HTTP POST request from the PCF is accepted, the NF service consumer shall acknowledge the receipt of the event notification with a "204 No Content" response to HTTP POST request, as shown in figure 4.2.5.2-1, step 2.

If the HTTP POST request from the PCF is not accepted, the NF service consumer shall indicate in the response to HTTP POST request the cause for the rejection as specified in clause 5.7.

If the feature "ES3XX" is supported, and the NF service consumer determines the received HTTP POST request needs to be redirected, the NF service consumer shall send an HTTP redirect response as specified in clause 6.10.9 of 3GPP TS 29.500 [5].

4.2.5.3 Notification about application session context termination

This procedure is invoked by the PCF to notify the NF service consumer that the application session context is no longer valid, as defined in 3GPP TS 23.501 [2], 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

Figure 4.2.5.3-1 illustrates the notification about application session context termination.

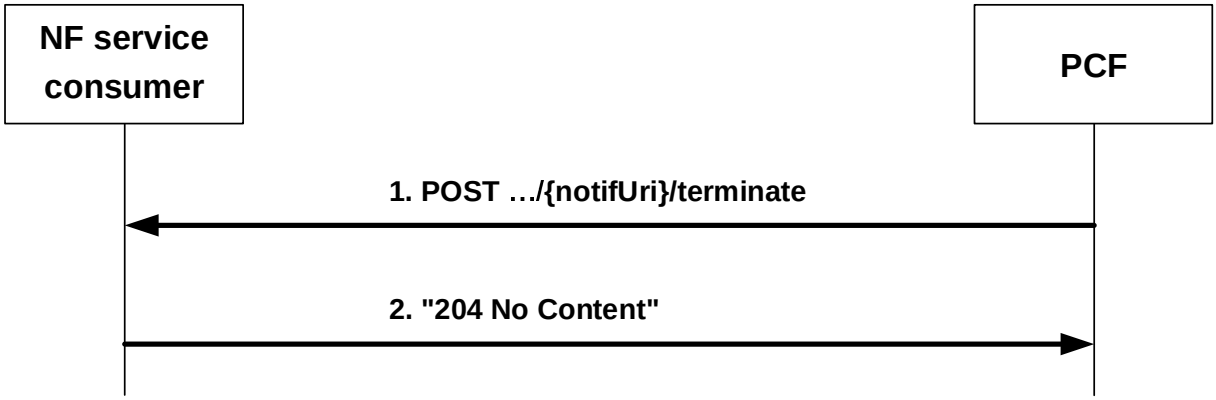


Figure 4.2.5.3-1: Notification about application session context termination

When the PCF determines that the AF application session context is no longer valid, the PCF shall invoke the Npcf_PolicyAuthorization_Notify service operation by sending the HTTP POST request (as shown in figure 4.2.5.3-1, step 1) using the notification URI received in the "Individual Application Session Context" context creation, as specified in clause 4.2.2 and clause 4.2.6.3, and appending the "terminate" segment path at the end of the URI, to trigger the NF service consumer to request the application session context termination (see clause 4.2.4.2). The PCF shall provide in the body of the HTTP POST request the "TerminationInfo" data type including:

- the Individual Application Session Context resource identifier related to the termination notification in the "resUri" attribute; and
- the application session context termination cause in the "termCause" attribute of the "TerminationCause" data type, indicating:

  i) "PDU_SESSION_TERMINATION" when the PCF received from the SMF the indication of SM Policy Context termination without a specific PDU session release cause value;

  ii) "ALL_SDF_DEACTIVATION" when the PCF received from the SMF the indication that all the SDFs of the Individual Application Session Context resource are deactivated or all resource allocation of an Individual Application Session Context fails because other reasons than "PS_TO_CS_HAN";

  iii) "PS_TO_CS_HO" if the "IMS_SBI" feature is supported and the PCF received from the SMF:

    a) the PDU session release cause value "PS_TO_CS_HO"; or

    b) the failure code value "PS_TO_CS_HAN" for all the SDFs of the Individual Application Session Context resource;

  iv) "INSUFICIENT_SERVER_RESOURCES" when the PCF is overloaded;

  v) "INSUFFICIENT_QOS_FLOW_RESOURCES" when the PCF received that the maximum number of QoS flows for the PDU session is reached or there was a QoS flow resource limitation error;

  vi) "SPONSORED_DATA_CONNECTIVITY_DISALLOWED" when the PCF detects that due to operator policy the UE accessing the sponsored data connectivity is disallowed;

  vii) "REQUEST_QOS_NOT_SUPPORTED_IN_PLMN" when the PCF detects that the requested QoS for an existing AF application session context is not supported in the current serving PLMN, and if the "VPLMNErrorRep" feature is supported;

  viii) "UE_ADDR_RELEASE" if the "ReleasedUEaddrReport" feature is supported and when the PCF received from the SMF the released UE address as described in clause 4.2.5.31;

ix) "SMF_FAILURE" if the "SmfFailureDetection" feature is supported and if the PCF detected due to operator configuration that the SMF has failed;

x) "REFLECTIVE_QOS_NOT_SUPPORTED_IN_UE" if the "TrafficCharChange" feature is supported and when the PCF detects the UE does not support reflective QoS.

Upon the reception of the HTTP POST request from the PCF requesting the application session context termination, the NF service consumer shall acknowledge that request by sending an HTTP response message with the corresponding status code.

If the HTTP POST request from the PCF is accepted, the NF service consumer shall acknowledge the receipt of the application session context termination request with a "204 No Content" response to HTTP POST request (as shown in figure 4.2.5.3-1, step 2) and shall invoke the Npcf_PolicyAuthorization_Delete service operation to the PCF as described in clause 4.2.4.

If the HTTP POST request from the PCF is not accepted, the NF service consumer shall indicate in the response to HTTP POST request the cause for the rejection as specified in clause 5.7.

If the feature "ES3XX" is supported, and the NF service consumer determines the received HTTP POST request needs to be redirected, the NF service consumer shall send an HTTP redirect response as specified in clause 6.10.9 of 3GPP TS 29.500 [5].

4.2.5.4 Notification about Service Data Flow QoS notification control

When the PCF gets the knowledge that one or more SDFs:

- cannot guarantee the GBR QoS targets; or
- can guarantee again the GBR QoS targets;

the PCF shall inform the NF service consumer accordingly if the AF has previously subscribed as described in clauses 4.2.2.6 and 4.2.3.6.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "QOS_NOTIF" in the "event" attribute; and
- the "qncReports" array with:

  a) the "notifType" attribute to indicate whether the GBR targets for the indicated SDFs are "NOT_GUARANTEED" or "GUARANTEED" again, or if the "ExtQoSR19" feature is supported, "NOT_GUARANTEED_DL" in DL direction or "NOT_GUARANTEED_UL" in UL direction;

  b) the identification of the affected service flows (if not all the flows are affected) encoded in the "flows" attribute if applicable; and

  c) if the "AuthorizationWithRequiredQoS" feature or the "AltSerReqsWithIndQoS" feature as defined in clause 5.8 is supported, the reference to the Alternative Service Requirement corresponding alternative QoS parameter set if received from the SMF within the "altSerReq" attribute. When the "altSerReq" attribute is omitted and the "notifType" attribute is NOT_GUARANTEED, or if the "ExtQoSR19" feature is supported, "NOT_GUARANTEED_DL" and/or "NOT_GUARANTEED_UL", it indicates that the lowest priority alternative service requirement could not be fulfilled.

When the "AuthorizationWithRequiredQoS" and "AltQoSProfilesSupportReport" features as defined in clause 5.8 are supported, and the AF included during the media component provisioning the "altSerReqs" attribute for the concerned media components(s), or the "AltSerReqsWithIndQoS" and "AltQoSProfilesSupportReport" features are supported and the AF included during media component provisioning the "altSerReqsData" attribute for the concerned media component(s), if the PCF receives from the SMF the indication that the GBR QoS targets cannot be guaranteed and the indication that alternative QoS profiles are not supported in the NG-RAN where the UE is currently located as specified in in 3GPP TS 29.512 [8], the PCF may include within the QosNotificationControlInfo data structure the "altSerReqNotSuppInd" attribute set to true. When the Alternative QoS profiles are supported by the NG-RAN where

the UE is currently located, the PCF may omit or set the "altSerReqNotSuppInd" attribute to false, as indicated by the SMF.

If "MediaComponentVersioning" feature is supported, and if the content version was included when the corresponding media component was provisioned, the "flows" attribute shall also contain the "contVers" attribute including the content version(s) of the media components. The PCF shall include more than one entry in the "contVers" attribute for the same media component if the PCF has received multiple content versions as described in clause 4.2.6.2.14 in 3GPP TS 29.512 [8].

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF. The NF service consumer may also update the AF application session context information by sending an HTTP PATCH request to the PCF.

Signalling flows for Service Data Flow QoS notification control are presented in 3GPP TS 29.513 [7].

4.2.5.5 Notification about Service Data Flow Deactivation

When the PCF gets the knowledge that one or more SDFs have been deactivated, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed as described in clauses 4.2.2.7 and 4.2.3.7.

When not all the service data flows within the AF application session context are affected, the PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include within the "evNotifs" attribute an event of "AfEventNotification" data type indicating the matched event ("FAILED_RESOURCES_ALLOCATION" if the resources could not be allocated or "UE_TEMPORARILY_UNAVAILABLE" if the UE was temporarily unavailable) in the "event" attribute and the deactivated service data flows (if not all the flows are affected) encoded in the "flows" attribute.

NOTE 1: If the PCF detects that the PCC rules related to an AF application session context cannot be installed or modified because there is a temporary network failure (e.g. SGW failed according to clause B.3.3.3 or B.3.4.9 of 3GPP TS 29.512 [8]) and if requested by the AF, the PCF can notify the AF of the event "FAILED_RESOURCES_ALLOCATION".

NOTE 2: For the services bound to a MA PDU session with the steering mode is "REDUNDANT", the "FAILED_RESOURCES_ALLOCATION" event is only triggered when the network could not allocate resources over none of the available multiple access types (see clause 4.2.6.2.17 in 3GPP TS 29.512 [8]).

If the "MediaComponentVersioning" feature is supported, and if the content version was included when the corresponding media component was provisioned as described in clause 4.2.5.8, the PCF shall also include in the "flows" attribute the "contVers" attribute with the content version(s) of the media components.

If the "RAN-NAS-Cause" feature is supported and the PCF received the RAN-NAS release cause and access network information from the SMF, the PCF shall provide in the "EventsNotification" data type of the HTTP POST request:

- in case of 3GPP access, the user location information in the "eutraLocation" or in the "nrLocation" attribute in the "ueLoc" attribute, if available;
- in case of untrusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if available, as follows:
  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute;
  b) the UDP source port or the TCP source port in the "portNumber" and "protocol" attributes, if available; and
  c) if the "WLAN_Location" feature is supported, the WLAN location information encoded in the "twapId" attribute, if available, that shall consist of:
     i. the SSID in the "ssId" attribute;
     ii. the BSSID the "bssId" attribute if available; and
     iii. the civic address in the "civicAddress" attribute if available;

NOTE 3: When the UE reaches the ePDG via a NAT, the combination of UE local IP address and the UE source port is needed for lawful interception purposes. The UE source port may be either a UDP or a TCP port, and it is indicated in the "protocol" attribute.

- in case of trusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if available, as follows:

  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available; and

  b) the UDP source port in the "portNumber" attribute if available; and

NOTE 4: The UDP protocol can be used between the UE and the TNGF to enable NAT traversal.

  c) either the TNAP identifier encoded in the "tnapId" attribute or the TWAP identifier encoded in the "twapId" attribute. The TNAP identifier and the TWAP identifier shall consist of:

     i. the SSID in the "ssId" attribute;

     ii. the BSSID the "bssId" attribute if available; and

     iii. the civic address in the "civicAddress" attribute if available;

- the serving network identity i.e. the PLMN Identifier (the PLMN network code and the country code) or the SNPN Identifier (the PLMN Identifier and the NID) in the "plmnId" attribute, if user location information is not available in any access;

- the UE timezone in the "ueTimeZone" attribute if available;

- the RAN and/or NAS release cause in the "ranNasRelCauses" attribute, if available; and

- if the "UeSatUeComm" feature is supported, the serving satellite identifier within the "servSatId" attribute if available, when the UE is accessible via regenerative satellite access.

NOTE 5: The PCF forwards both 3GPP and non-3GPP access UE locations in the "ueLoc" attribute when both UE locations are provided by the SMF as defined in 3GPP TS 29.512 [8].

The PCF shall include in the "evNotifs" attribute, together with the event "FAILED_RESOURCES_ALLOCATION", an event of the "AfEventNotification" data type with the "event" attribute set to the value "RAN_NAS_CAUSE".

The PCF shall include more than one entry in the "contVers" attribute for the same media component if the PCF has received multiple content versions as described in clause 4.2.6.2.14 in 3GPP TS 29.512 [8].

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF. The NF service consumer may also update the AF application session context information by sending an HTTP PATCH request to the PCF.

When all the service data flows within the AF session are affected, the PCF shall inform the NF service consumer by sending a notification about application session context termination as defined in clause 4.2.5.3.

Signalling flows for Service Data Flow Deactivation cases are presented in 3GPP TS 29.513 [7].

4.2.5.6 Reporting usage for sponsored data connectivity

When "SponsoredConnectivity" is supported, the NF service consumer enabled sponsored data connectivity and the NF service consumer provided usage thresholds for such sponsor to the PCF, the PCF shall report accumulated usage to the NF service consumer using the Npcf_PolicyAuthorization_Notify service operation when:

- the PCF detects that the usage threshold provided by the NF service consumer has been reached; or

- the NF service consumer disables the sponsored data connectivity.

The PCF shall notify the NF service consumer of the accumulated usage by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- an event of the "AfEventNotification" data type in the "evNotifs" attribute with the matched event "USAGE_REPORT" in the "event" attribute; and

- the accumulated usage, corresponding to the usage since the last report to the AF, encoded in the "usgRep" attribute.

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF. The NF service consumer may terminate the AF session sending an HTTP POST as described in clause 4.2.4.2 or update the AF application session context information by providing a new usage threshold sending an HTTP PATCH request to the PCF as described in clause 4.2.3.5 or an HTTP PUT request to the PCF as described in clause 4.2.6.4.

NOTE: Once the accumulated usage is reported by the PCF to the AF, the monitoring will not start until the PCF receives the new threshold from the NF service consumer and provides it to the SMF.

4.2.5.7 Void

4.2.5.8 Notification about resources allocation outcome

When the PCF becomes aware that the resources associated to service information for one or more SDFs have been allocated, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "SUCCESSFUL_RESOURCES_ALLOCATION" event as described in clauses 4.2.2.10 and 4.2.3.10. The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2. The PCF shall include in the "evNotifs" attribute an entry with the "event" attribute set to "SUCCESSFUL_RESOURCES_ALLOCATION" and (if not all the flows are affected) the identification of the related media components in the "flows" attribute. If the "MediaComponentVersioning" feature is supported, the PCF shall also include in the "flows" attribute the "contVers" attribute with the content version(s) of the media components if the content version was included when the corresponding media component was provisioned.

If the "AuthorizationWithRequiredQoS" feature or the "AltSerReqsWithIndQoS" feature as defined in clause 5.8 is supported, when the PCF becomes aware that the resources associated to service information for one or more SDFs have been allocated and additionally receives the alternative QoS parameter set(s), the PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2. The PCF shall include:

- an entry in the "evNotifs" attribute with the "event" attribute set to "SUCCESSFUL_RESOURCES_ALLOCATION"; and

- the "succResourcAllocReports" attribute with the reference to the Alternative Service Requirement corresponding alternative QoS parameter set within the "altSerReq" attribute and the identification of the related media components in the "flows" attribute. If the "MediaComponentVersioning" feature is supported, the PCF shall also include in the "flows" attribute the "contVers" attribute with the content version(s) of the media components if the content version was included when the corresponding media component was provisioned.

When the PCF becomes aware that the resources associated to service information for one or more SDFs cannot be allocated, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "FAILED_RESOURCES_ALLOCATION" event as described in clauses 4.2.2.10 and 4.2.3.10. The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2. The PCF shall include:

- an entry in the "evNotifs" attribute with the "event" attribute set to "FAILED_RESOURCES_ALLOCATION"; and

- the "failedResourcAllocReports" attribute with the active/inactive status of the PCC rules related to certain media components encoded in the "mcResourcStatus" attribute, and (if not all the flows are affected) the identification of the related media components in the "flows" attribute. If the "MediaComponentVersioning" feature is supported, the PCF shall also include in the "flows" attribute the "contVers" attribute with the content version(s) of the media components if the content version was included when the corresponding media component was provisioned.

NOTE 1: For the services bound to a MA PDU session with the steering mode is "REDUNDANT", the "FAILED_RESOURCES_ALLOCATION" event is only triggered when the network could not allocate resources over none of the available multiple access types (see clause 4.2.6.2.17 in 3GPP TS 29.512 [8]).

When the feature "UEUnreachable" is supported and if the PCF becomes aware that the UE is temporarily unavailable and thus the resources associated to service information for one or more SDFs are not allocated, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "UE_TEMPORARILY_UNAVAILABLE" event as described in clauses 4.2.2.10 and 4.2.3.10. The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2. The PCF shall include:

- an entry in the "evNotifs" attribute with the "event" attribute set to "UE_TEMPORARILY_UNAVAILABLE",
- the "failedResourcAllocReports" attribute with the active/inactive status of the PCC rules related to certain media components encoded in the "mcResourcStatus" attribute, and (if not all the flows are affected) the identification of the related media components in the "flows" attribute. If the "MediaComponentVersioning" feature is supported, the PCF shall also include in the "flows" attribute the "contVers" attribute with the content version(s) of the media components if the content version was included when the corresponding media component was provisioned; and
- the "retryAfter" attribute within the "evNotifs" attribute if this information was received from the SMF.

The PCF shall include more than one entry in the "contVers" attribute for the same media component if the PCF has received multiple content versions as described in clause 4.2.6.2.14 in 3GPP TS 29.512 [8].

NOTE 2: The NF service consumer will use the content version to identify the media component version that failed or succeeded when multiple provisions of the same media component occur in a short period of time. How the NF service consumer handles such situations is out of scope of this specification.

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF.

Signalling flows for resource allocation outcome are presented in 3GPP TS 29.513 [7].

4.2.5.9 Void

4.2.5.10 Notification of signalling path status

When the PCF is notified of the loss or release of resources associated to the PCC rules corresponding with AF signalling IP flows, the PCF shall inform the NF service consumer about the loss of the signalling transmission path if the NF service consumer has previously subscribed as described in clause 4.2.6.7.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include within the "evNotifs" attribute an event of "AfEventNotification" data type indicating the matched event "FAILED_RESOURCES_ALLOCATION" in the "event" attribute and the deactivated IP flow encoded in the "flows" attribute.

If the "RAN-NAS-Cause" feature is supported and the PCF received the RAN-NAS release cause and/or access network information from the SMF, the PCF shall provide in the "EventsNotification" data type in the HTTP POST request:

- in case of 3GPP access, the user location information in the "eutraLocation" or in the "nrLocation" attribute in the "ueLoc" attribute, if available;
- in case of untrusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if available, as follows:
  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute; and
  b) the UDP source port or the TCP source port in the "portNumber" and "protocol" attributes, if available; and
  c) if the "WLAN_Location" feature is supported, the WLAN location information encoded in the "twapId" attribute, if available, that shall consist of:
     i. the SSID in the "ssId" attribute;
     ii. the BSSID the "bssId" attribute if available; and

iii. the civic address in the "civicAddress" attribute if available;

NOTE 1: When the UE reaches the ePDG via a NAT, the combination of UE local IP address and the UE source port is needed for lawful interception purposes. The UE source port may be either a UDP or a TCP port, and it is indicated in the "protocol" attribute.

- in case of trusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if available, as follows:

  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available; and

  b) the UDP source port in the "portNumber" attribute if available; and

NOTE 2: The UDP protocol can be used between the UE and the TNGF to enable NAT traversal.

  c) either the TNAP identifier encoded in the "tnapId" attribute or the TWAP identifier encoded in the "twapId" attribute. The TNAP identifier and the TWAP identifier shall consist of:

     i. the SSID in the "ssId" attribute;

     ii. the BSSID the "bssId" attribute if available; and

     iii. the civic address in the "civicAddress" attribute if available;

- the serving network identity i.e. the PLMN Identifier (the PLMN network code and the country code) or the SNPN Identifier (the PLMN Identifier and the NID) in the "plmnId" attribute, if user location information is not available in any access;

- the UE timezone in the "ueTimeZone" attribute if available;

- the RAN and/or NAS release cause in the "ranNasRelCauses" attribute, if available; and

- if the "UeSatUeComm" feature is supported, the serving satellite identifier within the "servSatId" attribute if available, when the UE is accessible via regenerative satellite access.

NOTE 3: The PCF forwards both 3GPP and non-3GPP access UE locations in the "ueLoc" attribute when both UE locations are provided by the SMF as defined in 3GPP TS 29.512 [8].

The PCF shall include in the "evNotifs" attribute, together with the event "FAILED_RESOURCES_ALLOCATION", an event of the "AfEventNotification" data type with the "event" attribute set to the value "RAN_NAS_CAUSE".

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF.

4.2.5.11 Reporting access network information

This procedure is used by the PCF to report the access network information (i.e. user location and/or user timezone information) to the NF service consumer when the "NetLoc" feature is supported and to report serving satellite identifier within the access network information, if available and when the "UeSatUeComm" feature is also supported.

When the PCF receives the access network information from the SMF, the PCF shall include the "EventsNotification" data type in the body of the HTTP POST request message sent to the NF service consumer as described in clause 4.2.5.2. The PCF shall include in the "EventsNotification" data type:

- in case of 3GPP access, the user location information in the "eutraLocation" or in the "nrLocation" attribute in the "ueLoc" attribute, if available and required;

- in case of untrusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:

  a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available;

  b) the UDP source port or the TCP source port in the "portNumber" and "protocol" attributes, if available; and

  c) if the "WLAN_Location" feature is supported, the WLAN location information encoded in the "twapId" attribute, if available, that shall consist of:

i. the SSID in the "ssId" attribute;

ii. the BSSID the "bssId" attribute if available; and

iii. the civic address in the "civicAddress" attribute if available;

NOTE 1: When the UE reaches the ePDG via a NAT, the combination of UE local IP address and the UE source port is needed for lawful interception purposes. The UE source port may be either a UDP or a TCP port, and it is indicated in the "protocol" attribute.

- in case of trusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:

a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available; and

b) the UDP source port in the "portNumber" attribute if available; and

NOTE 2: The UDP protocol can be used between the UE and the TNGF to enable NAT traversal.

c) either the TNAP identifier encoded in the "tnapId" attribute or the TWAP identifier encoded in the "twapId" attribute. The TNAP identifier and the TWAP identifier shall consist of:

i. the SSID in the "ssId" attribute;

ii. the BSSID the "bssId" attribute if available; and

iii. the civic address in the "civicAddress" attribute if available;

- if user location was required, the time when it was last known in the "ueLocTime" attribute if available;

NOTE 3: The PCF derives the value of the "ueLocTime" attribute from the "userLocationInfoTime" attribute received from the SMF as specified in 3GPP TS 29.512 [8].

- the serving network identity i.e. the PLMN Identifier (the PLMN network code and the country code) or the SNPN Identifier (the PLMN Identifier and the NID) in the "plmnId" attribute, if user location information is required but not available in any access;

- the UE timezone in the "ueTimeZone" attribute if required and available; and/or

NOTE 4: The PCF forwards both 3GPP and non-3GPP access UE locations in the "ueLoc" attribute when both UE locations are provided by the SMF as defined in 3GPP TS 29.512 [8].

- if the "UeSatUeComm" feature is supported, the serving satellite identifier in the "servSatId" attribute if available, when the UE is accessible via regenerative satellite access.

When the PCF receives from the SMF that the access network does not support access network information report, the PCF shall include the "noNetLocSupp" attribute set to "ANR_NOT_SUPPORTED", "TZR_NOT_SUPPORTED", "UE_SAT_NOT_SUPPORTED" or "LOC_NOT_SUPPORTED" value received from the SMF in the "EventsNotification" data type in the HTTP POST request.

The PCF shall also include an event of the "AfEventNotification" data type in the "evNotifs" attribute with the "event" attribute set to the value "ANI_REPORT".

NOTE 5: The PCF receives the access network information from the SMF if it is previously requested by the NF service consumer or at PDU session termination or at the termination of all the service data flows of the AF session.

The PCF shall not invoke the Npcf_PolicyAuthorization_Notify service operation with the "event" attribute set to the value "ANI_REPORT" to report to the NF service consumer any subsequently received access network information, unless the NF service consumer sends a new request for access network information.

4.2.5.12 Notification about Out of Credit

If the "IMS_SBI" feature is supported and if the PCF becomes aware that there is no credit available in the CHF for one or more SDFs, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "OUT_OF_CREDIT" event as described in clauses 4.2.2.22 and 4.2.3.22.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- in the "evNotifs" attribute an entry with the "event" attribute set to the value "OUT_OF_CREDIT"; and
- the "outOfCredReports" attribute containing in each entry of the "OutOfCreditInformation" data type the credit information for one or more service data flows. The "OutOfCreditInformation" data type shall contain the termination action in the "finUnitAct" attribute, and the identification of the affected service data flows (if not all the flows are affected) encoded in the "flows" attribute.

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request by sending an HTTP response message as described in clause 4.2.5.2.

4.2.5.13 Notification about TSC user plane node management information and/or port management information detection, Individual Application Session Context exists

During the lifetime of a PDU session enabling Time Sensitive Communication, Time Synchronization and/or Deterministic Networking, the SMF may send to the PCF TSC user plane node management information of the UPF/NW-TT within a UMIC and/or, when the DS-TT or the NW-TT functions are used, a PMIC from the DS-TT port and/or one or more PMIC(s) from the one or more NW-TT ports.

If the "TimeSensitiveNetworking" or "TimeSensitiveCommunication"feature is supported and if the PCF becomes aware that updated TSC user plane node information is available, e.g., a UMIC and/or PMIC(s) have been received, for the SM Policy Association bound to the Individual Application Session Context resource,, the PCF shall inform the NF service consumer (i.e., the TSN AF or the TSCTSF) accordingly, if the NF service consumer has previously subscribed as described in clause 4.2.2.31.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include in the "evNotifs" attribute an entry with the "event" attribute set to the value "TSN_BRIDGE_INFO", and the "tsnBridgeManCont" attribute and/or the "tsnPortManContDstt" attribute and/or the "tsnPortManContNwtts" attribute as received from the SMF if the PCF is aware that a UMIC and/or a DS-TT PMIC and/or one or more NW-TT PMIC(s) are available or updated.

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request as specified in clause 4.2.5.2.

The NF service consumer may use the received UMIC and/or the received DS-TT PMIC and/or NW-TT PMIC(s) and the local configuration to construct the DS-TT port and or NW-TT port management information required to interwork with the external network (e.g. TSN).

If port management information shall be sent as a response of the received notification, the NF service consumer triggers the Npcf_PolicyAuthorization_Update service operation to send the port management information to the PCF as specified in clause 4.2.3. The NF service consumer delivers to the PCF the derived port management information containers as described in clause 4.2.3.25.

And/or if TSC user plane node management information shall be sent as a response of the received notification, the NF service consumer includes the UMIC in the Npcf_PolicyAuthorization_Update service operation as described in clause 4.2.3.25.

4.2.5.14 Notification about Service Data Flow QoS Monitoring control

When the PCF gets the information about real-time measurements of QoS monitoring parameters for one or more SDFs from the SMF (e.g. for QoS monitoring for packet delay, uplink packet delay(s), downlink packet delay(s) and/or round trip delay(s) or if the feature "PacketDelayFailureReport" is supported, indicator of packet delay measurement failure) the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed as described in clauses 4.2.2.23 and 4.2.3.23 and 4.2.6.8.

The PCF shall notify the NF service consumer of the QoS monitoring events by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "QOS_MONITORING" in the "event" attribute; and
- for QoS monitoring for packet delay, one or more entries of the "qosMonReports" array, where each entry shall contain:
  1. the identification of the affected service flows (if not all the flows are affected) encoded in the "flows" attribute if applicable; and:
  2. the received packet delay measurement:
     a) the uplink packet delays within the "ulDelays" attribute; and/or
     b) the downlink packet delays within the "dlDelays" attribute; and/or
     c) the round trip packet delays within the "rtDelays" attribute; or
     d) if the feature "PacketDelayFailureReport" is supported, the packet delay measurement failure indicator within the"pdmf" attribute; and/or

NOTE: The SMF reports one UL, DL and/or round-trip packet delay measurement for each periodic and/or event-triggered report as described in 3GPP TS 29.512 [8]. I.e, the PCF can include only one element within the "ulDelays", "dlDelays", and/or "rtDelays" array(s) respectively, each one with the received report from the SMF for the UL, DL and/or round trip delay(s).

- if the feature "EnQoSMon" is supported, to report data rate measurements, one or more entries of the "qosMonDatRateReps" array, where each entry shall contain:
  1. the identification of the affected service flows (if not all the flows are affected) encoded in the "flows" attribute if applicable; and
  2. the received data rate measurement:
     a) one data rate measurement for the UL within the "ulDataRate" attribute; and/or
     b) one data rate measurement for the DL within the "dlDataRate" attribute.
- if the feature "EnQoSMon" is supported, for QoS monitoring report for congestion information, one or more entries of the "congestReports" array, where each entry shall contain:
  1. the identification of the affected service flows (if not all the flows are affected) encoded in the "flows" attribute if applicable;
  2. the received congestion measurement:
     a) the uplink congestion information measurement within the "ulConInfo" attribute; and/or
     b) the downlink congestion information measurement within the "dlConInfo" attribute.

4.2.5.15 Report of EPS Fallback

When "EPSFallbackReport" feature is supported and the PCF becomes aware of the EPS Fallback for the resources requested for a particular service information (voice media type), the PCF shall inform the NF service consumer if the NF service consumer has previously subscribed as described in clauses 4.2.2.30 and 4.2.3.29.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "EPS_FALLBACK" in the "event" attribute.

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF.

4.2.5.16 Notification about TSC user plane node Information, no Individual Application Session Context exists

During PDU session establishment of a PDU session enabling Time Sensitive Communications, Time Synchronization and Deterministic Networking, the PCF may receive from the SMF TSC user plane node information as specified in clauses 4.2.2.19 and 4.2.3.23 of 3GPP TS 29.512 [8].

If the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported and if the PCF becomes aware that TSC user plane node information for an external network (e.g. TSN) is available, but there is no "Individual Application Session Context" resource bound to the SM Policy Association updated with TSC user plane node related information, the PCF shall inform the NF service consumer (i.e. TSN AF or TSCTSF) about the detection of a TSC user plane node information in the context of a PDU session by sending a notification request:

- to the request URI locally configured in the PCF for the NF service consumer; or
- if the request URI for the TSCTSF is not locally configured in the PCF, to the notification URI registered by the TSCTSF in the NRF as default notification subscription for time sensitive communication, time synchronization and deterministic networking notifications, and retrieved from NRF by the PCF using the discovery service, as specified in 3GPP TS 29.510 [27] for the PDU session DNN/S-NSSAI.

NOTE 1: PCF configuration of TSN AF needs to ensure that the notification is addressed to a TSN AF that connects to the same external network the UPF/NW-TT connects to. How it is achieved is implementation specific. It can be based e.g. on dedicated DNN/S-NSSAI combinations or on the received TSC user plane node information.

NOTE 2: It is assumed that there is only one TSCTSF for a given DNN/S-NSSAI in this release of the specification.

Figure 4.2.5.16-1 illustrates the notification about TSC user plane node information when there is no Individual Application Session Context bound to the SM Policy Association.

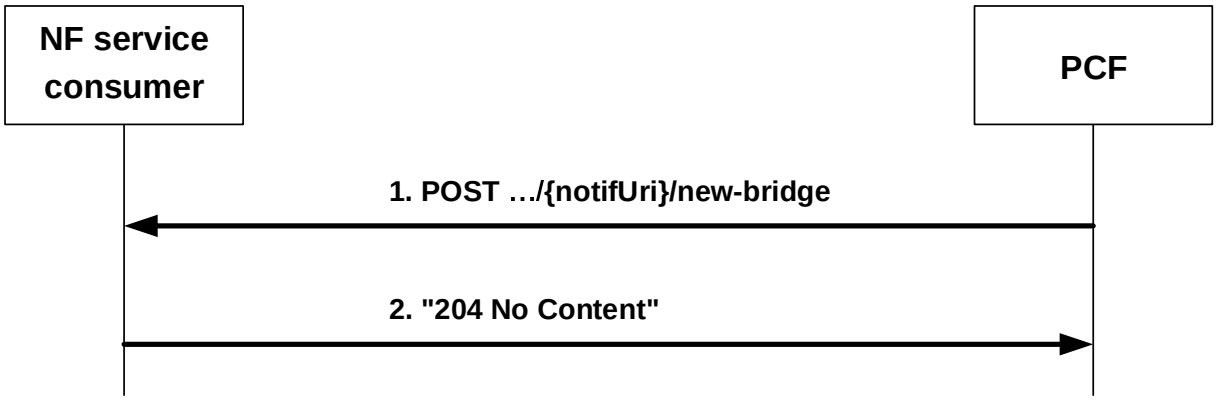


Figure 4.2.5.16-1: Notification about TSC user plane node Information, no AF session context exists

When the PCF determines that the AF application session context does not exist for the SM Policy Association that detected new port information and a notification URI for the NF service consumer can be determined, the PCF shall invoke the Npcf_PolicyAuthorization_Notify service operation by sending the HTTP POST request (as shown in figure 4.2.5.16-1, step 1) using the notification URI locally configured in the PCF or, retrieved from NRF, and appending the "new-bridge" segment path at the end of the URI, to trigger the NF service consumer (i.e. TSN AF or TSCTSF) to request the creation of an Individual Application Session Context resource to handle the TSC user plane node detected in the context of a PDU session, configuring ports and TSC user plane node management information, and providing the corresponding TSCAI input containers and TSC traffic QoS related data (see clauses 4.2.2.2, 4.2.2.24, 4.2.2.25 and 4.2.2.31).

The PCF shall provide in the body of the HTTP POST request the "PduSessionTsnBridge" data type including TSC user plane node information as follows:

- the "tsnBridgeInfo" attribute as received from the SMF;
- when DS-TT and/or NW-TT functionality is used, the "tsnPortManContDstt" attribute and/or "tsnPortManContNwtts" attribute as received from the SMF, if available;
- in case of Deterministic Networking, for the device side port, and if the "MTU_Size" feature is supported, the MTU size for IPv4 and the MTU size for IPv6 encoded in the "mtuIpv4" and "mtuIpv6" attributes respectively within the "tsnBridgeInfo" attribute as received from the SMF, if available; and
- when the "TimeSensitiveCommunication" feature is supported and for a PDU session of IP type, the UE IPv4 address within the "ueIpv4Addr" attribute or the UE IPv6 prefix within the "ueIpv6AddrPrefix", the DNN within the "dnn" attribute, the S-NSSAI within the "snssai" attribute and, if available, the domain identity within the "ipDomain" attribute if UE IPv4 address is provided.

NOTE 3: In the case of IP overlapping, the DNN, S-NSSAI and domain identity, if available, are required for session binding in the PCF. Domain identity applies as defined in clause 4.2.2.2.

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request.

With the received information, the NF service consumer (i.e. TSN AF or TSCTSF) shall immediately trigger the creation of an Individual Application Session Context resource to handle in this association the configuration of the new TSC user plane node in the context of this PDU session, as described in clauses 4.2.2.2, 4.2.2.24, 4.2.2.25 and 4.2.2.31.

NOTE 4: For the time synchronization service, the subscription to UE availability for time-synchronization service can occur after the PDU Session establishment has been completed in 5GS. Similarly, for the AF session with required QoS, the indication of the required QoS and TSC Assistance Container information can occur after the completion of the PDU session establishment. In such cases, the PCF sends the notification to the TSCTSF about the detection of a TSC user plane node information during PDU session establishment, and the TSCTSF could defer the creation of the related "Individual Application Session Context" till the reception of the subscription to UE availability for time synchronization or the AF session with required QoS occurs, as specified in 3GPP TS 29.513 [7].

The NF service consumer (i.e. TSN AF or TSCTSF) may use the received TSC user plane node information and/or the received DS-TT port management information container and/or NW-TT port management information containers and the local configuration to construct the DS-TT port and or NW-TT port management information required to interwork with the external network.

4.2.5.17 Notification about Reallocation of Credit

If the "IMS_SBI" and the "ReallocationOfCredit" features are supported and if the PCF becomes aware that there is credit reallocated for one or more SDFs after a former out of credit indication, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "REALLOCATION_OF_CREDIT" event as described in clauses 4.2.2.34 and 4.2.3.32.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include in the "evNotifs" attribute an entry with:

- the "event" attribute set to the value "REALLOCATION_OF_CREDIT"; and
- the SDFs that are impacted as consequence of the reallocation of credit condition encoded in the "flows" attribute.

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request by sending an HTTP response message as described in clause 4.2.5.2.

4.2.5.18 Notification of MPS for DTS Outcome

When the MPSforDTS feature is supported and the PCF is informed about the successful default QoS update, the PCF shall notify the NF service consumer as described in clause 4.2.5.2, if the NF service consumer has previously subscribed to the "SUCCESSFUL_QOS_UPDATE" event as described in clauses 4.2.2.12.2 and 4.2.3.12. The PCF shall notify the NF service consumer by including within the "evNotifs" attribute, an entry with the "event" attribute set to "SUCCESSFUL_QOS_UPDATE".

When the MPSforDTS feature is supported and the PCF is informed about the failure of a default QoS update, the PCF shall notify the NF service consumer as described in clause 4.2.5.2, if the NF service consumer has previously subscribed to the "FAILED_QOS_UPDATE" event as described in clauses 4.2.2.12.2 and 4.2.3.12. The PCF shall notify the NF service consumer by including within the "evNotifs" attribute, an entry with the "event" attribute set to "FAILED_QOS_UPDATE".

4.2.5.19 Notification about Application Detection Information

When the "ApplicationDetectionEvents" feature is supported, when the PCF gets the knowledge that the traffic of the indicated application started or stopped, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed as described in clause 4.2.6.9.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include, for the detected application(s)'s traffic:

- within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "APP_DETECTION" in the "event" attribute; and
- the "adReports" array, which for each detected application's traffic shall include:
  a) the "adNotifType" attribute to indicate whether the detection is about the start of the application's traffic encoded as the "APP_START" value, or about the stop of the application's traffic encoded as the "APP_STOP" value; and
  b) the application identifier within the "afAppId" attribute.

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF.

Signalling flows for the notification of application detection information are presented in 3GPP TS 29.513 [7].

NOTE: When the NF service consumer receives the notifications for multiple applications, the NF service consumer (e.g. the PCF for the UE) can determine which logic to apply (e.g. which AM policy to apply) based on local configuration and operator policy.

In this release of the specification application detection applies only to the application(s) with IP traffic.

4.2.5.20 Notification about satellite backhaul category changes

When the PCF gets the knowledge that there is a change of the backhaul used for the PDU session between satellite backhaul categories or between a satellite and a non-satellite backhaul category, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed as described in clauses 4.2.2.35 and 4.2.3.33 and 4.2.6.10.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

When the "SatelliteBackhaul" feature is supported, the PCF shall include within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "SAT_CATEGORY_CHG" in the "event" attribute, and within the "satBackhaulCategory" attribute the received satellite backhaul category or the indication of non-satellite backhaul.When the "EnSatBackhaulCatChg" feature is supported and the PCF received the dynamic satellite backhaul category, the PCF shall include the dynamic satellite backhaul category within the "satBackhaulCategory" attribute.

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF. The NF service consumer may also update the AF application session context information by sending an HTTP PATCH request to the PCF.

4.2.5.21 Notification about UP change enforcement failure

If the "RoutingReqOutcome" feature is supported and if the PCF becomes aware that the enforcement of the UP path change fails (as specified in clause 4.2.6.2.6.2 of 3GPP TS 29.512 [8]), the PCF shall inform the NF service consumer

accordingly if the NF service consumer has previously subscribed to the "UP_PATH_CHG_FAILURE" event as described in clauses 4.2.2.8 and 4.2.3.8.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include in the "evNotifs" attribute an entry with the "event" attribute set to the value "UP_PATH_CHG_FAILURE".

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request by sending an HTTP response message as described in clause 4.2.5.2.

4.2.5.22 Notification about PDU session established/terminated events

If the PCF becomes aware that the SM Policy Association contains the callback URI of the PCF for a UE then, the PCF shall inform the NF service consumer (i.e. the PCF for a UE) about:

- the PDU session establishment, when the PCF receives the callback URI of the PCF for a UE from the SMF; and
- the PDU session termination, when the PCF receives the SM Policy Association termination from the SMF;

by sending a notification request to the received callback URI of the PCF for a UE.

Figure 4.2.5.22-1 illustrates the notification about PDU session established/terminated events.

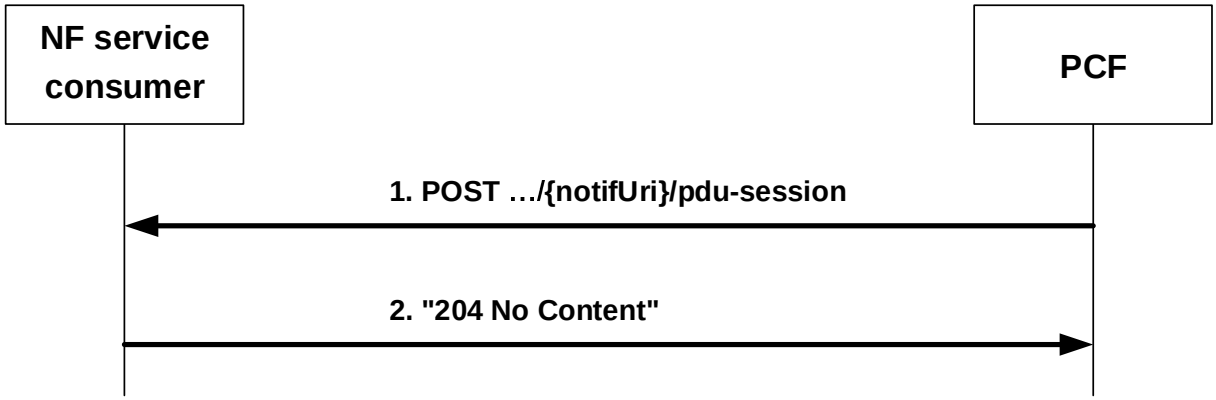


Figure 4.2.5.22-1: Notification about PDU session established/terminated events

When the PCF becomes aware that a SM Policy Association is receiving the callback URI of the PCF for a UE, or becomes aware that the SM Policy Association that is terminating contains the callback URI of the PCF for a UE, the PCF shall invoke the Npcf_PolicyAuthorization_Notify service operation by sending an HTTP POST request (as shown in figure 4.2.5.22-1, step 1) using the callback URI contained in the SM Policy Association and appending the "pdu-session" path segment at the end of the URI.

NOTE: The PCF includes in the notification request a Routing Binding Indication as specified in 3GPP TS 29.500 [5], clause 6.12 if an SBA binding indication relative to the PCF for a UE is available in the SM Policy Association together with the callback URI of the PCF for a UE.

The PCF shall provide in the body of the HTTP POST request the PduSessionEventNotification data type, which shall include an indication of PDU session establishment/termination as follows:

- the "evNotif" attribute, of "AfEventNotification" data type, which shall include the "PDU_SESSION_STATUS" event within the "event" attribute;
- the SUPI of the PDU session within the "supi" attribute;
- the served UE address as the identification of the reported PDU session:
  i. for IP type PDU sessions, the IP address (IPv4 or IPv6) of the UE in the "ueIpv4" or "ueIpv6" attribute; and
  ii. for Ethernet type PDU sessions, the MAC address of the UE in the "ueMac" attribute;

- whether the PDU session is established or terminated within the "status" attribute; and
- when the "status" attribute indicates "ESTABLISHED":
  i. the PCF addressing information where the NF service consumer (i.e. PCF for a UE) may send the subscription request to notification about the detected application traffic in the "pcfInfo" attribute; and
  ii. the context information of the related PDU session, i.e., the DNN withing the "dnn" attribute, the S-NSSAI within the "snssai" attribute and the GPSI within the "gpsi" attribute, if available.

Upon the reception of the HTTP POST request from the PCF, and if the request is accepted, the NF service consumer (i.e. PCF for a UE) shall acknowledge that request by sending an HTTP response message with a "204 No Content" status code as described in figure 4.2.5.22-1, step 2.

The NF service consumer (i.e. PCF for a UE) may use the notified PCF address(es) and SBA binding indication, if available, to subscribe with the PCF for a PDU session to the detection of application(s) traffic, as described in clause 4.2.6.9.

4.2.5.23 Notification about extra UE addresses

When the "ExtraUEaddrReport" feature is supported, and the PCF becomes aware that the there is a change in the list of IP addresses or address ranges allocated to the PDU session due to framed routes or IPv6 prefix delegation (as described in 3GPP TS 29.512 [8], clauses 4.2.4.2 and C.3.4.1), the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "EXTRA_UE_ADDR" event as described in clauses 4.2.2.36, 4.2.3.35 and 4.2.6.12. The report shall include the actual list of IPv4 addresses or list of IPv6 prefixes as currently allocated.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include in the "evNotifs" array an entry with the "event" attribute set to the value "EXTRA_UE_ADDR" and:

- the actual list of IPv4 addresses within the "ipv4AddrList" attribute, if there was a change in the one or more IPv4 framed routes associated to the PDU session; or
- the actual list of IPv6 prefixes allocated to the UE within the "ipv6PrefixList" attribute, if there was a change in the one or more IPv6 framed routes associated to the PDU session, or in the one or more IPv6 prefixes allocated to the PDU session.

NOTE: In this release of the specification the IP Framed Routes are received by the PCF during PDU session establishment and remain unchanged during the lifetime of the PDU session.

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request by sending an HTTP response message as described in clause 4.2.5.2.

4.2.5.24 Notification about BAT offset

If the "EnTSCAC" feature is supported, upon receiving the BAT offset and optionally the adjusted periodicity, the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "BAT_OFFSET_INFO" event as described in clauses 4.2.2.40 and 4.2.3.24.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- in the "evNotifs" attribute an entry with the "event" attribute set to the value "BAT_OFFSET_INFO";
- the "batOffsetInfo" attribute containing the offset of the BAT and the optionally adjusted periodicity. The "BatOffsetInfo" data type shall contain the BAT offset of the arrival time in the "ranBatOffsetNotif" attribute, and the optionally adjusted periodicity of the data bursts encoded in the "adjPeriod" attribute; and
- the identification of the affected service flows (if not all the flows are affected) encoded in the "flows" attribute if applicable.

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request by sending an HTTP response message as described in clause 4.2.5.2.

4.2.5.25 Notification about URSP rule enforcement information

When the "URSPEnforcement" feature is supported, when the PCF gets from the UE via the SMF the report of URSP rule enforcement information from associated URSP rule(s) as defined in 3GPP TS 29.512 [8], the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed as described in clause 4.2.6.14.

The PCF shall notify the NF service consumer by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include, for the notification about URSP rule enforcement information:

- within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "URSP_ENF_INFO" in the "event" attribute;
- within the "urspEnfRep" attribute, the URSP rule enforcement information received from the UE as defined in 3GPP TS 29.512 [8]; and
- if not previously reported, the SSC mode within the "sscMode" attribute, the UE requested DNN, if available, within the "ueReqDnn" attribute, and the UE requested PDU Session Type within the "ueReqPduSessionType" attribute.

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF.

Signalling flows for the notification about URSP rule enforcement information are presented in 3GPP TS 29.513 [7].

4.2.5.26 Notification about Packet Delay Variation

When the PCF gets the information about real-time measurements of QoS parameters for one or more SDFs from the SMF (e.g. for QoS monitoring for packet delay, uplink packet delay(s), downlink packet delay(s) and/or round trip delay(s)), the PCF calculates the uplink, downlink and/or round trip Packet Delay Variations based on the received uplink, downlink or round trip QoS Monitoring result, and reports to the NF service consumer accordingly if the NF service consumer has previously subscribed to Packet Delay Variation reports as described in clauses 4.2.2.41 and 4.2.3.40.

The PCF shall notify the NF service consumer of the Packet Delay Variation events by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "PACK_DEL_VAR" in the "event" attribute; and
- the "pdvMonReports" array with:
  a) the identification of the affected service flows (if not all the flows are affected) encoded in the "flows" attribute if applicable; and
  b) the uplink packet delay variation measurement(s) within the "ulPdv" attribute;
  c) the downlink packet delay variation measurement(s) within the "dlPdv" attribute; and/or
  d) the round trip packet delay variation measurement(s) within the "rtPdv" attribute.

4.2.5.27 Notification about 5GS support for Policy Control for L4S

When the "L4S" feature is supported, the NF service consumer provided the explicit indication of ECN marking for L4S support for the provided UL and/or DL SDF(s) as described in clauses 4.2.2.43 and 4.2.3.42, and the PCF gets the knowledge that there is a change of the 5GS support for ECN marking for L4S for the indicated SDF(s), the PCF, may notify the NF service consumer about the change of 5GS support by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event, "L4S_SUPP", in the "event" attribute and the "l4sReports" array. In each entry of the "l4sReports" array, the PCF shall include the indication that ECN marking for L4S is not available or is available again within the "notifType" attribute and the SDFs that are impacted as consequence of change of 5GS availability condition for ECN marking for L4S encoded in the "flows" attribute.

When the NF service consumer receives the HTTP POST request, it shall acknowledge the request by sending a "204 No Content" response to the PCF.

4.2.5.28 Notification about Round-Trip delay over two QoS flows

Upon receiving the delay information per QoS flow, e.g. the UL and DL of one service data flow are separated into two QoS flows, or the UL direction of a service data flow and the DL direction of another service data flow are mapped to two QoS flows, from the SMF (e.g. for QoS monitoring for packet delay, uplink packet delay(s), downlink packet delay(s) and/or round trip delay(s)), the PCF calculates the Round-Trip delay and reports to the NF service consumer accordingly if the NF service consumer has previously subscribed to Round-Trip delay over two QoS flows monitoring reports as described in clauses 4.2.2.44 and 4.2.3.43.

The PCF shall notify the NF service consumer of the Round-Trip delay over two QoS flows monitoring events by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- within the "evNotifs" attribute an event entry of the "AfEventNotification" data type with the matched event "RT_DELAY_TWO_QOS_FLOWS" in the "event" attribute; and
- the "qosMonReports" array with:
  a) the round trip packet delay measurement(s) within the "rtDelays" attribute.

NOTE: How the PCF calculates the requested Round-Trip delay over two QoS flows from the packet delays of each individual QoS Flow is not specified in this specification.

4.2.5.29 Event notification for AF requested QoS for a UE or group of UE(s) not identified by UE address(es)

An NF service consumer (e.g., AF, NEF) may subscribe to event(s) reporting implicitly via the UDR as specified in 3GPP TS 29.519 [53] (see also clause 4.15.6.14 of 3GPP TS 23.502 [3] and clause 6.1.3.28 of 3GPP TS 23.503 [4]) when provisioning/updating AF requested QoS for a UE or a group of UE(s) not identified by UE address(es).

When the PCF determines that event(s) occurred (e.g., as defined in clause 4.2.6.2.23 of 3GPP TS 29.512 [8]), the PCF shall invoke the Npcf_PolicyAuthorization_Notify service operation as defined in clause 4.2.5.2 with the following differences:

- The "{notifUri}" variable part of the request URI of the HTTP POST request shall contain the callback URI included within the "notifUri" attribute contained within the "evSubsc" attribute of the corresponding AF Requested QoS Data obtained from the UDR as specified in 3GPP TS 29.519 [53].
- The "EventsNotification" data type shall include within the "{appSessionId}" variable part of the "<apiSpecificResourceUriPart>" component (see clause 5.1) of the "evSubsUri" attribute, the notification correlation identifier contained within the "notifCorreId" attribute included within the "evSubsc" attribute of the corresponding AF Requested QoS Data also obtained from the UDR as specified in 3GPP TS 29.519 [53].

4.2.5.30 Notification about network support of QoS Monitoring

When the "QoSMonCapRepo" feature is supported, the NF service consumer provided the subscription to QoS Monitoring Capability report as described in clauses 4.2.2.46, 4.2.3.45 and/or 4.2.6.15, and the PCF gets the knowledge that there is a change of the QoS Monitoring Capability, the PCF notifies the NF service consumer about the change of the QoS Monitoring Capability by including the "EventsNotification" data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include the "evNotifs" attribute with an event entry with the "QOS_MON_CAP_REPO" value in the "event" attribute and the "qosMonCapRepos" attribute. In each entry of the "qosMonCapRepos" attribute, the PCF shall

include the indication of whether QoS monitoring is supported or not within the "capReport" attribute and the corresponding QoS Monitoring Capability type within the "capType" attribute.

NOTE: Current release of this specification only supports the reporting of QoS Monitoring Capability type for Packet delay, available bitrate and congestion. Each of the QoS Monitoring Capability can be active or not active independently.

4.2.5.31 Notification about released the application session context

When the "ReleasedUEaddrReport" feature is supported, and the AF application session context that was created as described in clause 4.2.2 or in clause 4.2.6.3 is released from the existing PDU session (as described in 3GPP TS 29.512 [8]), the PCF determines that the AF application session context is no longer valid, then the PCF shall inform the NF service consumer by sending a notification about application session context termination as defined in clause 4.2.5.3.

4.2.5.32 Notification about Data Rate Limitation Information

When the "RateLimitReport" feature is supported, and the PCF determines the Data Rate Limitation information has changed based on the local policies and the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "RATE_LIMIT_INFO_REPO" event as described in clauses 4.2.2.50 and 4.2.3.49.

The PCF shall notify the NF service consumer by including the EventsNotification data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include in the "evNotifs" array an entry with the "event" attribute set to the value "RATE_LIMIT_INFO_REPO" and the "rateLimitRepoUl" and/or "rateLimitRepoDl" attribute within RateLimitRepo data type including the maximum bitrate for uplink and/or downlink that is authorized for non-GBR service data flows.

Upon the reception of the HTTP POST request from the PCF, the NF service consumer shall acknowledge that request by sending an HTTP response message as described in clause 4.2.5.2.

4.2.5.33 Notification about UP path management events

If the "UeSatUeComm" feature is supported and if the PCF receives a UP path management event notification from the SMF indicating that the subscribed UP path change event has been detected as defined in 3GPP TS 29.508 [13], the PCF shall inform the NF service consumer accordingly if the NF service consumer has previously subscribed to the "UP_PATH_CHANGE" event as described in clause B.21.

The PCF shall notify the NF service consumer by including the EventsNotification data type in the body of the HTTP POST request as described in clause 4.2.5.2.

The PCF shall include:

- in the "evNotifs" attribute an entry with the "event" attribute set to the value "UP_PATH_CHANGE";
- type of notification ("EARLY" or "LATE") in the "dnaiChgType" attribute; and
- source DNAI and target DNAI in the "sourceDnai" attribute and "targetDnai" attribute if the DNAI changed.

4.2.5.34 Reporting 5GC network functions health monitoring information

This procedure is used by the PCF to report to the NF service consumer whether the PDU session is affected by a 5GC network functions (i.e. SMF or UPF) failure and whether recovery of the PDU session has been prioritized, when the "CnHealthMonitor" feature is supported.

When the PCF receives the 5GC network functions health monitoring status for this PDU Session from SMF. the PCF shall include the "EventsNotification" data type in the body of the HTTP POST request message sent to the NF service consumer as described in clause 4.2.5.2. The PCF shall include in the "EventsNotification" data type the "sessionRecovStatus" attribute containing the information received from the SMF.

If SMF does not answer or the PCF receives from the SMF a response message with an HTTP "404 Not Found" status code:

- the PCF shall include the "sessionRecovStatus" set to "NOT_FOUND" value inside the EventsNotification data type included in the body of the HTTP POST request message sent to NF service consumer; or

- if the "SmfFailureDetection" feature is supported, the PCF may send an Npcf_PolicyAuthorization_Notify to NF service consumer, including "TerminationInfo" data type set to "SMF_FAILURE" within the body of the HTTP POST request, as described in clause 4.2.5.3.

The PCF shall also include an event of the "AfEventNotification" data type in the "evNotifs" attribute with the "event" attribute set to the value "CN_HEALTH_MONITOR".

The PCF shall not invoke the Npcf_PolicyAuthorization_Notify service operation with the "event" attribute set to the value "CN_HEALTH_MONITOR" to report to the NF service consumer any 5GC Network Functions Health Monitoring information subsequently received from SMF, unless the NF service consumer sends a new request to get this information.

Once the NF service consumer is notified, it shall invoke the Npcf_PolicyAuthorization_Delete operation in PCF to delete the "Individual Application Session Context" resource by sending an HTTP DELETE request message.

4.2.6 Npcf_PolicyAuthorization_Subscribe service operation

4.2.6.1 General

The Npcf_PolicyAuthorization_Subscribe service operation enables NF service consumers handling of subscription to events for the existing application session context. Subscription to events for the existing application session context shall be created:

- within the application session context establishment procedure by invoking the Npcf_PolicyAuthorization_Create service operation, as described in clause 4.2.2; or

- within the application session context modification procedure by invoking the Npcf_PolicyAuthorization_Update service operation, as described in clause 4.2.3; or

- by invoking the Npcf_PolicyAuthorization_Subscribe service operation for the existing application session context, as described in clause 4.2.6.2.

The following procedures using the Npcf_PolicyAuthorization_Subscribe service operation is supported:

- Handling of subscription to events for the existing application session context.

- Initial subscription to events without provisioning of service information.

- Subscription to usage monitoring of sponsored data connectivity.

- Request of access network information.

- Subscription to notification of signalling path status.

- Subscription to Service Data Flow QoS Monitoring Information.

- Subscription to application detection notifications.

- Subscription to satellite backhaul category changes.

- Subscription to the report of extra UE addresses.

- Subscription to Service Data Flow QoS Monitoring multi-modal services.

- Subscription to URSP rule enforcement information.

- Subscription to the report of network support for QoS Monitoring.

4.2.6.2 Handling of subscription to events for the existing application session context

This procedure is used to create a subscription to events for the existing AF application session context bound to the corresponding PDU session or to modify an existing subscription, as defined in 3GPP TS 23.501 [2], 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

Figure 4.2.6.2-1 illustrates the creation of events subscription information using HTTP PUT method.

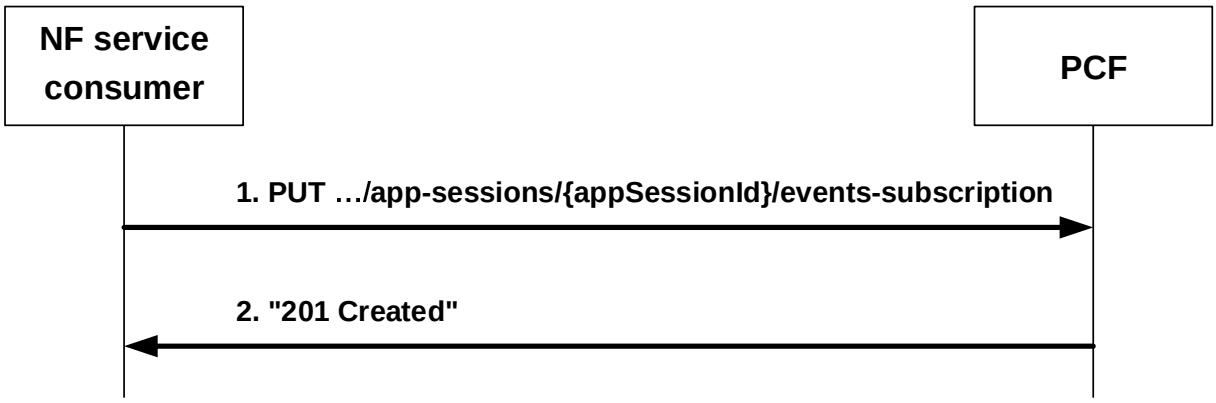


Figure 4.2.6.2-1: Creation of events subscription information using HTTP PUT

Figure 4.2.6.2-2 illustrates the modification of events subscription information using HTTP PUT method.

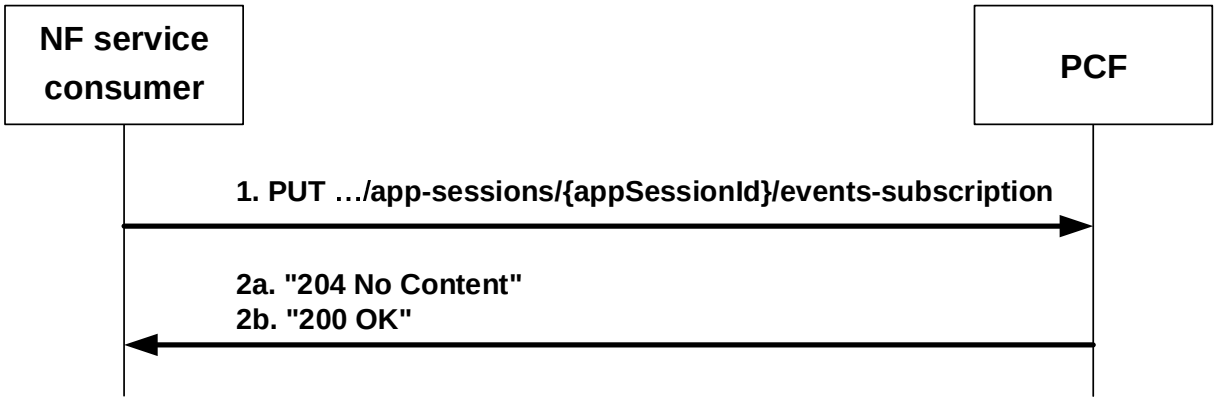


Figure 4.2.6.2-2: Modification of events subscription information using HTTP PUT

When the NF service consumer decides to create a subscription to one or more events for the existing application session context or to modify an existing subscription previously created by itself at the PCF, the NF service consumer shall invoke the Npcf_PolicyAuthorization_Subscribe service operation by sending the HTTP PUT request to the resource URI representing the "Events Subscription" sub-resource in the PCF, as shown in figure 4.2.6.2-1, step 1 and figure 4.2.6.2-2, step 1. The NF service consumer shall provide in the "EventsSubscReqData" data type of the body of the HTTP PUT request:

- the "events" attribute with the list of events to be subscribed; and
- the "notifUri" attribute that includes the Notification URI to indicate to the PCF where to send the notification of the subscribed events.

NOTE 1: The "notifUri" attribute within the EventsSubscReqData data structure can be modified to request that subsequent notifications are sent to a new NF service consumer.

Upon the reception of the HTTP PUT request from the NF service consumer, the PCF shall decide whether the received HTTP PUT request is accepted.

If the HTTP PUT request from the NF service consumer is rejected, the PCF shall indicate in the HTTP response the cause for the rejection as specified in clause 5.7.

If the feature "ES3XX" is supported, and the PCF determines the received HTTP PUT request needs to be redirected, the PCF shall send an HTTP redirect response as specified in clause 6.10.9 of 3GPP TS 29.500 [5].

If the PCF accepted the HTTP PUT request to create a subscription to events, the PCF shall create the "Events Subscription" sub-resource and shall send the HTTP response message to the NF service consumer as shown in figure 4.2.6.2-1, step 2. The PCF shall include in the "201 Created" response:

- a Location header field that shall contain the URI of the created "Events Subscription" sub-resource i.e. "{apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}/events-subscription"; and

- a response body with the "EventsSubscPutData" data type, that contains the attributes of the "EventsSubscReqData" data type, representing the created "Events Subscription" sub-resource.

If the PCF determines that one or more of the subscribed events are already met in the PCF, the PCF may also include the attributes of the "EventsNotification" data type within the "EventsSubscPutData" data type to notify about the already met events in the PCF.

If the PCF accepted the HTTP PUT request to modify the events subscription, the PCF shall modify the "Events Subscription" sub-resource and shall send to the NF service consumer:

- the HTTP "204 No Content" response (as shown in figure 4.2.6.2-2, step 2a); or

- the HTTP "200 OK" response (as shown in figure 4.2.6.2-2, step 2b) including in the "EventsSubscPutData" data type the updated representation of the "Events Subscription" sub-resource encoded within the attributes of the "EventsSubscReqData" data type and, if one or more of the updated subscribed events are already met in the PCF, the notification of these events by including the attributes of the "EventsNotification" data type.

The PCF shall include in the EventsNotification data type:

- if the NF service consumer subscribed to the "PLMN_CHG" event in the HTTP PUT request, an entry of the "evNotifs" attribute including the "event" attribute set to "PLMN_CHG" and the "plmnId" attribute including the PLMN Identifier or the SNPN Identifier if the PCF has previously requested to be updated with this information in the SMF; and

NOTE 2: The SNPN Identifier consists of the PLMN Identifier and the NID.

NOTE 3: Handover between non-equivalent SNPNs, and between SNPN and PLMN is not supported. When the UE is operating in SNPN access mode, the trigger reports changes of equivalent SNPNs.

- if the NF service consumer subscribed to the "ACCESS_TYPE_CHANGE" event in the HTTP PUT request, an entry of the "evNotifs" attribute including the "event" attribute set to "ACCESS_TYPE_CHANGE" and:

  i. the "accessType" attribute including the access type, and the "ratType" attribute including the RAT type when applicable for the notified access type; and

  ii. if the "ATSSS" feature is supported, the "addAccessInfo" attribute with the additional access type information if available, where the access type is encoded in the "accessType" attribute, and the RAT type is encoded in the "ratType" attribute when applicable for the notified access type; and

NOTE 4: For a MA PDU session, if the "ATSSS" feature is not supported by the NF service consumer the PCF includes the "accessType" attribute and the "ratType" attribute with a currently active combination of access type and RAT type (when applicable for the notified access type). When both 3GPP and non-3GPP accesses are available, the PCF includes the information corresponding to the 3GPP access.

  iii. the "anGwAddr" attribute including access network gateway address when available,

  if the PCF has previously requested to be updated with this information in the SMF.

Based on the received subscription information from the NF service consumer, the PCF may create a subscription to event notifications or may modify the existing subscription to event notifications, for a related PDU session from the SMF, as described in 3GPP TS 29.512 [8].

4.2.6.3 Initial subscription to events without provisioning of service information

The NF service consumer may subscribe with the PCF to events notification without providing service information.

NOTE 1: This service operation is intended to create a resource that enables to handle subscription to events without provisioning service information. For the scenarios where it is known the NF service consumer, after creating a subscription without service information, could require an application session context with the PCF with required service information, the NF service consumer needs to create an Individual Application Session context as described in clause 4.2.2.2.

Figure 4.2.6.3-1 illustrates the initial subscription to events without provisioning of service information.

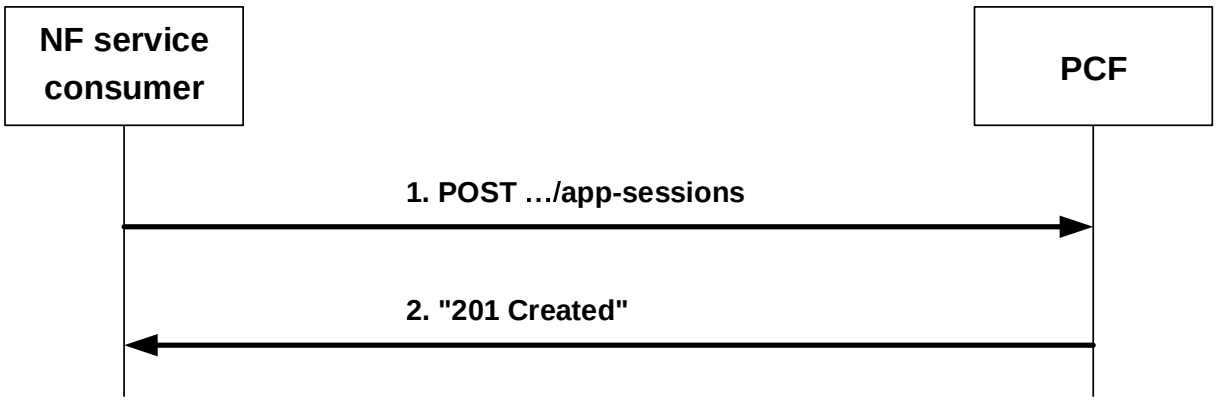


Figure 4.2.6.3-1: Initial Subscription to events without provisioning of service information

When a NF service consumer establishes an application session context with the PCF to subscribe to events and does not require PCC control for the related media, the NF service consumer shall invoke the Npcf_PolicyAuthorization_Subscribe service operation by sending the HTTP POST request to the resource URI representing the "Application Sessions" collection resource of the PCF, as shown in figure 4.2.6.3-1, step 1.

The NF service consumer shall include in the "ascReqData" attribute of the "AppSessionContext" data type in the content of the HTTP POST request:

- either the "ueMac" attribute containing the UE MAC address, or the "ueIpv4" attribute or "ueIpv6" attribute containing the UE IPv4 or the IPv6 address;
- the "notifUri" attribute containing the URI where the PCF shall request to the NF service consumer the deletion of the "Individual Application Session Context" resource"; and
- the "evSubsc" attribute of "EventsSubscReqData" data type to request the notification of certain user plane events. The NF service consumer shall include:
  a. the events to subscribe to in the "events" attribute; and
  b. the notification URI where to address the notification of the met events within the "notifUri" attribute.

The NF service consumer may provide in the "AppSessionContext" data type the DNN in the "dnn" attribute, SUPI in the "supi" attribute or other information if available.

If the PCF cannot successfully fulfil the received HTTP POST request due to the internal PCF error or due to the error in the HTTP POST request, the PCF shall send the HTTP error response as specified in clause 5.7.

Otherwise, when the PCF receives the HTTP POST request from the NF service consumer, the PCF shall apply session binding as described in 3GPP TS 29.513 [7]. The PCF identifies the PDU session for which the HTTP POST request applies as described in clause 4.2.2.2.

If the PCF fails in executing session binding, the PCF shall reject the Npcf_PolicyAuthorization_Subscribe service operation with an HTTP "500 Internal Server Error" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to "PDU_SESSION_NOT_AVAILABLE".

The information required for session binding (UE MAC address, or UE Ipv4 or IPv6 address, DNN, SUPI and other available information, such as S-NSSAI and/or IPv4 address domain identifier) is provisioned in the "Individual Application Session Context" resource. The events subscription is provisioned in the "Events Subscription" sub-resource.

Based on the received subscription information from the NF service consumer, the PCF may create a subscription to event notifications for a related PDU session from the SMF, as described in 3GPP TS 29.512 [8].

If the PCF created the "Events Subscription" sub-resource within the "Individual Application Session Context" resource, the PCF shall send to the NF service consumer a "201 Created" response to the HTTP POST request, as shown in figure 4.2.6.3-1, step 2. The PCF shall include in the "201 Created" response:

- a Location header field; and
- an "AppSessionContext" data type in the content.

The Location header field shall contain the URI of the created events subscription sub-resource i.e. "{apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}/events-subscription".

The "AppSessionContext" data type content shall contain the representation of the created "Individual Application Session Context" resource and "Events Subscription" sub-resource.

The PCF shall include in the "evsNotif" attribute:

- if the NF service consumer subscribed to the event "PLMN_CHG" in the HTTP POST request, an entry of the "evNotifs" attribute including the "event" attribute set to "PLMN_CHG" and the "plmnId" attribute including the PLMN Identifier or the SNPN Identifier if the PCF has previously requested to be updated with this information in the SMF;

NOTE 2: The SNPN Identifier consists of the PLMN Identifier and the NID.

NOTE 3: Handover between non-equivalent SNPNs, and between SNPN and PLMN is not supported. When the UE is operating in SNPN access mode, the trigger reports changes of equivalent SNPNs.

- if the NF service consumer subscribed to the event "ACCESS_TYPE_CHANGE" in the HTTP POST request, an entry of the "evNotifs" attribute including the "event" attribute set to "ACCESS_TYPE_CHANGE" and:

  i. the "accessType" attribute including the access type, and the "ratType" attribute including the RAT type when applicable for the notified access type; and

  ii. if the "ATSSS" feature is supported, the "addAccessInfo" attribute with the additional access type information if available, where the access type is encoded in the "accessType" attribute, and the RAT type is encoded in the "ratType" attribute when applicable for the notified access type; and

NOTE 4: For a MA PDU session, if the "ATSSS" feature is not supported by the NF service consumer the PCF includes the "accessType" attribute and the "ratType" attribute with a currently active combination of access type and RAT type (if applicable for the notified access type). When both 3GPP and non-3GPP accesses are available, the PCF includes the information corresponding to the 3GPP access.

  iii. the "anGwAddr" attribute including access network gateway address when available,

  if the PCF has previously requested to be updated with this information in the SMF; and

- if the "IMS_SBI" feature is supported and if the NF service consumer subscribed to the event "CHARGING_CORRELATION" in the HTTP POST request, an entry of the "evNotifs" attribute with the "event" attribute set to "CHARGING_CORRELATION" and may include the "anChargIds" attribute containing the access network charging identifier(s) and the "anChargAddr" attribute containing the access network charging address.

NOTE 5: Due to the resource structure, as result of the Npcf_PolicyAuthorization_Subscribe service operation using POST methods, the PCF creates an Individual Application Session context resource which can only be deleted via Npcf_PolicyAuthorization_Delete service operation.

4.2.6.4 Subscription to usage monitoring of sponsored data connectivity

This procedure is used by a NF service consumer to subscribe with the PCF to usage monitoring of sponsored data connectivity or to provide updated usage thresholds for the existing application session context, when the "Sponsored Connectivity" feature is supported.

The NF service consumer shall include in the HTTP PUT request message described in clause 4.2.6.2 the "EventsSubscReqData" data type, that shall contain:

- the "events" attribute with a new entry of the "AfEventSubscription" data type with the "event" attribute set to "USAGE_REPORT"; and
- the "usgThres" attribute with the usage thresholds to apply.

The PCF shall reply to the NF service consumer as described in clause 4.2.6.2.

4.2.6.5 Void

4.2.6.6 Request of access network information

This procedure is used by a NF service consumer to request the PCF to report the access network information (i.e. user location and/or user timezone information) without providing service information when the "NetLoc" feature is supported and to report serving satellite identifier within the access network information, if available and when the "UeSatUeComm" feature is supported.

The NF service consumer can request access network information without providing service information:

- at initial subscription to events, using the HTTP POST request message as described in clause 4.2.6.3; and
- at modification of the subscription to events, using the HTTP PUT request message as described in clause 4.2.6.2.

The NF service consumer shall include in the HTTP request message:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
  a) the "event" attribute set to the value "ANI_REPORT"; and
  b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information and/or if the "UeSatUeComm" feature is supported, identifier of the serving satellite of the UE.

When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature, the PCF shall respond to the NF service consumer including in the "EventsNotification" data type the "noNetLocSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

If the PCF supports "SmfFailureDetection" feature and the PCF due to operator configuration detects that the SMF has failed as defined in 3GPP TS 23.380 [39], and when the AF initiates an Npcf_PolicyAuthorization_Subscribe service operation to request access network information, the PCF shall reject the request with an HTTP "403 Forbidden" response including the ExtendedProblemDetails data structure containing the ProblemDetails data structure with the "cause" attribute set to "TEMPORARY_NETWORK_FAILURE".

The PCF shall reply to the NF service consumer with the HTTP POST response as described in clause 4.2.6.3 and with the HTTP PUT response as described in clause 4.2.6.2.

4.2.6.7 Subscription to notification of signalling path status

When the feature "IMS_SBI" is supported, this procedure is used by a NF service consumer to subscribe to notifications of the status of the AF signalling transmission path.

The NF service consumer shall create a new "Individual Application Session Context" resource with the PCF for the AF signalling using the Npcf_PolicyAuthorization_Create service operation.

The NF service consumer shall provide:

- the IP address (IPv4 or IPv6) of the UE in the "ueIpv4" or "ueIpv6" attribute;
- within the "evSubsc" attribute, the "event" attribute set to "FAILED_RESOURCES_ALLOCATION"; and

- a media component within the "medComponents" attribute including:
  - the "medCompN" attribute set to "0"; and
  - a single media subcomponent within the "medSubComps" attribute with:
    - the "flowUsage" attribute set to the value "AF_SIGNALLING"; and
    - if the procedures for NF service consumer provisioning of AF signalling flow information do not apply, the "fNum" attribute set to "0".

When the "fNum" attribute is set to "0", the rest of attributes within the related media component and media subcomponent shall not be used.

The PCF shall perform session binding as described in 3GPP TS 29.513 [7] and shall reply to the NF service consumer as described in clause 4.2.6.3.

PCC rules related to AF signalling IP flows should be provisioned to SMF using the corresponding procedures specified at 3GPP TS 29.512 [8] at an earlier stage (e.g. typically at the establishment of the QoS flow dedicated for AF signalling IP flows). The PCF may install the corresponding dynamic PCC rules for the AF signalling IP flows if none has been installed before.

NOTE 1: Well-known ports (e.g. 3GPP TS 24.229 [32] for SIP) or wildcard ports can be used by PCF to derive the dynamic PCC for the AF signalling IP flows.

If the "Individual Application Session Context" resource is only used for subscription to notification of AF signalling path status, the NF service consumer may cancel the subscription to notifications of the status of the AF signalling transmission path removing the resource as described in clause 4.2.4.2.

NOTE 2: The "Individual Application Session Context" resource created for the AF signalling can also be used when the NF service consumer requests notifications of access type change, access network information for SMS over IP and/or when the NF service consumer provisions AF Signalling Flow Information.

4.2.6.8 Subscription to Service Data Flow QoS Monitoring Information

This procedure is used by NF service consumer to subscribe and/or modify the PCF subscription at AF session level for notification about real-time measurements of QoS monitoring parameters for a QoS flow (i.e. when the AF request involves only flows with the same QoS requirements), e.g. packet delay between UPF and UE, when the "QoSMonitoring" feature is supported.

NOTE 1: When the subscription to QoS Monitoring is for multiple QoS flows, i.e, when the AF-session can contain multiple media components, the subscription for QoS monitoring can only be indicated within the corresponding "medSubComps" entry and the Npcf_PolicyAuthorization_Subscribe service operation is not supported.

The NF service consumer shall include in the HTTP PUT request message described in clause 4.2.6.2 the "EventsSubscReqData" data type, that shall contain:

- to create a subscription to notifications of QoS monitoring report(s) the provisions specified in clause 4.2.2.23.1 apply;

NOTE 2: When the "congestMon" and "avlBitRateMon" attributes are included, only the "EVENT_DETECTION" reporting frequency is applicable.

- to remove a subscription to QoS monitoring information for packet delay:
  a) shall include the "events" array containing an array that shall omit the corresponding entry with the "event" attribute value "QOS_MONITORING"; and
  b) when the "notifMethod" of the removed entry is "EVENT_DETECTION", it shall omit the "qosMon" and/or "qosMonDatRate" and/or "congestMon" and/or "avlBitRateMon" attributes;
  c) shall omit the "reqQosMonParams";
  d) if the feature "ExposureToEAS" is supported, shall omit the "directNotifInd" attribute;

- when the "EnQoSMon" feature is supported, to remove a subscription to QoS monitoring:

  a) shall include the "events" array and shall omit the corresponding entry(ies) with the "event" attribute value "QOS_MONITORING" and "qosMonParamType" attribute, if applicable;

  b) if the notification method of the removed entry is "EVENT_DETECTION", it shall omit the "qosMon" and/or "qosMonDatRate" and/or "congestMon" and/or "avlBitRateMon" attribute(s); and

  c) if the "directNotifInd" attribute and/or the "avrgWndw" attribute was previously provided, it shall omit the "directNotifInd" attribute and/or the "avrgWndw" attribute.

The NF service consumer shall include other events related information that shall remain unchanged.

As result of this action, the PCF shall set the appropriate subscription to QoS monitoring information for the corresponding active PCC rule(s) as described in 3GPP TS 29.512 [8].

If the AF provided an indication of direct event notification, and the PCF determines that the QoS Monitoring reports cannot be notified directly (e.g. the AF requests for monitoring packet delay variation or round trip packet delay when UL and DL are on different service data flows and the information cannot be consolidated in the QoS monitoring policy in the PCC rule), the PCF generates a successful response to the AF and indicates that direct event notification is not possible by including within the "servAuthInfo" attribute the value "DIRECT_NOTIF_NOT_POSSIBLE", as described in clause 4.2.2.23.1.

The PCF shall reply to the NF service consumer as described in clause 4.2.6.2.

4.2.6.9 Subscription to application detection notification

This procedure is used by a NF service consumer to request the PCF the subscription to application (e.g. start, stop) detection notifications, if the "ApplicationDetectionEvents" feature is supported.

The NF service consumer can request the subscription to notification of application detection events without providing service information:

- at initial subscription to events, using the HTTP POST request message as described in clause 4.2.6.3; and
- at modification of the subscription to events, using the HTTP PUT request message as described in clause 4.2.6.2.

The NF service consumer shall include:

- To subscribe to notifications about the detection of the start/stop of one or more application's traffic the "evSubsc" attribute within the POST request as described in clause 4.2.6.3, with:

  a. the "events" array, including an event with the "event" attribute value set to "APP_DETECTION"; and

  b. the "afAppIds" attribute, with the list of AF application identifier(s) that refer to the applications' traffic to detect.

- To modify the subscription to notifications of application's traffic detection the "EventsSubscReqData" data type within the PUT request as described in clause 4.2.6.2, including an event with the "event" attribute value set to "APP_DETECTION" and an updated list of AF application identifiers within the "afAppIds" attribute.
- To remove the subscription to notifications about the start and stop of the applications traffic, an "events" array within the PUT request as described in clause 4.2.6.2, without including any event with the "event" attribute value "APP_DETECTION" and omitting the "afAppIds" attribute.

The PCF shall reply to the NF service consumer with the HTTP POST response as described in clause 4.2.6.3 and with the HTTP PUT response as described in clause 4.2.6.2.

The PCF shall set the appropriate subscription to Application Detection for the corresponding PCC rule(s) as described in 3GPP TS 29.512 [8].

In this release of the specification application detection applies only to the application(s) with IP traffic.

4.2.6.10 Subscription to satellite backhaul category changes

When the feature "SatelliteBackhaul"is supported, this procedure is used by NF service consumer to subscribe and/or modify the PCF subscription to receive a notification when the satellite backhaul category changes and when the backhaul category changes between satellite backhaul and non-satellite backhaul. When the feature "EnSatBackhaulCatChg" is supported, the subscription is also used to receive a notification when the satellite backhaul category change comprises a dynamic satellite backhaul category.

The NF service consumer shall include in the HTTP PUT request message described in clause 4.2.6.2 the "EventsSubscReqData" data type, or in the HTTP POST request message described in clause 4.2.6.3 the "evSubsc" attribute, that shall contain the "events" array, with a new entry with the "event" attribute set to "SAT_CATEGORY_CHG" to indicate the creation of a subscription to backhaul category changes.

The NF service consumer shall include other events related information that shall remain unchanged.

As result of this action, the PCF shall set the appropriate subscription to satellite backhaul changes for the PDU session as described in in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer as described in clause 4.2.6.2 or in clause 4.2.6.3. The PCF shall include the EventsNotification data type with an entry in the "evNotifs" array with the "event" attribute set to "SAT_CATEGORY_CHG" and the "satBackhaulCategory" attribute including the satellite backhaul category or the indication of non-satellite backhaul if the PCF has previously subscribed with the SMF to changes in this information.

4.2.6.11 Subscription to TSC user plane node related events

When the "TimeSensitiveNetworking" or "TimeSensitiveCommunication" feature is supported, this procedure is used by the NF service consumer to subscribe and/or modify the PCF subscription to received notification when TSC user plane node information is updated.

The NF service consumer shall include in the HTTP PUT request message described in clause 4.2.6.2 the "EventsSubscReqData" data type, or in the HTTP POST request message described in clause 4.2.6.3 the "evSubsc" attribute, that shall contain the "events" array, with a new entry with the "event" attribute set to the value "TSN_BRIDGE_INFO" to subscribe to the reception of TSC user plane node information.

The NF service consumer shall include other events related information that shall remain unchanged.

As result of this action, the PCF shall set the corresponding subscription to the report of TSC user plane node management information and port management information for the corresponding PDU session as described in 3GPP TS 29.512 [8].

The PCF shall reply to the NF service consumer in clause 4.2.6.2 or in clause 4.2.6.3. The PCF shall include the EventsNotification data type with an entry in the "evNotifs" array with the "event" attribute set to attribute set to "TSN_BRIDGE_INFO" and the "tsnBridgeManCont" attribute and/or the "tsnPortManContDstt" attribute and/or the "tsnPortManContNwtts" attribute as received from the SMF, if not previously reported.

As result of this action, the PCF shall subscribe to the report of TSC user plane node management information and port management information for the corresponding PDU session, if not previously subscribed, as described in 3GPP TS 29.512 [8].

4.2.6.12 Subscription to the report of extra UE addresses

When the feature "ExtraUEaddrReport" is supported, the subscription to the report of extra UE addresses is used to report about the extra IP addresses or address ranges allocated to the PDU session due to framed routes or IPv6 prefix delegation. The report shall include the actual list of IPv4 addresses or list of IPv6 prefixes as currently allocated.

NOTE: In case of Deterministic Networking, the 5GS DetNet Node, as described in 3GPP TS 23.501 [2], can forward via its device side interface IP packets destined not only to the UE's IP address or prefix but also to a range of IPv4 addresses or IPv6 IP prefixes according to one or more Framed Routes or prefixes delegated to the UE by IPv6 prefix delegation.

The NF service consumer shall include in the HTTP PUT request message described in clause 4.2.6.2 the "EventsSubscReqData" data type, or in the HTTP POST request message described in clause 4.2.6.3 the "evSubsc"

attribute, that shall contain the "events" array, with a new entry with the "event" attribute set to the value "EXTRA_UE_ADDR".

The NF service consumer shall include other events related information that shall remain unchanged, if applicable.

The PCF shall reply to the NF service consumer with the HTTP POST response as described in clause 4.2.6.3 and with the HTTP PUT response as described in clause 4.2.6.2.

If the PCF received from the SMF the framed routes as described in 3GPP TS 29.512 [8], clause 4.2.2.2, or the PCF received updated information of the one or more IPv6 prefixes allocated to the UE as described in 3GPP TS 29.512 [8], clauses 4.2.4.2, 5.6.2.19, and C.3.4.1, the PCF shall include in the response the EventsNotification data type with an entry in the "evNotifs" array with the "event" attribute set to "EXTRA_UE_ADDR" and:

- the actual list of IPv4 addresses within the "ipv4AddrList" attribute, if one or more IPv4 framed routes are associated to the PDU session and are available in the PCF; or
- the actual list of IPv6 prefixes allocated to the UE within the "ipv6PrefixList" attribute, if one or more IPv6 framed routes are associated to the PDU session and are available in the PCF, or if the PCF keeps updated IPv6 prefix(es) information.

As result of this action, the PCF shall set the appropriate subscription to the report of UE IP addresses, if not previously subscribed, as described in 3GPP TS 29.512 [8].

4.2.6.13 Void

4.2.6.14 Subscription to notifications about URSP rule enforcement information

This procedure is used by a NF service consumer to request the PCF the subscription to URSP rule enforcement information notification, if the "URSPEnforcement" feature is supported.

The NF service consumer may request the subscription to notification of URSP rule enforcement information event without providing service information:

- at initial subscription to events, using the HTTP POST request message as described in clause 4.2.6.3; and
- at modification of the subscription to events, using the HTTP PUT request message as described in clause 4.2.6.2.

The NF service consumer shall include:

- To subscribe to notifications about URSP rule enforcement information, the "evSubsc" attribute within the POST request as described in clause 4.2.6.3, with the "events" array, including an event with the "event" attribute value set to "URSP_ENF_INFO"; and
- To remove the subscription to notifications about URSP rule enforcement information, an "events" array within the PUT request as described in clause 4.2.6.2, without including any event with the "event" attribute value "URSP_ENF_INFO".

The PCF shall reply to the NF service consumer with the HTTP POST response as described in clause 4.2.6.3 and If URSP rule enforcement information corresponding to the subscription is available, the PCF shall include in the POST response (within the "evsNotif" attribute) and in the PUT response (within the EventsNotification data type), the received URSP rule enforcement information within the "urspEnfRep" attribute, the SSC mode within the "sscMode" attribute, the UE requested DNN, if available, within the "ueReqDnn" attribute, and the UE requested PDU Session Type within the "ueReqPduSessionType" attribute, and the "URSP_ENF_INFO" event in the "event" attribute of the "evNotifs" array as defined in clause 4.2.5.25.

4.2.6.15 Subscription to the report of network support for QoS Monitoring

When the feature "QoSMonCapRepo" is supported, the NF service consumer may subscribe to receive notifications from the PCF of whether the requested QoS monitoring policy is not supported or is supported again by including within the "events" array an entry with the "event" attribute set to "QOS_MON_CAP_REPO" and the monitoring type(s) within the "capTypes" attribute. The NF service consumer may delete a subscription to QoS monitoring support

report by providing the "events" array and omitting an entry with the "event" attribute set to "QOS_MON_CAP_REPO".

The PCF shall reply to the NF service consumer as described in clause 4.2.6.2.

4.2.7 Npcf_PolicyAuthorization_Unsubscribe service operation

4.2.7.1 General

The Npcf_PolicyAuthorization_Unsubscribe service operation enables NF service consumers to remove subscription to all subscribed events for the existing application session context. Subscription to events for the exsiting application session context shall be removed:

- by invoking the Npcf_PolicyAuthorization_Unsubscribe service operation for the existing application session context, as described in clause 4.2.7.2; or
- within the application session context modification procedure by invoking the Npcf_PolicyAuthorization_Update service operation, as described in clause 4.2.3; or
- within the application session context termination procedure by invoking the Npcf_PolicyAuthorization_Delete service operation, as described in clause 4.2.4.

The following procedure using the Npcf_PolicyAuthorization_Unsubscribe service operation is supported:

- Unsubscription to events for an application session context.

4.2.7.2 Unsubscription to events for an application session context

This procedure is used to unsubscribe to all subscribed events for the existing AF application session context, as defined in 3GPP TS 23.501 [2], 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

Figure 4.2.7.2-1 illustrates the unsubscription to events using the HTTP DELETE method.

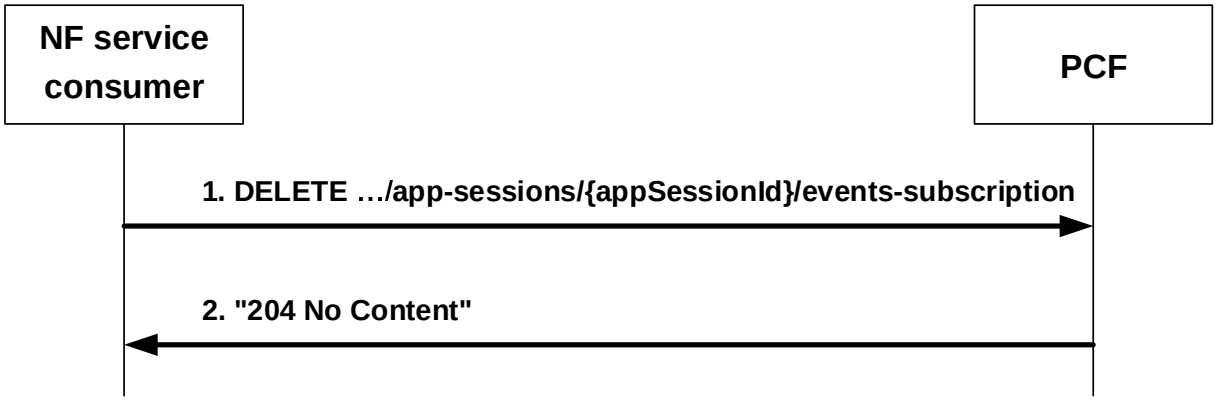


Figure 4.2.7.2-1: Removal of events subscription information using HTTP DELETE

When the NF service consumer decides to unsubscribe to all subscribed events for the existing application session context, the NF service consumer shall invoke the Npcf_PolicyAuthorization_Unsubscribe service operation by sending the HTTP DELETE request message to the resource URI representing the "Events Subscription" sub-resource in the PCF, as shown in figure 4.2.7.2-1, step 1.

Upon the reception of the HTTP DELETE request message from the NF service consumer, the PCF shall decide whether the received HTTP request message is accepted.

If the HTTP DELETE request message from the NF service consumer is accepted, the PCF shall delete "Events Subscription" sub-resource and shall send to the NF service consumer a HTTP "204 No Content" response message. The PCF may delete the existing subscription to event notifications for the related PDU session from the SMF as described in 3GPP TS 29.512 [8].

If the HTTP DELETE request message from the NF service consumer is rejected, the PCF shall indicate in the HTTP response message the cause for the rejection as specified in clause 5.7.

5 Npcf_PolicyAuthorization Service API

5.1 Introduction

The Npcf_PolicyAuthorization Service shall use the Npcf_PolicyAuthorization API.

The API URI of the Npcf_PolicyAuthorization API shall be:

{apiRoot}/<apiName>/<apiVersion>

The request URIs used in each HTTP requests from the NF service consumer towards the PCF shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [6], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [6].
- The <apiName> shall be "npcf-policyauthorization".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 5.3.

5.2 Usage of HTTP

5.2.1 General

HTTP/2, IETF RFC 9113 [9], shall be used as specified in clause 5.2 of 3GPP TS 29.500 [5].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [5].

The OpenAPI [11] specification of HTTP messages and content bodies for the Npcf_PolicyAuthorization service is contained in Annex A.

5.2.2 HTTP standard headers

5.2.2.1 General

See clause 5.2.2 of 3GPP TS 29.500 [5] for the usage of HTTP standard headers.

5.2.2.2 Content type

JSON, IETF RFC 8259 [10], shall be used as content type of the HTTP bodies specified in the present specification, as specified in clause 5.4 of 3GPP TS 29.500 [5]. The use of the JSON format shall be signalled by the content type "application/json".

JSON object used in the HTTP PATCH request shall be encoded according to "JSON Merge Patch" and shall be signalled by the content type "application/merge-patch+json", as defined in IETF RFC 7396 [21].

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 9457 [24].

5.2.3 HTTP custom headers

The Npcf_PolicyAuthorization API shall support mandatory HTTP custom header fields specified in clause 5.2.3.2 of 3GPP TS 29.500 [5] and may support optional HTTP custom header fields specified in clause 5.2.3.3 of 3GPP TS 29.500 [5].

In this Release of the specification, no specific custom headers are defined for the Npcf_PolicyAuthorization API.

5.3 Resources

5.3.1 Resource Structure

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 5.3.1-1 depicts the resource URIs structure for the Npcf_PolicyAuthorization API.

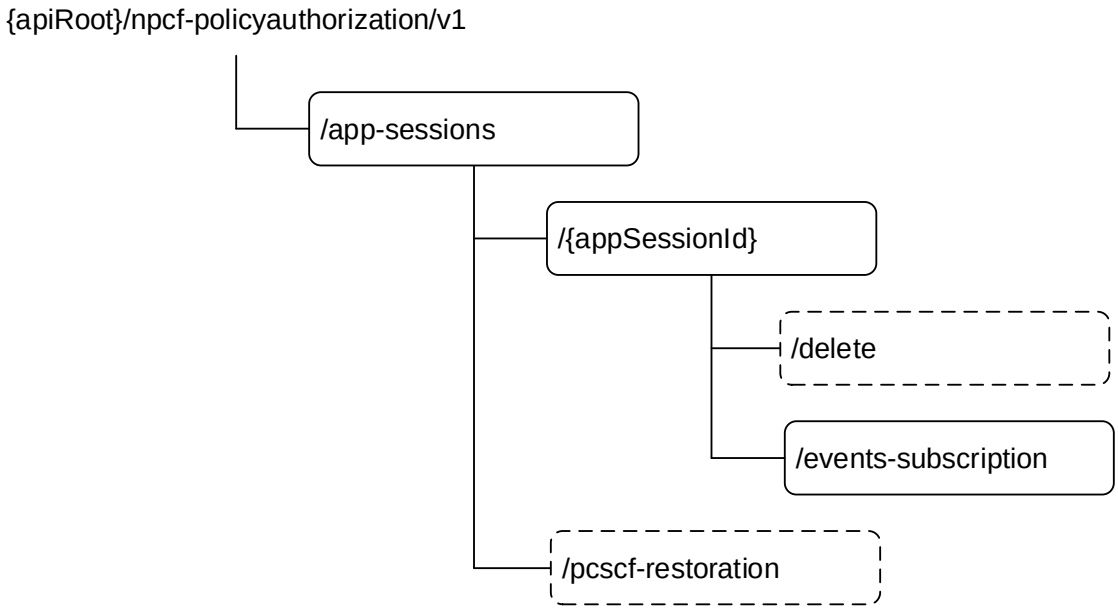


Figure 5.3.1-1: Resource URI structure of the Npcf_PolicyAuthorization API

Table 5.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 5.3.1-1: Resources and methods overview

Resource name	Resource URI	HTTP method or custom operation	Description
Application Sessions	/app-sessions	POST	Npcf_PolicyAuthorization_Create. Creates a new Individual Application Session Context resource and may create the child Events Subscription sub-resource.
	/app-sessions/pcscf-restoration	PcscfRestoration (POST)	P-CSCF restoration. It indicates that P-CSCF restoration needs to be performed.
Individual Application Session Context	/app-sessions/{appSessionId}	PATCH	Npcf_PolicyAuthorization_Update. Updates an existing Individual Application Session Context resource. It can also update an Events Subscription sub-resource.
		GET	Reads an existing Individual Application Session Context resource.
	/app-sessions/{appSessionId}/delete	delete (POST)	Npcf_PolicyAuthorization_Delete. Deletes an existing Individual Application Session Context resource and the child Events Subscription sub-resource.
Events Subscription	/app-sessions/{appSessionId} /events-subscription	PUT	Npcf_PolicyAuthorization_Subscribe. Creates a new Events Subscription sub-resource or modifies an existing Events Subscription sub-resource.
		DELETE	Npcf_PolicyAuthorization_Unsubscribe. Deletes an Events Subscription sub-resource.

5.3.2 Resource: Application Sessions (Collection)

5.3.2.1 Description

The Application Sessions resource represents all application session contexts that exist in the Npcf_PolicyAuthorization service at a given PCF instance.

5.3.2.2 Resource definition

Resource URI: **{apiRoot}/npcf-policyauthorization/v1/app-sessions**

This resource shall support the resource URI variables defined in table 5.3.2.2-1.

Table 5.3.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 5.1

5.3.2.3 Resource Standard Methods

5.3.2.3.1 POST

This method shall support the URI query parameters specified in table 5.3.2.3.1-1.

Table 5.3.2.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.2.3.1-2 and the response data structures and response codes specified in table 5.3.2.3.1-3.

Table 5.3.2.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
AppSessionContext	M	1	Contains the information for the creation of a new Individual Application Session Context resource.

Table 5.3.2.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AppSessionContext	M	1	201 Created	Successful case. The creation of an Individual Application Session Context resource is confirmed and a representation of that resource is returned.
n/a			303 See Other	The result of the HTTP POST request would be equivalent to the existing Application Session Context. The HTTP response shall contain a Location header field set to the URI of the existing individual Application Session Context resource.
ProblemDetails	O	0..1	400 Bad Request	(NOTE 2)
ExtendedProblemDetails	O	0..1	403 Forbidden	(NOTE 2)
ProblemDetails	O	0..1	500 Internal Server Error	(NOTE 2)
NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the POST method shall also apply. NOTE 2: Failure cases are described in clause 5.7.				

Table 5.3.2.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}

Table 5.3.2.3.1-5: Headers supported by the 303 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the existing individual Application Session Context resource.

Table 5.3.2.3.1-6: Headers supported by the 403 Response Code on this resource

Name	Data type	P	Cardinality	Description
Retry-After	string or integer	M	1	Indicates the time the NF service consumer has to wait before making a new request.

5.3.2.4 Resource Custom Operations

5.3.2.4.1 Overview

Table 5.3.2.4.1-1: Custom operations

Operation name	Custom operation URI	Mapped HTTP method	Description
PcscfRestoration	/app-sessions/pcscf-restoration	POST	The P-CSCF Restoration custom operation invokes P-CSCF restoration. It does not create an Individual Application Session Context resource.

5.3.2.4.2 Operation: PcscfRestoration

5.3.2.4.2.1 Description

5.3.2.4.2.2 Operation Definition

This custom operation invokes P-CSCF restoration in the PCF and does not create an Individual Application Session Context resource.

This operation shall support the request data structure specified in table 5.3.2.4.2.2-1 and the response data structure and response codes specified in table 5.3.2.4.2.2-2.

Table 5.3.2.4.2.2-1: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
PcscfRestorationRequestData	O	0..1	P-CSCF restoration data to be sent by the NF service consumer to request the P-CSCF restoration to the PCF.

Table 5.3.2.4.2.2-2: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case. The P-CSCF restoration has been successfully invoked.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during P-CSCF restoration. Applicable if the feature "ES3XX" is supported. (NOTE 3)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during P-CSCF restoration. Applicable if the feature "ES3XX" is supported. (NOTE 3)
ProblemDetails	O	0..1	500 Internal Server Error	(NOTE 2)

NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the POST method shall also apply.
NOTE 2: Failure cases are described in subclause 5.7.
NOTE 3: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).

Table 5.3.2.4.2.2-3: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the request is redirected.

Table 5.3.2.4.2.2-4: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

5.3.3 Resource: Individual Application Session Context (Document)

5.3.3.1 Description

The Individual Application Session Context resource represents a single application session context that exists in the Npcf_PolicyAuthorization service.

5.3.3.2 Resource definition

Resource URI: **{apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}**

This resource shall support the resource URI variables defined in table 5.3.2.2-1.

Table 5.3.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 5.1
appSessionId	string	Identifies an application session context formatted according to IETF RFC 3986 [19].

5.3.3.3 Resource Standard Methods

5.3.3.3.1 GET

This method shall support the URI query parameters specified in table 5.3.3.3.1-1.

Table 5.3.3.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.3.3.1-2 and the response data structures and response codes specified in table 5.3.3.3.1-3.

Table 5.3.3.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.3.3.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AppSessionContext	M	1	200 OK	A representation of an Individual Application Session Context resource is returned.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during Individual Application Session Context retrieval. Applicable if the feature "ES3XX" is supported. (NOTE 3)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during Individual Application Session Context retrieval. Applicable if the feature "ES3XX" is supported. (NOTE 3)
ProblemDetails	O	0..1	404 Not Found	(NOTE 2)
NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the GET method shall also apply. NOTE 2: Failure cases are described in clause 5.7. NOTE 3: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).				

Table 5.3.3.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

Table 5.3.3.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

5.3.3.3.2 PATCH

This method shall support the URI query parameters specified in table 5.3.3.3.2-1.

Table 5.3.3.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.3.3.2-2 and the response data structures and response codes specified in table 5.3.3.3.2-3.

Table 5.3.3.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
AppSessionContextUpdateDataPatch	M	1	Contains the modification(s) to apply to the Individual Application Session Context resource.

Table 5.3.3.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AppSessionContext	M	1	200 OK	Successful case. The Individual Application Session Context resource was modified and a representation of that resource is returned.
n/a			204 No Content	Successful case. The Individual Application session context resource was modified.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during Individual Application Session Context modification. Applicable if the feature "ES3XX" is supported. (NOTE 3)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during Individual Application Session Context modification. Applicable if the feature "ES3XX" is supported. (NOTE 3)
ProblemDetails	O	0..1	400 Bad Request	(NOTE 2)
ExtendedProblemDetails	O	0..1	403 Forbidden	(NOTE 2)
ProblemDetails	O	0..1	404 Not Found	(NOTE 2)

NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the PATCH method shall also apply.
NOTE 2: Failure cases are described in clause 5.7.
NOTE 3: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).

Table 5.3.3.3.2-4: Headers supported by the 403 Response Code on this resource

Name	Data type	P	Cardinality	Description
Retry-After	string or integer	M	1	Indicates the time the NF service consumer has to wait before making a new request.

Table 5.3.3.3.2-5: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

Table 5.3.3.3.2-6: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located in an alternative PCF (service) instance.
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the request is redirected.

5.3.3.4 Resource Custom Operations

5.3.3.4.1 Overview

Table 5.3.3.4.1-1: Custom operations

Operation name	Custom operation URI	Mapped HTTP method	Description
delete	/app-sessions/{appSessionId}/delete	POST	Npcf_PolicyAuthorization_Delete. Deletes an existing Individual Application Session Context resource and the child Events Subscription sub-resource.

5.3.3.4.2 Operation: delete

5.3.3.4.2.1 Description

5.3.3.4.2.2 Operation Definition

This custom operation deletes an existing Individual Application Session Context resource and the child Events Subscription sub-resource in the PCF.

This operation shall support the request data structures specified in table 5.3.3.4.2.2-1 and the response data structure and response codes specified in table 5.3.3.4.2.2-2.

Table 5.3.3.4.2.2-1: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
EventsSubscReqData	O	0..1	Events subscription information to be sent by the NF service consumer to request event notification when the Individual Application Session Context resource is deleted.

Table 5.3.3.4.2.2-2: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case. The Individual Application session context resource was deleted.
AppSessionContext	M	1	200 OK	Successful case. The Individual Application Session Context resource was deleted and a partial representation of that resource containing event notification information is returned.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during Individual Application Session Context termination. Applicable if the feature "ES3XX" is supported. (NOTE 3)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during Individual Application Session Context termination. Applicable if the feature "ES3XX" is supported. (NOTE 3)
ProblemDetails	O	0..1	404 Not Found	(NOTE 2)

NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the POST method shall also apply.
NOTE 2: Failure cases are described in clause 5.7.
NOTE 3: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).

Table 5.3.3.4.2.2-3: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

Table 5.3.3.4.2.2-4: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the request is redirected.

5.3.4 Resource: Events Subscription (Document)

5.3.4.1 Description

The Events Subscription sub-resource represents a subscription to events for an application session context that exists in the Npcf_PolicyAuthorization service.

NOTE: When an NF service consumer requests subscription to event(s) for different flows of an application session context, the subscription to events is represented within the "medSubComps" attribute of the Individual Application Session Context resource, and does not exist within the Events Subscription sub-resource.

5.3.4.2 Resource definition

Resource URI: **{apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}/events-subscription**

This resource shall support the resource URI variables defined in table 5.3.4.2-1.

Table 5.3.4.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 5.1
appSessionId	string	Identifies an application session context formatted according to IETF RFC 3986 [19].

5.3.4.3 Resource Standard Methods

5.3.4.3.1 PUT

This method shall support the URI query parameters specified in table 5.3.4.3.1-1.

Table 5.3.4.3.1-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.4.3.1-2 and the response data structures and response codes specified in table 5.3.4.3.1-3.

Table 5.3.4.3.1-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
EventsSubscReqData	M	1	Contains information for the modification of the Events Subscription sub-resource.

Table 5.3.4.3.1-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
EventsSubscPutData	M	1	201 Created	Successful case. The Events Subscription sub-resource was created. The properties of the EventsSubscReqData data type shall be included. The properties of the EventsNotification data type shall be included when the notification for one or more created events is already available in the PCF.
EventsSubscPutData	M	1	200 OK	Successful case. The Events Subscription sub-resource was modified and a representation of that sub-resource is returned. The properties of the EventsSubscReqData data type shall be included. The properties of the EventsNotification data type shall be included when the notification for one or more updated events is already available in the PCF.
n/a			204 No Content	Successful case. The Events Subscription sub-resource was modified.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during Events Subscription modification. Applicable if the feature "ES3XX" is supported. (NOTE 3)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during Events Subscription modification. Applicable if the feature "ES3XX" is supported. (NOTE 3)
ProblemDetails	O	0..1	403 Forbidden	(NOTE 2)
ProblemDetails	O	0..1	404 Not Found	(NOTE 2)

NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the PUT method shall also apply.
NOTE 2: Failure cases are described in clause 5.7.
NOTE 3: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).

Table 5.3.4.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}/events-subscription

Table 5.3.4.3.1-5: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

Table 5.3.4.3.1-6: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

5.3.4.3.2 DELETE

This method shall support the URI query parameters specified in table 5.3.4.3.2-1.

Table 5.3.4.3.2-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.4.3.2-2 and the response data structures and response codes specified in table 5.3.4.3.2-3.

Table 5.3.4.3.2-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.3.4.3.2-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case. The Events Subscription sub-resource was deleted.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during Events Subscription termination. Applicable if the feature "ES3XX" is supported. (NOTE 3)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during Events Subscription termination. Applicable if the feature "ES3XX" is supported. (NOTE 3)
ProblemDetails	O	0..1	404 Not Found	(NOTE 2)
NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the DELETE method shall also apply. NOTE 2: Failure cases are described in clause 5.7. NOTE 3: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).				

Table 5.3.4.3.2-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

Table 5.3.4.3.2-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI of the resource located in an alternative PCF (service) instance towards which the request is redirected. For the case where the request is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the request is redirected.

5.3.3.4 Resource Custom Operations

None.

5.4 Custom Operations without associated resources

No custom operation is defined in this Release of the specification.

5.5 Notifications

5.5.1 General

Notifications shall comply to clause 6.2 of 3GPP TS 29.500 [5] and clause 4.6.2.3 of 3GPP TS 29.501 [6].

Table 5.5.1-1: Notifications overview

Notification	Callback URI	HTTP method or custom operation	Description (service operation)
Event Notification	{notifUri}/notify	notify (POST)	PCF event notification.
Termination Request	{notifUri}/terminate	terminate (POST)	Request for termination of an Individual Application Session Context.
Notification about new 5GS Bridge	{notifUri}/new-bridge	new-bridge (POST)	Notification about new 5GS Bridge
Notification about PDU session events	{notifUri}/pdu-session	pdu-session (POST)	Notification about PDU session events not bound to an Individual Application Session Context.

5.5.2 Event Notification

5.5.2.1 Description

The Event Notification is used by the PCF to report one or several observed application session context events to the NF service consumer that has subscribed to such notifications, via the Events Subscription sub-resource for explicit subscriptions or, via the UDR for implicit subscriptions.

NOTE: The "callbacks" definition for the "eventNotification" in the OpenAPI description in Annex A.2 is used for the definition of the notification request also when the notification is triggered without an explicit subscription to events (e.g., clause 4.2.5.29).

5.5.2.2 Target URI

The Callback URI **"{notifUri}/notify"** shall be used with the callback URI variables defined in table 5.5.2.2-1.

Table 5.5.2.2-1: Callback URI variables

Name	Data type	Definition
notifUri	Uri	The Notification URI as assigned within the Events Subscription sub-resource and described within the EventsSubscReqData type (see table 5.6.2.6-1) for explicit subscriptions. For implicit subscriptions, the Notification URI is assigned via the provisioning of the corresponding application data in the UDR (e.g., as specified in clause 4.2.5.29 (see also 3GPP TS 29.519 [53])).

5.5.2.3 Standard Methods

5.5.2.3.1 POST

This method shall support the URI query parameters specified in table 5.5.2.3.1-1.

Table 5.5.2.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.5.2.3.1-2 and the response data structures and response codes specified in table 5.5.2.3.1-3.

Table 5.5.2.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
EventsNotification	M	1	Provides Information about observed events.

Table 5.5.2.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The receipt of the Notification is acknowledged.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during event notification. Applicable if the feature "ES3XX" is supported. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during event notification. Applicable if the feature "ES3XX" is supported. (NOTE 2)
NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the POST method shall also apply. NOTE 2: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).				

Table 5.5.2.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the notification request is redirected.

Table 5.5.2.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the notification request is redirected.

5.5.3 Termination Request

5.5.3.1 Description

The Termination Request is used by the PCF to request the NF service consumer the deletion of the Individual Application Session Context resource.

5.5.3.2 Target URI

The Callback URI **"{notifUri}/terminate"** shall be used with the callback URI variables defined in table 5.5.3.2-1.

Table 5.5.3.2-1: Callback URI variables

Name	Data type	Definition
notifUri	Uri	The Notification Uri as assigned within the Individual Application Session Context resource and described within the AppSessionContextReqData Data type (see table 5.6.2.3-1).

5.5.3.3 Standard Methods

5.5.3.3.1 POST

This method shall support the URI query parameters specified in table 5.5.3.3.1-1.

Table 5.5.3.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.5.3.3.1-2 and the response data structures and response codes specified in table 5.5.3.3.1-3.

Table 5.5.3.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
TerminationInfo	M	1	Provides information about the deletion of the resource.

Table 5.5.3.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The receipt of the Notification is acknowledged.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during event notification. Applicable if the feature "ES3XX" is supported. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during event notification. Applicable if the feature "ES3XX" is supported. (NOTE 2)
NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the POST method shall also apply. NOTE 2: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).				

Table 5.5.3.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the notification request is redirected.

Table 5.5.3.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the notification request is redirected.

5.5.4 Detected 5GS Bridge for a PDU session

5.5.4.1 Description

The detected TSC user plane node for a PDU session operation is used by the PCF to notify the NF service consumer about the detection of TSC user plane node information in the context of a PDU session and to trigger in the NF service consumer (i.e. TSN AF or TSCTSF) the creation of a new Individual Application Session Context to associate it with the detected TSC user plane node for the PDU session.

The PCF shall use the locally configured notification URI of the NF service consumer (i.e. TSN AF or TSCTSF) or the notification URI of the NF service consumer (i.e. TSCTSF) discovered via Nnrf_NFDiscovery service as defined in 3GPP TS 29.510 [27], if not configured, as request URI of the notification request. The "callback" definition in the OpenAPI specification is associated to the "ApplicationSessions" resource.

5.5.4.2 Target URI

The Callback URI **"{notifUri}/new-bridge"** shall be used with the callback URI variables defined in table 5.5.4.2-1.

Table 5.5.4.2-1: Callback URI variables

Name	Data type	Definition
notifUri	Uri	It is locally configured in the PCF or discovered via Nnrf_NFDiscovery service.

5.5.4.3 Standard Methods

5.5.4.3.1 POST

This method shall support the URI query parameters specified in table 5.5.4.3.1-1.

Table 5.5.4.3.1-1: URI query parameters supported by the POST method on this resource

Name		Data type	P	Cardinality	Description
n/a					

This method shall support the request data structures specified in table 5.5.4.3.1-2 and the response data structures and response codes specified in table 5.5.4.3.1-3.

Table 5.5.4.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
PduSessionTsnBridge	M	1	Provides information about the UP node of the reported PDU session.

Table 5.5.4.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The receipt of the notification is acknowledged.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during PDU session TSC user plane node notification. Applicable if the feature "ES3XX" is supported. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during PDU session TSC user plane node notification. Applicable if the feature "ES3XX" is supported. (NOTE 2)
NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the POST method shall also apply. NOTE 2: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).				

Table 5.5.4.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the notification request is redirected.

Table 5.5.4.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target PCF (service) instance towards which the notification request is redirected.

5.5.5 Notification about PDU session event

5.5.5.1 Description

The Notification about PDU session events not bound to an Individual Application Session Context (eventNotificationPduSession) is used by the PCF to notify the NF service consumer (e.g., the PCF for a UE) about the PDU session events.

The PCF shall use the NF service consumer (e.g. PCF for a UE) callback URI implicitly subscribed (e.g. contained in the SM Policy Association of the related PDU session) as request URI of the notification request, and append the "pdu-session" segment path at the end of the URI. The "callback" definition in the OpenAPI specification is associated to the "ApplicationSessions" resource.

5.5.5.2 Target URI

The Callback URI **"{notifUri}/pdu-session"** shall be used with the callback URI variables defined in table 5.5.5.2-1.

Table 5.5.5.2-1: Callback URI variables

Name	Data type	Definition
notifUri	Uri	It is the PCF for a UE callback URI stored in the SM Policy Association.

5.5.5.3 Standard Methods

5.5.5.3.1 POST

This method shall support the URI query parameters specified in table 5.5.5.3.1-1.

Table 5.5.5.3.1-1: URI query parameters supported by the POST method on this resource

Name		Data type	P	Cardinality	Description
n/a					

This method shall support the request data structures specified in table 5.5.5.3.1-2 and the response data structures and response codes specified in table 5.5.5.3.1-3.

Table 5.5.5.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
PduSessionEventNotification	M	1	Provides information about the PDU session related event implicitly subscribed.

Table 5.5.5.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The receipt of the Notification is acknowledged.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
NOTE 1: In addition, the HTTP status codes which are specified as mandatory in table 5.2.7.1-1 of 3GPP TS 29.500 [5] for the POST method shall also apply. NOTE 2: The RedirectResponse data structure may be provided by an SCP (see clause 6.10.9.1 of 3GPP TS 29.500 [5]).				

Table 5.5.5.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the notification request is redirected.

Table 5.5.5.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains an alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification is redirected. For the case where the notification is redirected to the same target via a different SCP, refer to clause 6.10.9.1 of 3GPP TS 29.500 [5].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the notification request is redirected.

5.6 Data Model

5.6.1 General

This clause specifies the application data model supported by the API.

Table 5.6.1-1 specifies the data types defined for the Npcf_PolicyAuthorization service based interface protocol.

Table 5.6.1-1: Npcf_PolicyAuthorization specific Data Types

Data type	Section defined	Description	Applicability
AcceptableServiceInfo	5.6.2.30	Acceptable service information.	
AccessNetChargingIdentifier	5.6.2.32	Contains the access network charging identifier.	IMS_SBI
AddFlowDescriptionInfo	5.6.2.55	Contains additional flow description information, as the flow label and the IPsec SPI.	AddFlowDescriptionInformation
AfAppId	5.6.3.2	Contains an AF application identifier.	
AfEvent	5.6.3.7	Represents an event to notify to the NF service consumer.	
AfEventNotification	5.6.2.11	Represents the notification of an event.	
AfEventSubscription	5.6.2.10	Represents the subscription to events.	
AfHeaderHandlingControlInfo	5.6.2.62	Represents the header handling control information.	HeaderHandling
AfNotifMethod	5.6.3.8	Represents the notification methods that can be subscribed for an event.	
AfRequestedData	5.6.3.18	Represents the information the NF service consumer requested to be exposed.	IMS_SBI
AfRoutingRequirement	5.6.2.13	Describes the routing requirements for the application traffic flows.	InfluenceOnTrafficRouting
AfRoutingRequirementRm	5.6.2.24	This data type is defined in the same way as the "AfRoutingRequirement" data type, but with the OpenAPI "nullable: true" property.	InfluenceOnTrafficRouting
AfSfcRequirement	5.6.2.49	Describes the requirements to steer the traffic to a pre-configured chain of service functions on N6-LAN.	SFC
AlternativeServiceRequirementsData	5.6.2.47	Contains alternative QoS related parameter sets.	AltSerReqsWithIndQoS
AnGwAddress	5.6.2.20	Carries the control plane address of the access network gateway.	
AppDetectionReport	5.6.2.44	Indicates the start or stop of the detected application traffic and the detected AF application identifier.	ApplicationDetectionEvents
AppDetectionNotifType	5.6.3.23	Represents the types of reports bound to the notification of application detection information.	ApplicationDetectionEvents
AppSessionContext	5.6.2.2	Represents an Individual Application Session Context resource.	
AppSessionContextReqData	5.6.2.3	Represents the Individual Application Session Context resource data received in an HTTP POST request message.	
AppSessionContextRespData	5.6.2.4	Represents the Individual Application Session Context resource data produced by the server and returned in an HTTP response message.	
AppSessionContextUpdateData	5.6.2.5	Describes the modifications to the "ascReqData" property of an Individual Application Session Context resource.	
AppSessionContextUpdateDataPatch	5.6.2.43	Describes the modifications to an Individual Application Session Context resource	PatchCorrection
AspId	5.6.3.2	Contains an identity of an application service provider.	SponsoredConnectivity
BatOffsetInfo	5.6.2.50	Contains the offset of the BAT and the optionally adjusted periodicity.	EnTSCAC
CapabilityReport	5.6.2.60	Contains information about whether a QoS monitoring capability is supported or not for the corresponding capability type defined in the attribute "capType".	QoSMonCapRepo
CodecData	5.6.3.2	Contains a codec related information.	
ContentVersion	5.6.3.2	Represents the version of a media component.	MediaComponentVersioning
DirectNotificationReport	5.6.2.57	Represents the QoS monitoring parameter that is not authorized to be directly notified for the indicated flows.	EnQoSMon
DurationMilliSec	5.6.3.2	Indicates the time interval in units of milliseconds.	PowerSaving

Data type	Section defined	Description	Applicability
DurationMilliSecRm	5.6.3.2	This data type is defined in the same way as the "DurationMilliSec" data type, but with the OpenAPI "nullable: true" property.	PowerSaving
EthFlowDescription	5.6.2.17	Defines a packet filter for an Ethernet flow.	
EventsNotification	5.6.2.9	Describes the notification about the events occurred within an Individual Application Session Context resource.	
EventsSubscPutData	5.6.2.42	Identifies the events the application subscribes to within an Events Subscription sub-resource data. It may also include the attributes of the notification about the events already met at the time of subscription. It is represented as a non-exclusive list of two data types: EventsSubscReqData and EventsNotification.	
EventsSubscReqData	5.6.2.6	Identifies the events the application subscribes to within an Individual Application Session Context resource.	
EventsSubscReqDataRm	5.6.2.25	Describes the possible modifications to Events Subscription Data. This data type is defined in the same way as the "EventsSubscReqData" data type, but: - with the OpenAPI "nullable: true" property; and - with individual attribute(s) defined as removable as specified in clause 5.6.2.25.	
ExtendedProblemDetails	5.6.2.29	Data type that extends ProblemDetails.	
FlowDescription	5.6.3.2	Defines a packet filter for an IP flow.	
Flows	5.6.2.21	Identifies the flows related to a media component.	
FlowStatus	5.6.3.12	Describes whether the IP flow(s) are enabled or disabled.	
FlowUsage	5.6.3.14	Describes the flow usage of the flows described by a media subcomponent.	
HeaderHandlingAction	5.6.3.28	Represents the type of header handling actions.	HeaderHandling
HeaderHandlingActionRequest	5.6.2.63	Represents the header handling action related information.	HeaderHandling
HeaderHandlingCond	5.6.3.29	Represents the header handling conditions to perform the header actions.	HeaderHandling
HeaderHandlingReporting	5.6.2.65	Indicates that reporting is requested for the performed header handling actions.	HeaderHandling
L4sNotifType	5.6.3.26	Indicates whether the ECN marking for L4S support for the indicated SDFs is "NOT_AVAILABLE" or "AVAILABLE" again.	L4S
L4sSupport	5.6.2.56	Indicates whether the ECN marking for L4S is available in 5GS for the indicated service data flows.	L4S
MaxDataBurstVol	5.6.3.2	Indicates the Maximum Data Burst Volume value.	ExtQoSR19
MediaComponent	5.6.2.7	Contains service information for a media component of an AF session.	
MediaComponentRm	5.6.2.26	Describes the possible modifications to a Media Component. This data type is defined in the same way as the "MediaComponent" data type, but: - with the OpenAPI "nullable: true" property; and - with individual attribute(s) defined as removable as specified in clause 5.6.2.26.	
MediaComponentResourcesStatus	5.6.3.13	Indicates whether the media component is active or inactive.	
MediaSubComponent	5.6.2.8	Contains the requested bitrate and filters for the set of IP flows identified by their common flow identifier.	

Data type	Section defined	Description	Applicability
MediaSubComponentRm	5.6.2.27	Describes the possible modifications to a Media Subcomponent. This data type is defined in the same way as the "MediaSubComponent" data type, but: - with the OpenAPI "nullable: true" property; and - with individual attribute(s) defined as removable as specified in clause 5.6.2.27.	
MediaType	5.6.3.3	Indicates the media type of a media component.	
MpsAction	5.6.3.22	Indicates whether it is an invocation, a revocation or an invocation with authorization of the MPS for DTS service.	MPSforDTS, MPSforMessaging
MpxMediaInfo	5.6.2.61	Represents the (S)RTP multiplexed media identification information.	MpxMedia
MultiModalId	5.6.3.2	Contains a multi-modal service identifier.	MultiMedia
NotifCap	5.6.3.27	Contains information about whether the notified capability is supported or not supported.	QoSMonCapRepo
NotifCapType	5.6.3.31	Contains information about which type of QoS Monitoring capability report is applied.	QoSMonCapRepo
OnPathN6Method	5.6.3.30	Represents the on-path N6 signaling method.	OnPathN6MediaInfo
OnPathN6SigInfo	5.6.2.64	Represents the on-path N6 signaling information, with the OpenAPI "nullable: true" property.	OnPathN6MediaInfo
OutOfCreditInformation	5.6.2.33	Indicates the service data flows without available credit and the corresponding termination action.	IMS_SBI
PcfAddressingInfo	5.6.2.46	Contains PCF address information.	
PcscfRestorationRequestData	5.6.2.36	Indicates P-CSCF restoration.	PCSCF-Restoration-Enhancement
PcSessionRecoveryStatus	5.6.3.32	Describes whether the session is being restored or the session is not found and requires to be re-established.	CnHealthMonitor
PduSessionEventNotification	5.6.2.45	Represents PDU session related event reporting information.	
PduSessionStatus	5.6.3.24	Indicates whether the PDU session is established or terminated.	
PduSessionTsnBridge	5.6.2.40	Contains the TSC user plane node Information and DS-TT port and/or NW-TT ports management information of a new detected TSC user plane node in the context of a new PDU session.	TimeSensitiveNetworking
PdvMonitoringReport	5.6.2.53	Packet Delay Variation reporting information.	EnQoSMon
PeriodicityRange	5.6.2.48	Contains the acceptable range (which is formulated as lower bound and upper bound of the periodicity of the start two bursts in reference to the external GM) or acceptable periodicity value(s) (which is formulated as a list of values for the periodicity).	EnTSCAC
PreemptionControlInformation	5.6.3.19	Pre-emption control information.	MCPTT-Preemption
PreemptionControlInformationRm	5.6.3.21	This data type is defined in the same way as the "PreemptionControlInformation" data type, but with the OpenAPI "nullable: true" property.	MCPTT-Preemption
PrioritySharingIndicator	5.6.3.20	Priority sharing indicator.	PrioritySharing
QosMonitoringInformation	5.6.2.34	QoS monitoring information (e.g. UL, DL or round trip packet delay).	QoSMonitoring

Data type	Section defined	Description	Applicability
QosMonitoringInformationRm	5.6.2.41	Describes the possible modifications to QoS monitoring data. This data type is defined in the same way as the "QosMonitoringInformation" data type, but: - with the OpenAPI "nullable: true" property; and - when the "EnQoSMon" feature is supported, with individual attribute(s) defined as removable as specified in clause 5.6.2.41	QoSMonitoring
QosMonitoringReport	5.6.2.37	Contains QoS monitoring reporting information.	QoSMonitoring
QosNotificationControlInfo	5.6.2.15	Indicates whether the QoS targets related to certain media component are not guaranteed or are guaranteed again.	
QosNotifType	5.6.3.9	Indicates type of notification for QoS Notification Control.	
RateLimitRepo	5.6.2.66	Indicates the maximum data rate authorized for the non-GBR service data flow(s).	RateLimitReport
RequiredAccessInfo	5.6.3.15	Indicates the access network information required for an AF session.	NetLoc
ReservPriority	5.6.3.4	Indicates the reservation priority.	
ResourcesAllocationInfo	5.6.2.14	Indicates the status of the PCC rule(s) related to certain media component.	
RttFlowReference	5.6.2.58	Contains the shared key with the media subcomponent that shares the subscription to round trip time measurements or shares the RT Latency requirement in the complementary direction.	EnQoSMon, RTLatency
RttFlowReferenceRm	5.6.2.59	This data type is defined in the same way as the "RttFlowReference" data type, but with the OpenAPI "nullable: true" property.	RTLatency EnQoSMon
ServAuthInfo	5.6.3.5	Indicates the result of the Policy Authorization service request from the NF service consumer.	
ServiceInfoStatus	5.6.3.16	Preliminary or final service information status.	IMS_SBI
ServiceUrn	5.6.3.2	Service URN.	IMS_SBI
SipForkingIndication	5.6.3.17	Describes if several SIP dialogues are related to an "Individual Application Session Context" resource.	IMS_SBI
SpatialValidity	5.6.2.16	Describes the spatial validity of an NF service consumer request for influencing traffic routing.	InfluenceOnTrafficRouting SFC
SpatialValidityRm	5.6.2.28	This data type is defined in the same way as the "SpatialValidity" data type, but with the OpenAPI "nullable: true" property.	InfluenceOnTrafficRouting SFC HeaderHandling
SponId	5.6.3.2	Contains an Identity of a sponsor.	SponsoredConnectivity
SponsoringStatus	5.6.3.6	Represents whether sponsored data connectivity is enabled or disabled/not enabled.	SponsoredConnectivity
TemporalValidity	5.6.2.22	Indicates the time interval during which the NF service consumer request is to be applied.	InfluenceOnTrafficRouting HeaderHandling
TerminationCause	5.6.3.10	Indicates the cause for requesting the deletion of the Individual Application Session Context resource.	
TerminationInfo	5.6.2.12	Includes information related to the termination of the Individual Application Session Context resource.	
TosTrafficClass	5.6.3.2	Contains the IPv4 Type-of-Service or the IPv6 Traffic-Class field and the ToS/Traffic Class mask field.	
TosTrafficClassRm	5.6.3.2	This data type is defined in the same way as the "TosTrafficClass" data type, but with the OpenAPI "nullable: true" property.	

Data type	Section defined	Description	Applicability
TscPriorityLevel	5.6.3.2	Priority of TSC Flows	TimeSensitiveNetworking
TscPriorityLevelRm	5.6.3.2	This data type is defined in the same way as the "TscPriorityLevel" data type, but with the OpenAPI "nullable: true" property	TimeSensitiveNetworking
TscaiInputContainer	5.6.2.39	TSCAI Input information container.	TimeSensitiveNetworking
TsnQosContainer	5.6.2.35	TSC traffic QoS parameters.	TimeSensitiveNetworking
TsnQosContainerRm	5.6.2.38	This data type is defined in the same way as the "TsnQosContainer" data type, but with the OpenAPI "nullable: true" property.	TimeSensitiveNetworking
UeIdentityInfo	5.6.2.31	Represents 5GS-Level UE Identities.	IMS_SBI
UplinkDownlinkSupport	5.6.3.25	Represents whether a capability is supported for the UL, the DL or both UL and DL service data flows	L4S

Table 5.6.1-2 specifies data types re-used by the Npcf_PolicyAuthorization service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Npcf_PolicyAuthorization service based interface.

Table 5.6.1-2: Npcf_PolicyAuthorization re-used Data Types

Data type	Reference	Comments	Applicability
AccNetChargingAddress	3GPP TS 29.512 [8]	Indicates the IP address of the network entity within the access network performing charging.	IMS_SBI
AccessType	3GPP TS 29.571 [12]	The identification of the type of access network.	
AccumulatedUsage	3GPP TS 29.122 [15]	Accumulated Usage.	SponsoredConnectivity
AdditionalAccessInfo	3GPP TS 29.512 [8]	Indicates the combination of additional Access Type and RAT Type for MA PDU session	ATSSS
AfSigProtocol	3GPP TS 29.512 [8]	Represents the protocol used for signalling between the UE and the NF service consumer.	ProvAFsignalFlow
ApplicationChargingId	3GPP TS 29.571 [12]	Application provided charging identifier allowing correlation of charging information.	IMS_SBI
AverWindow	3GPP TS 29.571 [12]	Averaging Window.	EnQoSMon ExtQoSR19
AverWindowRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "AverWindow" data type, but with the OpenAPI "nullable: true" property.	EnQoSMon
BdtReferenceId	3GPP TS 29.122 [15]	Identifies transfer policies.	
BitRate	3GPP TS 29.571 [12]	Specifies bitrate in kbits per second.	
BitRateRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "BitRate" data type, but with the OpenAPI "nullable: true" property.	
BridgeManagementContainer	3GPP TS 29.512 [8]	Contains TSC user plane node management information.	TimeSensitiveNetworking
ChargingId	3GPP TS 29.571 [12]	Charging identifier allowing correlation of charging information.	IMS_SBI
DateTime	3GPP TS 29.571 [12]	String with format "date-time" as defined in OpenAPI Specification [11].	InfluenceOnTrafficRouting, TimeSensitiveNetworking
Dnai	3GPP TS 29.571 [12]	Identifies a DNAI.	UeSatUeComm
DnaiChangeType	3GPP TS 29.571 [12]	Describes the types of DNAI change.	UeSatUeComm
Dnn	3GPP TS 29.571 [12]	Data Network Name.	
DurationSec	3GPP TS 29.571 [12]	Identifies a period of time in units of seconds.	TimeSensitiveNetworking, EnhancedSubscriptionToNotification, SimultConnectivity
DurationSecRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "DurationSec" data type, but with the OpenAPI "nullable: true" property.	SimultConnectivity
EasIpReplacementInfo	3GPP TS 29.571 [12]	Contains EAS IP replacement information for a Source and a Target EAS.	EASIPreplacement
FinalUnitAction	3GPP TS 32.291 [22]	Indicates the action to be taken when the user's account cannot cover the service cost.	
Float	3GPP TS 29.571 [12]	Number with format "float" as defined in OpenAPI Specification [11].	FLUS
FloatRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "Float" data type, but with the OpenAPI "nullable: true" property.	FLUS
FlowDirection	3GPP TS 29.512 [8]	Flow Direction.	
Fqdn	3GPP TS 29.571 [12]	Contains a FQDN	
ExtMaxDataBurstVol	3GPP TS 29.571 [12]	Represents the Maximum Burst Size. Minimum = 4096. Maximum = 2000000.	TimeSensitiveNetworking
ExtMaxDataBurstVolRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "ExtMaxDataBurstVol" data type, but with the OpenAPI "nullable: true" property.	TimeSensitiveNetworking
Gpsi	3GPP TS 29.571 [12]	Identifies the GPSI.	
IpAddr	3GPP TS 29.571 [12]	Contains an IP address.	OnPathN6MediaInfo

Data type	Reference	Comments	Applicability
Ipv4Addr	3GPP TS 29.571 [12]	Identifies an IPv4 address.	
Ipv4AddrMask	3GPP TS 29.571 [12]	IPv4 address mask	ExtraUEaddrReport
Ipv6Addr	3GPP TS 29.571 [12]	Identifies an IPv6 address.	
IpEndPoint	3GPP TS 29.510 [27]	Contains a NF IPv4 and/or IPv6 end points.	
MacAddr48	3GPP TS 29.571 [12]	MAC Address.	
MaxDataBurstVol	3GPP TS 29.571 [12]	Represents the Maximum Burst Size. Minimum = 1. Maximum = 4095.	SmallBurstSizes
MaxDataBurstVolRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "MaxDataBurstVol" data type of 3GPP TS 29.571 [12], but with the OpenAPI "nullable: true" property.	SmallBurstSizes
Metadata	3GPP TS 29.571 [12]	This datatype contains opaque information for the service functions in the N6-LAN that is provided by AF and transparently sent to UPF.	SFC
NetLocAccessSupport	3GPP TS 29.512 [8]	Indicates the access network does not support the report of the requested access network information.	NetLoc
NullValue	3GPP TS 29.571 [12]	JSON's null value, used as an explicit value of an enumeration.	MCPTT-Preemption
PacketDelBudget	3GPP TS 29.571 [12]	Packet Delay Budget.	TimeSensitiveNetworking
PacketDelBudgetRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "PacketDelBudget" data type, but with the OpenAPI "nullable: true" property	TimeSensitiveNetworking
PacketErrRate	3GPP TS 29.571 [12]	String representing Packet Error Rate (see clauses 5.7.3.5 and 5.7.4 of 3GPP TS 23.501 [2]), expressed as a "*scalar* x 10-k" where the scalar and the *exponent k are each encoded as one decimal digit*. Pattern: '^([0-9]E-[0-9])$' Examples: Packer Error Rate $4x10^{-6}$ shall be encoded as "4E-6". Packer Error Rate 10^{-2} shall be encoded as "1E-2".	ExtQoS
PacketErrRateRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "PacketErrRate" data type, but with the OpenAPI "nullable: true" property.	ExtQoS
PacketLossRateRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "PacketLossRate" data type, but with the OpenAPI "nullable: true" property.	CHEM
PduSessionType	3GPP TS 29.571 [12]	Contains the PDU Session Type	URSPEnforcement
PduSetQosPara	3GPP TS 29.571 [12]	PDU Set related QoS parameters.	PDUSetHandling
PduSetQosParaRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "PduSetQosPara" data type, but with the OpenAPI "nullable: true" property.	PDUSetHandling
Pei	3GPP TS 29.571 [12]	Identifies the PEI.	IMS_SBI
PlmnIdNid	3GPP TS 29.571 [12]	Identifies the network: the PLMN Identifier (the mobile country code and the mobile network code) or the SNPN Identifier (the PLMN Identifier and the NID).	
PreemptionCapability	3GPP TS 29.571 [12]	Pre-emption capability.	MCPTT-Preemption
PreemptionVulnerability	3GPP TS 29.571 [12]	Pre-emption vulnerability.	MCPTT-Preemption
PreemptionCapabilityRm	3GPP TS 29.571 [12]	It is defined in the same way as the "PreemptionCapability" data type, but with the OpenAPI "nullable: true" property.	MCPTT-Preemption
PreemptionVulnerabilityRm	3GPP TS 29.571 [12]	It is defined in the same way as the "PreemptionVulnerability" data type, but with the OpenAPI "nullable: true" property.	MCPTT-Preemption

Data type	Reference	Comments	Applicability
PresenceInfo	3GPP TS 29.571 [12]	Represents an area of interest, e.g. a Presence Reporting Area.	InfluenceOnTrafficRouting
PortManagementContainer	3GPP TS 29.512 [8]	Contains port management information for a related port.	TimeSensitiveNetworking
ProblemDetails	3GPP TS 29.571 [12]	Contains a detailed information about an error.	
ProtocolDescription	3GPP TS 29.571 [12]	Represents Protocol description of the media flow	PDUSetHandling PowerSaving
ProtocolDescriptionRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "ProtocolDescription" data type, but with the OpenAPI "nullable: true" property.	PDUSetHandling PowerSaving
QosMonitoringParamType	3GPP TS 29.512 [8]	Contains the QoS monitoring parameter to be monitored.	EnQoSMon
RanNasRelCause	3GPP TS 29.512 [8]	Indicates RAN and/or NAS release cause code information.	RAN-NAS-Cause
RatType	3GPP TS 29.571 [12]	RAT Type.	
RedirectResponse	3GPP TS 29.571 [12]	Contains redirection related information.	ES3XX
ReportingSuggestionInformation	3GPP TS 29.564 [61]	Contains reporting suggestion information.	HeaderHandling
RequestedQosMonitoringParameter	3GPP TS 29.512 [8]	Indicate the QoS information to be monitored, e.g. UL packet delay, DL packet delay or round trip packet delay between the UE and the UPF is to be monitored when the QoS Monitoring for packet delay is enabled for the service data flow.	QoSMonitoring
RouteToLocation	3GPP TS 29.571 [12]	Identifies routes to locations of applications.	InfluenceOnTrafficRouting
SatelliteBackhaulCategory	3GPP TS 29.571 [12]	Indicates the satellite or non-satellite backhaul category	SatelliteBackhaul
SatelliteId	3GPP TS 29.571 [12]	Unique identifier of a satellite.	UeSatUeComm
SimConnFailEvent	3GPP TS 29.512 [8]	Represents the subscription information to be delivered to the SMF for the simultaneous connectivity failure event reporting.	SimConnFailure
Snssai	3GPP TS 29.571 [12]	Identifies the S-NSSAI.	
SscMode	3GPP TS 29.571 [12]	Service and session continuity mode.	URSPEnforcement
Supi	3GPP TS 29.571 [12]	Identifies the SUPI.	
SupportedFeatures	3GPP TS 29.571 [12]	Used to negotiate the applicability of the optional features defined in table 5.8-1.	
TimeWindow	3GPP TS 29.122 [15]	Time window identified by a start time and a stop time.	EnTSCAC
TrafficCorrelationInfo	3GPP TS 29.519 [53]	Contains the information for traffic correlation.	CommonEASDNAI
TimeZone	3GPP TS 29.571 [12]	Time Zone.	NetLoc
TraffRouteReqOutcomeEvent	3GPP TS 29.512 [8]	Represents the subscription information to be delivered to the SMF for the traffic routing requirements installation outcome event reporting.	TraffRouteReqOutcome
TsnBridgeInfo	3GPP TS 29.512 [8]	TSC user plane node information.	TimeSensitiveNetworking
UeReachabilityStatus	3GPP TS 29.512 [8]	Represents the UE Reachability Status.	UEUnreachable
Uint32	3GPP TS 29.571 [12]	Unsigned 32-bit integers, i.e. only value 0 and 32-bit integers above 0 are permissible.	ResourceSharing
Uint32Rm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "Uint32" data type, but with the OpenAPI "nullable: true" property.	ResourceSharing
Uinteger	3GPP TS 29.571 [12]	Unsigned Integer, i.e. only value 0 and integers above 0 are permissible. Minimum = 0.	TimeSensitiveNetworking
UintegerRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "Uint32" data type, but with the OpenAPI "nullable: true" property.	AF_latency, QoSMonitoring

Data type	Reference	Comments	Applicability
UpPathChgEvent	3GPP TS 29.512 [8]	Contains the subscription information to be delivered to SMF for the UP path management events.	InfluenceOnTrafficRouting
Uri	3GPP TS 29.571 [12]	String providing an URI.	
UriRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "Uri" data type, but with the OpenAPI "nullable: true" property.	HeaderHandling
UrspEnforcementInfo	3GPP TS 29.512 [8]	Contains the URSP rule enforcement information from the UE.	URSPEnforcement
UsageThreshold	3GPP TS 29.122 [15]	Usage Thresholds.	SponsoredConnectivity
UsageThresholdRm	3GPP TS 29.122 [15]	This data type is defined in the same way as the "UsageThreshold" data type, but with the OpenAPI "nullable: true" property.	SponsoredConnectivity
UserLocation	3GPP TS 29.571 [12]	User Location(s).	NetLoc

5.6.2 Structured data types

5.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

5.6.2.2 Type AppSessionContext

Table 5.6.2.2-1: Definition of type AppSessionContext

Attribute name	Data type	P	Cardinality	Description	Applicability
ascReqData	AppSessionContextReqData	C	0..1	Identifies the service requirements of an Individual Application Session Context. It shall be present in HTTP POST request messages for the creation of the resource and may be included in the HTTP response messages.	
ascRespData	AppSessionContextRespData	C	0..1	Describes the authorization data of an Individual Application Session Context created by the PCF. It may be present in the HTTP response messages.	
evsNotif	EventsNotification	O	0..1	Describes information related to the notification of events.	

5.6.2.3 Type AppSessionContextReqData

Table 5.6.2.3-1: Definition of type AppSessionContextReqData

Attribute name	Data type	P	Cardinality	Description	Applicability
afAppId	AfAppId	O	0..1	AF application identifier.	
afChargId	ApplicationChargingId	O	0..1	AF charging identifier. This information may be used for charging correlation with QoS flow.	IMS_SBI
afReqData	AfRequestedData	O	0..1	Represents the NF service consumer requested data to be exposed.	IMS_SBI
afRoutReq	AfRoutingRequirement	C	0..1	Indicates the AF traffic routing requirements. It shall be included if InfluenceOnTrafficRouting feature is supported.	InfluenceOnTrafficRouting
afSfcReq	AfSfcRequirement	O	0..1	Describes the AF requirements to steer the traffic to a pre-configured chain of service functions on N6-LAN.	SFC
afHdrReq	AfHeaderHandlingControlInfo	O	0..1	Indicates the AF handling of payload headers requirements.	HeaderHandling
aspId	AspId	C	0..1	Application service provider identity. It shall be included if "SponsoredConnectivity" feature is supported.	SponsoredConnectivity
bdtRefId	BdtReferenceId	O	0..1	Reference to a transfer policy negotiated for background data traffic.	
dnn	Dnn	C	0..1	Data Network Name, a full DNN with both the Network Identifier and Operator Identifier, or a DNN with the Network Identifier only. It shall be present when the "afRoutReq" attribute is present. (NOTE 2)	
evSubsc	EventsSubscReqData	O	0..1	Identifies the events the application subscribes to at creation of an Individual Application Session Context resource.	
ipDomain	string	O	0..1	Indicates the IPv4 address domain information that assists session binding.	
mcpttId	string	O	0..1	Indicates that the created Individual Application Session Context resource relates to an MCPTT session prioritized call. It includes either one of the namespace values used for MCPTT (see IETF RFC 8101 [42]) and it may include the name of the MCPTT service provider.	MCPTT
mcVideoId	string	O	0..1	Indicates that the created Individual Application Session Context resource relates to an MCVideo session prioritized call. It includes either one of the namespace values used for MCPTT (see IETF RFC 8101 [42]) and it may include the name of the MCVideo service provider.	MCVideo
medComponents	map(MediaComponent)	O	1..N	Media Component information. The key of the map is the attribute "medCompN".	
mpsAction	MpsAction	O	0..1	Indicates a request to invoke an MPS action.	MPSforDTS, MPSforMessaging
mpsId	string	O	0..1	Indicates that the created Individual Application Session Context resource relates to an MPS service. It contains the national variant for MPS service name.	

Attribute name	Data type	P	Cardinality	Description	Applicability
mcsId	string	O	0..1	Indicates that the created Individual Application Session Context resource relates to an MCS service. It contains the national variant for MCS service name.	
preemptControlInfo	PreemptionControlInformation	O	0..1	Pre-emption control information.	MCPTT-Preemption
resPrio	ReservPriority	O	0..1	Indicates the reservation priority.	
servInfStatus	ServiceInfoStatus	O	0..1	Indicates whether the service information is preliminary or final. When the attribute is not provided the default value is "FINAL".	IMS_SBI
notifUri	Uri	M	1	Notification URI for Application Session Context termination requests.	
servUrn	ServiceUrn	O	0..1	Service URN.	IMS_SBI
sliceInfo	Snssai	O	0..1	Identifies the S-NSSAI.	
sponId	SponId	C	0..1	Sponsor identity. It shall be included if "SponsoredConnectivity" feature is supported.	SponsoredConnectivity
sponStatus	SponsoringStatus	O	0..1	Indication of whether sponsored connectivity is enabled or disabled/not enabled. The absence of the attribute indicates that the sponsored connectivity is enabled.	SponsoredConnectivity
supi	Supi	O	0..1	Subscription Permanent Identifier.	
gpsi	Gpsi	O	0..1	Generic Public Subscription Identifier.	
suppFeat	SupportedFeatures	M	1	Contains the list of supported features among the ones defined in clause 5.8. This attribute shall be provided only when feature negotiation needs to take place.	
ueIpv4	Ipv4Addr	C	0..1	The IPv4 address of the served UE. (NOTE 1)	
ueIpv6	Ipv6Addr	C	0..1	The IPv6 address of the served UE. (NOTE 1)	
ueMac	MacAddr48	C	0..1	The MAC address of the served UE. When the feature "TimeSensitiveNetworking" is supported this attribute represents the DS-TT port MAC address. (NOTE 1)	
tsnBridgeManCont	BridgeManagementContainer	O	0..1	Transports TSC user plane node management information.	TimeSensitiveNetworking
tsnPortManContDstt	PortManagementContainer	O	0..1	Transports port management information for the DS-TT port.	TimeSensitiveNetworking
tsnPortManContNwtts	array(PortManagementContainer)	O	1..N	Transports port management information for one or more NW-TT ports.	TimeSensitiveNetworking
tscNotifUri	Uri	O	0..1	Notification address of the TSCTSF or TSN AF receiving the TSC management information.	ExposureToTSC
tscNotifCorreId	string	O	0..1	Correlation identifier for TSC management information notifications. It shall be provided if the “tscNotifUri” attribute is provided.	ExposureToTSC
multiModalId	MultiModalId	O	0..1	Multi-modal Service Identifier	MultiMedia
qosDuration	DurationSec	O	0..1	Contains the QoS duration to transfer data transmission (e.g., AI/ML transmission). The minimum value of the QoS duration is 60 sec.	QoSTiming_5G

Attribute name	Data type	P	Cardinality	Description	Applicability
qosInactInt	DurationSec	O	0..1	Contains the QoS inactivity interval for the given data transmission (e.g., AI/ML transmission). The minimum value of the QoS duration is 60 sec	QoSTiming_5G

NOTE 1: Only one of the served UE addressing parameters (the IPv4 address or the IPv6 address or MAC address) shall always be included.
NOTE 2: The PCF uses the DNN as received from the NF service consumer without applying any transformation (e.g. during session binding). To successfully perform DNN matching, in a specific deployment a DNN shall always be encoded either with the full DNN (e.g., because there are multiple Operator Identifiers for a Network Identifier) or the DNN Network Identifier only. The NF service consumer may include the DNN Operator Identifier based on local configuration.

5.6.2.4 Type AppSessionContextRespData

Table 5.6.2.4-1: Definition of type AppSessionContextRespData

Attribute name	Data type	P	Cardinality	Description	Applicability
servAuthInfo	ServAuthInfo	O	0..1	Indicates additional information related with the result of the authorization for a service request.	
directNotifReports	array(DirectNotificationReport)	C	1.. N	Indicates the QoS monitoring parameter(s) that cannot be directly notified for the indicated flows. It shall be provided when the "servAuthInfo" attribute is set to "DIRECT_NOTIF_NOT_POSSIBLE".	EnQoSMon
ueIds	array(UeIdentityInfo)	O	1..N	Represents the 5GS-Level UE identities available for an AF session context.	IMS_SBI
suppFeat	SupportedFeatures	C	0..1	Contains the list of supported features among the ones defined in clause 5.8. This attribute shall be provided only when feature negotiation needs to take place.	

NOTE: The "ueIds" attribute needs to include only a single array element.

5.6.2.5 Type AppSessionContextUpdateData

Table 5.6.2.5-1: Definition of type AppSessionContextUpdateData

Attribute name	Data type	P	Cardinality	Description	Applicability
afAppId	AfAppId	O	0..1	AF application identifier.	
afRoutReq	AfRoutingRequirementRm	O	0..1	Indicates the AF traffic routing requirements.	InfluenceOnTrafficRouting
afSfcReq	AfSfcRequirement	O	0..1	Indicates the AF requirements on steering traffic to a pre-configured chain of service functions on N6-LAN.	SFC
afHdrReq	AfHeaderHandlingControlInfo	O	0..1	Indicates the AF handling of payload headers requirements.	HeaderHandling
aspId	AspId	O	0..1	Application service provider identity.	SponsoredConnectivity
bdtRefId	BdtReferenceId	O	0..1	Reference to a transfer policy negotiated for background data traffic.	
evSubsc	EventsSubscReqDataRm	O	0..1	Identifies the events the application subscribes to at modification of an Individual Application Session Context resource.	
mcpttId	string	O	0..1	Indicates that the updated Individual Application Session Context resource relates to an MCPTT session prioritized call. It includes either one of the namespace values used for MCPTT (see IETF RFC 8101 [42]) and it may include the name of the MCPTT service provider.	MCPTT
mcVideoId	string	O	0..1	Indicates that the updated Individual Application Session Context resource relates to an MCVideo session prioritized call. It includes either one of the namespace values used for MCPTT (see IETF RFC 8101 [42]) and it may include the name of the MCVideo service provider.	MCVideo
medComponents	map(MediaComponentRm)	O	1..N	Media Component information. The key of the map is the "medCompN" attribute.	
mpsAction	MpsAction	O	0..1	Indicates a request to invoke or revoke an MPS action.	MPSforDTS, MPSforMessaging
mpsId	string	O	0..1	Indicates that the modified Individual Application Session Context resource relates to an MPS service. It contains the national variant for MPS service name.	
mcsId	string	O	0..1	Indicates that the updated Individual Application Session Context resource relates to an MCS service. It contains the national variant for MCS service name.	
preemptControlInfo	PreemptionControlInformationRm	O	0..1	Preemption control information.	MCPTT-Preemption
resPrio	ReservPriority	O	0..1	Indicates the reservation priority.	
servInfStatus	ServiceInfoStatus	O	0..1	Indicates whether the service information is preliminary or final.	IMS_SBI
sipForkInd	SipForkingIndication	O	0..1	Describes if several SIP dialogues are related to an "Individual Application Session Context" resource.	IMS_SBI
sponId	SponId	O	0..1	Sponsor identity.	SponsoredConnectivity
sponStatus	SponsoringStatus	O	0..1	Indication of whether sponsored connectivity is enabled or disabled/not enabled.	SponsoredConnectivity
tsnBridgeManCont	BridgeManagementContainer	O	0..1	Transports TSC user plane node management information.	TimeSensitiveNetworking

Attribute name	Data type	P	Cardinality	Description	Applicability
tsnPortManContDstt	PortManagementContainer	O	0..1	Transports port management information for the DS-TT port.	TimeSensitiveNetworking
tsnPortManContNwtts	array(PortManagementContainer)	O	1..N	Transports port management information for one or more NW-TT ports.	TimeSensitiveNetworking
tscNotifUri	Uri	O	0..1	Notification address of the TSCTSF or TSN AF receiving the TSC management information.	ExposureToTSC
tscNotifCorreId	string	O	0..1	Correlation identifier for TSC management information notifications. It shall be provided if the “tscNotifUri” attribute is provided.	ExposureToTSC
qosDuration	DurationSecRm	O	0..1	Contains the QoS duration to transfer data transmission (e.g., AI/ML transmission). The minimum value of the QoS duration is 60 second.	QoSTiming_5G
qosInactInt	DurationSecRm	O	0..1	Contains the QoS inactivity interval for the given data transmission (e.g., AI/ML transmission). The minimum value of the QoS inactivity interval duration is 60 second.	QoSTiming_5G

5.6.2.6 Type EventsSubscReqData

Table 5.6.2.6-1: Definition of type EventsSubscReqData

Attribute name	Data type	P	Cardinality	Description	Applicability
events	array(AfEventSubscription)	M	1..N	Subscribed Events.	
notifUri	Uri	O	0..1	Notification URI.	
reqQosMonParams	array(RequestedQosMonitoringParameter)	O	1..N	Indicates the QoS information to be monitored (e.g. UL packet delay, DL packet delay, and/or round trip packet delay) when the QoS Monitoring is enabled for the service data flow. If the "EnQoSMon" feature is supported, it may also indicate the UL/DL congestion information, and/or the UL/DL data rate to be monitored. If the "EnQoSMon_v2" feature is supported, it may also indicate the UL/DL available bitrate to be monitored. It shall be present when the event "QOS_MONITORING" is subscribed. (NOTE 2)	QoSMonitoring
qosMon	QosMonitoringInformation	O	0..1	Packet delay threshold(s) information. It shall be present when the event "QOS_MONITORING" is subscribed and event based packet delay measurements are required. Only the "repThreshDl", "repThreshUl" and/or "repThreshRp" attributes may be present within the "QosMonitoringInformation" data type.	QoSMonitoring
qosMonDatRate	QosMonitoringInformation	O	0..1	Contains the data rate measurements information (data rate thresholds) for the subscribed report. It shall be present when the event "QOS_MONITORING" is subscribed and event based data rate measurements are required. Only the "repThreshDatRateDl", and/or the "repThreshDatRateUl" attributes may be present within the "QosMonitoringInformation" data type.	EnQoSMon
pdvReqMonParams	array(RequestedQosMonitoringParameter)	O	1..N	Indicates the Packet Delay Variation to be monitored, e.g. UL packet delay variation, DL packet delay variation and/or round trip packet delay variation between the UE and the UPF is to be monitored. Only the "DOWNLINK", "UPLINK" or "ROUNDTRIP" value may be present within the "RequestedQosMonitoringParameter" data type.	EnQoSMon
pdvMon	QosMonitoringInformation	O	0..1	Packet Delay Variation information for the subscribed report. It may be present when the event "PACK_DEL_VAR" is subscribed. Only the "repThreshDl", "repThreshUl" and/or "repThreshRp" attributes may be present within the "QosMonitoringInformation" data type.	EnQoSMon
congestMon	QosMonitoringInformation	O	0..1	Congestion threshold for the subscribed report. It shall be present when the event "QOS_MONITORING" is subscribed and event based congestion measurements are required. Only the "conThreshDl" and/or the "conThreshUl" attributes may be present within the "QosMonitoringInformation" data type. (NOTE 2)	EnQoSMon

Attribute name	Data type	P	Cardinality	Description	Applicability
rttMon	QosMonitoringInformation	O	0..1	Contains the round-trip delay over two QoS flows information for the subscribed report. It may be present when the event "RT_DELAY_TWO_QOS_FLOWS" is subscribed. Only the "repThreshRp" attribute may be present within the "QosMonitoringInformation" data type.	EnQoSMon
rttFlowRef	RttFlowReference	O	0..1	Contains information about the UL and DL flows involved in the round-trip delay measurements over two QoS flows. It may only be present when the "rttMon" attribute is present. (NOTE 3)	EnQoSMon
avlBitRateMon	QosMonitoringInformation	O	0..1	Contains information about the available bitrate threshold for the subscribed report. It may be present when the event "QOS_MONITORING" is subscribed. Only the "avlBitrateUlThrs" and/or the "avlBitrateDlThrs" attributes may be present within the "QosMonitoringInformation" data type. (NOTE 2)	EnQoSMon_v2
reqAnis	array(RequiredAccessInfo)	C	1..N	Represents the required access network information. It shall be present when the event "ANI_REPORT" is subscribed.	NetLoc
usgThres	UsageThreshold	O	0..1	Includes the volume and/or time thresholds for sponsored data connectivity.	SponsoredConnectivity
notifCorreId	string	O	0..1	It is used to set the value of Notification Correlation ID in the corresponding notification.	EnhancedSubscriptionToNotification
afAppIds	array(AfAppId)	O	1..N	AF application identifier(s). It shall be present when the event "APP_DETECTION" is subscribed.	ApplicationDetectionEvents
directNotifInd	boolean	O	0..1	Indicates that the event notification of QoS Monitoring data is sent by the UPF to Local NEF or AF if it is included and set to true. It may be present when the event "QOS_MONITORING" is subscribed. The default value "false" shall apply, if the attribute is not present. (NOTE 1)	ExposureToEAS EnQoSMon
avrgWndw	AverWindow	O	0..1	Averaging window for the calculation of the data rate for the service data flow. It may be included when the "qosMonDatRate" attribute is included.	EnQoSMon

NOTE 1: When the "EnQoSMon" feature is supported, the "directNotifInd" attribute indicates whether direct event notification is requested for the QoS measurement(s) provided in the "reqQosMonParams" attribute.
NOTE 2: Only the "EVENT_DETECTION" reporting frequency in "notifMethod" attribute included in "events" attribute is applicable when the "reqQosMonParams" includes "DOWNLINK_CONGESTION" and/or "UPLINK_CONGESTION", or "DOWNLINK_AVAILABLE_BITRATE" and/or "UPLINK_AVAILABLE_BITRATE".
NOTE 3: The "rttFlowRef" attribute shall be present when the request contains two or more media subcomponents and more than one of them with the DL and UL flow description.

5.6.2.7 Type MediaComponent

Table 5.6.2.7-1: Definition of type MediaComponent

Attribute name	Data type	P	Cardinality	Description	Applicability
afAppId	AfAppId	O	0..1	Contains information that identifies the particular service the AF session belongs to.	
afRoutReq	AfRoutingRequirement	O	0..1	Indicates the AF traffic routing requirements.	InfluenceOnTrafficRouting
afSfcReq	AfSfcRequirement	O	0..1	Indicates the AF requirements on steering traffic to a pre-configured chain of service functions on N6-LAN.	SFC
afHdrReq	AfHeaderHandlingControlInfo	O	0..1	Indicates the AF handling of payload headers requirements.	HeaderHandling
qosReference	string	O	0..1	Identifies a pre-defined QoS information.	AuthorizationWithRequiredQoS
altSerReqs	array(string)	O	1..N	Ordered list of alternative service requirements that include a set of QoS references. The lower the index of the array for a given entry, the higher the priority.(NOTE 1)	AuthorizationWithRequiredQoS
altSerReqsData	array(AlternativeServiceRequirementsData)	O	1..N	Ordered list of alternative service requirements that include individual QoS parameter sets. The lower the index of the array for a given entry, the higher the priority. (NOTE 1)	AltSerReqsWithIndQoS
disUeNotif	boolean	O	0..1	Indicates to disable QoS flow parameters signalling to the UE when the SMF is notified by the NG-RAN of changes in the fulfilled QoS situation when it is included and set to "true". The fulfilled situation is either the QoS profile or an Alternative QoS Profile. The default value "false" shall apply, if the attribute is not present and has not been supplied previously.	DisableUENotification
contVer	ContentVersion	O	0..1	Represents the content version of a media component.	MediaComponentVersioning
desMaxLatency	Float	O	0..1	Indicates a maximum desirable transport level packet latency in milliseconds.	FLUS, QoSHint
desMaxLoss	Float	O	0..1	Indicates the maximum desirable transport level packet loss rate in percent (without "%" sign).	FLUS, QoSHint
flusId	string	O	0..1	Indicates that the media component is used for FLUS media. It is derived from the media level attribute "a=label:" (see IETF RFC 4574 [50]) obtained from the SDP body. It contains the string after "a=label:" starting with "flus" and may be followed by more characters as described in 3GPP TS 26.238 [51].	FLUS
medCompN	integer	M	1	Identifies the media component number, and it contains the ordinal number of the media component.	
medSubComps	map(MediaSubComponent)	O	1..N	Contains the requested bitrate and filters for the set of service data flows identified by their common flow identifier. The key of the map is the attribute "fNum". (NOTE 3)	
medType	MediaType	O	0..1	Indicates the media type of the service.	
marBwUl	BitRate	O	0..1	Maximum requested bandwidth for the Uplink.	
marBwDl	BitRate	O	0..1	Maximum requested bandwidth for the Downlink.	
maxPacketLossRateDl	PacketLossRateRm	O	0..1	Indicates the downlink maximum rate for lost packets that can be tolerated for the service data flow.	CHEM

Attribute name	Data type	P	Cardinality	Description	Applicability
maxPacketLossRateUl	PacketLossRateRm	O	0..1	Indicates the uplink maximum rate for lost packets that can be tolerated for the service data flow.	CHEM
maxSuppBwDl	BitRate	O	0..1	Maximum supported bandwidth for the Downlink.	IMS_SBI
maxSuppBwUl	BitRate	O	0..1	Maximum supported bandwidth for the Uplink.	IMS_SBI
minDesBwDl	BitRate	O	0..1	Minimum desired bandwidth for the Downlink.	IMS_SBI
minDesBwUl	BitRate	O	0..1	Minimum desired bandwidth for the Uplink.	IMS_SBI
mirBwUl	BitRate	O	0..1	Minimum requested bandwidth for the Uplink.	
mirBwDl	BitRate	O	0..1	Minimum requested bandwidth for the Downlink.	
fStatus	FlowStatus	O	0..1	Indicates whether the status of the service data flows is enabled, or disabled.	
preemptCap	PreemptionCapability	O	0..1	Defines whether the media flow may get resources that were already assigned to another media flow with a lower priority level. It may be included together with "prioSharingInd" for ARP decision.	MCPTT-Preemption
preemptVuln	PreemptionVulnerability	O	0..1	Defines whether the media flow may lose the resources assigned to it in order to admit a media flow with higher priority level. It may be included together with "prioSharingInd" for ARP decision.	MCPTT-Preemption
prioSharingInd	PrioritySharingIndicator	O	0..1	Indicates that the media flow is allowed to use the same ARP as media flows belonging to other "Individual Application Session Context" resources bound to the same PDU session.	PrioritySharing
resPrio	ReservPriority	O	0..1	Indicates the reservation priority.	
rrBw	BitRate	O	0..1	Indicates the maximum required bandwidth in bits per second for RTCP receiver reports within the session component as specified in IETF RFC 3556 [37]. The bandwidth contains all the overhead coming from the IP-layer and the layers above, i.e. IP, UDP and RTCP.	IMS_SBI
rsBw	BitRate	O	0..1	Indicates the maximum required bandwidth in bits per second for RTCP sender reports within the session component as specified in IETF RFC 3556 [37]. The bandwidth contains all the overhead coming from the IP-layer and the layers above, i.e. IP, UDP and RTCP.	IMS_SBI

Attribute name	Data type	P	Cardinality	Description	Applicability
sharingKeyDl	Uint32	O	0..1	Identifies which media components share resources in the downlink direction. If resource sharing applies between media components across "Individual Application Session Context" resources for the same PDU session, the same value of the "sharingKeyDl" attribute shall be used. If resource sharing does not apply among media components across "Individual Application Session Context" resources for the same PDU session, a different value for the "sharingKeyDl" attribute shall be used.	ResourceSharing
sharingKeyUl	Uint32	O	0..1	Identifies which media components share resources in the uplink direction. If resource sharing applies between media components across "Individual Application Session Context" resources for the same PDU session, the same value of the "sharingKeyUl" attribute shall be used. If resource sharing does not apply among media components across "Individual Application Session Context" resources for the same PDU session, a different value for the "sharingKeyUl" attribute shall be used.	ResourceSharing
codecs	array(CodecData)	O	1..2	Indicates the codec data.	
tsnQos	TsnQosContainer	O	0..1	Transports QoS parameters for TSC traffic.	TimeSensitiveNetworking
tscaiInputUl	TscaiInputContainer	O	0..1	Transports TSCAI input parameters for TSC traffic at the ingress interface of the DS-TT/UE (uplink flow direction). (NOTE 2)	TimeSensitiveNetworking
tscaiInputDl	TscaiInputContainer	O	0..1	Transports TSCAI input parameters for TSC traffic at the ingress of the NW-TT (downlink flow direction). (NOTE 2)	TimeSensitiveNetworking
tscaiTimeDom	Uinteger	O	0..1	Indicates the (g)PTP domain that the (TSN)AF is located in.	TimeSensitiveCommunication
capBatAdaptation	boolean	O	0..1	Indicates the capability for AF to adjust the burst sending time, when it is supported and set to "true". The default value is "false" if omitted. (NOTE 2)	EnTSCAC
rTLatencyInd	boolean	O	0..1	Indicates the service data flow needs to meet the Round-Trip (RT) latency requirement of the service, when it is included and set to "true". The default value is "false" if omitted. (NOTE 4, NOTE 5)	RTLatency
pdb	PacketDelBudget	O	0..1	Indicates an upper bound for the time that a packet may be delayed between the UE and the PSA UPF.	RTLatency
rTLatencyIndCorreId	RttFlowReference	O	0..1	Identifies which Media Components contribute to the RT Latency requirement for two service data flows. (NOTE 4)	RTLatency
pduSetQosDl	PduSetQosPara	O	0..1	PDU Set QoS parameter(s) for the downlink direction.	PDUSetHandling
pduSetQosUl	PduSetQosPara	O	0..1	PDU Set QoS parameter(s) for the uplink direction.	PDUSetHandling

Attribute name	Data type	P	Cardinality	Description	Applicability
protoDescDl	ProtocolDescription	O	0..1	Downlink Protocol description for PDU Set identification, the detection of the end of data burst indication, the detection of the Data Burst Size marking indication, TTNB indication and/or indication of whether MoQ or UDP-option is used to carry media related information.	PDUSetHandling PowerSaving TrafficCharChange OnPathN6MediaInfo
protoDescUl	ProtocolDescription	O	0..1	Uplink Protocol description for PDU Set identification in UPF.	PDUSetHandling
periodUl	DurationMilliSec	O	0..1	Indicates the time period between the start of the two data bursts in units of milliseconds in Uplink direction.	PowerSaving
periodDl	DurationMilliSec	O	0..1	Indicates the time period between the start of the two data bursts in units of milliseconds in Downlink direction.	PowerSaving
l4sInd	UplinkDownlinkSupport	O	0..1	Indicates whether ECN marking for L4S support is supported for the UL, the DL or both, UL and DL. (NOTE 3)	L4S
datBurstSizeInd	boolean	O	0..1	Indicates the Data Burst Size marking for the DL service data flow is supported when it is included and set to "true". The default value is "false" if omitted.	TrafficCharChange
timetoNextBurstInd	boolean	O	0..1	Indicates the Time to Next Burst for the DL service data flow is supported, when it is included and set to "true". The default value is "false" if omitted.	TrafficCharChange
onPathN6SigInfo	OnPathN6SigInfo	O	0..1	Contains the on-path N6 signaling information for delivering media related information.	OnPathN6MediaInfo
expTranInd	boolean	O	0..1	Expedited Transfer Indication for the downlink traffic to enable expedited data transfer with reflective QoS for the Non-GBR service data flow. - "true": the expedited data transfer of larger payload for XR application is enabled for the flow. - "false": the expedited data transfer of larger payload for XR application is not enabled for the flow. The default value is "false" if omitted.	TrafficCharChange
ulBrRecInd	boolean	O	0..1	Indicates whether the RAN-controlled UL Bitrate Recommendation indication is supported or not. - Set to "true": the RAN-controlled UL Bitrate Recommendation Indication is supported. - Set to "false": the RAN-controlled UL Bitrate Recommendation Indication is not supported - The default value "false" is used if this attribute is omitted.	ExtQoSR19

NOTE 1: The attributes "altSerReqs" and "altSerReqsData" are mutually exclusive. Of the two, only the attribute "altSerReqs" may be provided if the attribute "qosReference" is provided, while only the attribute "altSerReqsData" may be provided if the attribute "qosReference" is not provided.
NOTE 2: The "burstArrivalTimeWnd" attribute, within the "tscaiInputUl" and/or "tscaiInputDl" attributes, and the "capBatAdaptation attribute are mutually exclusive.
NOTE 3: Within the MediaComponent entry, the NF service consumer may include either the indication of L4S support within the "l4sInd" attribute or the request for congestion measurements within the "evSubsc" attribute included in one or more entries of the "medSubComps" attribute, but the indication of L4S and the subscription to congestion monitoring shall not be provided simultaneously.
NOTE 4: The "rTLatencyInd" attribute and the "rTLatencyIndCorreId" attribute are mutually exclusive.
NOTE 5: If more than one "medSubComps" attributes are present, the PCF selects the media subcomponent and derive the PCC rule for RT latency control.

All IP flows within a "MediaSubComponent" data type are permanently disabled by supplying "FlowStatus" data type with a deletion indication.

Bandwidth information and the "fStatus" attribute provided within the MediaComponent applies to all those IP flows within the media component, for which no corresponding information is being provided within the "medSubComps" attribute. As defined in 3GPP TS 29.513 [7], the bandwidth information within the media component level "marBwUl" and "marBwDl" attributes applies separately to each media subcomponent except for media subcomponents with a "flowUsage" attribute with the value "RTCP". The mapping of bandwidth information for RTCP media subcomponent is defined in 3GPP TS 29.513 [7] clause 7.3.3.

5.6.2.8 Type MediaSubComponent

Table 5.6.2.8-1: Definition of type MediaSubComponent

Attribute name	Data type	P	Cardinality	Description	Applicability
afSigProtocol	AfSigProtocol	O	0..1	Indicates the protocol used for signalling between the UE and the NF service consumer. It may be included only if the "flowUsage" attribute is set to the value "AF_SIGNALLING".	ProvAFsignalFlow
ethfDescs	array(EthFlowDescription)	O	1..2	Contains the flow description for the Uplink and/or Downlink Ethernet flows.	
fNum	integer	M	1	Identifies the ordinal number of the service data flow.	
fDescs	array(FlowDescription)	O	1..2	Contains the flow description for the Uplink and/or Downlink IP flows.	
addInfoFlowDescs	array(AddFlowDescriptionInfo)	O	1..2	Represents additional flow description information (flow label and IPsec SPI) per Uplink and/or Downlink IP flows represented in the "fDescs" attribute.	AddFlowDescriptionInformation
fStatus	FlowStatus	O	0..1	Indicates whether the status of the service data flows is enabled or disabled.	
flowUsage	FlowUsage	O	0..1	Flow usage of the flows (e.g. RTCP, AF signalling).	
marBwUl	BitRate	O	0..1	Maximum requested bandwidth for the Uplink.	
marBwDl	BitRate	O	0..1	Maximum requested bandwidth for the Downlink.	
tosTrCl	TosTrafficClass	O	0..1	Type of Service or Traffic Class.	
evSubsc	EventsSubscReqData	O	0..1	Identifies the events the application subscribes to at creation of a media component. (NOTE 1, NOTE 2)	EnQoSMon, L4S
mpxMediaUlInfos	array(MpxMediaInfo)	O	1..N	Contains the (S)RTP Multiplexed Media Identification Information for the Uplink IP flows.	MpxMedia
mpxMediaDlInfos	array(MpxMediaInfo)	O	1..N	Contains the (S)RTP Multiplexed Media Identification Information for the Downlink IP flows.	MpxMedia

NOTE 1: If attribute "evSubsc" is present, one or more of the following IEs within EventsSubscReqData data type may be included: "events", "notifUri", "reqQosMonParams", "qosMon", "qosMonDatRate", "pdvReqMonParams", "pdvMon", "congestMon", "notifCorreId", "rttMon", "directNotifInd", "avrgWndw". In addition, when the attribute "events" is present, only the following Enumeration "AfEvent" may be included: "QOS_MONITORING", "PACK_DEL_VAR", "RT_DELAY_TWO_QOS_FLOWS", "L4S_SUPP".

NOTE 2: Within the MediaSubComponent entry, the NF service consumer may include the subscription for congestion measurements within the "evSubsc" attribute only if the "l4sInd" attribute is not included within the corresponding MediaComponent entry.

The bit rate information and flow status information provided within the "MediaSubComponent" data type takes precedence over information provided within "MediaComponent" data type.

All service data flows within a "MediaSubComponent" data type are permanently disabled by supplying "FlowStatus" data type with a deletion indication.

If the "EnQoSMon" feature is supported, and the NF service consumer includes the attribute "evSubsc" in the "MediaSubComponent" data type with a subscription to a specific event, then the "evSubsc" attribute within the "AppSessionContextReqData" data type shall not include a subscription to notifications for that specific event. In this case, the PCF shall use the value of the "notifUri" attribute included within the "evSubsc" attribute in the "MediaSubComponent" data type as target URI of the HTTP POST request for that specific event notification.

NOTE: The NF service consumer can provide different values per media subcomponent for the "notifUri" attribute and/or "notifCorreId" attribute, e.g. to identify to the media subcomponent of a received report.

5.6.2.9 Type EventsNotification

Table 5.6.2.9-1: Definition of type EventsNotification

Attribute name	Data type	P	Cardinality	Description	Applicability
adReports	array(AppDetection Report)	C	1..N	Includes the detected application report. It shall be present when the notified event is "APP_DETECTION".	ApplicationDet ectionEvents
accessType	AccessType	C	0..1	Includes the access type. It shall be present when the notified event is "ACCESS_TYPE_CHANGE".	
addAccessInfo	AdditionalAccessInf o	O	0..1	Indicates the additional combination of Access Type and RAT Type available for MA PDU session. It may be present when the notified event is "ACCESS_TYPE_CHANGE" and the PDU session is a Multi-Access PDU session.	ATSSS
relAccessInfo	AdditionalAccessInf o	O	0..1	Indicates the released combination of Access Type and RAT Type previously available for MA PDU session. It may be present when the notified event is "ACCESS_TYPE_CHANGE" and the PDU session is a Multi-Access PDU session.	ATSSS
anChargAddr	AccNetChargingAdd ress	O	0..1	Includes the access network charging address. It shall be present if available when the notified event is "CHARGING_CORRELATION".	IMS_SBI
anChargIds	array(AccessNetCh argingIdentifier)	C	1..N	Includes the access network charging identifier(s). It shall be present when the notified event is "CHARGING_CORRELATION".	IMS_SBI
anGwAddr	AnGwAddress	O	0..1	Access network Gateway Address. It carries the IP address of the ePDG used as IPSec tunnel endpoint with the UE for EPC/ePDG and 5GS interworking. It shall be present, if applicable, when the notified event is "ACCESS_TYPE_CHANGE".	
l4sReports	array(L4sSupport)	C	1..N	ECN marking for L4S support information. It shall be present when the notified event is "L4S_SUPP".	L4S
evSubsUri	Uri	M	1	The Events Subscription URI. Identifies the Events Subscription sub-resource that triggered the notification. (NOTE 1, NOTE 5)	
evNotifs	array(AfEventNotific ation)	M	1..N	Notifications about individual events.	
failedResourcAllo cReports	array(ResourcesAllo cationInfo)	C	1..N	Indicates the status of the PCC rule(s) related to certain failed media components. It shall be included when the event trigger is "FAILED_RESOURCES_ALLOCATION".	
succResourcAlloc Reports	array(ResourcesAllo cationInfo)	O	1..N	Indicates the alternative service requirement the NG-RAN can guarantee to certain media components. It may be included when the event trigger is "SUCCESSFUL_RESOURCES_ALLOC ATION".	Authorization WithRequired QoS
noNetLocSupp	NetLocAccessSupp ort	O	0..1	Indicates the access network does not support the report of the requested access network information.	NetLoc
outOfCredReports	array(OutOfCreditInf ormation)	C	1..N	Out of credit information per service data flow. It shall be present when the notified event is "OUT_OF_CREDIT".	IMS_SBI

Attribute name	Data type	P	Cardinality	Description	Applicability
plmnId	PlmnIdNid	C	0..1	PLMN Identifier or the SNPN Identifier. It shall be present when the notified event is "PLMN_CHG" or, if location information is required but is not available when the notified event is "ANI_REPORT". It shall be present if available when the notified event is "RAN_NAS_CAUSE". (NOTE 2)	
qncReports	array(QosNotificationControlInfo)	C	1..N	QoS notification control information. It shall be present when the notified event is "QOS_NOTIF".	
qosMonReports	array(QosMonitoringReport)	C	1..N	QoS Monitoring reporting information. It shall be present when the notified event is "QOS_MONITORING" and packet delay measurements are available. Only the "ulDelays", "dlDelays" and/or "rtDelays" attributes, or the "pdmf" attribute in QosMonitoringReport may be present.	QoSMonitoring
qosMonDatRateReps	array(QosMonitoringReport)	C	1..N	QoS Monitoring reporting information with data rate measurements. It shall be present when the notified event is "QOS_MONITORING" and data rate measurements are available. Only the "ulDataRate" and/or "dlDataRate" attributes in QosMonitoringReport may be present.	EnQoSMon
congestReports	array(QosMonitoringReport)	C	1..N	Congestion information. It shall be present when the notified event is "QOS_MONITORING" and congestion information is available. Only the "ulConInfo" and/or "dlConInfo" attributes in QosMonitoringReport may be present.	EnQoSMon
pdvMonReports	array(PdvMonitoringReport)	C	1..N	Packet Delay Variation information. It shall be present when the notified event is "PACK_DEL_VAR".	EnQoSMon
rttMonReports	array(QosMonitoringReport)	C	1..N	The measurement result of Round-Trip delay over two QoS flows. It shall be present when the notified event is "RT_DELAY_TWO_QOS_FLOWS". Only the "rtDelays" attribute in QosMonitoringReport may be present.	EnQoSMon
qosMonCapRepos	map(CapabilityReport)	C	1..N	QoS monitoring is not supported or QoS monitoring is supported again. The key of the map is the attribute "capType". It shall be present when the notified event is "QOS_MON_CAP_REPO".	QoSMonCapRepo
ranNasRelCauses	array(RanNasRelCause)	C	1..N	RAN-NAS release cause. It shall be present if available when the notified event is "RAN_NAS_CAUSE".	RAN-NAS-Cause
ratType	RatType	O	0..1	RAT type. It shall be present, if applicable, when the notified event is "ACCESS_TYPE_CHANGE".	

Attribute name	Data type	P	Cardinality	Description	Applicability
satBackhaulCategory	SatelliteBackhaulCategory	C	0..1	Indicates the satellite or non-satellite backhaul category of the PDU session. It shall be present, if applicable, when the notified event is "SAT_CATEGORY_CHG". If the "EnSatBackhaulCatChg" feature is supported, the different dynamic satellite backhaul categories may also be provided.	SatelliteBackhaul
ueLoc	UserLocation	O	0..1	E-UTRA, or NR, and/or non-3GPP trusted and untrusted access user location information. "n3gppTai" and "n3IwfId" attributes within the "N3gaLocation" data type shall not be supplied. It shall be present if required and available when the notified event is "ANI_REPORT". It shall be present if available when the notified event is "RAN_NAS_CAUSE". (NOTE 3) (NOTE 4)	NetLoc, RAN-NAS-Cause
ueLocTime	DateTime	O	0..1	Contains the NTP time at which the UE was last known to be in the location. (NOTE 3)	NetLoc
ueTimeZone	TimeZone	O	0..1	UE time zone. It shall be present if required and available when the notified event is "ANI_REPORT". It shall be present if available when the notified event is "RAN_NAS_CAUSE".	NetLoc, RAN-NAS-Cause
usgRep	AccumulatedUsage	C	0..1	Indicates the measured volume and/or time for sponsored data connectivity. It shall be present when the notified event is "USAGE_REPORT".	SponsoredConnectivity
urspEnfRep	UrspEnforcementInfo	C	0..1	Includes the URSP rule enforcement information received from a UE from associated URSP rule(s). It shall be present when the notified event is "URSP_ENF_INFO".	URSPEnforcement
sscMode	SscMode	O	0..1	SSC Mode of the PDU session. It may be present when the notified event is "URSP_ENF_INFO".	URSPEnforcement
ueReqDnn	Dnn	O	0..1	UE requested DNN. It may be present when the notified event is "URSP_ENF_INFO".	URSPEnforcement
ueReqPduSessionType	PduSessionType	O	0..1	UE requested PDU session Type. It may be present when the notified event is "URSP_ENF_INFO".	URSPEnforcement
tsnBridgeManCont	BridgeManagementContainer	O	0..1	Transports TSC user plane node management information.	TimeSensitiveNetworking
tsnPortManContDstt	PortManagementContainer	O	0..1	Transports port management information for the DS-TT port.	TimeSensitiveNetworking
tsnPortManContNwtts	array(PortManagementContainer)	O	1..N	Transports port management information for one or more NW-TT ports.	TimeSensitiveNetworking
ipv4AddrList	array(Ipv4AddrMask)	O	1..N	List of Framed Route information of IPv4.	ExtraUEaddrReport
ipv6PrefixList	array(Ipv6Prefix)	O	1..N	List of Framed Route information of IPv6 or list of IPv6 address prefixes of the served UE.	ExtraUEaddrReport
batOffsetInfo	BatOffsetInfo	C	0..1	The offset of the BAT and the optionally adjusted periodicity. It shall be present if available when the notified event is "BAT_OFFSET_INFO".	EnTSCAC

Attribute name	Data type	P	Cardinality	Description	Applicability
ueReachStatus	UeReachabilityStatus	C	0..1	Contains the UE reachability Status. This attribute shall be present only when the notified event is "UE_REACH_STATUS_CH".	UEUnreachable
retryAfter	Uinteger	O	0..1	Contains the estimated time duration (expressed in units of seconds) during which the UE is unreachable. This attribute may be present only when the "ueReachStatus" attribute is present and set to "UNREACHABLE".	UEUnreachable
servSatId	SatelliteId	C	0..1	Indicates UE serving satellite identifier. It shall be present if required and available when the notified event is "ANI_REPORT". It shall be present if available when the notified event is "RAN_NAS_CAUSE".	UeSatUeComm
rateLimitRepo	RateLimitRepo	O	0..1	Indicates the maximum data rate authorized for the non-GBR service data flow(s). It shall be present if required and available when the notified event is "RATE_LIMIT_INFO_REPO".	RateLimitReport
dnaiChgType	DnaiChangeType	C	0..1	Identifies the type of notification regarding UP path management event. This attribute shall be present when the "subscribedEvent" attribute is set to "UP_PATH_CHANGE".	UeSatUeComm
sourceDnai	Dnai	C	0..1	Source DN Access Identifier. Shall be included for event "UP_PATH_CHANGE" if the DNAI changed.	UeSatUeComm
targetDnai	Dnai	C	0..1	Target DN Access Identifier. Shall be included for event "UP_PATH_CHANGE" if the DNAI changed.	UeSatUeComm
sessionRecovStatus	PcSessionRecoveryStatus	C	0..1	Contains the 5GC Session Recovery Status. This attribute shall be present only when the notified event is "CN_HEALTH_MONITOR".	CnHealthMonitor

NOTE 1: Either the complete resource URI included in the "evSubsUri" attribute or the "apiSpecificResourceUriPart" component (see clause 5.1) of the resource URI included in the "evSubsUri" attribute may be used by the NF service consumer for the identification of the Individual Application Session Context resource related to the notification.
NOTE 2: The SNPN Identifier consists of the PLMN Identifier and the NID.
NOTE 3: Whether the "ueLoc" attribute also encodes the age of location is implementation specific.
NOTE 4: When the "ueLoc" attribute contains both, the 3GPP and the non-3GPP UE location, the "ueLocTime" attribute contains the age of the last known 3GPP UE location.
NOTE 5: For event notifications of implicit subscriptions, the content of "evSubsUri" attribute shall be set based on the corresponding application data in the UDR (e.g., clause 4.2.5.29).

5.6.2.10 Type AfEventSubscription

Table 5.6.2.10-1: Definition of type AfEventSubscription

Attribute name	Data type	P	Cardinality	Description	Applicability
event	AfEvent	M	1	Subscribed Event.	
notifMethod	AfNotifMethod	O	0..1	If notifMethod is not supplied, the default value "EVENT_DETECTION" applies.	
repPeriod	DurationSec	O	0..1	Indicates the time interval between successive event notifications. It is supplied for notification method "PERIODIC". If the feature "PacketDelayFailureReport" is supported, it also indicates the time interval at which a measurement failure needs to be reported if no measurement result is provided. It is supplied for notification methods "PERIODIC" and "EVENT_DETECTION".	EnhancedSubscriptionToNotification PacketDelayFailureReport
waitTime	DurationSec	O	0..1	Indicates the minimum waiting time between subsequent reports. Only applicable when the notification is set to "EVENT_DETECTION".	EnhancedSubscriptionToNotification
qosMonParamType	QosMonitoringParamType	C	0..1	Indicates the type of QoS monitoring parameter when the "event" attribute indicates "QOS_MONITORING". It may be omitted when the QoS monitoring parameter is Packet Delay, otherwise it shall be provided when the "EnQoSMon" feature is supported.	EnQoSMon
capTypes	array(NotifCapType)	C	1..N	Indicates the requested QoS Monitoring Capability type(s). This attribute shall be present if the "QOS_MON_CAP_REPO" event is subscribed within the "event" attribute.	QoSMonCapRepo

5.6.2.11 Type AfEventNotification

Table 5.6.2.11-1: Definition of type AfEventNotification

Attribute name	Data type	P	Cardinality	Description	Applicability
event	AfEvent	M	1	Notified Event.	
flows	array(Flows)	O	1..N	Affected Service Data Flows.	
retryAfter	Uinteger	O	0..1	Indicates the estimate on how long it will take before it can be considered the paging procedure as completed. It may be provided when the event attribute indicates UE_TEMPORARILY_UNAVAILABLE. The value shall be in seconds.	UEUnreachable

5.6.2.12 Type TerminationInfo

Table 5.6.2.12-1: Definition of type TerminationInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
termCause	TerminationCause	M	1	Indicates the cause for requesting the deletion of the Individual Application Session Context resource.	
resUri	Uri	M	1	Identifies the Individual Application Session Context that triggered the termination notification. (NOTE)	
NOTE: Either the complete resource URI included in the "resUri" attribute or the "apiSpecificResourceUriPart" component (see clause 5.1) of the resource URI included in the "resUri" attribute may be used by the NF service consumer for the identification of the Individual Application Session Context resource related to the termination notification.					

5.6.2.13 Type AfRoutingRequirement

Table 5.6.2.13-1: Definition of type AfRoutingRequirement

Attribute name	Data type	P	Cardinality	Description	Applicability
appReloc	boolean	O	0..1	Indication of application relocation possibility. When it is included and set to "true", it indicates that the application cannot be relocated once a location of the application is selected by the 5GC. The default value is "false".	InfluenceOnTrafficRouting
routeToLocs	array(RouteToLocation)	O	1..N	A list of traffic routes to applications locations.	InfluenceOnTrafficRouting
spVal	SpatialValidity	O	0..1	Indicates where the traffic routing requirements apply. The absence of this attribute indicates no spatial restrictions.	InfluenceOnTrafficRouting
tempVals	array(TemporalValidity)	O	1..N	Indicates the time interval(s) during which the NF service consumer request is to be applied.	InfluenceOnTrafficRouting
upPathChgSub	UpPathChgEvent	O	0..1	Subscription to UP path management events.	InfluenceOnTrafficRouting UeSatUeComm
outcomeSub	TraffRouteReqOutcomeEvent	O	0..1	Contains the subscription to the traffic routing requirements installation outcome event.	TraffRouteReqOutcome
simConnFailSub	SimConnFailEvent	O	0..1	Contains the subscription information for the simultaneous connectivity failure event. (NOTE) It may only be provided if the "simConnInd" attribute is provided and set to "true".	SimConnFailure
addrPreserInd	boolean	O	0..1	Indicates whether UE IP address should be preserved. This attribute shall set to "true" if preserved, otherwise, set to "false". Default value is false if omitted.	URLLC
simConnInd	boolean	O	0..1	Indication of simultaneous connectivity temporarily maintained for the source and target PSA. If it is included and set to "true", temporary simultaneous connectivity should be kept. The default value "false" applies, if the attribute is not present and has not been supplied previously.	SimultConnectivity
simConnTerm	DurationSec	C	0..1	Indication of the minimum time interval to be considered for inactivity of the traffic routed via the source PSA during the edge re-location procedure. It may be included when the "simConnInd" attribute is set to true.	SimultConnectivity
maxAllowedUpLat	Uinteger	O	0..1	Indicates the target user plane latency in units of milliseconds.	AF_latency
easIpReplaceInfos	array(EasIpReplacementInfo)	O	1..N	Contains EAS IP replacement information.	EASIPreplacement
easRedisInd	boolean	O	0..1	Indicates the EAS rediscovery is required for the application if it is included and set to "true". Default value is "false" if omitted. The indication shall be invalid after it was applied unless it is provided again.	EASDiscovery
tfcCorreInfo	TrafficCorrelationInfo	O	0..1	Contains the information for traffic correlation.	CommonEASDNAI

Attribute name	Data type	P	Cardinality	Description	Applicability
candDnaiInd	boolean	O	0..1	Indication of whether the candidate DNAI(s) are requested to be reported or not. - "true": the candidate DNAI(s) for the PDU session are requested to be reported. - "false": the candidate DNAI(s) for the PDU session are not requested to be reported. The default value is "false" if omitted.	EnCommonDnai
n6DelayInd	boolean	O	0..1	Indication whether the N6 delay measurement is requested to be considered or not. - If it is included and set to "true", the N6 delay measurement is requested to be considered. - If it is included and set to "false", the N6 delay measurement is not requested to be considered. - The default value "false" applies, if the attribute is not present and has not been supplied previously.	N6DelayMeasurement
NOTE: When the Simultaneous Connectivity succeeds, no related notifications will be sent.					

5.6.2.14 Type ResourcesAllocationInfo

Table 5.6.2.14-1: Definition of type ResourcesAllocationInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
mcResourcStatus	MediaComponentResourcesStatus	C	0..1	Indicates the status of the PCC rule(s) related to the media components identified by the "flows" attribute.	
flows	array(Flows)	C	1..N	Identification of the flows. It shall be included if "MediaComponentVersioning" feature is supported. When "MediaComponentVersioning" feature is not supported, if no flows are provided, the status in the "mcResourcStatus" applies for all flows within the AF session.	
altSerReq	string	O	0..1	When present, indicates the alternative service requirement the NG-RAN can guarantee for the indicated "flows".	AuthorizationWithRequiredQoS
NOTE: The "mcResourcStatus" attribute shall be included if AuthorizationWithRequiredQoS feature is not supported.					

5.6.2.15 Type QosNotificationControlInfo

Table 5.6.2.15-1: Definition of type QosNotificationControlInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
notifType	QosNotifType	M	1	Indicates whether the GBR targets for the indicated SDFs are "NOT_GUARANTEED" or "GUARANTEED" again. When the "ExtQoSR19" feature is supported, the direction information may be provided if the QoS targets of one or more SDFs are not being guaranteed.	
flows	array(Flows)	C	1..N	Identification of the flows. It shall be included if "MediaComponentVersioning" feature is supported. When "MediaComponentVersioning" feature is not supported, if no flows are provided, the notification in the "notifType" applies for all flows within the AF session.	
altSerReq	string	O	0..1	Indicates the alternative service requirement the NG-RAN can guarantee. It contains a QoS reference or the reference to the alternative individual QoS related parameters (see "altQosParamSetRef" attribute in Table 5.6.2.47). When it is omitted and the "notifType" attribute is "NOT_GUARANTEED", if the "ExtQoSR19" feature is supported, "NOT_GUARANTEED_DL" and/or "NOT_GUARANTEED_UL", it indicates that the lowest priority alternative service requirement could not be fulfilled in the corresponding directions by the NG-RAN.	AuthorizationWithRequiredQoS
altSerReqNotSupInd	boolean	O	0..1	It may be set to true when the "notifType" attribute is "NOT_GUARANTEED", if the "ExtQoSR19" feature is supported, "NOT_GUARANTEED_DL" and/or "NOT_GUARANTEED_UL" to indicate that alternative service requirements are not supported in the corresponding directions by NG-RAN. The default value false shall apply if the attribute is not present. It may be used when the AuthorizationWithRequiredQoS feature or the AltSerReqsWithIndQoS feature is supported.	AltQoSProfilesSupportReport

5.6.2.16 Type SpatialValidity

Table 5.6.2.16-1: Definition of type SpatialValidity

Attribute name	Data type	P	Cardinality	Description	Applicability
presenceInfoList	map(PresenceInfo)	M	1..N	Defines the presence information provisioned by the NF service consumer. The "presenceState" attribute within the "PresenceInfo" data type shall not be supplied. The "praId" attribute within the PresenceInfo data type shall also be the key of the map.	InfluenceOnTrafficRouting

5.6.2.17 Type EthFlowDescription

Table 5.6.2.17-1: Definition of type EthFlowDescription

Attribute name	Data type	P	Cardinality	Description	Applicability
destMacAddr	MacAddr48	O	0..1	Destination MAC address.	
ethType	string	M	1	A two-octet string that represents the Ethertype, as described in IEEE 802.3 [16] and IETF RFC 7042 [18] in hexadecimal representation. Each character in the string shall take a value of "0" to "9" or "A" to "F" and shall represent 4 bits. The most significant character representing the 4 most significant bits of the ethType shall appear first in the string, and the character representing the 4 least significant bits of the ethType shall appear last in the string.	
fDesc	FlowDescription	C	0..1	Contains the flow description for the Uplink or Downlink IP flow. It shall be present when the Ethertype is IP. (NOTE 3)	
fDir	FlowDirection	O	0..1	Contains the packet filter direction. Only the "DOWNLINK" or "UPLINK" value is applicable. (NOTE 2)	
sourceMacAddr	MacAddr48	O	0..1	Source MAC address.	
vlanTags	array(string)	O	1..2	Customer-VLAN and/or Service-VLAN tags containing the VID, PCP/DEI fields as defined in IEEE 802.1Q [17] and IETF RFC 7042 [18]. The first/lower instance in the array stands for the Customer-VLAN tag and the second/higher instance in the array stands for the Service-VLAN tag. Each field is encoded as a two-octet string in hexadecimal representation. Each character in the string shall take a value of "0" to "9" or "A" to "F" and shall represent 4 bits. The most significant character representing the PCP/DEI field shall appear first in the string, followed by character representing the 4 most significant bits of the VID field, and the character representing the 4 least significant bits of the VID field shall appear last in the string. If only Service-VLAN tag is provided, empty string for Customer-VLAN tag shall be provided.	
srcMacAddrEnd	MacAddr48	O	0..1	Source MAC address end. If this attribute is present, the sourceMacAddr attribute specifies the source MAC address start. E.g. srcMacAddrEnd with value 00-10-A4-23-3E-FE and sourceMacAddr with value 00-10-A4-23-3E-02 means all MAC addresses from 00-10-A4-23-3E-02 up to and including 00-10-A4-23-3E-FE.	MacAddressRange
destMacAddrEnd	MacAddr48	O	0..1	Destination MAC address end. If this attribute is present, the destMacAddr attribute specifies the destination MAC address start.	MacAddressRange

Attribute name	Data type	P	Cardinality	Description	Applicability
NOTE 1: The "srcMacAddrEnd" attribute may only be present if the "sourceMacAddr" attribute is present; the "destMacAddrEnd" attribute may only be present if the "destMacAddr" attribute is present. NOTE 2: If the "UPLINK" is included within the "fDir" attribute, the "sourceMacAddr" attribute and "srcMacAddrEnd" attribute (if MacAddressRange feature is supported) contain the UE address and "destMacAddr" attribute and "destMacAddrEnd" attribute (if MacAddressRange feature is supported)contain the remote address; otherwise if the "DOWNLINK" is included within the "fDir" attribute or the "fDir" attribute is never provided, the "sourceMacAddr" attribute and "srcMacAddrEnd" attribute (if MacAddressRange feature is supported) contain the remote address and "destMacAddr" attribute and "destMacAddrEnd" attribute (if MacAddressRange feature is supported) contain the UE address. NOTE 3: The direction of the "fDesc" attribute shall be set to "in" if the "UPLINK" is included within the "fDir" attribute; the direction of the "fDesc" attribute shall be set to "out" if the "DOWNLINK" is included within the "fDir" attribute or the "fDir" attribute is never provided.					

5.6.2.18 Void

5.6.2.19 Void

5.6.2.20 Type AnGwAddress

Table 5.6.2.20-1: Definition of type AnGwAddress

Attribute name	Data type	P	Cardinality	Description	Applicability
anGwIpv4Addr	Ipv4Addr	O	0..1	Includes the IPv4 address of the access network gateway control node.	
anGwIpv6Addr	Ipv6Addr	O	0..1	Includes the IPv6 address of the access network gateway control node.	
NOTE: At least one address of the access network gateway control node (the IPv4 address or the IPv6 address or both if both addresses are available) shall be included.					

5.6.2.21 Type Flows

Table 5.6.2.21-1: Definition of type Flows

Attribute name	Data type	P	Cardinality	Description	Applicability
contVers	array(ContentVersion)	C	1..N	Represents the content version of the content of a media component. If "MediaComponentVersioning" feature is supported, the content version shall be included if it was included when the corresponding media component was provided or modified.	MediaComponentVersioning
fNums	array(integer)	O	1..N	Indicates the service data flows via their flow identifier. If no flow identifier is supplied, the Flows data type refers to all the flows matching the media component number.	
medCompN	integer	M	1	Identifies the media component number, and it contains the ordinal number of the media component.	

5.6.2.22 Type TemporalValidity

Table 5.6.2.22-1: Definition of type TemporalValidity

Attribute name	Data type	P	Cardinality	Description	Applicability
startTime	DateTime	O	0..1	Indicates the time from which the traffic routing requirements start to apply. The absence of this attribute indicates the traffic routing requirements apply immediately.	InfluenceOnTraffic Routing
stopTime	DateTime	O	0..1	Indicates the time when the traffic routing requirements cease to apply. The absence of this attribute indicates the traffic routing requirements do not cease at any time.	InfluenceOnTraffic Routing

5.6.2.23 Void

5.6.2.24 Type AfRoutingRequirementRm

This data type is defined in the same way as the "AfRoutingRequirement" data type, but:

- with the OpenAPI "nullable: true" property;
- the removable attribute "spVal" is defined with the data type "SpatialValidityRm", and "maxAllowedUpLat" is defined with the data type "UintegerRm"; and
- the removable attributes "tempVals", "routeToLocs", "addrPreserInd", "simConnInd", "simConnTerm", "easIpReplaceInfos" and "n6DelayInd" are defined as nullable in the OpenAPI.

Table 5.6.2.24-1: Definition of type AfRoutingRequirementRm

Attribute name	Data type	P	Cardinality	Description	Applicability
appReloc	boolean	O	0..1	Indication of application relocation possibility. When it is set to "true", it indicates that the application cannot be relocated once a location of the application is selected by the 5GC.	InfluenceOnTrafficRouting
routeToLocs	array(RouteToLocation)	O	1..N	A list of traffic routes to applications locations.	InfluenceOnTrafficRouting
spVal	SpatialValidityRm	O	0..1	Indicates where the traffic routing requirements apply.	InfluenceOnTrafficRouting
tempVals	array(TemporalValidity)	O	1..N	Indicates the time interval(s) during which the NF service consumer request is to be applied.	InfluenceOnTrafficRouting
upPathChgSub	UpPathChgEvent	O	0..1	Subscription to UP path management events.	InfluenceOnTrafficRouting UeSatUeComm
simConnFailSub	SimConnFailEvent	O	0..1	Contains the subscription information for the simultaneous connectivity failure event. It may only be provided when the simultaneous connectivity temporarily maintained for the source and target PSA is required.	SimConnFailure
addrPreserInd	boolean	O	0..1	Indicates whether UE IP address should be preserved.	URLLC
simConnInd	boolean	O	0..1	Indication of simultaneous connectivity temporarily maintained for the source and target PSA. If it is included and set to "true", temporary simultaneous connectivity should be kept.	SimultConnectivity
simConnTerm	DurationSecRm	C	0..1	Indication of the minimum time interval to be considered for inactivity of the traffic routed via the source PSA during the edge re-location procedure.	SimultConnectivity
maxAllowedUpLat	UintegerRm	O	0..1	Indicates the target user plane latency in units of milliseconds.	AF_latency
easIpReplaceInfos	array(EasIpReplacementInfo)	O	1..N	Contains EAS IP replacement information.	EASIPreplacement
easRedisInd	boolean	O	0..1	Indicates the EAS rediscovery is required for the application if it is included and set to "true". Default value is "false" if omitted. The indication shall be invalid after it was applied unless it is provided again.	EASDiscovery
tfcCorreInfo	TrafficCorrelationInfo	O	0..1	Contains the information for traffic correlation.	CommonEASDNAI
candDnaiInd	boolean	O	0..1	Indication of whether the candidate DNAI(s) are requested to be reported or not. - "true": the candidate DNAI(s) for the PDU session are requested to be reported. - "false": the candidate DNAI(s) for the PDU session are not requested to be reported.	EnCommonDnai
n6DelayInd	boolean	O	0..1	Indication whether the N6 delay measurement is requested to be considered or not. - If it is included and set to "true", the N6 delay measurement is requested to be considered. - If it is included and set to "false", the N6 delay measurement is not requested to be considered.	N6DelayMeasurement

5.6.2.25 Type EventsSubscReqDataRm

This data type is defined in the same way as the "EventsSubscReqData" data type, but:

- with the OpenAPI "nullable: true" property;
- the removable attribute "usgThres" is defined with the removable data type "UsageThresholdRm"; the removable attribute "avrgWndw" is defined with the removable data type "AverWindowRm"; the removable attribute "rttFlowRef" is defined with the removable data type "RttFlowReferenceRm"; and removable attributes "qosMon", "qosMonDatRate", "congestMon", "pdvMon", "rttMon" and "avlBitRateMon" are defined with the removable data type "QosMonitoringInformationRm"; and
- the "reqQosMonParams","pdvReqMonParams" and "directNotifInd" attributes are defined with the "nullable: true" property.

Table 5.6.2.25-1: Definition of type EventsSubscReqDataRm

Attribute name	Data type	P	Cardinality	Description	Applicability
events	array(AfEventSubscription)	M	1..N	Subscribed Events.	
notifUri	Uri	O	0..1	Notification URI.	
reqQosMonParams	array(RequestedQosMonitoringParameter)	O	1..N	Indicates the QoS information to be monitored, e.g. UL packet delay, DL packet delay, round trip packet delay. If the "EnQoSMon" feature is supported, it may indicate, e.g., congestion information, and/or data rate are to be monitored when the QoS Monitoring is enabled for the service data flow. If the "EnQoSMon_v2" feature is supported, it may also indicate the UL/DL available bitrate to be monitored. It may be present when the event "QOS_MONITORING" is subscribed. If the "EnQoSMon" feature is supported, the "reqQosMonParams" attribute may be set to value null, otherwise it shall not be set to value null.	QoSMonitoring
qosMon	QosMonitoringInformationRm	O	0..1	Packet delay threshold(s) information. It may be present when the event "QOS_MONITORING" is subscribed and event based packet delay measurements are required. Only the "repThreshDl", "repThreshUl" and/or "repThreshRp" attributes may be present within the "QosMonitoringInformationRm" data type.	QoSMonitoring
qosMonDatRate	QosMonitoringInformationRm	O	0..1	Contains the data rate measurements information for the subscribed report. It may be present when the event "QOS_MONITORING" is subscribed and event-based data rate measurements are required. Only the "repThreshDatRateDl", and/or the "repThreshDatRateUl" attributes may be present within the "QosMonitoringInformationRm" data type.	EnQoSMon
pdvReqMonParams	array(RequestedQosMonitoringParameter)	O	1..N	Indicates the Packet Delay Variation to be monitored, e.g. UL packet delay, DL packet delay and/or round trip packet delay between the UE and the UPF is to be monitored. Only the "DOWNLINK", "UPLINK" or "ROUNDTRIP" value may be present within the "RequestedQosMonitoringParameter" data type.	EnQoSMon
pdvMon	QosMonitoringInformationRm	O	0..1	Packet Delay Variation information for the subscribed report. It may be present when the event "PACK_DEL_VAR" is subscribed. Only the "repThreshDl", "repThreshUl" and/or "repThreshRp" attributes may be present within the "QosMonitoringInformationRm" data type.	EnQoSMon

Attribute name	Data type	P	Cardinality	Description	Applicability
congestMon	QosMonitoringInformationRm	O	0..1	Congestion threshold for the subscribed report. It may be present when the event "QOS_MONITORING" is subscribed and event-based congestion measurements are required. Only the "conThreshDl" and/or the "conThreshUl" attributes may be present within the "QosMonitoringInformationRm" data type.(NOTE 3)	EnQoSMon
rttMon	QosMonitoringInformationRm	O	0..1	Contains the round-trip delay over two QoS flows information for the subscribed report. It may be present when the event "RT_DELAY_TWO_QOS_FLOWS" is subscribed.	EnQoSMon
rttFlowRef	RttFlowReferenceRm	O	0..1	Contains information about the UL and DL flows involved in the round-trip delay measurements over two QoS flows.	EnQoSMon
avlBitRateMon	QosMonitoringInformationRm	O	0..1	Contains information about the available bitrate threshold for the subscribed report. It may be present when the event "QOS_MONITORING" is subscribed. Only the "avlBitrateUlThrs" and/or the "avlBitrateDlThrs" attributes may be present within the "QosMonitoringInformationRm" data type. (NOTE 3)	EnQoSMon_v2
reqAnis	array(RequiredAccessInfo)	C	1..N	Represents the required access network information. It shall be present when the event "ANI_REPORT" is subscribed. (NOTE 1)	NetLoc
usgThres	UsageThresholdRm	O	0..1	Includes the volume and/or time thresholds for sponsored data connectivity.	SponsoredConnectivity
notifCorreId	string	O	0..1	It is used to set the value of Notification Correlation ID in the corresponding notification.	EnhancedSubscriptionToNotification
directNotifInd	boolean	C	0..1	Indicates that the event notification of QoS Monitoring data is sent by the UPF to Local NEF or AF if it is included and set to true. It may be present when the event "QOS_MONITORING" is subscribed. (NOTE 2)	ExposureToEAS EnQoSMon
avrgWndw	AverWindowRm	O	0..1	Averaging window for the calculation of the data rate for the service data flow	EnQoSMon

NOTE 1: "ANI_REPORT" is the one-time reported event and thus the attribute "reqAnis" is not defined as removable attribute (i.e. with the removable data type "RequiredAccessInfoRm"). Once the access network information is reported to the NF service consumer the subscription to this event is automatically terminated in the PCF and the related information is removed.

NOTE 2: When the "EnQoSMon" feature is supported, the "directNotifInd" attribute indicates whether direct event notification is requested for the QoS measurement(s) provided (or previously provided) in the "reqQosMonParams" attribute.

NOTE 3: Only the "EVENT_DETECTION" reporting frequency in "notifMethod" attribute included in "events" attribute is applicable when the "reqQosMonParams" includes "DOWNLINK_CONGESTION" and/or "UPLINK_CONGESTION", or "DOWNLINK_AVAILABLE_BITRATE" and/or "UPLINK_AVAILABLE_BITRATE".

5.6.2.26 Type MediaComponentRm

This data type is defined in the same way as the "MediaComponent" data type, but:

- with the OpenAPI "nullable: true" property; and
- the removable attributes "afRoutReq" is defined with the removable data type "AfRoutingRequirementRm"; "maxPacketLossRateDl" and "maxPacketLossRateUl" are defined with the removable data type "PacketLossRateRm"; "medSubComps" is defined with the removable data type "MediaSubComponentRm"; "preemptCap" is defined with the removable data type "PreemptionCapabilityRm"; "preemptVuln" is defined with the removable data type "PreemptionVulnerabilityRm"; "marBwDl", "marBwUl", "minDesBwDl", "minDesBwUl", "mirBwDl", "mirBwUl", "maxSuppBwDl", "maxSuppBwUl", "rrBw", "rsBw" are defined with the removable data type "BitRateRm"; "sharingKeyDl" and "sharingKeyUl" are defined with the removable data types "Uint32Rm", "tsnQos" is defined with the removable data type "TsnQosContainerRm"; "pduSetQosDl" and "pduSetQosUl" are defined with the removable data type "pduSetQosParaRm"; "desMaxLatency" and "desMaxLoss" are defined with the removable data type "FloatRm"; "protoDescDl" and "protoDescUl" are defined with the removable data type "ProtocolDescriptionRm"; the "afSfcReq" attributo with the removable data type "AfSfcRequirement", the removable attribute "pdb" with the removable data type "PacketDelBudgetRm", the removable attribute "rTLatencyIndCorreId" with the removable data type "RttFlowReferenceRm", the removable attribute "afHdrReq" with the removable data type "AfHeaderHandlingControlInfo"; the removable attributes "periodUl" and "periodDl" are defined with the removable data type "DurationMilliSecRm"; the removable attribute "onPathN6SigInfo" with the removable data type "OnPathN6SigInfo"; and
- the removable attributes "flusId", "qosReference", "altSerReqs", "altSerReqsData", "rTLatencyInd", "datBurstSizeInd", "timetoNextBurstInd", "expTranInd" and "ulBrRecInd" are defined with the property "nullable: true" in the OpenAPI.

Table 5.6.2.26-1: Definition of type MediaComponentRm

Attribute name	Data type	P	Cardinality	Description	Applicability
afAppId	AfAppId	O	0..1	Contains information that identifies the particular service the AF session belongs to.	
afRoutReq	AfRoutingRequirementRm	O	0..1	Indicates the AF traffic routing requirements.	InfluenceOnTrafficRouting
afSfcReq	AfSfcRequirement	O	0..1	Indicates the AF requirements on steering traffic to a pre-configured chain of service functions on N6-LAN.	SFC
afHdrReq	AfHeaderHandlingControlInfo	O	0..1	Indicates the AF handling of payload headers requirements.	HeaderHandling
qosReference	string	O	0..1	Identifies a pre-defined QoS information.	AuthorizationWithRequiredQoS
altSerReqs	array(string)	O	1..N	Ordered list of alternative service requirements that include a set of QoS references. The lower the index of the array for a given entry, the higher the priority. (NOTE 1)	AuthorizationWithRequiredQoS
altSerReqsData	array(AlternativeServiceRequirementsData)	O	1..N	Ordered list of alternative service requirements that include individual QoS parameter sets. The lower the index of the array for a given entry, the higher the priority. (NOTE 1)	AltSerReqsWithIndQoS
disUeNotif	boolean	O	0..1	Indicates to disable QoS flow parameters signalling to the UE when the SMF is notified by the NG-RAN of changes in the fulfilled QoS situation when it is included and set to "true". The fulfilled situation is either the QoS profile or an Alternative QoS Profile. The default value "false" shall apply, if the attribute is not present and has not been supplied previously.	DisableUENotification
contVer	ContentVersion	O	0..1	Represents the content version of a media component.	MediaComponentVersioning
desMaxLatency	FloatRm	O	0..1	Indicates a maximum desirable transport level packet latency in milliseconds.	FLUS, QoSHint
desMaxLoss	FloatRm	O	0..1	Indicates the maximum desirable transport level packet loss rate in percent (without "%" sign).	FLUS, QoSHint
flusId	string	O	0..1	Indicates that the media component is used for FLUS media. It is derived from the media level attribute "a=label:" (see IETF RFC 4574 [50]) obtained from the SDP body. It contains the string after "a=label:" starting with "flus" and may be followed by more characters as described in 3GPP TS 26.238 [51].	FLUS
maxPacketLossRateDl	PacketLossRateRm	O	0..1	Indicates the downlink maximum rate for lost packets that can be tolerated for the service data flow.	CHEM
maxPacketLossRateUl	PacketLossRateRm	O	0..1	Indicates the uplink maximum rate for lost packets that can be tolerated for the service data flow.	CHEM
medCompN	integer	M	1	Identifies the media component number, and it contains the ordinal number of the media component.	
medSubComps	map(MediaSubComponentRm)	O	1..N	Contains the requested bitrate and filters for the set of service data flows identified by their common flow identifier. The key of the map is the attribute "fNum". (NOTE 3)	
medType	MediaType	O	0..1	Indicates the media type of the service.	

Attribute name	Data type	P	Cardinality	Description	Applicability
marBwUl	BitRateRm	O	0..1	Maximum requested bandwidth for the Uplink.	
marBwDl	BitRateRm	O	0..1	Maximum requested bandwidth for the Downlink.	
maxSuppBwDl	BitRateRm	O	0..1	Maximum supported bandwidth for the Downlink.	IMS_SBI
maxSuppBwUl	BitRateRm	O	0..1	Maximum supported bandwidth for the Uplink.	IMS_SBI
minDesBwDl	BitRateRm	O	0..1	Minimum desired bandwidth for the Downlink.	IMS_SBI
minDesBwUl	BitRateRm	O	0..1	Minimum desired bandwidth for the Uplink.	IMS_SBI
mirBwUl	BitRateRm	O	0..1	Minimum requested bandwidth for the Uplink.	
mirBwDl	BitRateRm	O	0..1	Minimum requested bandwidth for the Downlink.	
fStatus	FlowStatus	O	0..1	Indicates whether the status of the service data flows is enabled, or disabled.	
preemptCap	PreemptionCapabilityRm	O	0..1	Defines whether the media flow may get resources that were already assigned to another media flow with a lower priority level.	MCPTT-Preemption
preemptVuln	PreemptionVulnerabilityRm	O	0..1	Defines whether the media flow may lose the resources assigned to it in order to admit a media flow with higher priority level.	MCPTT-Preemption
prioSharingInd	PrioritySharingIndicator	O	0..1	Indicates that the media flow is allowed to use the same ARP as media flows belonging to other "Individual Application Session Context" resources bound to the same PDU session.	PrioritySharing
resPrio	ReservPriority	O	0..1	Indicates the reservation priority.	
rrBw	BitRateRm	O	0..1	Indicates the maximum required bandwidth in bits per second for RTCP receiver reports within the session component as specified in IETF RFC 3556 [37]. The bandwidth contains all the overhead coming from the IP-layer and the layers above, i.e. IP, UDP and RTCP.	IMS_SBI
rsBw	BitRateRm	O	0..1	Indicates the maximum required bandwidth in bits per second for RTCP sender reports within the session component as specified in IETF RFC 3556 [37]. The bandwidth contains all the overhead coming from the IP-layer and the layers above, i.e. IP, UDP and RTCP.	IMS_SBI
codecs	array(CodecData)	O	1..2	Indicates the codec data.	

Attribute name	Data type	P	Cardinality	Description	Applicability
sharingKeyDl	Uint32Rm	O	0..1	Identifies which media components share resources in the downlink direction. If resource sharing applies between media components across "Individual Application Session Context" resources for the same PDU session, the same value of the "sharingKeyDl" attribute shall be used. If resource sharing does not apply among media components across "Individual Application Session Context" resources for the same PDU session, a different value for the "sharingKeyDl" attribute shall be used. If resource sharing does no longer apply for this media component, the "sharingKeyDl" attribute shall be set to "null".	ResourceSharing
sharingKeyUl	Uint32Rm	O	0..1	Identifies which media components share resources in the uplink direction. If resource sharing applies between media components across "Individual Application Session Context" resources for the same PDU session, the same value of the "sharingKeyUl" attribute shall be used. If resource sharing does not apply among media components across "Individual Application Session Context" resources for the same PDU session, a different value for the "sharingKeyUl" attribute shall be used. If resource sharing does no longer apply for this media component, the "sharingKeyUl" attribute shall be set to "null".	ResourceSharing
tsnQos	TsnQosContainerRm	O	0..1	Transports QoS parameters for TSC traffic.	TimeSensitiveNetworking
tscaiInputUl	TscaiInputContainer	O	0..1	Transports TSCAI input parameters for TSC traffic at the ingress interface of the DS-TT/UE (uplink flow direction).	TimeSensitiveNetworking
tscaiInputDl	TscaiInputContainer	O	0..1	Transports TSCAI input parameters for TSC traffic at the ingress of the NW-TT (downlink flow direction).	TimeSensitiveNetworking
tscaiTimeDom	Uinteger	O	0..1	Indicates the (g)PTP domain that the (TSN)AF is located in.	TimeSensitiveCommunication
capBatAdaptation	boolean	O	0..1	Indicates the capability for AF to adjust the burst sending time, when it is supported and set to "true". (NOTE 2)	EnTSCAC
rTLatencyInd	boolean	O	0..1	Indicates the service data flow needs to meet the Round-Trip (RT) latency requirement of the service, when it is included and set to "true".	RTLatency
pdb	PacketDelBudgetRm	O	0..1	Indicates an upper bound for the time that a packet may be delayed between the UE and the PSA UPF.	RTLatency
rTLatencyIndCorreId	RttFlowReferenceRm	O	0..1	Identifies which Media Components contribute to the RT Latency requirement for two service data flows.	RTLatency
pduSetQosDl	PduSetQosParaRm	O	0..1	PDU Set QoS parameter(s) for the downlink direction.	PDUSetHandling
pduSetQosUl	PduSetQosParaRm	O	0..1	PDU Set QoS parameter(s) for the uplink direction.	PDUSetHandling

Attribute name	Data type	P	Cardinality	Description	Applicability
protoDescDl	ProtocolDescriptionRm	O	0..1	Downlink Protocol description for PDU Set identification, the detection of the end of data burst indication, the detection of the Data Burst Size marking indication, TTNB indication, and/or indication of whether MoQ or UDP-option is used to carry media related information.	PDUSetHandling PowerSaving TrafficCharChange OnPathN6MediaInfo
protoDescUl	ProtocolDescriptionRm	O	0..1	Uplink Protocol description for PDU Set identification in UE.	PDUSetHandling
periodUl	DurationMilliSecRm	O	0..1	Indicates the time period between the start of the two data bursts in units of milliseconds in Uplink direction.	PowerSaving
periodDl	DurationMilliSecRm	O	0..1	Indicates the time period between the start of the two data bursts in units of milliseconds in Downlink direction.	PowerSaving
l4sInd	UplinkDownlinkSupport	O	0..1	When provided, it represents an explicit indication of whether ECN marking for L4S support is supported for the UL, the DL or both, UL and DL. It may be present when the media component is initially provided. (NOTE 3)	L4S
datBurstSizeInd	boolean	O	0..1	Indicates the Data Burst Size marking for the DL service data flow is supported when it is included and set to "true".	TrafficCharChange
timetoNextBurstInd	boolean	O	0..1	Indicates the Time to Next Burst for the DL service data flow is supported, when it is included and set to "true".	TrafficCharChange
onPathN6SigInfo	OnPathN6SigInfo	O	0..1	Contains the on-path N6 signaling information for delivering media related information.	OnPathN6MediaInfo
expTranInd	boolean	O	0..1	Expedited Transfer Indication for the downlink traffic to enable expedited data transfer with reflective QoS for the Non-GBR service data flow. - "true": the expedited data transfer of larger payload for XR application is enabled for the flow. - "false": the expedited data transfer of larger payload for XR application is not enabled for the flow.	TrafficCharChange
ulBrRecInd	boolean	O	0..1	Indicates whether the RAN-controlled UL Bitrate Recommendation indication is supported or not. - Set to "true": the RAN-controlled UL Bitrate Recommendation Indication is supported. - Set to "false": the RAN-controlled UL Bitrate Recommendation Indication is not supported.	ExtQoSR19

NOTE 1: The attributes "altSerReqs" and "altSerReqsData" are mutually exclusive.
NOTE 2: The "burstArrivalTimeWnd" attribute, within the "tscaiInputUl" and/or "tscaiInputDl" attributes, and the "capBatAdaptation" attribute are mutually exclusive.
NOTE 3: Within a MediaComponentRm entry, the NF service consumer may include either the indication of L4S support within the "l4sInd" attribute or the request for congestion measurements within the "evSubsc" attribute included in one or more entries of the "medSubComps" attribute. A MediaComponent entry within the Individual Application Session Context resource shall not contain simultaneously both, the indication of L4S support and the subscription to congestion monitoring.

5.6.2.27 Type MediaSubComponentRm

This data type is defined in the same way as the "MediaSubComponent" data type, but:

- with the OpenAPI "nullable: true" property;

- the removable attributes "marBwDl", "marBwUl", defined with the removable data type "BitRateRm"; the removable attribute "tosTrCl", defined with the removable data type "TosTrafficClassRm"; the removable attribute "evSubsc" defined with the removable data type "EventsSubscReqDataRm"; and

- the removable attributes "ethfDescs" and "fDescs" and "addInfoFlowDescs" are defined as nullable in the OpenAPI.

Table 5.6.2.27-1: Definition of type MediaSubComponentRm

Attribute name	Data type	P	Cardinality	Description	Applicability
afSigProtocol	AfSigProtocol	O	0..1	Indicates the protocol used for signalling between the UE and the NF service consumer. It may be included only if the "flowUsage" attribute is set to the value "AF_SIGNALLING".	ProvAFsignalFlow
ethfDescs	array(EthFlowDescription)	O	1..2	Contains the flow description for the Uplink and/or Downlink Ethernet flows.	
fNum	integer	M	1	Identifies the ordinal number of the IP flow.	
fDescs	array(FlowDescription)	O	1..2	Contains the flow description for the Uplink and/or Downlink IP flows.	
addInfoFlowDescs	array(AddFlowDescriptionInfo)	O	1..2	Represents additional flow description information (flow label and IPsec SPI) per Uplink and/or Downlink IP flows represented in the "fDescs" attribute.	AddFlowDescriptionInformation
fStatus	FlowStatus	O	0..1	Indicates whether the status of the service data flows is enabled or disabled.	
flowUsage	FlowUsage	O	0..1	Flow usage of the flows (e.g. RTCP, AF signalling).	
marBwUl	BitRateRm	O	0..1	Maximum requested bandwidth for the Uplink.	
marBwDl	BitRateRm	O	0..1	Maximum requested bandwidth for the Downlink.	
tosTrCl	TosTrafficClassRm	O	0..1	Type of Service or Traffic Class.	
evSubsc	EventsSubscReqDataRm	O	0..1	Identifies the events the application subscribes to at update of a media component. (NOTE 1, NOTE 2)	EnQoSMon, L4S
mpxMediaUlInfos	array(MpxMediaInfo)	O	1..N	Contains the (S)RTP Multiplexed Media Identification Information for the Uplink IP flows.	MpxMedia
mpxMediaDlInfos	array(MpxMediaInfo)	O	1..N	Contains the (S)RTP Multiplexed Media Identification Information for the Downlink IP flows.	MpxMedia

NOTE 1: If attribute "evSubsc" is present, one or more of the following IEs within EventsSubscReqDataRm data type may be included: "events", "notifUri", "reqQosMonParams", "qosMon", "qosMonDatRate", "pdvReqMonParams", "pdvMon", "congestMon", "notifCorreId", "rttMon", "directNotifInd", "avrgWndw". In addition, when the attribute "events" is present, only the following Enumeration "AfEvent" may be included: "QOS_MONITORING", "PACK_DEL_VAR", "RT_DELAY_TWO_QOS_FLOWS", "L4S_SUPP".

NOTE 2: Within a MediaSubComponentRm entry, the NF service consumer may include the subscription for congestion measurements within the "evSubsc" attribute only if the "l4sInd" attribute is not included within the corresponding MediaComponent entry.

5.6.2.28 Type SpatialValidityRm

This data type is defined in the same way as the "SpatialValidity" data type, but with the OpenAPI "nullable: true" property.

5.6.2.29 Type ExtendedProblemDetails

This data type is the "ProblemDetails" data type defined in 3GPP TS 29.571 [12] but extended with the attribute "acceptableServInfo" of data type "AcceptableServiceInfo".

Table 5.6.2.29-1: Definition of type ExtendedProblemDetails

Attribute name	Data type	P	Cardinality	Description	Applicability
acceptableServInfo	AcceptableServiceInfo	O	0..1	Describes information related to the acceptable service information.	
NOTE: ExtendedProblemDetails data type also contains all the properties defined for ProblemDetails data type in 3GPP TS 29.571 [12].					

5.6.2.30 Type AcceptableServiceInfo

Table 5.6.2.30-1: Definition of type AcceptableServiceInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
accBwMedComps	map(MediaComponent)	O	1..N	Indicates the maximum bandwidth that shall be authorized by the PCF for each media component of the map. Each media component of the map shall only include the "medCompN" attribute and the "marBwDl" and/or "marBwUl" attributes indicating the maximum acceptable bandwidth, and/or, if the AcceptableQosDetails feature is supported, the "mirBwDl" and/or "mirBwUl" attributes indicating the acceptable guaranteed bandwidth, and/or the "tsnQos" attribute containing one or more of the acceptable maximum burst size, the acceptable packet error rate, the acceptable delay, and the acceptable priority. The key of the map is the media component number. (NOTE 1)	
marBwUl	BitRate	O	0..1	Maximum acceptable bandwidth for the Uplink for the AF session. (NOTE 1)	
marBwDl	BitRate	O	0..1	Maximum acceptable bandwidth for the Downlink for the AF session. (NOTE 1)	
mirBwUl	BitRate	O	0..1	Acceptable guaranteed bandwidth for the Uplink. (NOTE 1)	AcceptableQosDetails
mirBwDl	BitRate	O	0..1	Acceptable guaranteed bandwidth for the Downlink. (NOTE 1)	AcceptableQosDetails
tsnQos	TsnQosContainer	O	0..1	Contains acceptable values for one or more individual QoS parameters. (NOTE 1)	AcceptableQosDetails
addAccQosCombs	array(AcceptableServiceInfo)	O	1..N	Contains additional combinations of acceptable service information. (NOTE 2)	AcceptableQosDetails
NOTE 1: When the acceptable QoS parameters apply to one or more media components, only the "accBwMedComps" attribute shall be provided. When the acceptable QoS parameters apply to the whole AF session, only the "accBwMedComps" attributes shall not be present. NOTE 2: The attribute "addAccQosCombs" within each element of the array is not applicable.					

5.6.2.31 Type UeIdentityInfo

Table 5.6.2.31-1: Definition of type UeIdentityInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
gpsi	Gpsi	O	0..1	Represents the GPSI.	
pei	Pei	O	0..1	Represents the PEI.	
supi	Supi	O	0..1	Represents the SUPI.	
NOTE: At least one of the "gpsi", "supi" and "pei" attributes shall be present. More than one attribute may be present simultaneously.					

5.6.2.32 Type AccessNetChargingIdentifier

Table 5.6.2.32-1: Definition of type AccessNetChargingIdentifier

Attribute name	Data type	P	Cardinality	Description	Applicability
accNetChaIdValue	ChargingId	C	0..1	Contains a charging identifier. (NOTE 1)	
accNetChargIdString	string	C	0..1	A character string containing the charging identifier (see clause 5.1.9.1 of 3GPP TS 32.255 [35]). (NOTE 1)	AccNetChargId_String
flows	array(Flows)	O	1..N	Identifications of the flows transported within the corresponding QoS flow. If no flows are provided, the charging identifier applies for all flows within the AF session.	
NOTE 1: The "accNetChaIdValue" shall be used to encode the charging identifier when the charging identifier is within the Uint32 value range. The "accNetChargIdString" attribute shall be used to encode the charging identifier when the "AccNetChargId_String" feature is supported by the AF and the PCF and the charging identifier is out of the Uint32 range. NOTE 2: When the "AccNetChargId_String" feature is not supported and the value of the charging identifier is out of the ChargingId data type value range (Uint32) it is not possible to ensure a proper charging correlation using value of the "accNetChaIdValue" attribute.					

5.6.2.33 Type OutOfCreditInformation

Table 5.6.2.33-1: Definition of type OutOfCreditInformation

Attribute name	Data type	P	Cardinality	Description	Applicability
finUnitAct	FinalUnitAction	M	1	Indicates the termination action to be taken when the user's account cannot cover the service cost.	
flows	array(Flows)	O	1..N	Identifications of the flows without available credit. If no flows are provided, the termination action in "finUnitAct" attribute applies for all flows within the AF session.	

5.6.2.34 Type QosMonitoringInformation

Table 5.6.2.34-1: Definition of type QosMonitoringInformation

Attribute name	Data type	P	Cardinality	Description	Applicability
repThreshDl	integer	O	0..1	Indicates the threshold in units of milliseconds for DL packet delay. If the "EnQoSMon" feature is supported, and the "PACK_DEL_VAR" event is subscribed, it indicates the threshold for DL PDV measurement. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION". Minimum = 0.	
repThreshUl	integer	O	0..1	Indicates the threshold in units of milliseconds for UL packet delay. If the "EnQoSMon" feature is supported, and the "PACK_DEL_VAR" event is subscribed, it indicates the threshold for UL PDV measurement. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION". Minimum = 0.	
repThreshRp	integer	O	0..1	Indicates the threshold in units of milliseconds for round trip packet delay. If the "EnQoSMon" feature is supported and the "RT_DELAY_TWO_QOS_FLOWS" event is subscribed, it indicates the threshold for round trip delay of two QoS flows. If the "EnQoSMon" feature is supported, and the "PACK_DEL_VAR" event is subscribed, it indicates the threshold for round trip PDV measurement. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION". Minimum = 0.	
repThreshDatRateDl	BitRate	O	0..1	Indicates the threshold for DL data rate. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon
repThreshDatRateUl	BitRate	O	0..1	Indicates the threshold for UL data rate. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon
conThreshDl	Uinteger	O	0..1	Indicates the downlink threshold percentage of congestion reporting. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon
conThreshUl	Uinteger	O	0..1	Indicates the uplink threshold percentage of congestion reporting. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon

Attribute name	Data type	P	Cardinality	Description	Applicability
avlBitrateUlThrs	array(BitRate)	O	1..N	Indicates a list of thresholds for uplink available bitrate reporting. (NOTE)	EnQoSMon_v2
avlBitrateDlThrs	array(BitRate)	O	1..N	Indicates a list of thresholds for downlink available bitrate reporting. (NOTE)	EnQoSMon_v2
NOTE: The "avlBitrateUlThrs" and "avlBitrateDlThrs" attributes are only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION", and may contain the bitrate values, the value equal to the Requested Guaranteed Bitrate and the value equal to the Requested Maximum Bitrate (in case it is different).					

5.6.2.35 Type TsnQosContainer

Table 5.6.2.35-1: Definition of type TsnQosContainer

Attribute name	Data type	P	Cardinality	Description	Applicability
maxTscBurstSize	ExtMaxDataBurstVol	O	0..1	Maximum burst size of the TSC traffic in units of Bytes. Minimum = 4096, Maximum = 2000000. (NOTE 2)	
maxTscBurstSizeSmall	MaxDataBurstVol	O	0..1	Maximum burst size of the TSC traffic in units of Bytes. Minimum = 1, Maximum = 4095. (NOTE 2)	SmallBurstSizes
tscPackDelay	PacketDelBudget	O	0..1	Delay of the TSC traffic.	
maxPer	PacketErrRate	O	0..1	Packet error rate of the TSC traffic	ExtQoS
tscPrioLevel	TscPriorityLevel	O	0..1	TSC traffic priority in relation to other TSC and non-TSC traffic.	
NOTE 1: At least one of the attributes shall be present in an instance of the TsnQosContainer. NOTE 2: These attributes are mutually exclusive.					

5.6.2.36 Type PcscfRestorationRequestData

Table 5.6.2.36-1: Definition of type PcscfRestorationRequestData

Attribute name	Data type	P	Cardinality	Description	Applicability
dnn	Dnn	O	0..1	Data Network Name, a full DNN with both the Network Identifier and Operator Identifier, or a DNN with the Network Identifier only. (NOTE 2)	
ipDomain	string	O	0..1	Indicates the IPv4 address domain information that assists session binding.	
sliceInfo	Snssai	O	0..1	Identifies the S-NSSAI.	
supi	Supi	O	0..1	Subscription Permanent Identifier.	
ueIpv4	Ipv4Addr	C	0..1	The IPv4 address of the served UE. (NOTE 1)	
ueIpv6	Ipv6Addr	C	0..1	The IPv6 address of the served UE. (NOTE 1)	
NOTE 1: When present, only one of the served UE addressing parameters (the IPv4 address or the IPv6 address) shall always be included. NOTE 2: The PCF uses the DNN as received from the NF service consumer without applying any transformation (e.g. during session binding). To successfully perform DNN matching, in a specific deployment a DNN shall always be encoded either with the full DNN (e.g., because there are multiple Operator Identifiers for a Network Identifier) or the DNN Network Identifier only. The NF service consumer may include the DNN Operator Identifier based on local configuration.					

5.6.2.37 Type QosMonitoringReport

Table 5.6.2.37-1: Definition of type QosMonitoringReport

Attribute name	Data type	P	Cardinality	Description	Applicability
flows	array(Flows)	C	1..N	Identification of the flows. It shall be included if "MediaComponentVersioning" feature is supported. When "MediaComponentVersioning" feature is not supported, if no flows are provided, the packet delay applies for all flows within the AF session.	
ulDelays	array(integer)	O	1..N	Uplink packet delay in units of milliseconds. (NOTE)	
dlDelays	array(integer)	O	1..N	Downlink packet delay in units of milliseconds. (NOTE)	
rtDelays	array(integer)	O	1..N	Round trip delay in units of milliseconds. (NOTE) If the "EnQoSMon" feature is supported and the "RT_DELAY_TWO_QOS_FLOWS" event is subscribed, it indicates the round trip delay of multiple QoS flows.	
pdmf	boolean	O	0..1	Packet delay measurement failure indicator. When set to true, it indicates that a packet delay failure has occurred. Default value is false if omitted.	PacketDelayFailureReport EnQoSMon
ulDataRate	BitRate	O	0..1	Average data throughput in uplink direction.	EnQoSMon
dlDataRate	BitRate	O	0..1	Average data throughput in downlink direction].	EnQoSMon
ulConInfo	array(integer)	O	1..N	Percentage of congestion information in uplink direction. Minimum = 0, Maximum = 100. (NOTE)	EnQoSMon
dlConInfo	array(integer)	O	1..N	Percentage of congestion information in downlink direction. Minimum = 0, Maximum = 100. (NOTE)	EnQoSMon
NOTE: In this release of the specification one element may be included in the array, as specified in clause 4.2.5.14.					

5.6.2.38 Type TsnQosContainerRm

This data type is defined in the same way as the "TsnQosContainer" data type, but with the OpenAPI "nullable: true" property for each of the attributes as well as for the entire data type.

5.6.2.39 Type TscaiInputContainer

Table 5.6.2.39-1: Definition of type TscaiInputContainer

Attribute name	Data type	P	Cardinality	Description	Applicability
periodicity	Uinteger	O	0..1	Unsigned 64-bit integer identifying a period of time in units of microseconds, i.e. 0 to (2^64)-1. Minimum = 0. Maximum = 18446744073709551615. Identifications of the time period between the start of two bursts in reference to the external GM.	
burstArrivalTime	DateTime	O	0..1	Indicates the arrival time of the data burst in reference to the external GM.	
surTimeInNumMsg	Uinteger	O	0..1	Unsigned 32-bit integer indicates the survival time in terms of maximum number of messages an application can survive without any burst. A message is equivalent to a burst, i.e. 0 to (2^32)-1. Minimum = 0. Maximum = 4294967295.	TimeSensitiveCommunication
surTimeInTime	Uinteger	O	0..1	Unsigned 64-bit integer indicates the survival time in terms of time units of microseconds an application can survive without any burst, i.e. 0 to (2^64)-1. Minimum = 0. Maximum = 18446744073709551615.	TimeSensitiveCommunication
periodicityRange	PeriodicityRange	O	0..1	Contains the acceptable time period range between the start of two bursts in reference to the external GM or the acceptable periodicity value(s). (NOTE 1)	EnTSCAC
burstArrivalTimeWnd	TimeWindow	O	0..1	Contains the acceptable earliest and latest arrival time of the data burst in reference to the external GM. The start time contains the earliest arrival time, and the stop time contains the latest arrival time. (NOTE 2)	EnTSCAC

NOTE 1: The attribute "periodicityRange" may be only present together with the "periodicity" attribute when the "burstArrivalTime" attribute and "burstArrivalTimeWnd" attribute are present.
NOTE 2: The "burstArrivalTimeWnd" attribute may only present when the "burstArrivalTime" is present.

5.6.2.40 Type PduSessionTsnBridge

Table 5.6.2.40-1: Definition of type PduSessionTsnBridge

Attribute name	Data type	P	Cardinality	Description	Applicability
tsnBridgeInfo	TsnBridgeInfo	M	1	Reports the TSC user plane node information.	
tsnBridgeManCont	BridgeManagementContainer	O	0..1	Transports TSC user plane node management information.	
tsnPortManContDstt	PortManagementContainer	O	0..1	Transports port management information for the DS-TT port.	
tsnPortManContNwtts	array(PortManagementContainer)	O	1..N	Transports port management information for one or more NW-TT ports.	
ueIpv4Addr	Ipv4Addr	O	0..1	It represents the identifier of the PDU session related to the reported UP node information, and contains the UE IPv4 address. It might be present for PDU sessions of IP type. (NOTE 1)	TimeSensitiveCommunication
dnn	Dnn	O	0..1	The DNN of the PDU session, a full DNN with both the Network Identifier and Operator Identifier, or a DNN with the Network Identifier only. (NOTE 2)	TimeSensitiveCommunication
snssai	Snssai	O	0..1	Identifies the S-NSSAI.	TimeSensitiveCommunication
ipDomain	string	O	0..1	IPv4 address domain identifier.	TimeSensitiveCommunication
ueIpv6AddrPrefix	Ipv6Prefix	O	0..1	It represents the identifier of the PDU session related to the reported UP node information, and contains the UE IPv6 address prefix. It might be present for PDU sessions of IP type. (NOTE 1)	TimeSensitiveCommunication
NOTE 1: For PDU sessions of IP type, either the ueIpv4Addr or the ueIpv6AddrPrefix shall be present in this release of the specification. NOTE 2: The PCF includes the DNN of the PDU session as received from the SMF without applying any transformation. To successfully perform DNN matching, in a specific deployment a DNN shall always be encoded either with the full DNN (e.g., because there are multiple Operator Identifiers for a Network Identifier) or the DNN Network Identifier only. The NF service consumer may include the DNN Operator Identifier based on local configuration.					

5.6.2.41 Type QosMonitoringInformationRm

This data type is defined in the same way as the "QosMonitoringInformation" data type, but:

- with the OpenAPI "nullable: true" property;
- the removable attributes "repThreshDatRateDl", "repThreshDatRateUl", "avlBitrateUlThrs" and "avlBitrateDlThrs" are defined with the removable data type "BitRateRm", and the removable attributes "conThreshDl" and "conThreshUl" are defined with the removable data type "UintegerRm"; and
- the attributes "repThreshDl", "repThreshUl" and "repThreshRp" are defined with the OpenAPI "nullable: true" property.

Table 5.6.2.41-1: Definition of type QosMonitoringInformationRm

Attribute name	Data type	P	Cardinality	Description	Applicability
repThreshDl	integer	O	0..1	Indicates the threshold in units of milliseconds for DL packet delay. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION". Minimum = 0. (NOTE 1)	
repThreshUl	integer	O	0..1	Indicates the threshold in units of milliseconds for UL packet delay. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION". Minimum = 0. (NOTE 1)	
repThreshRp	integer	O	0..1	Indicates the threshold in units of milliseconds for round trip packet delay. If the "EnQoSMon" feature is supported and the "RT_DELAY_TWO_QOS_FLOWS" event is subscribed, it indicates the round trip delay of multiple QoS flows. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION". Minimum = 0. (NOTE 1)	
repThreshDatRateDl	BitRateRm	O	0..1	Indicates the threshold for DL data rate. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon
repThreshDatRateUl	BitRateRm	O	0..1	Indicates the threshold for UL data rate. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon
conThreshDl	UintegerRm	O	0..1	Indicates the downlink threshold percentage of congestion reporting. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon
conThreshUl	UintegerRm	O	0..1	Indicates the uplink threshold percentage of congestion reporting. Only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION".	EnQoSMon
avlBitrateUlThrs	array(BitRateRm)	O	1..N	Indicates a list of thresholds for uplink available bitrate reporting. (NOTE 2)	EnQoSMon_v2
avlBitrateDlThrs	array(BitRateRm)	O	1..N	Indicates a list of thresholds for downlink available bitrate reporting. (NOTE 2)	EnQoSMon_v2

NOTE 1: If the "EnQoSMon" feature is supported, "repThreshDl" attribute, the "repThreshUl" attribute and/or the "repThreshRp" attribute may be set to value null, otherwise they shall not be set to value null.

NOTE 2: The "avlBitrateUlThrs" and "avlBitrateDlThrs" attributes are only applicable when the "notifMethod" attribute is not supplied or the "notifMethod" is supplied and set to "EVENT_DETECTION", and may contain the bitrate values, the value equal to the Requested Guaranteed Bitrate and the value equal to the Requested Maximum Bitrate (in case it is different).

5.6.2.42 Type EventsSubscPutData

Table 5.6.2.42-1: Definition of type EventsSubscPutData

Data Type	P	Cardinality	Description	Applicability
EventsSubscReqData	C	0..1	Identifies the events the application subscribes to and represents the Events Subscription sub-resource data. It shall be present in the response to PUT requests as specified in table 5.3.4.3.1-3.	
EventsNotification	C	0..1	Describes the notification about the events already met at the time of subscription. It shall be present if available.	
NOTE: EventsSubscPutData data type is represented as a non-exclusive list of two data types: EventsSubscReqData and EventsNotification.				

5.6.2.43 Type AppSessionContextUpdateDataPatch

Table 5.6.2.43-1: Definition of type AppSessionContextUpdateDataPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
ascReqData	AppSessionContext UpdateData	O	0..1	Describes the requested update to the services requirements of an Individual Application Session Context.	

5.6.2.44 Type AppDetectionReport

Table 5.6.2.44-1: Definition of type AppDetectionReport

Attribute name	Data type	P	Cardinality	Description	Applicability
adNotifType	AppDetectionNotifTyp e	M	1	Indicates whether the report is about the detection of application start or application stop.	
afAppId	AfAppId	M	1	It indicates the application identifier of the detected traffic.	

5.6.2.45 Type PduSessionEventNotification

Table 5.6.2.45-1: Definition of PduSessionEventNotification

Attribute name	Data type	P	Cardinality	Description	Applicability
evNotif	AfEventNotification	M	1	Indicates the reported event (e.g. "PDU_SESSION_STATUS").	
supi	Supi	C	0..1	Contains the SUPI of the PDU session. It shall be present for the "PDU_SESSION_STATUS" event.	
ueIpv4	Ipv4Addr	C	0..1	The IPv4 address of the served UE for the reported PDU session. (NOTE 1)	
ueIpv6	Ipv6Addr	C	0..1	The IPv6 address of the served UE for the reported PDU session. (NOTE 1)	
ueMac	MacAddr48	C	0..1	The MAC address of the served UE for the reported PDU session. (NOTE 1)	
status	PduSessionStatus	C	0..1	It shall be present for the "PDU_SESSION_STATUS" event. Indicates whether the PDU session is "ESTABLISHED" or "TERMINATED".	
pcfInfo	PcfAddressingInfo	C	0..1	Contains PCF addressing information. It may be present for the "PDU_SESSION_STATUS" event. It shall be included when the PDU session operation is "ESTABLISHED".	
dnn	Dnn	C	0..1	Contains the DNN of the PDU session. It shall be included when the event is "PDU_SESSION_STATUS" and the PDU session operation is "ESTABLISHED". (NOTE 2)	
snssai	Snssai	C	0..1	Contains the S-NSSAI of the PDU session. It shall be included when the event is "PDU_SESSION_STATUS" and the PDU session operation is "ESTABLISHED".	
gpsi	Gpsi	O	0..1	Contains the GPSI of the PDU session. It shall be included, if available, when the event is "PDU_SESSION_STATUS" and the PDU session operation is "ESTABLISHED".	

NOTE 1: Only one of the served UE addressing parameters (the IPv4 address or the IPv6 address or MAC address) shall always be included.

NOTE 2: The PCF includes the DNN of the PDU session as received from the SMF without applying any transformation. To successfully perform DNN matching, in a specific deployment a DNN shall always be encoded either with the full DNN (e.g., because there are multiple Operator Identifiers for a Network Identifier) or the DNN Network Identifier only. The NF service consumer may include the DNN Operator Identifier based on local configuration.

5.6.2.46 Type PcfAddressingInfo

Table 5.6.2.46-1: Definition of type PcfAddressingInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
pcfFqdn	Fqdn	C	0..1	FQDN of the PCF hosting the Npcf_PolicyAuthorization service. It shall be provided if available. (NOTE)	
pcfIpEndPoints	array(IpEndPoint)	C	1..N	IP end points of the PCF hosting the Npcf_PolicyAuthorization service. It shall be provided if available. (NOTE)	
bindingInfo	string	O	0..1	This IE shall be present, if available. When present, this IE shall contain the Binding indications of the PCF indicated by the pcfFqdn IE and/or pcfIpEndPoints IE, and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of 3GPP TS 29.500 [5], without the header name and including only binding indications for "nf-instance" or "nf-set" binding levels.	
NOTE: The pcfFqdn and/or the pcfIpEndPoints shall always be included.					

5.6.2.47 Type AlternativeServiceRequirementsData

Table 5.6.2.47-1: Definition of type AlternativeServiceRequirementsData

Attribute name	Data type	P	Cardinality	Description	Applicability
altQosParamSetRef	string	M	1	It contains a reference to the alternative individual QoS related parameter(s) included in this set. The value of this attribute shall only be used in QoS notification control information (see "altSerReq" attribute in Table 5.6.2.15) to indicate the alternative individual QoS related parameters that can be guaranteed (if any).	
gbrUl	BitRate	O	0..1	Indicates the guaranteed bandwidth in uplink. (NOTE 1)	
gbrDl	BitRate	O	0..1	Indicates the guaranteed bandwidth in downlink. (NOTE 1)	
pdb	PacketDelBudget	O	0..1	Unsigned integer. It indicates the Packet Delay Budget expressed in milliseconds. (NOTE 1)	
per	PacketErrRate	O	0..1	It indicates the Packet Error Rate. (NOTE 1)	ExtQoS
averWindow	AverWindow	O	0..1	Indicates the Averaging Window.	ExtQoSR19
maxDataBurstVol	MaxDataBurstVol	O	0..1	Indicates the Maximum Data Burst Volume.	ExtQoSR19
pduSetQosDl	PduSetQosParaRm	O	0..1	Contains the PDU Set QoS parameters which are used to support PDU Set based QoS handling in the downlink direction. (NOTE 2)	ExtQoSR19
pduSetQosUl	PduSetQosParaRm	O	0..1	Contains the PDU Set QoS parameters which are used to support PDU Set based QoS handling in the uplink direction. (NOTE 2)	ExtQoSR19

NOTE 1: If the "ExtQoSR19" feature is not supported, at least one of the "pdb" attribute, the "per" attribute (if the ExtQoS feature is supported), and the combination of the "gbrUl" and "gbrDl" attributes, shall be provided.
NOTE 2: Only the "pduSetDelayBudget" and "pduSetErrRate" attributes within the "pduSetQosDl" and "pduSetQosUl" attributes are applicable and shall be present if the "pduSetDelayBudget" and "pduSetErrRate" attributes in "pduSetQosDl" and/or "pduSetQosUl" attribute(s) within the "medComponents" attribute are provided.

5.6.2.48 Type PeriodicityRange

Table 5.6.2.48-1: Definition of type PeriodicityRange

Attribute name	Data type	P	Cardinality	Description	Applicability
lowerBound	Uinteger	C	0..1	Unsigned 64-bit integer identifying a period of time in units of microseconds, i.e. 0 to (2^64)-1. Minimum = 0. Maximum = 18446744073709551615. Indicates the acceptable lower bound of the periodicity of the start two bursts in reference to the external GM. (NOTE 1)	
upperBound	Uinteger	C	0..1	Unsigned 64-bit integer identifying a period of time in units of microseconds, i.e. 0 to (2^64)-1. Minimum = 0. Maximum = 18446744073709551615. Indicates the acceptable upper bound of the periodicity of the start two bursts in reference to the external GM. (NOTE 1)	
periodicVals	array(Uinteger)	C	1..N	Contains the acceptable periodicity value(s). (NOTE 1) (NOTE 2)	
addPeriodicVals	array(Uinteger)	C	1..N	Contains the acceptable periodicity value(s) in units of microseconds. (NOTE 2)	
NOTE 1: Either the "periodicVals" attribute, the "addPeriodicVals" attribute, or both the "lowerBound" attribute and the "upperBound" attribute shall be present. NOTE 2: The "periodicVals" and "addPeriodicVals" attributes are mutually exclusive.					

5.6.2.49 Type AfSfcRequirement

Table 5.6.2.49-1: Definition of type AfSfcRequirement

Attribute name	Data type	P	Cardinality	Description	Applicability
sfcIdDl	string	C	0..1	Indicates the identifier of the pre-configured service function chain for downlink traffic. (NOTE)	
sfcIdUl	string	C	0..1	Indicates the identifier of the pre-configured service function chain for uplink traffic. (NOTE)	
spVal	SpatialValidityRm	O	0..1	Indicates the spatial validity where the traffic routing requirements apply. The absence of this attribute indicates no spatial restrictions.	
metadata	Metadata	O	0..1	Contains the opaque information for the service functions in the N6-LAN that is provided by AF and transparently sent to UPF.	
NOTE: Either one of "sfcIdDl", "sfcIdUl" attribute or both shall be present when initially provided.					

5.6.2.50 Type BatOffsetInfo

Table 5.6.2.50-1: Definition of type BatOffsetInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
ranBatOffsetNotif	integer	M	1	Indicates the BAT offset of the arrival time of the data burst in units of milliseconds.	
adjPeriod	Uinteger	O	0..1	Contains the adjusted periodicity in units of milliseconds of the data bursts in reference to the external GM.	
flows	array(Flows)	C	1..N	Identification of the flows. If no flows are provided, the BAT offset applies for all flows of the AF session.	

5.6.2.51 Void

5.6.2.52 Void

5.6.2.53 Type PdvMonitoringReport

Table 5.6.2.53-1: Definition of type PdvMonitoringReport

Attribute name	Data type	P	Cardinality	Description	Applicability
flows	array(Flows)	O	1..N	Identification of the flows.	
ulPdv	integer	O	0..1	Uplink packet delay variations in units of milliseconds.	
dlPdv	integer	O	0..1	Downlink packet delay variations in units of milliseconds.	
rtPdv	integer	O	0..1	Round trip packet delay variations in units of milliseconds.	

5.6.2.54 Void

5.6.2.55 Type AddFlowDescriptionInfo

Table 5.6.2.55-1: Definition of AddFlowDescriptionInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
spi	string	O	0..1	4 octet string, representing the security parameter index of the IPSec packet in hexadecimal representation. Each character in the string shall take a value of "0" to "9" or "A" to "F" and shall represent 4 bits. One example is that of a TFT packet filter as defined in 3GPP TS 24.008 [36].	
flowLabel	string	O	0..1	3-octet string, representing the IPv6 flow label header field in hexadecimal representation. Each character in the string shall take a value of "0" to "9" or "A" to "F" and shall represent 4 bits. One example is that of a TFT packet filter as defined in 3GPP TS 24.008 [36].	
flowDir	FlowDirection	O	0..1	Contains the packet filter direction to which the IPsec SPI and/or flow label applies. Only the "DOWNLINK" or "UPLINK" value is applicable. Default value is "DOWNLINK".	

5.6.2.56 Type L4sSupport

Table 5.6.2.56-1: Definition of type L4sSupport

Attribute name	Data type	P	Cardinality	Description	Applicability
notifType	L4sNotifType	M	1	Indicates whether the ECN marking for L4S for the indicated SDFs are "NOT_AVAILABLE" or "AVAILABLE" again.	
flows	array(Flows)	O	1..N	Identification of the flows. If no flows are provided, the notification in the "notifType" applies for all flows within the AF session.	

5.6.2.57 Type DirectNotificationReport

Table 5.6.2.57-1: Definition of type DirectNotificationReport

Attribute name	Data type	P	Cardinality	Description	Applicability
qosMonParamType	QosMonitoringParamType	M	1	Indicates the QoS monitoring parameter that cannot be directly notified.	
flows	array(Flows)	O	1..N	Identification of the flows. If no flows are provided, the direct notification cannot be provided for all flows with QoS monitoring subscription within the AF session.	

5.6.2.58 Type RttFlowReference

Table 5.6.2.58-1: Definition of type RttFlowReference

Attribute name	Data type	P	Cardinality	Description	Applicability
flowDir	FlowDirection	O	0..1	It contains the flow direction to which the round-trip time measurement/the RT latency requirement over two service data flow applies. Only the "DOWNLINK" or "UPLINK" value is applicable. If omitted, the request applies to the flow direction of the flow described in the "fDescs" attribute of the MediaSubComponent entry.	
sharedKey	Uint32	M	1	Identifies the media subcomponents or media components of the Individual Application Session Context resource that contribute to the round trip time measurements or RT latency requirement.	

5.6.2.59 Type RttFlowReferenceRm

This data type is defined in the same way as the "RttFlowReference" data type, but with the OpenAPI "nullable:true" property.

5.6.2.60 Type CapabilityReport

Table 5.6.2.60-1: Definition of type CapabilityReport

Attribute name	Data type	P	Cardinality	Description	Applicability
capReport	NotifCap	M	1	Indicates whether the QoS Monitoring Capability for the indicated type within the "capType" attribute is not supported or supported again.	
capType	NotifCapType	M	1	Indicates the QoS Monitoring Capability type.	

5.6.2.61 Type MpxMediaInfo

Table 5.6.2.61-1: Definition of type MpxMediaInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
ssrcId	Uint32	C	0..1	Contains the synchronization source as defined in IETF RFC 3550 [58]. (NOTE 1)	
payloadType	integer	C	0..1	Contains the Payload Type value in the RTP header of RTP packets as defined in IETF RFC 3550 [58]. When this attribute is present, the multiplex media information instance may be applicable only to (S)RTP traffic. Minimum = 0. Maximum = 127. (NOTE 1), (NOTE 2), (NOTE 3)	
identificationTag	string	C	0..1	Contains the media identification in terms of identification-tag as defined in IETF RFC 9143 [59]. (NOTE 1)	
rtpSdesHdrExtId	Uinteger	C	0..1	Contains the RTP SDES Header Extension ID for MID as defined in clause 9.1 of IETF RFC 9143 [59] and IETF RFC 8285 [60]. This attribute may be present only when identification-tag attribute is present. For SDP example of RTP SDES header extension ID: a=extmap:**1** urn:ietf:params:rtp-hdrext:sdes:mid extmap attribute's value "1" has to be transmitted as RTP SDES Header Extension ID for MID. (NOTE 1)	
rtcpPacketType	integer	C	0..1	Contains the Packet Type value of RTCP packets. When this attribute is present, the multiplex media information instance may be applicable only to (S)RTCP traffic. Minimum = 0. Maximum = 255. (NOTE 1), (NOTE 2), (NOTE 3)	

NOTE 1: At least one of the "ssrcId", "payloadType", "rtcpPacketType", "identificationTag" and the combination of "identificationTag" and "rtpSdesHdrExtId" attributes shall be present.
NOTE 2: These attributes are mutually exclusive.
NOTE 3: End System shall ensure the RTP type and RTCP type are distinguishable as specified in IETF RFC 5761 [31].

5.6.2.62 Type AfHeaderHandlingControlInfo

Table 5.6.2.62-1: Definition of type AfHeaderHandlingControlInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
hDetectionReference	string	M	1	Indicates header detection reference to a configuration pre-configured in a UPF for performing the header handling actions.	
hDetectionSuppInfo	string	O	0..1	Contains dynamic information not pre-configured in a UPF and is transparently passed to the UPF to be applied to detect the headers. Its format is unspecified and operator specific.	
notifUri	UriRm	O	0..1	Contains a URI indicating the notification target address as a header handling reporting endpoint.	
notifId	string	O	0..1	Contains a Notification Correlation ID indicating the notification destination as a header handling reporting endpoint.	
spVal	SpatialValidityRm	O	0..1	Indicates where the traffic routing requirements apply. The absence of this attribute indicates no spatial restrictions.	
tempVals	array(TemporalValidity)	O	1..N	Indicates the time interval(s) during which the NF service consumer request is to be applied.	
hHndlgUl	array(HeaderHandlingActionRequest)	O	1..N	Represents the list of header handling action request parameters in the uplink direction. If multiple elements are provided within the "hHndlgUl" attribute, then the hInfo value within each "HeaderHandlingActionRequest" data type shall be unique, if available.	
hHndlgDl	array(HeaderHandlingActionRequest)	O	1..N	Represents the list of header handling action request parameters in the downlink direction. If multiple elements are provided within the "hHndlgDl" attribute, then the hInfo value within each "HeaderHandlingActionRequest" data type shall be unique, if available.	

5.6.2.63 Type: HeaderHandlingActionRequest

Table 5.6.2.63-1: Definition of type HeaderHandlingActionRequest

Attribute name	Data type	P	Cardinality	Description	Applicability
hHndlgCtrlRef	string	O	0..1	Indicates reference to a header handling action related information that is pre-configured in the UPF, and the corresponding header handling behaviour has to be agreed as part of the SLA. An operator may choose to set this reference to the same value as the attribute "hDetectionReference" in the "HeaderHandlingControInfo" data type.	
hHndlgAction	array(HeaderHandlingAction)	O	1..N	Contains the action to be performed on a specific header field that is identified by the attribute "hInfo" and the attribute "hVal", if it is provided. The execution order is relevant to build a request that has the intended impact on the traffic. For example, a header that is removed cannot be replaced. If multiple header handing actions are provided then it shall follow the order: DETECT, REMOVE, REPLACE, and INSERT.	
hInfo	string	O	0..1	A reference to a configuration pre-configured in the UPF that defines how to identify or build a specific header field for which to perform the header handling action.	
hVal	string	O	0..1	Contains a string providing the value of the specific header field relevant for the header handing action. This attribute shall be provided if "hHndlgAction" attribute is provided and one of the values is set to REPLACE.	
hHndlgCond	HeaderHandlingCond	O	0..1	Contains the condition to apply the header handling action.	
hHndlgRep	HeaderHandlingReporting	O	0..1	Indicates that reporting is requested for the performed Header Handling Actions.	
NOTE: If "hHndlgCtrlRef" attribute is present, any other attributes included within the "HeaderHandlingActionRequest" data type shall overwrite the related information pre-configured in the UPF.					

5.6.2.64 Type OnPathN6SigInfo

Table 5.6.2.64-1: Definition of type OnPathN6SigInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
onPathN6Method	OnPathN6Method	M	1	indicates the supported on-path N6 signaling method.	
asProxyAddr	IpAddr	C	0..1	Represents the AS proxy address. The attribute shall be present when the attribute "onPathN6Method" is set to "CONNECT_UDP".	

5.6.2.65 Type: HeaderHandlingReporting

Table 5.6.2.65-1: Definition of type HeaderHandlingReporting

Attribute name	Data type	P	Cardinality	Description	Applicability
notifFlag	boolean	O	0..1	Indicates whether reporting is requested for the performed Header Handling Actions. - Set to "true": the reporting is requested for the performed Header Handling. - Set to "false": the reporting is not requested for the performed Header Handling. - Default value is "false" if omitted and has not been supplied previously.	
repSuggInfo	ReportingSuggestionInformation	O	0..1	Indicates whether the reporting information is delay tolerant.	
oneTimeInd	boolean	O	0..1	Indicates whether only the reporting of a first occurrence of the action per packet flow is needed. - Set to "true": the reporting applies only to the first occurrence. - Set to "false": the reporting applies to all occurrences. - Default value is "false" if omitted and has not been supplied previously. It may be present if the "notifFlag" attribute is set to "true".	

5.6.2.66 Type RateLimitRepo

Table 5.6.2.33-1: Definition of type RateLimitRepo

Attribute name	Data type	P	Cardinality	Description	Applicability
rateLimitRepoUl	array(BitRate)	O	1..N	Indicates the maximum uplink data rate authorized for the non-GBR service data flow(s) as indicated in the attribute "flows". If no flows are provided, the maximum data rate applies for all the flows.	
rateLimitRepoDl	array(BitRate)	O	1..N	Indicates report the maximum downlink data rate authorized for the non-GBR service data flow(s) as indicated in the attribute "flows". If no flows are provided, the maximum data rate applies for all the flows.	
flows	array(Flows)	O	1..N	Identifications of the non-GBR service data flows.	

5.6.3 Simple data types and enumerations

5.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

5.6.3.2 Simple data types

The simple data types defined in table 5.6.3.2-1 shall be supported.

Table 5.6.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
AfAppId	string	Contains an AF application identifier.	
AspId	string	Contains an identity of an application service provider.	SponsoredConnectivity
CodecData	string	Contains codec related information. Refer to clause 5.3.7 of 3GPP TS 29.214 [20] for encoding.	
ContentVersion	integer	Unsigned 64-bit integer that indicates the version of some content, as e.g. the content of a media component. The content version shall be unique for the content and for the lifetime of that content. (NOTE)	MediaComponentVersioning
DurationMilliSec	integer	Indicates the time interval in units of milliseconds.	PowerSaving
DurationMilliSecRm	integer	This data type is defined in the same way as the "DurationMilliSec" data type, but with the OpenAPI "nullable: true" property.	PowerSaving
FlowDescription	string	Defines a packet filter for an IP flow. It contains an IPFilterRule according to clause 4.3 of IETF RFC 6733 [52]. Refer to clause 5.3.8 of 3GPP TS 29.214 [20] for encoding.	
MaxDataBurstVol	integer	Unsigned integer that indicates the Maximum Data Burst Volume value, expressed in Bytes. Minimum = 1. Maximum = 2000000.	ExtQoSR19
MultiModalId	string	Contains a multi-modal service identifier.	MultiMedia
SponId	string	Contains an identity of a sponsor.	SponsoredConnectivity
ServiceUrn	string	Indicates that an AF session is used for Emergency traffic. It contains values of the service URN and it may include subservices, as defined in IETF RFC 5031 [34] or registered at IANA. The string "urn:service:" in the beginning of the URN shall be omitted and all subsequent text shall be included. Examples of valid values of the ServiceUrn data structure are "sos", "sos.fire", "sos.police" and "sos.ambulance".	IMS_SBI
TosTrafficClass	string	2-octet string, where each octet is encoded in hexadecimal representation. The first octet contains the IPv4 Type-of-Service or the IPv6 Traffic-Class field and the second octet contains the ToS/Traffic Class mask field. Each character in the string shall take a value of "0" to "9" or "A" to "F" and shall represent 4 bits. One example is that of a TFT packet filter as defined in 3GPP TS 24.008 [36].	
TosTrafficClassRm	string	This data type is defined in the same way as the "TosTrafficClass" data type, but with the OpenAPI "nullable: true" property.	
TscPriorityLevel	integer	Indicates the TSC traffic Priority Level, within the range 1 to 8. Values are ordered in decreasing order of priority, i.e. with 1 as the highest priority and 8 as the lowest priority.	TimeSensitiveNetworking
TscPriorityLevelRm	integer	This data type is defined in the same way as the "TscPriorityLevel" data type, but with the OpenAPI "nullable: true" property.	TimeSensitiveNetworking
NOTE: The method of assigning content versions is implementation specific.			

5.6.3.3 Enumeration: MediaType

The enumeration "MediaType" represents the media type of a media component.

Table 5.6.3.3-1: Enumeration MediaType

Enumeration value	Description	Applicability
AUDIO	The type of media is audio.	
VIDEO	The type of media is video.	
DATA	The type of media is data.	
APPLICATION	The type of media is application data.	
CONTROL	The type of media is control.	
TEXT	The type of media is text.	
MESSAGE	The type of media is message	
OTHER	Other type of media.	

5.6.3.4 Enumeration: ReservPriority

The enumeration "ReservPriority" represents the reservation priority. The lowest priority shall be indicated with the "PRIO_1" value, the next after the lowest with the "PRIO_2" value, and so on up to the highest priority which shall be indicated with "PRIO_16".

Table 5.6.3.4-1: Enumeration ReservPriority

Enumeration value	Description	Applicability
PRIO_1		
PRIO_2		
PRIO_3		
PRIO_4		
PRIO_5		
PRIO_6		
PRIO_7		
PRIO_8		
PRIO_9		
PRIO_10		
PRIO_11		
PRIO_12		
PRIO_13		
PRIO_14		
PRIO_15		
PRIO_16		

5.6.3.5 Enumeration: ServAuthInfo

The enumeration "ServAuthInfo" represents the result of the Npcf_PolicyAuthorization service request from the NF service consumer.

Table 5.6.3.5-1: Enumeration ServAuthInfo

Enumeration value	Description	Applicability
TP_NOT_KNOWN	Indicates the transfer policy is not known.	
TP_EXPIRED	Indicates the transfer policy has expired.	
TP_NOT_YET_OCURRED	Indicates the time window of the transfer policy has not yet occurred.	
ROUT_REQ_NOT_AUTHORIZED	Indicates the AF influence on traffic routing request is not allowed for the concerned PDU session.	RoutingReqOutcome
DIRECT_NOTIF_NOT_POSSIBLE	Indicates that direct notification for QoS monitoring is not applied.	EnQoSMon
MPX_MEDIA_NOT_SUPPORTED_IN_UE	Indicates the NF service consumer requested differentiated QoS for multiplexed media flows is not supported in the UE.	MpxMedia

5.6.3.6 Enumeration: SponsoringStatus

The enumeration "SponsoringStatus" represents whether the sponsored data connectivity is enabled or disabled/not-enabled.

Table 5.6.3.6-1: Enumeration SponsoringStatus

Enumeration value	Description	Applicability
SPONSOR_DISABLED	Sponsored data connectivity is disabled or not enabled.	SponsoredConnectivity
SPONSOR_ENABLED	Sponsored data connectivity is enabled.	SponsoredConnectivity

5.6.3.7 Enumeration: AfEvent

The enumeration "AfEvent" represents the traffic events the PCF can notify to the NF service consumer.

Table 5.6.3.7-1: Enumeration AfEvent

Enumeration value (NOTE 1)	Description	Applicability
ACCESS_TYPE_CHANGE	Access type change.	
ANI_REPORT	Access Network Information Report requested.	NetLoc
APP_DETECTION	Application detection report is requested.	ApplicationDetectionEvents
CHARGING_CORRELATION	Access Network Charging Correlation Information.	IMS_SBI
UP_PATH_CHG_FAILURE	Indicates that the enforcement of the AF required routing requirements (i.e. DNAI change) failed.	RoutingReqOutcome
L4S_SUPP	Indicates whether ECN marking for L4S is not available or available again in 5GS.	L4S
EPS_FALLBACK	Indicates the rejection of the establishment of the QoS flow for the requested voice media type in 5GS and the subsequent fallback to EPS.	EPSFallbackReport
EXTRA_UE_ADDR	Indicates the report of extra IP addresses or address ranges allocated for the given PDU session resulting from framed routes or IPv6 prefix delegation.	ExtraUEaddrReport
FAILED_QOS_UPDATE	Indicates that the invocation/revocation indication included in the mpsAction requested by the NF service consumer has failed.	MPSforDTS
FAILED_RESOURCES_ALLOCATION	Indicates that one or more of the SDFs of an Individual Application Session Context are deactivated at the SMF. It also indicates that the resources requested for a particular service information cannot be successfully allocated. (NOTE 2)	
OUT_OF_CREDIT	Out of credit. (NOTE 2)	IMS_SBI
PDU_SESSION_STATUS	Indicates the status of the PDU session (established/terminated). It only applies to notifications to the PCF for a UE as specified in clause 4.2.5.22.	
PLMN_CHG	This trigger indicates PLMN change.	
QOS_NOTIF	The GBR QoS targets of a SDF are not guaranteed or are guaranteed again.	
QOS_MONITORING	Indicates PCF to enable Qos Monitoring for the Service Data Flow.	QoSMonitoring
QOS_MON_CAP_REPO	Indicates that the PCF notifies about whether QoS monitoring is not supported or is supported again.	QoSMonCapRepo
RAN_NAS_CAUSE	This trigger indicates RAN-NAS release cause information is available in the PCF from the SMF. This event does not require explicit subscription.	RAN-NAS-Cause
REALLOCATION_OF_CREDIT	Credit has been reallocated after a former out of credit indication. (NOTE 2)	IMS_SBI, ReallocationOfCredit
SAT_CATEGORY_CHG	Indicates that the SMF has detected a change between different satellite backhaul category, or non-satellite backhaul.	SatelliteBackhaul
SUCCESSFUL_QOS_UPDATE	Indicates that the invocation/revocation indication included in the mpsAction requested by the NF service consumer has been successful.	MPSforDTS
SUCCESSFUL_RESOURCES_ALLOCATION	Indicates that the resources requested for particular service information have been successfully allocated. (NOTE 2)	
TSN_BRIDGE_INFO	5GS Bridge information (UMIC and/or PMIC(s)) received by the PCF from the SMF.	TimeSensitiveNetworking
USAGE_REPORT	Volume and/or time usage for sponsored data connectivity.	SponsoredConnectivity
UE_REACH_STATUS_CH	Indicates that there is a change in the UE reachability status.	UEUnreachable
BAT_OFFSET_INFO	BAT offset and the optionally adjusted periodicity received by the PCF from the SMF.	EnTSCAC
URSP_ENF_INFO	Request to forward UE reporting of URSP enforcement information from associated URSP rule(s).	URSPEnforcement
PACK_DEL_VAR	Indicates Packet Delay Variation is enabled for the SDF.	EnQoSMon
RT_DELAY_TWO_QOS_FLOWS	Indicates PCF to enable Qos Monitoring for the Round-trip delay measurement over two QoS flows	EnQoSMon
RATE_LIMIT_INFO_REPO	Indicates the report of the data rate limitation information for uplink and/or downlink according to the local policies at PCF.	RateLimitReport
UP_PATH_CHANGE	Indicates that the UP path changes for the PDU session.	UeSatUeComm

Enumeration value (NOTE 1)	Description	Applicability
CN_HEALTH_MONITOR	Indicates the report of checking whether the PDU session is being restored or does not exist in SMF.	CnHealthMonitor
UE_TEMPORARILY_UNAVAILABLE	Indicates that the resources associated with the service information cannot be allocated because the SMF has reported the UE as temporarily unreachable.	UEUnreachable
NOTE 1: The subscription to events applies at AF session level, i.e., to all the media components/subcomponents of the Individual Application Session Context resource, unless otherwise specified in the AF event definition. NOTE 2: To ensure the event reports the requested information for all the media components of the Individual Application Session Context resource, the event should be subscribed during the initial provisioning of the service information. When the event is subscribed after the initial provisioning of the service information, it is unknown the status for the unmodified service information previously provisioned, and in this case, only future status changes may be reported.		

5.6.3.8 Enumeration: AfNotifMethod

The enumeration "AfNotifMethod" represents the notification methods that can be subscribed by an NF service consumer.

Table 5.6.3.8-1: Enumeration AfNotifMethod

Enumeration value	Description	Applicability
EVENT_DETECTION	Event is reported whenever the event is met and the subscription is alive.	
ONE_TIME	Events are reported once the event is met and are not reported again unless the AF refreshes the subscription.	
PERIODIC	The notification is periodically sent.	EnhancedSubscriptionToNotification

5.6.3.9 Enumeration: QosNotifType

The enumeration "QosNotifType" represents the types of reports bound to the notification of QoS Notification Control.

Table 5.6.3.9-1: Enumeration QosNotifType

Enumeration value	Description	Applicability
GUARANTEED	The QoS targets of one or more SDFs are guaranteed again. When the "ExtQoSR19" feature is supported, the QoS targets of one or more SDFs are guaranteed again in both DL and UL directions.	
NOT_GUARANTEED	The QoS targets of one or more SDFs are not being guaranteed. When the "ExtQoSR19" feature is supported, the QoS targets of one or more SDFs are not being guaranteed in both DL and UL directions.	
NOT_GUARANTEED_DL	The QoS targets of one or more SDFs are not being guaranteed in DL direction.	ExtQoSR19
NOT_GUARANTEED_UL	The QoS targets of one or more SDFs are not being guaranteed in UL direction.	ExtQoSR19

5.6.3.10 Enumeration: TerminationCause

The enumeration "TerminationCause" represents the types of causes the PCF can report when requesting to the NF service consumer the deletion of the "Individual Application Session Context" resource.

Table 5.6.3.10-1: Enumeration TerminationCause

Enumeration value	Description	Applicability
ALL_SDF_DEACTIVATION	All the SDFs of an Individual Application Session Context are deactivated at the SMF. It also indicates the case that the all resource allocation of an Individual Application Session Context fails.	
PDU_SESSION_TERMINATION	The PDU session is terminated.	
PS_TO_CS_HO	Indication of PS to CS handover is received from the SMF.	IMS_SBI
INSUFFICIENT_SERVER_RESOURCES	Indicates that the server is overloaded and needs to release the Individual Application Session Context resource.	
INSUFFICIENT_QOS_FLOW_RESOURCES	Indicates that the QoS flow has been deactivated due to insufficient QoS flow resources (e.g. the maximum number of QoS flows for the PDU session is reached).	
SPONSORED_DATA_CONNECTIVITY_DISALLOWED	Indicates that due to operator policy (e.g. disallowing the UE accessing the sponsored data connectivity in the roaming case) the Individual Application Session Context resource needs to be terminated.	
REQUEST_QOS_NOT_SUPPORTED_IN_PLMN	Indicates that the Individual Application Session Context resource needs to be terminated because the requested QoS associated to that session is not supported in the current serving PLMN.	VPLMNErrorRep
UE_ADDR_RELEASE	Indicates that due to released UE Address for the Individual Application Session Context resource from the SMF, the Individual Application Session Context resource needs to be terminated.	ReleasedUEaddrReport
REFLECTIVE_QOS_NOT_SUPPORTED_IN_UE	Indicates that the Individual Application Session Context resource needs to be terminated because the requested expedite data transfer with reflective QoS is not supported in the UE.	TrafficCharChange
SMF_FAILURE	Indicates the report of the SMF failure is detected due to operator configuration.	SmfFailureDetection

5.6.3.11 Void

5.6.3.12 Enumeration: FlowStatus

The enumeration "FlowStatus" represents whether the service data flow(s) are enabled or disabled.

Table 5.6.3.12-1: Enumeration FlowStatus

Enumeration value	Description	Applicability
ENABLED-UPLINK (NOTE)	Indicates to enable associated uplink service data flow(s) and to disable associated downlink service data flow(s).	
ENABLED-DOWNLINK (NOTE)	Indicates to enable associated downlink service data flow(s) and to disable associated uplink service data flow(s).	
ENABLED	Indicates to enable all associated service data flow(s) in both directions.	
DISABLED	Indicates to disable all associated service data flow(s) in both directions.	
REMOVED	Indicates to remove all associated service data flow(s). The IP Filters for the associated service data flow(s) shall be removed. The associated service data flows shall not be taken into account when deriving the authorized QoS.	
NOTE: The enumeration value does not follow the related naming convention (i.e. "UPPER_WITH_UNDERSCORE") defined in clause 5.1.4 of 3GPP TS 29.501 [6]. This enumeration value is however kept as currently defined in this specification for backward compatibility considerations.		

5.6.3.13 Enumeration: MediaComponentResourcesStatus

The enumeration "MediaComponentResourcesStatus" indicates whether the PCC rule(s) related to certain media component are active or inactive.

Table 5.6.3.13-1: Enumeration MediaComponentResourcesStatus

Enumeration value	Description	Applicability
ACTIVE	Indicates that the PCC rule(s) related to certain media component are active.	
INACTIVE	Indicates that the PCC rule(s) related to certain media component are inactive.	

5.6.3.14 Enumeration: FlowUsage

The enumeration "FlowUsage" represents the flow usage of the flows described by a media subcomponent.

Table 5.6.3.14-1: Enumeration FlowUsage

Enumeration value	Description	Applicability
NO_INFO	This value is used to indicate that no information about the usage of the IP flow is being provided. This is the default value.	
RTCP	This value is used to indicate that an IP flow is used to transport RTCP.	
AF_SIGNALLING	This value is used to indicate that the IP flow is used to transport AF Signalling Protocols (e.g. SIP/SDP).	IMS_SBI

NOTE: A NF service consumer can choose not to identify RTCP flows, e.g. in order to avoid that RTCP flows are always enabled by the server.

5.6.3.15 Enumeration: RequiredAccessInfo

The enumeration "RequiredAccessInfo" represents the access network information required for the "Individual Application Session Context" resource.

Table 5.6.3.15-1: Enumeration RequiredAccessInfo

Enumeration value	Description	Applicability
USER_LOCATION	Indicates that the user location information shall be reported.	
UE_TIME_ZONE	Indicates that the user timezone shall be reported.	
UE_SAT_INFO	Indicates that the user serving satellite identifier shall be reported.	UeSatUeComm

5.6.3.16 Enumeration: ServiceInfoStatus

The enumeration "ServiceInfoStatus" represents whether the NF service consumer provided service information is preliminary or final.

Table 5.6.3.16-1: Enumeration ServiceInfoStatus

Enumeration value	Description	Applicability
FINAL	This value is used to indicate that the service has been fully negotiated between the two ends and service information provided is the result of that negotiation.	
PRELIMINARY	This value is used to indicate that the service information that the AF has provided to the PCF is preliminary and needs to be further negotiated between the two ends (e.g. for IMS when the service information is sent based on the SDP offer).	

5.6.3.17 Enumeration: SipForkingIndication

The enumeration "SipForkingIndication" describes if several SIP dialogues are related to an "Individual Application Session Context" resource.

Table 5.6.3.17-1: Enumeration SipForkingIndication

Enumeration value	Description	Applicability
SINGLE_DIALOGUE	This value is used to indicate that the "Individual Application Session Context" resource relates to a single SIP dialogue. This is the default value.	
SEVERAL_DIALOGUES	This value is used to indicate that the "Individual Application Session Context" resource relates to several SIP dialogues.	

5.6.3.18 Enumeration: AfRequestedData

The enumeration "AfRequestedData" represents the information the NF service consumer requested to be exposed.

Table 5.6.3.18-1: Enumeration AfRequestedData

Enumeration value	Description	Applicability
UE_IDENTITY	Indicates that the NF service consumer requests the PCF to provide the 5GS-level UE identities (SUPI, GPSI, PEI) available for that PDU session.	

5.6.3.19 Enumeration: PreemptionControlInformation

The enumeration "PreemptionControlInformation" represents how to perform pre-emption among multiple potential media flow candidates of same priority.

Table 5.6.3.19-1: Enumeration PreemptionControlInformation

Enumeration value	Description	Applicability
MOST_RECENT	Indicates the most recent added flow is to be pre-empted.	
LEAST_RECENT	Indicates the least recent added flow is to be pre-empted.	
HIGHEST_BW	Indicates the highest bandwidth flow is to be pre-empted.	

5.6.3.20 Enumeration: PrioritySharingIndicator

The enumeration "PrioritySharingIndicator" represents whether the media component is enabled or disabled for priority sharing with other media components which are assigned the same 5QI and belong to other "Individual Application Session Context" resource bound to the same PDU session.

Table 5.6.3.20-1: Enumeration PrioritySharingIndicator

Enumeration value	Description	Applicability
ENABLED	Indicates the media component is allowed to share ARP with other medias which are assigned the same 5QI and belong to other "Individual Application Session Context" resources bound to the same PDU session.	
DISABLED	Indicates the media component is not allowed to share ARP with other media components. This is the default value when omitted.	

5.6.3.21 Enumeration: PreemptionControlInformationRm

This data type is defined in the same way as the "PreemptionControlInformation" data type but also allows null value (specified as "NullValue" data type).

5.6.3.22 Enumeration: MpsAction

The enumeration "MpsAction" indicates the type of action for an MPS request.

Table 5.6.3.22-1: Enumeration MpsAction

Enumeration value	Description	Applicability
DISABLE_MPS_FOR_DTS	Disable MPS for DTS.	
ENABLE_MPS_FOR_DTS	Enable MPS for DTS.	
AUTHORIZE_AND_ENABLE_MPS_FOR_DTS	Check the UE's MPS subscription and enable MPS for DTS.	
AUTHORIZE_AND_ENABLE_MPS_FOR_AF_SIGNALLING	Check the UE's MPS subscription and enable MPS for AF signalling.	AuthorizationForMpsSignalling
DISABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING	Disable MPS for Messaging for AF signalling	MPSforMessaging
ENABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING	Enable MPS for Messaging for AF signalling	MPSforMessaging

5.6.3.23 Enumeration: AppDetectionNotifType

The enumeration "AppDetectionNotifType" represents the types of reports bound to the notification of application detection information.

Table 5.6.3.23-1: Enumeration AppDetectionNotifType

Enumeration value	Description	Applicability
APP_START	The start of application's traffic is detected.	
APP_STOP	The stop of application's traffic is detected.	

5.6.3.24 Enumeration: PduSessionStatus

The enumeration "PduSessionStatus" represents the notification is about PDU session established or terminated.

Table 5.6.3.24-1: Enumeration PduSessionStatus

Enumeration value	Description	Applicability
ESTABLISHED	The PDU session is established.	
TERMINATED	The PDU session is terminated.	

5.6.3.25 Enumeration: UplinkDownlinkSupport

The enumeration "UplinkDownlinkSupport" represents whether a capability is supported for the UL, the DL or both, UL and DL.

Table 5.6.3.25-1: Enumeration UplinkDownlinkSupport

Enumeration value	Description	Applicability
UL	ECN marking for L4S supported for the UL.	
DL	ECN marking for L4S supported for the DL.	
UL_DL	ECN marking for L4S supported for the UL and the DL	

5.6.3.26 Enumeration: L4sNotifType

The enumeration "L4sNotifType" represents unavailability or availability again of the ECN marking for L4S support in 5GS.

Table 5.6.3.26-1: Enumeration L4sNotifType

Enumeration value	Description	Applicability
AVAILABLE	The ECN marking for L4S of one or more SDFs is available again.	
NOT_AVAILABLE	The ECN marking for L4S of one or more SDFs is not available.	

5.6.3.27 Enumeration: NotifCap

Table 5.6.3.27-1: Enumeration NotifCap

Enumeration value	Description	Applicability
NOT_SUPPORTED	The capability is not supported.	
SUPPORTED	The capability is supported.	

5.6.3.28 Enumeration: HeaderHandlingAction

The enumeration HeaderHandlingAction represents the type of header handling actions. It shall comply with the provisions defined in table 5.6.3.28-1.

Table 5.6.3.28-1: Enumeration HeaderHandlingAction

Enumeration value	Description
DETECT	Indicates that the request is for the detection of a header field.
REMOVE	Indicates that the request is for the removal of a header field.
REPLACE	Indicates that the request is for the replacement of information in a header field.
INSERT	Indicates that the request is for the addition of a header field.

5.6.3.29 Enumeration: HeaderHandlingCond

The enumeration HeaderHandlingCond represents the condition to apply the header handling action. It shall comply with the provisions defined in table 5.6.3.29-1.

Table 5.6.3.29-1: Enumeration HeaderHandlingCond

Enumeration value	Description
EVERY_MATCH	Indicates that the header handling action is applied to every match.
FIRST_MATCH	Indicates that the header handling action is applied only to the first match.

5.6.3.30 Enumeration: OnPathN6Method

The enumeration OnPathN6Method represents the supported on-path N6 signaling method(s). It shall comply with the provisions defined in table 5.6.3.30-1.

Table 5.6.3.30-1: Enumeration OnPathN6Method

Enumeration value	Description
CONNECT_UDP	Support of on-path N6 signaling with method connect UDP.

5.6.3.31 Enumeration: NotifCapType

Table 5.6.3.31-1: Enumeration NotifCapType

Enumeration value	Description	Applicability
PACKET_DELAY	Indicates that the QoS Monitoring Capability type is Packet delay.	
CONGESTION	Indicates that the QoS Monitoring Capability type is Congestion information.	
AVAILABLE_BITRATE	Indicates that the QoS Monitoring Capability type is Available bitrate.	

5.6.3.32 Enumeration: PcSessionRecoveryStatus

The enumeration "PcSessionRecoveryStatus" contains the session recovery indication and describes whether the session is being restored or the session is not found.

Table 5.6.3.32-1: Enumeration PcSessionRecoveryStatus

Enumeration value	Description	Applicability
REST_REQUEST	Information about whether PDU session is being restored is requested.	
REST_TRIGGERED	The PDU session is being restored.	
REST_NOT_TRIGGERED	The PDU session exists and it is not being restored.	
NOT_FOUND	The PDU session was not found.	

5.7 Error handling

5.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of 3GPP TS 29.500 [5].

For the Npcf_PolicyAuthorization API, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [6].

Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [5] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [5].

In addition, the requirements in the following clauses shall apply.

5.7.2 Protocol Errors

In this Release of the specification, there are no additional protocol errors applicable for the Npcf_PolicyAuthorization API.

5.7.3 Application Errors

The application errors defined for the Npcf_PolicyAuthorization API are listed in table 5.7.3-1.

Table 5.7.3-1: Application errors

Application Error	HTTP status code	Description	Applicability
INVALID_SERVICE_INFORMATION	400 Bad Request	The HTTP request is rejected because the service information is invalid or insufficient for the PCF to perform the requested action, e.g. invalid media type or invalid QoS reference. (NOTE 1)	
FILTER_RESTRICTIONS	400 Bad Request	The HTTP request is rejected because the IP flow descriptions cannot be handled by the PCF because the restrictions defined in clause 5.3.8 of 3GPP TS 29.214 [20] are not observed. (NOTE 1)	
DUPLICATED_AF_SESSION	400 Bad Request	The HTTP request is rejected because the new Individual Application Session Context relates to an AF session with another related active Individual Application Session Context, e.g. if the AF provided the same AF charging identifier for this new Individual Application Session Context that is already in use for the other ongoing Individual Application Session Context. (NOTE 2)	
REQUESTED_SERVICE_NOT_AUTHORIZED	403 Forbidden	The service information provided in the request is rejected. (NOTE 7)	
REQUESTED_SERVICE_TEMPORARILY_NOT_AUTHORIZED	403 Forbidden	The service information provided in the request is temporarily rejected. (NOTE 2)	
UNAUTHORIZED_SPONSORED_DATA_CONNECTIVITY	403 Forbidden	The request for sponsored data connectivity is not authorized. (NOTE 3)	
UNAUTHORIZED_NON_EMERGENCY_SESSION	403 Forbidden	The PCF rejects a new AF session context setup because the session binding function associated a non-Emergency IMS session to a PDU session established to an Emergency DNN.	
TEMPORARY_NETWORK_FAILURE	403 Forbidden	The PCF rejects new or modified service information because there is a temporary failure in the access network (e.g. the SGW has failed)	
INVALID_SIGNALING_PATH	403 Forbidden	The PCF rejects the creation of the Individual Application Session Context resource because the NF that invoked the service request is invalid (i.e. the NEF sent the request while the PCF expected the TSCTSF to do so). (NOTE 8)	
REQUEST_QOS_NOT_SUPPORTED_IN_PLMN	403 Forbidden	The PCF rejects new or modified QoS because the requested QoS are not supported in the current serving PLMN where UE is registered.	
REFLECTIVE_QOS_NOT_SUPPORTED_IN_UE	403 Forbidden	The HTTP request is rejected because the request for expedited data transfer with reflective QoS is not supported in the UE.	TrafficCharChange
MPX_MEDIA_NOT_SUPPORTED_IN_UE	403 Forbidden	The HTTP request is rejected because the request for differentiated QoS for multiplexed media flows is not supported in the UE.	MpxMedia
APPLICATION_SESSION_CONTEXT_NOT_FOUND	404 Not Found	The HTTP request is rejected because the specified Individual Application Session Context does not exist. (NOTE 4)	

Application Error	HTTP status code	Description	Applicability
PDU_SESSION_NOT_AVAILABLE	500 Internal Server Error	The PCF failed in executing session binding. (NOTE 5)	

NOTE 1: This application error is included in the response to the POST request (see clauses 4.2.2.2 and 4.2.2.5) and to the PATCH request (see clauses 4.2.3.2 and 4.2.3.5).
NOTE 2: This application error is included in the response to the POST request (see clause 4.2.2.2) and to the PATCH request (see clause 4.2.3.2).
NOTE 3: This application error is included in the response to the POST request (see clause 4.2.2.5) and to the PATCH request (see clause 4.2.3.5).
NOTE 4: This application error is included in the responses to the GET, PATCH and delete custom operation requests to the Individual Application Session Context resource, and to the PUT and DELETE requests to the Events Subscription resource.
NOTE 5: This application error is included in the response to the POST request (see clauses 4.2.2.2, 4.2.6.3 and 4.2.2.27).
NOTE 6: Including a "ProblemDetails" data structure with the "cause" attribute in the HTTP response is optional unless explicitly mandated in the service operation clauses.
NOTE 7: This application error is included in the response to the POST request (see clauses 4.2.2.2, 4.2.2.5, and 4.2.2.12) and to the PATCH request (see clauses 4.2.3.2, 4.2.3.5 and 4.2.3.12).
NOTE 8: This application error is included in the response to the POST request (see clause 4.2.2.2).

5.8 Feature negotiation

The optional features in table 5.8-1 are defined for the Npcf_PolicyAuthorization API. They shall be negotiated using the extensibility mechanism defined in clause 6.6.2 of 3GPP TS 29.500 [5].

When requesting the PCF to create an Individual Application Session Context resource the NF service consumer shall indicate the optional features the NF service consumer supports for the Npcf_PolicyAuthorization service by including the "suppFeat" attribute in the "AppSessionContextReqData" data type of the HTTP POST request.

The PCF shall determine the supported features for the created Individual Application Session Context resource as specified in clause 6.6.2 of 3GPP TS 29.500 [5]. The PCF shall indicate the supported features in the HTTP response confirming the creation of the Individual Application Session Context resource by including the "suppFeat" attribute in the "AppSessionContextRespData" data type.

Table 5.8-1: Supported Features

Feature number	Feature Name	Description
1	InfluenceOnTrafficRouting	Indicates support of Application Function influence on traffic routing. If the PCF supports this feature, the NF service consumer may influence SMF routing to applications or subscribe to notifications of UP path management for the traffic flows of an active PDU session.
2	SponsoredConnectivity	Indicates support of sponsored data connectivity. If the PCF supports this feature, the NF service consumer may provide sponsored data connectivity to the SUPI.
3	MediaComponentVersioning	Indicates the support of the media component versioning.
4	URLLC	Indicates support of Ultra-Reliable Low-Latency Communication (URLLC) requirements, i.e. AF application relocation acknowledgement and UE address(es) preservation. The InfluenceOnTrafficRouting feature shall be supported in order to support this feature.
5	IMS_SBI	Indicates support of the communication with the 5GC IMS NF service consumer via Service Based Interfaces.
6	NetLoc	Indicates the support of access network information reporting.
7	ProvAFsignalFlow	This indicates support for the feature of provisioning of AF signalling flow information as described in clauses 4.2.2.16 and 4.2.3.17. If the PCF supports this feature the NF service consumer may provision AF signalling flow information. NOTE: This feature is used by the IMS Restoration Procedures to provide to the SMF the address of the P-CSCF selected by the UE, refer to 3GPP TS 23.380 [39]. The IMS_SBI feature shall be supported in order to support this feature.
8	ResourceSharing	This feature indicates the support of resource sharing across several "Individual Application Session Context" resources. The IMS_SBI feature shall be supported in order to support this feature.
9	MCPTT	This feature indicates the support of Mission Critical Push To Talk services as described in 3GPP TS 24.379 [41].
10	MCVideo	This feature indicates the support of Mission Critical Video services as described in 3GPP TS 24.281 [43].
11	PrioritySharing	This feature indicates that Priority Sharing is supported as described in 3GPP TS 23.503 [4], clause 6.1.3.15.
12	MCPTT-Preemption	This feature indicates the support of service pre-emption based on the information provided by the NF service consumer. It requires that both PrioritySharing and MCPTT features are also supported.
13	MacAddressRange	Indicates the support of a set of MAC addresses with a specific range in the traffic filter.
14	RAN-NAS-Cause	This feature indicates the support for the release cause code information from the access network.
15	EnhancedSubscriptionToNotification	Indicates the support of: - Subscription to periodic notifications. - Definition of a waiting time between the reporting of two event triggered events. - Indication of whether the event has to be reported at PDU Session termination. - Notification Correlation Id for a subscription to an event.
16	QoSMonitoring	Indicates the support of QoS monitoring functionality and the report of packet delay monitoring. This feature requires the support of the EnhancedSubscriptionToNotification feature.
17	AuthorizationWithRequiredQoS	Indicates support of policy authorization for the AF session with required QoS.
18	TimeSensitiveNetworking	Indicates that the 5G System is integrated within the external network as a TSN bridge.
19	PCSCF-Restoration-Enhancement	This feature indicates support of P-CSCF Restoration Enhancement. It is used for the PCF and the P-CSCF to indicate if they support P-CSCF Restoration Enhancement.
20	CHEM	This feature indicates the support of Coverage and Handover Enhancements for Media (CHEM).

Feature number	Feature Name	Description
21	FLUS	This feature indicates the support of FLUS functionality as described in 3GPP TS 26.238 [51].
22	EPSFallbackReport	This feature indicates the support of the report of EPS Fallback as defined in clauses 4.2.2.30, 4.2.3.29 and 4.2.5.15.
23	ATSSS	Indicates the support of the report of the multiple access types of a MA PDU session.
24	QoSHint	This feature indicates the support of specific QoS hint parameters as described in 3GPP TS 26.114 [30], clause 6.2.10.
25	ReallocationOfCredit	This feature indicates the support of notifications of reallocation of credits events. It requires the support of IMS_SBI feature.
26	ES3XX	Extended Support for 3xx redirections. This feature indicates the support of redirection for any service operation, according to Stateless NF procedures as specified in clauses 6.5.3.2 and 6.5.3.3 of 3GPP TS 29.500 [5] and according to HTTP redirection principles for indirect communication, as specified in clause 6.10.9 of 3GPP TS 29.500 [5].
27	DisableUENotification	Indicates the support of disabling QoS flow parameters signalling to the UE when the SMF is notified by the NG-RAN of changes in the fulfilled QoS situation. This feature requires that the AuthorizationWithRequiredQoS featute is also supported.
28	PatchCorrection	Indicates support of the correction to the PATCH method: When this feature is not supported, the interoperability between a NF service consumer and the PCF can only be ensured when it is not required the update of the Individual Application Session Context resource.
29	MPSforDTS	Indicates support for MPS for DTS as described in clauses 4.2.2.12.2 and 4.2.3.12.
30	ApplicationDetectionEvents	This feature indicates the support of the subscription to notifications of the detection of the start and stop of an application's traffic.
31	TimeSensitiveCommunication	Indicates that the 5G System is integrated within the external network as a TSC user plane node to enable Time Sensitive Communication, Time Synchronization and Deterministic Networking. This feature requires that the TimeSensitiveNetworking feature is also supported.
32	ExposureToEAS	This feature indicates the support of the indication of direct event notification of QoS monitoring events from the UPF to the Local NEF or AF in 5GC. This indication requires that the QoSMonitoring feature is supported.
33	SatelliteBackhaul	Indicates the support of the report of the satellite or non-satellite backhaul category of the PDU session.
34	RoutingReqOutcome	Indicates the support of: - the report of UP path change failures; and - the indication of whether AF routing requirements are applied. It requires the support of InfluenceOnTrafficRouting feature.
35	EASDiscovery	This feature indicates the support of EAS (re)discovery.
36	AltSerReqsWithIndQoS	Indicates the support of provisioning Alternative Service Requirements with individual QoS parameters. This feature requires that the AuthorizationWithRequiredQoS feature is also supported.
37	SimultConnectivity	This feature indicates the support of the indication of temporary simultaneous connectivity over source and target PSA at edge relocation. This indication requires that the InfluenceOnTrafficRouting feature is supported.
38	EASIPreplacement	This feature indicates the support of provisioning of EAS IP replacement info. This support requires that InfluenceOnTrafficRouting feature is also supported
39	AccNetChargId_String	This feature indicates the support of long character strings as access network charging identifier.
40	WLAN_Location	This feature indicates the support of the report of the WLAN location information received from the ePDG/EPC, if available. It is only applicable to EPS interworking scenarios as described in 3GPP TS 29.512 [8], Annex B.

Feature number	Feature Name	Description
41	AF_latency	This feature indicates support for edge relocation considering user plane latency.
42	UEUnreachable	This feature indicates the support for the reporting of UE temporary unavailable.
43	AltQoSProfilesSupportReport	This feature indicates the support of the report of whether Alternative QoS parameters are supported by NG-RAN. This feature requires that AuthorizationWithRequiredQoS feature is also supported.
44	PacketDelayFailureReport	Indicates the support of packet delay failure report as part of QoS Monitoring procedures. This feature requires that QoSMonitoring feature is supported.
45	EnTSCAC	Indicates the support of extensions to TSCAC and the RAN feedback for BAT offset and adjusted periodicity. This feature requires that the TimeSensitiveCommunication feature is also supported.
46	SignalingPathValidation	This feature indicates the support of the validation of the NF type that originates the Npcf_PolicyAuthorization_Create request.
47	ExtQoS	This feature indicates the support for the extensions to the QoS mechanisms.
48	CommonEASDNAI	This feature controls the support of the common EAS/DNAI selection. This feature requires that the InfluenceOnTrafficRouting feature is also supported.
49	SFC	This feature indicates support of Service Function Chaining functionality.
50	MultiMedia	This feature indicates the support of multi-modal or multimedia communication service. This feature acts as a basic functional block for extended reality (XR) and interactive media services.
51	EnSatBackhaulCatChg	This feature indicates the support also of the report of the dynamic satellite backhaul category of the PDU session. This feature requires the support of SatelliteBackhaul feature.
52	MTU_Size	This feature indicates the support of the report of the MTU size of the device side port. This feature requires that the TimeSensitiveCommunication feature is also supported.
53	ExtraUEaddrReport	This feature indicates the support of the report of additional IP addresses or address ranges allocated for the given PDU session resulting from framed routes or IPv6 prefix delegation.
54	AuthorizationForMpsSignalling	This feature indicates support for use of the "mpsAction" attribute to signal that the UE's MPS subscription shall be checked by the PCF prior to enabling MPS for AF signalling.
55	ExposureToTSC	This feature indicates the support of the direct event notification of TSC management information from the UPF to the TSCTSF or TSN AF in 5GC. This feature requires that the TimeSensitiveCommunication feature is also supported.
56	URSPEnforcement	This feature indicates the support of awareness of URSP rule enforcement
57	AddFlowDescriptionInformation	This feature indicates support for use e.g. of additional flow description parameters, as the flow label and the IPSec SPI.
58	QoSTiming_5G	This feature indicates the support of QoS timing information for the transfer and support of data transmission (e.g., AI/ML traffic transmission).
59	PDUSetHandling	This feature indicates the support of PDU Set handling. This feature may be used for eXtended Reality (XR) and interactive media services.
60	RTLatency	This feature indicates the support of Round-Trip latency. This feature may be used for eXtended Reality (XR) and interactive media services.

Feature number	Feature Name	Description
61	EnQoSMon	This feature indicates the support of enhanced QoS monitoring functionality, i.e. the enhancement of packet delay QoS monitoring, and/or, the report of the congestion information, and/or, the RTT delay over two QoS flows, and/or, the data rate information, and/or, the Packet Delay Variation monitoring. This feature requires that the QoSMonitoring feature is supported. In order to support the report of packet delay measurement failure, the PacketDelayFailureReport feature also requires to be supported.
62	PowerSaving	This feature indicates the support of UE Power Saving management in multi modal traffic as described in clause 4.2.2.42.
63	L4S	This feature indicates the support of the AF indication of ECN marking for L4S support.
64	QoSMonCapRepo	This feature indicates the support of the subscription to notifications about network support of QoS Monitoring for packet delay, available bitrate and/or congestion.
65	MPSforMessaging	Indicates support for MPS for Messaging as described in clauses 4.2.2.12.4 and 4.2.3.13.
66	VPLMNErrorRep	This feature indicates the support of failure due to QoS not supported by the current serving PLMN.
67	TraffRouteReqOutcome	This feature indicates the support of the subscription to the traffic routing requirements installation outcome event reporting of SMF. This feature requires that the InfluenceOnTrafficRouting feature is supported.
68	MpxMedia	This feature indicates the support of uniquely identifying each media flow of multiplexed media with the (S)RTP Multiplexed Media Identification Information.
69	TrafficCharChange	This feature indicates the support of dynamically changing traffic characteristics, including: - the handling of Data Burst Size marking indication. - the handling of Time to Next Burst Indication. - the handling of Expedite Data Transfer with reflective QoS Indication for the Non-GBR flows.
70	N6DelayMeasurement	This feature indicates the support of considering N6 delay measurement for traffic influence.
71	ReleasedUEaddrReport	Indicates the support of: - Indication of PCF triggered AF application session context termination due to released UE address from the SMF.
72	UeSatUeComm	This feature indicates the support of reporting about serving satellite identity for UE-Satellite-UE communication in IMS. This feature requires that the IMS_SBI feature is supported. - In order to support of access network information reporting, the NetLoc feature also requires to be supported. - In order to support for the release cause code information from the access network, the RAN-NAS-Cause feature also requires to be supported.
73	HeaderHandling	This feature indicates the support of the header handling functionality. This feature enables the following functionality: - the support of provisioning of Header Handling Control information for handling of Payload Headers.
74	OnPathN6MediaInfo	This feature indicates the support of deliver media related information for encrypted traffic, including: - Using on-path N6 signaling method to deliver media related information for encrypted traffic.
75	RateLimitReport	This feature indicates the support of the AF request the 5GS to expose the data rate limitation information which is determined by the PCF according to the local policy.

Feature number	Feature Name	Description
76	EnCommonDnai	This feature indicates support of providing requests to report the candidate DNAI(s) of the PDU Session.
77	AcceptableQosDetails	This feature indicates the support of providing detailed information about the QoS that the PCF can authorize in error responses of not authorized requests.
78	ExtQoSR19	This feature indicates the enhancements on the support for the extensions to the QoS mechanisms, including: - the support of PDU Set QoS parameters in Alternative QoS Profile when the PDUSetHandling feature is supported. - the support of Averaging Window and Maximum Data Burst Volume parameters. - the support of reporting the QoS targets for the indicated SDFs can no longer be guaranteed or can be guaranteed in the certain direction. - the support of RAN-Controlled UL Bitrate Recommendation Indication for the media flow. This feature requires that AltSerReqsWithIndQoS feature is also supported.
79	SimConnFailure	This feature indicates the support of Simultaneous Connectivity failure events. It requires that the SimultConnectivity feature is also supported.
80	EnQoSMon_v2	This feature indicates the enhancements on the QoS monitoring functionality, including: - the reporting of available bitrate for a GBR QoS Flow. This feature requires that the EnQoSMon feature is supported.
81	SmfFailureDetection	This feature indicates the support of P-CSCF receiving SMF failure notification from PCF. This feature requires that the IMS_SBI feature is supported.
82	CnHealthMonitor	This feature indicates the support of the reporting of session recovery status in case of 5GC network functions failure.
83	SmallBurstSizes	This feature indicates the support of provisioning the Maximum Burst sizes that are smaller than 4096 bytes. This feature requires that the "TimeSensitiveNetworking" feature is supported.

5.9 Security

As indicated in 3GPP TS 33.501 [25] and 3GPP TS 29.500 [5], the access to the Npcf_PolicyAuthorization API, based on local configuration, may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [26]), using the "Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [27]) plays the role of the authorization server.

If OAuth2 authorization is used, an NF service consumer, prior to consuming services offered by the Npcf_PolicyAuthorization API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [27], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF service consumer used for discovering the Npcf_PolicyAuthorization service.

The Npcf_PolicyAuthorization API the following scopes for OAuth2 authorization as described in 3GPP TS 29.501 [6], clause 4.x.

Table 5.9-1: OAuth2 scopes defined in Npcf_PolicyAuthorization API

Scope	Description
"npcf-policyauthorization"	Access to the PCF Policy Authorization API
"npcf-policyauthorization:policy-auth-mgmt"	Access to service operations applying to PCF Policy Authorization for creation, updation, deletion, retrieval

Annex A (normative): OpenAPI specification

A.1 General

The present Annex contains an OpenAPI [11] specification of HTTP messages and content bodies used by the Npcf_PolicyAuthorization API.

This Annex shall take precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API.

NOTE: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification file contained in this 3GPP Technical Specification are available on a Git-based repository that uses the GitLab software version control system (see clause 5B of the 3GPP TR 21.900 [28] and clause 5.3.1 of the 3GPP TS 29.501 [6] for further information).

A.2 Npcf_PolicyAuthorization API

```
openapi: 3.0.0

info:
  title: Npcf_PolicyAuthorization Service API
  version: 1.4.1
  description: |
    PCF Policy Authorization Service.
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: 3GPP TS 29.514 V19.6.0; 5G System; Policy Authorization Service; Stage 3.
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.514/'

servers:
  - url: '{apiRoot}/npcf-policyauthorization/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
    - npcf-policyauthorization

paths:
  /app-sessions:
    post:
      summary: Creates a new Individual Application Session Context resource
      operationId: PostAppSessions
      tags:
        - Application Sessions (Collection)
      security:
        - {}
        - oAuth2ClientCredentials:
          - npcf-policyauthorization
        - oAuth2ClientCredentials:
          - npcf-policyauthorization
          - npcf-policyauthorization:policy-auth-mgmt
      requestBody:
        description: Contains the information for the creation the resource.
        required: true
        content:
          application/json:
            schema:
```

```
        $ref: '#/components/schemas/AppSessionContext'
responses:
  '201':
    description: Successful creation of the resource
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/AppSessionContext'
    headers:
      Location:
        description: >
          Contains the URI of the created individual application session context resource,
          according to the structure
          {apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}
          or the URI of the created events subscription sub-resource,
          according to the structure
          {apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}
          /events-subscription
        required: true
        schema:
          type: string
  '303':
    description: >
      See Other. The result of the HTTP POST request would be equivalent to the existing
      Application Session Context.
    headers:
      Location:
        description: >
          Contains the URI of the existing individual Application Session Context resource.
        required: true
        schema:
          type: string
  '400':
    $ref: 'TS29571_CommonData.yaml#/components/responses/400'
  '401':
    $ref: 'TS29571_CommonData.yaml#/components/responses/401'
  '403':
    description: Forbidden
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ExtendedProblemDetails'
    headers:
      Retry-After:
        description: >
          Indicates the time the AF has to wait before making a new request. It can be a
          non-negative integer (decimal number) indicating the number of seconds the AF
          has to wait before making a new request or an HTTP-date after which the AF can
          retry a new request.
        schema:
          anyOf:
            - type: integer
            - type: string
  '404':
    $ref: 'TS29571_CommonData.yaml#/components/responses/404'
  '411':
    $ref: 'TS29571_CommonData.yaml#/components/responses/411'
  '413':
    $ref: 'TS29571_CommonData.yaml#/components/responses/413'
  '415':
    $ref: 'TS29571_CommonData.yaml#/components/responses/415'
  '429':
    $ref: 'TS29571_CommonData.yaml#/components/responses/429'
  '500':
    $ref: 'TS29571_CommonData.yaml#/components/responses/500'
  '502':
    $ref: 'TS29571_CommonData.yaml#/components/responses/502'
  '503':
    $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  default:
    $ref: 'TS29571_CommonData.yaml#/components/responses/default'
callbacks:
  terminationRequest:
    '{$request.body#/ascReqData/notifUri}/terminate':
      post:
        requestBody:
          description: >
            Request of the termination of the Individual Application Session Context.
```

```
          required: true
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/TerminationInfo'
        responses:
          '204':
            description: The receipt of the notification is acknowledged.
          '307':
            $ref: 'TS29571_CommonData.yaml#/components/responses/307'
          '308':
            $ref: 'TS29571_CommonData.yaml#/components/responses/308'
          '400':
            $ref: 'TS29571_CommonData.yaml#/components/responses/400'
          '401':
            $ref: 'TS29571_CommonData.yaml#/components/responses/401'
          '403':
            $ref: 'TS29571_CommonData.yaml#/components/responses/403'
          '404':
            $ref: 'TS29571_CommonData.yaml#/components/responses/404'
          '411':
            $ref: 'TS29571_CommonData.yaml#/components/responses/411'
          '413':
            $ref: 'TS29571_CommonData.yaml#/components/responses/413'
          '415':
            $ref: 'TS29571_CommonData.yaml#/components/responses/415'
          '429':
            $ref: 'TS29571_CommonData.yaml#/components/responses/429'
          '500':
            $ref: 'TS29571_CommonData.yaml#/components/responses/500'
          '502':
            $ref: 'TS29571_CommonData.yaml#/components/responses/502'
          '503':
            $ref: 'TS29571_CommonData.yaml#/components/responses/503'
          default:
            $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  eventNotification:
    '{$request.body#/ascReqData/evSubsc/notifUri}/notify':
      post:
        requestBody:
          description: Notification of an event occurrence in the PCF.
          required: true
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/EventsNotification'
        responses:
          '204':
            description: The receipt of the notification is acknowledged.
          '307':
            $ref: 'TS29571_CommonData.yaml#/components/responses/307'
          '308':
            $ref: 'TS29571_CommonData.yaml#/components/responses/308'
          '400':
            $ref: 'TS29571_CommonData.yaml#/components/responses/400'
          '401':
            $ref: 'TS29571_CommonData.yaml#/components/responses/401'
          '403':
            $ref: 'TS29571_CommonData.yaml#/components/responses/403'
          '404':
            $ref: 'TS29571_CommonData.yaml#/components/responses/404'
          '411':
            $ref: 'TS29571_CommonData.yaml#/components/responses/411'
          '413':
            $ref: 'TS29571_CommonData.yaml#/components/responses/413'
          '415':
            $ref: 'TS29571_CommonData.yaml#/components/responses/415'
          '429':
            $ref: 'TS29571_CommonData.yaml#/components/responses/429'
          '500':
            $ref: 'TS29571_CommonData.yaml#/components/responses/500'
          '502':
            $ref: 'TS29571_CommonData.yaml#/components/responses/502'
          '503':
            $ref: 'TS29571_CommonData.yaml#/components/responses/503'
          default:
            $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  detected5GsBridgeForPduSession:
```

```
    '{$request.body#/ascReqData/evSubsc/notifUri}/new-bridge':
      post:
        requestBody:
          description: Notification of a new TSC user plane node detected in the PCF.
          required: true
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PduSessionTsnBridge'
        responses:
          '204':
            description: The receipt of the notification is acknowledged.
          '307':
            $ref: 'TS29571_CommonData.yaml#/components/responses/307'
          '308':
            $ref: 'TS29571_CommonData.yaml#/components/responses/308'
          '400':
            $ref: 'TS29571_CommonData.yaml#/components/responses/400'
          '401':
            $ref: 'TS29571_CommonData.yaml#/components/responses/401'
          '403':
            $ref: 'TS29571_CommonData.yaml#/components/responses/403'
          '404':
            $ref: 'TS29571_CommonData.yaml#/components/responses/404'
          '411':
            $ref: 'TS29571_CommonData.yaml#/components/responses/411'
          '413':
            $ref: 'TS29571_CommonData.yaml#/components/responses/413'
          '415':
            $ref: 'TS29571_CommonData.yaml#/components/responses/415'
          '429':
            $ref: 'TS29571_CommonData.yaml#/components/responses/429'
          '500':
            $ref: 'TS29571_CommonData.yaml#/components/responses/500'
          '502':
            $ref: 'TS29571_CommonData.yaml#/components/responses/502'
          '503':
            $ref: 'TS29571_CommonData.yaml#/components/responses/503'
          default:
            $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  eventNotificationPduSession:
    '{$request.body#/ascReqData/evSubsc/notifUri}/pdu-session':
      post:
        requestBody:
          description: Notification of PDU session established or terminated.
          required: true
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PduSessionEventNotification'
        responses:
          '204':
            description: The receipt of the notification is acknowledged.
          '307':
            $ref: 'TS29571_CommonData.yaml#/components/responses/307'
          '308':
            $ref: 'TS29571_CommonData.yaml#/components/responses/308'
          '400':
            $ref: 'TS29571_CommonData.yaml#/components/responses/400'
          '401':
            $ref: 'TS29571_CommonData.yaml#/components/responses/401'
          '403':
            $ref: 'TS29571_CommonData.yaml#/components/responses/403'
          '404':
            $ref: 'TS29571_CommonData.yaml#/components/responses/404'
          '411':
            $ref: 'TS29571_CommonData.yaml#/components/responses/411'
          '413':
            $ref: 'TS29571_CommonData.yaml#/components/responses/413'
          '415':
            $ref: 'TS29571_CommonData.yaml#/components/responses/415'
          '429':
            $ref: 'TS29571_CommonData.yaml#/components/responses/429'
          '500':
            $ref: 'TS29571_CommonData.yaml#/components/responses/500'
          '502':
            $ref: 'TS29571_CommonData.yaml#/components/responses/502'
          '503':
```

```
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
              default:
                $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /app-sessions/pcscf-restoration:
    post:
      summary: "Indicates P-CSCF restoration and does not create an Individual Application Session
Context"
      operationId: PcscfRestoration
      tags:
        - PCSCF Restoration Indication
      requestBody:
        description: PCSCF Restoration Indication.
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PcscfRestorationRequestData'
      responses:
        '204':
          description: The deletion is confirmed without returning additional data.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /app-sessions/{appSessionId}:
    get:
      summary: "Reads an existing Individual Application Session Context"
      operationId: GetAppSession
      tags:
        - Individual Application Session Context (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - npcf-policyauthorization
        - oAuth2ClientCredentials:
          - npcf-policyauthorization
          - npcf-policyauthorization:policy-auth-mgmt
      parameters:
        - name: appSessionId
          description: String identifying the resource.
          in: path
          required: true
          schema:
            type: string
      responses:
        '200':
          description: A representation of the resource is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AppSessionContext'
        '307':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: "Modifies an existing Individual Application Session Context"
    operationId: ModAppSession
    tags:
      - Individual Application Session Context (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - npcf-policyauthorization
      - oAuth2ClientCredentials:
        - npcf-policyauthorization
        - npcf-policyauthorization:policy-auth-mgmt
    parameters:
      - name: appSessionId
        description: String identifying the resource.
        in: path
        required: true
        schema:
          type: string
    requestBody:
      description: Modification of the resource.
      required: true
      content:
        application/merge-patch+json:
          schema:
            $ref: '#/components/schemas/AppSessionContextUpdateDataPatch'
    responses:
      '200':
        description: >
          Successful modification of the resource and a representation of that resource is
          returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AppSessionContext'
      '204':
        description: The successful modification.
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        description: Forbidden
        content:
          application/problem+json:
            schema:
              $ref: '#/components/schemas/ExtendedProblemDetails'
        headers:
          Retry-After:
            description: >
              Indicates the time the AF has to wait before making a new request. It can be a
```

```
              non-negative integer (decimal number) indicating the number of seconds the AF has
              to wait before making a new request or an HTTP-date after which the AF can retry
              a new request.
            schema:
              anyOf:
                - type: integer
                - type: string
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    callbacks:
      eventNotification:
        '{$request.body#/ascReqData/evSubsc/notifUri}/notify':
          post:
            requestBody:
              description: Notification of an event occurrence in the PCF.
              required: true
              content:
                application/json:
                  schema:
                    $ref: '#/components/schemas/EventsNotification'
            responses:
              '204':
                description: The receipt of the notification is acknowledged
              '307':
                $ref: 'TS29571_CommonData.yaml#/components/responses/307'
              '308':
                $ref: 'TS29571_CommonData.yaml#/components/responses/308'
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '401':
                $ref: 'TS29571_CommonData.yaml#/components/responses/401'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
              default:
                $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /app-sessions/{appSessionId}/delete:
    post:
      summary: "Deletes an existing Individual Application Session Context"
      operationId: DeleteAppSession
      tags:
        - Individual Application Session Context (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - npcf-policyauthorization
```

```
      - oAuth2ClientCredentials:
        - npcf-policyauthorization
        - npcf-policyauthorization:policy-auth-mgmt
    parameters:
      - name: appSessionId
        description: String identifying the Individual Application Session Context resource.
        in: path
        required: true
        schema:
          type: string
    requestBody:
      description: >
        Deletion of the Individual Application Session Context resource, req notification.
      required: false
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/EventsSubscReqData'
    responses:
      '200':
        description: The deletion of the resource is confirmed and a resource is returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AppSessionContext'
      '204':
        description: The deletion is confirmed without returning additional data.
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /app-sessions/{appSessionId}/events-subscription:
    put:
      summary: "creates or modifies an Events Subscription subresource"
      operationId: updateEventsSubsc
      tags:
        - Events Subscription (Document)
      parameters:
        - name: appSessionId
          description: String identifying the Events Subscription resource.
          in: path
          required: true
          schema:
            type: string
      requestBody:
        description: Creation or modification of an Events Subscription resource.
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/EventsSubscReqData'
      responses:
        '201':
```

```
      description: >
        The creation of the Events Subscription resource is confirmed and its representation is
        returned.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/EventsSubscPutData'
      headers:
        Location:
          description: >
            Contains the URI of the created Events Subscription resource,
            according to the structure
            {apiRoot}/npcf-policyauthorization/v1/app-sessions/{appSessionId}/
            events-subscription
          required: true
          schema:
            type: string
    '200':
      description: >
        The modification of the Events Subscription resource is confirmed its representation is
        returned.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/EventsSubscPutData'
    '204':
      description: >
        The modification of the Events Subscription subresource is confirmed without returning
        additional data.
    '307':
      $ref: 'TS29571_CommonData.yaml#/components/responses/307'
    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    default:
      $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  callbacks:
    eventNotification:
      '{$request.body#/notifUri}/notify':
        post:
          requestBody:
            description: >
              Contains the information for the notification of an event occurrence in the PCF.
            required: true
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/EventsNotification'
          responses:
            '204':
              description: The receipt of the notification is acknowledged.
            '307':
              $ref: 'TS29571_CommonData.yaml#/components/responses/307'
            '308':
              $ref: 'TS29571_CommonData.yaml#/components/responses/308'
            '400':
              $ref: 'TS29571_CommonData.yaml#/components/responses/400'
```

```
                '401':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
                '500':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
                default:
                  $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    delete:
      summary: deletes the Events Subscription subresource
      operationId: DeleteEventsSubsc
      tags:
        - Events Subscription (Document)
      parameters:
        - name: appSessionId
          description: String identifying the Individual Application Session Context resource.
          in: path
          required: true
          schema:
            type: string
      responses:
        '204':
          description: >
            The deletion of the of the Events Subscription sub-resource is confirmed without
            returning additional data.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            npcf-policyauthorization: Access to the Npcf_PolicyAuthorization API
            npcf-policyauthorization:policy-auth-mgmt: >
              Access to service operations applying to PCF Policy Authorization for creation,
              updation, deletion, retrieval.

  schemas:
```

```
    AppSessionContext:
      description: Represents an Individual Application Session Context resource.
      type: object
      properties:
        ascReqData:
          $ref: '#/components/schemas/AppSessionContextReqData'
        ascRespData:
          $ref: '#/components/schemas/AppSessionContextRespData'
        evsNotif:
          $ref: '#/components/schemas/EventsNotification'

    AppSessionContextReqData:
      description: Identifies the service requirements of an Individual Application Session Context.
      type: object
      required:
        - notifUri
        - suppFeat
      oneOf:
        - required: [ueIpv4]
        - required: [ueIpv6]
        - required: [ueMac]
      properties:
        afAppId:
          $ref: '#/components/schemas/AfAppId'
        afChargId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationChargingId'
        afReqData:
          $ref: '#/components/schemas/AfRequestedData'
        afRoutReq:
          $ref: '#/components/schemas/AfRoutingRequirement'
        afSfcReq:
          $ref: '#/components/schemas/AfSfcRequirement'
        afHdrReq:
          $ref: '#/components/schemas/AfHeaderHandlingControlInfo'
        aspId:
          $ref: '#/components/schemas/AspId'
        bdtRefId:
          $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceId'
        dnn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        evSubsc:
          $ref: '#/components/schemas/EventsSubscReqData'
        mcpttId:
          description: Indication of MCPTT service request.
          type: string
        mcVideoId:
          description: Indication of MCVideo service request.
          type: string
        medComponents:
          type: object
          additionalProperties:
            $ref: '#/components/schemas/MediaComponent'
          minProperties: 1
          description: >
            Contains media component information. The key of the map is the medCompN attribute.
        multiModalId:
          $ref: '#/components/schemas/MultiModalId'
        ipDomain:
          type: string
        mpsAction:
          $ref: '#/components/schemas/MpsAction'
        mpsId:
          description: Indication of MPS service request.
          type: string
        mcsId:
          description: Indication of MCS service request.
          type: string
        preemptControlInfo:
          $ref: '#/components/schemas/PreemptionControlInformation'
        qosDuration:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
        qosInactInt:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
        resPrio:
          $ref: '#/components/schemas/ReservPriority'
        servInfStatus:
          $ref: '#/components/schemas/ServiceInfoStatus'
        notifUri:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      servUrn:
        $ref: '#/components/schemas/ServiceUrn'
      sliceInfo:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      sponId:
        $ref: '#/components/schemas/SponId'
      sponStatus:
        $ref: '#/components/schemas/SponsoringStatus'
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      gpsi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      ueIpv4:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
      ueIpv6:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'
      ueMac:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
      tsnBridgeManCont:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/BridgeManagementContainer'
      tsnPortManContDstt:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
      tsnPortManContNwtts:
        type: array
        items:
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
        minItems: 1
      tscNotifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      tscNotifCorreId:
        type: string
        description: >
          Correlation identifier for TSC management information notifications.

  AppSessionContextRespData:
    description: >
      Describes the authorization data of an Individual Application Session Context created by
      the PCF.
    type: object
    properties:
      servAuthInfo:
        $ref: '#/components/schemas/ServAuthInfo'
      directNotifReports:
        type: array
        items:
          $ref: '#/components/schemas/DirectNotificationReport'
        minItems: 1
        description: >
          QoS monitoring parameter(s) that cannot be directly notified for the indicated flows.
      ueIds:
        type: array
        items:
          $ref: '#/components/schemas/UeIdentityInfo'
        minItems: 1
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'

  AppSessionContextUpdateDataPatch:
    description: >
      Identifies the modifications to an Individual Application Session Context and/or the
      modifications to the sub-resource Events Subscription.
    type: object
    properties:
      ascReqData:
        $ref: '#/components/schemas/AppSessionContextUpdateData'

  AppSessionContextUpdateData:
    description: >
      Identifies the modifications to the "ascReqData" property of an Individual Application
      Session Context which may include the modifications to the sub-resource Events Subscription.
    type: object
    properties:
      afAppId:
        $ref: '#/components/schemas/AfAppId'
      afRoutReq:
```

```
        $ref: '#/components/schemas/AfRoutingRequirementRm'
      afSfcReq:
        $ref: '#/components/schemas/AfSfcRequirement'
      afHdrReq:
        $ref: '#/components/schemas/AfHeaderHandlingControlInfo'
      aspId:
        $ref: '#/components/schemas/AspId'
      bdtRefId:
        $ref: 'TS29122_CommonData.yaml#/components/schemas/BdtReferenceId'
      evSubsc:
        $ref: '#/components/schemas/EventsSubscReqDataRm'
      mcpttId:
        description: Indication of MCPTT service request.
        type: string
      mcVideoId:
        description: Indication of modification of MCVideo service.
        type: string
      medComponents:
        type: object
        additionalProperties:
          $ref: '#/components/schemas/MediaComponentRm'
        minProperties: 1
        description: >
          Contains media component information. The key of the map is the medCompN attribute.
      mpsAction:
        $ref: '#/components/schemas/MpsAction'
      mpsId:
        description: Indication of MPS service request.
        type: string
      mcsId:
        description: Indication of MCS service request.
        type: string
      preemptControlInfo:
        $ref: '#/components/schemas/PreemptionControlInformationRm'
      qosDuration:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSecRm'
      qosInactInt:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSecRm'
      resPrio:
        $ref: '#/components/schemas/ReservPriority'
      servInfStatus:
        $ref: '#/components/schemas/ServiceInfoStatus'
      sipForkInd:
        $ref: '#/components/schemas/SipForkingIndication'
      sponId:
        $ref: '#/components/schemas/SponId'
      sponStatus:
        $ref: '#/components/schemas/SponsoringStatus'
      tsnBridgeManCont:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/BridgeManagementContainer'
      tsnPortManContDstt:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
      tsnPortManContNwtts:
        type: array
        items:
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
        minItems: 1
      tscNotifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      tscNotifCorreId:
        type: string
        description: >
          Correlation identifier for TSC management information notifications.

  EventsSubscReqData:
    description: Identifies the events the application subscribes to.
    type: object
    required:
      - events
    properties:
      events:
        type: array
        items:
          $ref: '#/components/schemas/AfEventSubscription'
        minItems: 1
      notifUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      reqQosMonParams:
```

```
          type: array
          items:
            $ref:
'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/RequestedQosMonitoringParameter'
          minItems: 1
        qosMon:
          $ref: '#/components/schemas/QosMonitoringInformation'
        qosMonDatRate:
          $ref: '#/components/schemas/QosMonitoringInformation'
        pdvReqMonParams:
          type: array
          items:
            $ref:
'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/RequestedQosMonitoringParameter'
          minItems: 1
        pdvMon:
          $ref: '#/components/schemas/QosMonitoringInformation'
        congestMon:
          $ref: '#/components/schemas/QosMonitoringInformation'
        rttMon:
          $ref: '#/components/schemas/QosMonitoringInformation'
        avlBitRateMon:
          $ref: '#/components/schemas/QosMonitoringInformation'
        rttFlowRef:
          $ref: '#/components/schemas/RttFlowReference'
        reqAnis:
          type: array
          items:
            $ref: '#/components/schemas/RequiredAccessInfo'
          minItems: 1
        usgThres:
          $ref: 'TS29122_CommonData.yaml#/components/schemas/UsageThreshold'
        notifCorreId:
          type: string
        afAppIds:
          type: array
          items:
            $ref: '#/components/schemas/AfAppId'
          minItems: 1
        directNotifInd:
          type: boolean
          description: >
            Indicates whether the direct event notification is requested (true) or not (false) for
            the provided QoS monitoring parameters.
            Default value is false.
        avrgWndw:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AverWindow'

    EventsSubscReqDataRm:
      description: >
        This data type is defined in the same way as the EventsSubscReqData data type, but with
        the OpenAPI nullable property set to true.
      type: object
      required:
        - events
      properties:
        events:
          type: array
          items:
            $ref: '#/components/schemas/AfEventSubscription'
          minItems: 1
        notifUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        reqQosMonParams:
          type: array
          nullable: true
          items:
            $ref:
'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/RequestedQosMonitoringParameter'
          minItems: 1
        qosMon:
          $ref: '#/components/schemas/QosMonitoringInformationRm'
        qosMonDatRate:
          $ref: '#/components/schemas/QosMonitoringInformationRm'
        pdvReqMonParams:
          type: array
          nullable: true
          items:
```

```
            $ref:
'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/RequestedQosMonitoringParameter'
          minItems: 1
        pdvMon:
          $ref: '#/components/schemas/QosMonitoringInformationRm'
        congestMon:
          $ref: '#/components/schemas/QosMonitoringInformationRm'
        rttMon:
          $ref: '#/components/schemas/QosMonitoringInformationRm'
        rttFlowRef:
          $ref: '#/components/schemas/RttFlowReferenceRm'
        avlBitRateMon:
          $ref: '#/components/schemas/QosMonitoringInformationRm'
        reqAnis:
          type: array
          items:
            $ref: '#/components/schemas/RequiredAccessInfo'
          minItems: 1
        usgThres:
          $ref: 'TS29122_CommonData.yaml#/components/schemas/UsageThresholdRm'
        notifCorreId:
          type: string
        directNotifInd:
          type: boolean
          nullable: true
          description: >
            Indicates whether the direct event notification is requested (true) or not (false) for
            the provided and/or previously provided QoS monitoring parameters.
        avrgWndw:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AverWindowRm'
      nullable: true

    MediaComponent:
      description: Identifies a media component.
      type: object
      required:
        - medCompN
      allOf:
        - not:
            required: [altSerReqs,altSerReqsData]
        - not:
            required: [qosReference,altSerReqsData]
      properties:
        afAppId:
          $ref: '#/components/schemas/AfAppId'
        afRoutReq:
          $ref: '#/components/schemas/AfRoutingRequirement'
        afSfcReq:
          $ref: '#/components/schemas/AfSfcRequirement'
        afHdrReq:
          $ref: '#/components/schemas/AfHeaderHandlingControlInfo'
        qosReference:
          type: string
        disUeNotif:
          type: boolean
        altSerReqs:
          type: array
          items:
            type: string
          minItems: 1
        altSerReqsData:
          type: array
          items:
            $ref: '#/components/schemas/AlternativeServiceRequirementsData'
          minItems: 1
          description: >
            Contains alternative service requirements that include individual QoS parameter sets.
        contVer:
          $ref: '#/components/schemas/ContentVersion'
        codecs:
          type: array
          items:
            $ref: '#/components/schemas/CodecData'
          minItems: 1
          maxItems: 2
        desMaxLatency:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
        desMaxLoss:
```

```
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
    flusId:
      type: string
    fStatus:
      $ref: '#/components/schemas/FlowStatus'
    marBwDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    marBwUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    maxPacketLossRateDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketLossRateRm'
    maxPacketLossRateUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketLossRateRm'
    maxSuppBwDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    maxSuppBwUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    medCompN:
      type: integer
    medSubComps:
      type: object
      additionalProperties:
        $ref: '#/components/schemas/MediaSubComponent'
      minProperties: 1
      description: >
        Contains the requested bitrate and filters for the set of service data flows identified
        by their common flow identifier. The key of the map is the fNum attribute.
    medType:
      $ref: '#/components/schemas/MediaType'
    minDesBwDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    minDesBwUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    mirBwDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    mirBwUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    preemptCap:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PreemptionCapability'
    preemptVuln:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PreemptionVulnerability'
    prioSharingInd:
      $ref: '#/components/schemas/PrioritySharingIndicator'
    resPrio:
      $ref: '#/components/schemas/ReservPriority'
    rrBw:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    rsBw:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    sharingKeyDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint32'
    sharingKeyUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint32'
    tsnQos:
      $ref: '#/components/schemas/TsnQosContainer'
    tscaiInputDl:
      $ref: '#/components/schemas/TscaiInputContainer'
    tscaiInputUl:
      $ref: '#/components/schemas/TscaiInputContainer'
    tscaiTimeDom:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
    capBatAdaptation:
      type: boolean
      description: >
        Indicates the capability for AF to adjust the burst sending time, when it is supported
        and set to "true".
    rTLatencyInd:
      type: boolean
      description: >
        Indicates the service data flow needs to meet the Round-Trip (RT) latency requirement of
        the service, when it is included and set to "true".
    pdb:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketDelBudget'
    rTLatencyIndCorreId:
      $ref: '#/components/schemas/RttFlowReference'
    pduSetQosDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSetQosPara'
    pduSetQosUl:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSetQosPara'
      protoDescDl:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProtocolDescription'
      protoDescUl:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProtocolDescription'
      periodUl:
        $ref: '#/components/schemas/DurationMilliSec'
      periodDl:
        $ref: '#/components/schemas/DurationMilliSec'
      l4sInd:
        $ref: '#/components/schemas/UplinkDownlinkSupport'
      datBurstSizeInd:
        type: boolean
        description: >
          Indicates the Data Burst Size marking for the DL service data flow is supported if
          present and set to "true". The default value is "false" if omitted.
      timetoNextBurstInd:
        type: boolean
        description: >
          Indicates the Time to Next Burst for the DL service data flow is supported, when it is
          included and set to "true". The default value is "false" if omitted.
      onPathN6SigInfo:
        $ref: '#/components/schemas/OnPathN6SigInfo'
      expTranInd:
        type: boolean
        description: >
          Expedited Transfer Indication for the downlink traffic to enable expedited data transfer
          with reflective QoS for the Non-GBR service data flow. "true": the expedited data
          transfer of larger payload for XR application is enabled in the flow. "false": the
          expedited data transfer of larger payload for XR application is not enabled in the flow.
          The default value is "false" if omitted.
      ulBrRecInd:
        type: boolean
        description: >
          Indicates whether the RAN-controlled UL Bitrate Recommendation Indication is supported
          for the corresponding service data flow or not.
          - Set to "true": the RAN-controlled UL Bitrate Recommendation Indication is supported.
          - Set to "false": the RAN-controlled UL Bitrate Recommendation Indication is not
            supported
          - The default value "false" is used if this attribute is omitted.

  MediaComponentRm:
    description: >
      This data type is defined in the same way as the MediaComponent data type, but with the
      OpenAPI nullable property set to true.
    type: object
    required:
      - medCompN
    not:
      required: [altSerReqs,altSerReqsData]
    properties:
      afAppId:
        $ref: '#/components/schemas/AfAppId'
      afRoutReq:
        $ref: '#/components/schemas/AfRoutingRequirementRm'
      afSfcReq:
        $ref: '#/components/schemas/AfSfcRequirement'
      afHdrReq:
        $ref: '#/components/schemas/AfHeaderHandlingControlInfo'
      qosReference:
        type: string
        nullable: true
      altSerReqs:
        type: array
        items:
          type: string
        minItems: 1
        nullable: true
      altSerReqsData:
        type: array
        items:
          $ref: '#/components/schemas/AlternativeServiceRequirementsData'
        minItems: 1
        description: >
          Contains removable alternative service requirements that include individual QoS
          parameter sets.
        nullable: true
      disUeNotif:
```

```
    type: boolean
  contVer:
    $ref: '#/components/schemas/ContentVersion'
  codecs:
    type: array
    items:
      $ref: '#/components/schemas/CodecData'
    minItems: 1
    maxItems: 2
  desMaxLatency:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/FloatRm'
  desMaxLoss:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/FloatRm'
  flusId:
    type: string
    nullable: true
  fStatus:
    $ref: '#/components/schemas/FlowStatus'
  marBwDl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  marBwUl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  maxPacketLossRateDl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketLossRateRm'
  maxPacketLossRateUl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketLossRateRm'
  maxSuppBwDl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  maxSuppBwUl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  medCompN:
    type: integer
  medSubComps:
    type: object
    additionalProperties:
      $ref: '#/components/schemas/MediaSubComponentRm'
    minProperties: 1
    description: >
      Contains the requested bitrate and filters for the set of service data flows identified
      by their common flow identifier. The key of the map is the fNum attribute.
  medType:
    $ref: '#/components/schemas/MediaType'
  minDesBwDl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  minDesBwUl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  mirBwDl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  mirBwUl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  preemptCap:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/PreemptionCapabilityRm'
  preemptVuln:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/PreemptionVulnerabilityRm'
  prioSharingInd:
    $ref: '#/components/schemas/PrioritySharingIndicator'
  resPrio:
    $ref: '#/components/schemas/ReservPriority'
  rrBw:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  rsBw:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
  sharingKeyDl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint32Rm'
  sharingKeyUl:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint32Rm'
  tsnQos:
    $ref: '#/components/schemas/TsnQosContainerRm'
  tscaiInputDl:
    $ref: '#/components/schemas/TscaiInputContainer'
  tscaiInputUl:
    $ref: '#/components/schemas/TscaiInputContainer'
  tscaiTimeDom:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
  capBatAdaptation:
    type: boolean
    description: >
      Indicates the capability for AF to adjust the burst sending time, when it is supported
```

```
          and set to "true". The default value is "false" if omitted.
        rTLatencyInd:
          type: boolean
          nullable: true
          description: >
            Indicates the service data flow needs to meet the Round-Trip (RT) latency requirement of
            the service, when it is included and set to "true". The default value is "false" if
            omitted.
        pdb:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketDelBudgetRm'
        rTLatencyIndCorreId:
          $ref: '#/components/schemas/RttFlowReferenceRm'
        pduSetQosDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSetQosParaRm'
        pduSetQosUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSetQosParaRm'
        protoDescDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ProtocolDescriptionRm'
        protoDescUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ProtocolDescriptionRm'
        periodUl:
          $ref: '#/components/schemas/DurationMilliSecRm'
        periodDl:
          $ref: '#/components/schemas/DurationMilliSecRm'
        l4sInd:
          $ref: '#/components/schemas/UplinkDownlinkSupport'
        datBurstSizeInd:
          type: boolean
          nullable: true
          description: >
            Indicates the Data Burst Size marking for the DL service data flow is supported if
            present and set to "true".
        timetoNextBurstInd:
          type: boolean
          nullable: true
          description: >
            Indicates the Time to Next Burst for the DL service data flow is supported, when it is
            included and set to "true".
        onPathN6SigInfo:
          $ref: '#/components/schemas/OnPathN6SigInfo'
        expTranInd:
          type: boolean
          nullable: true
          description: >
            Expedited Transfer Indication for the downlink traffic to enable expedited data transfer
            with reflective QoS for the Non-GBR service data flow. "true": the expedited data
            transfer of larger payload for XR application is enabled in the flow. "false":
            the expedited data transfer of larger payload for XR application is not enabled in the
            flow.
        ulBrRecInd:
          type: boolean
          nullable: true
          description: >
            Indicates whether the RAN-controlled UL Bitrate Recommendation indication is supported
            for the corresponding service data flow or not.
            - Set to "true": the RAN-controlled UL Bitrate Recommendation Indication is supported.
            - Set to "false": the RAN-controlled UL Bitrate Recommendation Indication is not
              supported.
      nullable: true

    MediaSubComponent:
      description: Identifies a media subcomponent.
      type: object
      required:
        - fNum
      properties:
        afSigProtocol:
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/AfSigProtocol'
        ethfDescs:
          type: array
          items:
            $ref: '#/components/schemas/EthFlowDescription'
          minItems: 1
          maxItems: 2
        fNum:
          type: integer
        fDescs:
          type: array
```

```
        items:
          $ref: '#/components/schemas/FlowDescription'
        minItems: 1
        maxItems: 2
      addInfoFlowDescs:
        type: array
        items:
          $ref: '#/components/schemas/AddFlowDescriptionInfo'
        minItems: 1
        maxItems: 2
        description: >
          Represents additional flow description information (flow label and IPsec SPI)
          per Uplink and/or Downlink IP flows.
      fStatus:
        $ref: '#/components/schemas/FlowStatus'
      marBwDl:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
      marBwUl:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
      tosTrCl:
        $ref: '#/components/schemas/TosTrafficClass'
      flowUsage:
        $ref: '#/components/schemas/FlowUsage'
      evSubsc:
        $ref: '#/components/schemas/EventsSubscReqData'
      mpxMediaUlInfos:
        type: array
        items:
          $ref: '#/components/schemas/MpxMediaInfo'
        minItems: 1
        description: (S)RTP Multiplexed media identification information for the Uplink IP flow.
      mpxMediaDlInfos:
        type: array
        items:
          $ref: '#/components/schemas/MpxMediaInfo'
        minItems: 1
        description: (S)RTP Multiplexed media identification information for the Downlink IP flow.

  MediaSubComponentRm:
    description: >
      This data type is defined in the same way as the MediaSubComponent data type, but with the
      OpenAPI nullable property set to true. Removable attributes marBwDl and marBwUl are defined
      with the corresponding removable data type.
    type: object
    required:
      - fNum
    properties:
      afSigProtocol:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/AfSigProtocol'
      ethfDescs:
        type: array
        items:
          $ref: '#/components/schemas/EthFlowDescription'
        minItems: 1
        maxItems: 2
        nullable: true
      fNum:
        type: integer
      fDescs:
        type: array
        items:
          $ref: '#/components/schemas/FlowDescription'
        minItems: 1
        maxItems: 2
        nullable: true
      addInfoFlowDescs:
        type: array
        items:
          $ref: '#/components/schemas/AddFlowDescriptionInfo'
        minItems: 1
        maxItems: 2
        nullable: true
        description: >
          Represents additional flow description information (flow label and IPsec SPI)
          per Uplink and/or Downlink IP flows.
      fStatus:
        $ref: '#/components/schemas/FlowStatus'
      marBwDl:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
      marBwUl:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
      tosTrCl:
        $ref: '#/components/schemas/TosTrafficClassRm'
      flowUsage:
        $ref: '#/components/schemas/FlowUsage'
      evSubsc:
        $ref: '#/components/schemas/EventsSubscReqDataRm'
      mpxMediaUlInfos:
        type: array
        items:
          $ref: '#/components/schemas/MpxMediaInfo'
        minItems: 1
        description: (S)RTP Multiplexed media identification information for the Uplink IP flow.
        nullable: true
      mpxMediaDlInfos:
        type: array
        items:
          $ref: '#/components/schemas/MpxMediaInfo'
        minItems: 1
        description: (S)RTP Multiplexed media identification information for the Downlink IP flow.
        nullable: true
    nullable: true

  EventsNotification:
    description: Describes the notification of a matched event.
    type: object
    required:
      - evSubsUri
      - evNotifs
    properties:
      adReports:
        type: array
        items:
          $ref: '#/components/schemas/AppDetectionReport'
        minItems: 1
        description: Includes the detected application report.
      accessType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      addAccessInfo:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/AdditionalAccessInfo'
      relAccessInfo:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/AdditionalAccessInfo'
      anChargAddr:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/AccNetChargingAddress'
      anChargIds:
        type: array
        items:
          $ref: '#/components/schemas/AccessNetChargingIdentifier'
        minItems: 1
      anGwAddr:
        $ref: '#/components/schemas/AnGwAddress'
      l4sReports:
        type: array
        items:
          $ref: '#/components/schemas/L4sSupport'
        minItems: 1
      evSubsUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      evNotifs:
        type: array
        items:
          $ref: '#/components/schemas/AfEventNotification'
        minItems: 1
      failedResourcAllocReports:
        type: array
        items:
          $ref: '#/components/schemas/ResourcesAllocationInfo'
        minItems: 1
      succResourcAllocReports:
        type: array
        items:
          $ref: '#/components/schemas/ResourcesAllocationInfo'
        minItems: 1
      noNetLocSupp:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/NetLocAccessSupport'
      outOfCredReports:
```

```
    type: array
    items:
      $ref: '#/components/schemas/OutOfCreditInformation'
    minItems: 1
  plmnId:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnIdNid'
  qncReports:
    type: array
    items:
      $ref: '#/components/schemas/QosNotificationControlInfo'
    minItems: 1
  qosMonReports:
    type: array
    items:
      $ref: '#/components/schemas/QosMonitoringReport'
    minItems: 1
  qosMonDatRateReps:
    type: array
    items:
      $ref: '#/components/schemas/QosMonitoringReport'
    minItems: 1
  pdvMonReports:
    type: array
    items:
      $ref: '#/components/schemas/PdvMonitoringReport'
    minItems: 1
  congestReports:
    type: array
    items:
      $ref: '#/components/schemas/QosMonitoringReport'
    minItems: 1
  rttMonReports:
    type: array
    items:
      $ref: '#/components/schemas/QosMonitoringReport'
    minItems: 1
  qosMonCapRepos:
    type: object
    additionalProperties:
      $ref: '#/components/schemas/CapabilityReport'
    minProperties: 1
    description: >
      Contains the QoS monitoring is supported or not. It shall be present when
      the notified event is "QOS_MON_CAP_REPO". The key of the map is the attribute
      "capType".
  ranNasRelCauses:
    type: array
    items:
      $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/RanNasRelCause'
    minItems: 1
    description: Contains the RAN and/or NAS release cause.
  ratType:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
  satBackhaulCategory:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/SatelliteBackhaulCategory'
  ueLoc:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
  ueLocTime:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
  ueTimeZone:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
  usgRep:
    $ref: 'TS29122_CommonData.yaml#/components/schemas/AccumulatedUsage'
  urspEnfRep:
    $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/UrspEnforcementInfo'
  sscMode:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/SscMode'
  ueReqDnn:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
  ueReqPduSessionType:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionType'
  tsnBridgeManCont:
    $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/BridgeManagementContainer'
  tsnPortManContDstt:
    $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
  tsnPortManContNwtts:
    type: array
    items:
```

```
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
        minItems: 1
      ipv4AddrList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4AddrMask'
        minItems: 1
      ipv6PrefixList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
        minItems: 1
      batOffsetInfo:
        $ref: '#/components/schemas/BatOffsetInfo'
      ueReachStatus:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/UeReachabilityStatus'
      retryAfter:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      servSatId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SatelliteId'
      rateLimitRepo:
        $ref: '#/components/schemas/RateLimitRepo'
      dnaiChgType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DnaiChangeType'
      sourceDnai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnai'
      targetDnai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnai'
      sessionRecovStatus:
          $ref: '#/components/schemas/PcSessionRecoveryStatus'

  AfEventSubscription:
    description: Describes the event information delivered in the subscription.
    type: object
    required:
      - event
    properties:
      event:
        $ref: '#/components/schemas/AfEvent'
      notifMethod:
        $ref: '#/components/schemas/AfNotifMethod'
      repPeriod:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      waitTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      qosMonParamType:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/QosMonitoringParamType'
      capTypes:
        type: array
        items:
          $ref: '#/components/schemas/NotifCapType'
        minItems: 1
        description: >
          Contains the type(s) of QoS Monitoring capability report is applied.

  AfEventNotification:
    description: Describes the event information delivered in the notification.
    type: object
    required:
      - event
    properties:
      event:
        $ref: '#/components/schemas/AfEvent'
      flows:
        type: array
        items:
          $ref: '#/components/schemas/Flows'
        minItems: 1
      retryAfter:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'

  TerminationInfo:
    description: >
      Indicates the cause for requesting the deletion of the Individual Application Session
      Context resource.
    type: object
    required:
      - termCause
```

```
      - resUri
    properties:
      termCause:
        $ref: '#/components/schemas/TerminationCause'
      resUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'

  AfRoutingRequirement:
    description: Describes AF requirements on routing traffic.
    type: object
    properties:
      appReloc:
        type: boolean
      routeToLocs:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RouteToLocation'
        minItems: 1
      spVal:
        $ref: '#/components/schemas/SpatialValidity'
      tempVals:
        type: array
        items:
          $ref: '#/components/schemas/TemporalValidity'
        minItems: 1
      upPathChgSub:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/UpPathChgEvent'
      outcomeSub:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/TraffRouteReqOutcomeEvent'
      simConnFailSub:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/SimConnFailEvent'
      addrPreserInd:
        type: boolean
      simConnInd:
        type: boolean
        description: >
          Indicates whether simultaneous connectivity should be temporarily maintained for the
          source and target PSA.
      simConnTerm:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      easIpReplaceInfos:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/EasIpReplacementInfo'
        minItems: 1
        description: Contains EAS IP replacement information.
      easRedisInd:
        type: boolean
        description: Indicates the EAS rediscovery is required.
      maxAllowedUpLat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      tfcCorreInfo:
        $ref: 'TS29519_Application_Data.yaml#/components/schemas/TrafficCorrelationInfo'
      candDnaiInd:
        type: boolean
        description: Indicates whether candidate DNAI(s) are requested to be reported.
      n6DelayInd:
        type: boolean
        description: >
          Indication of whether the N6 delay measurement is requested to be considered or not.
          The N6 delay measurement is requested to be considered if it is set to true.
          The N6 delay measurement is not requested to be considered if it is set to false.
          The default value is false.

  AfSfcRequirement:
    description: Describes AF requirements on steering traffic to N6-LAN.
    type: object
    properties:
      sfcIdDl:
        type: string
        description: Reference to a pre-configured SFC for downlink traffic.
        nullable: true
      sfcIdUl:
        type: string
        description: Reference to a pre-configured SFC for uplink traffic.
        nullable: true
      spVal:
        $ref: '#/components/schemas/SpatialValidityRm'
```

```
      metadata:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Metadata'
    nullable: true

  SpatialValidity:
    description: Describes explicitly the route to an Application location.
    type: object
    required:
      - presenceInfoList
    properties:
      presenceInfoList:
        type: object
        additionalProperties:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
        minProperties: 1
        description: >
          Defines the presence information provisioned by the AF. The praId attribute within the
          PresenceInfo data type is the key of the map.

  SpatialValidityRm:
    description: >
      This data type is defined in the same way as the SpatialValidity data type, but with the
      OpenAPI nullable property set to true.
    type: object
    required:
      - presenceInfoList
    properties:
      presenceInfoList:
        type: object
        additionalProperties:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
        minProperties: 1
        description: >
          Defines the presence information provisioned by the AF. The praId attribute within the
          PresenceInfo data type is the key of the map.
    nullable: true

  AfRoutingRequirementRm:
    description: >
      This data type is defined in the same way as the AfRoutingRequirement data type, but with
      the OpenAPI nullable property set to true and the spVal and tempVals attributes defined as
      removable.
    type: object
    properties:
      appReloc:
        type: boolean
      routeToLocs:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RouteToLocation'
        minItems: 1
        nullable: true
      spVal:
        $ref: '#/components/schemas/SpatialValidityRm'
      tempVals:
        type: array
        items:
          $ref: '#/components/schemas/TemporalValidity'
        minItems: 1
        nullable: true
      upPathChgSub:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/UpPathChgEvent'
      simConnFailSub:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/SimConnFailEvent'
      addrPreserInd:
        type: boolean
        nullable: true
      simConnInd:
        type: boolean
        nullable: true
        description: >
          Indicates whether simultaneous connectivity should be temporarily maintained for the
          source and target PSA.
      simConnTerm:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSecRm'
      easIpReplaceInfos:
        type: array
        items:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/EasIpReplacementInfo'
        minItems: 1
        description: Contains EAS IP replacement information.
        nullable: true
      easRedisInd:
        type: boolean
        description: Indicates the EAS rediscovery is required.
      maxAllowedUpLat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/UintegerRm'
      tfcCorreInfo:
        $ref: 'TS29519_Application_Data.yaml#/components/schemas/TrafficCorrelationInfo'
      candDnaiInd:
        type: boolean
        description: Indicates whether candidate DNAI(s) are requested to be reported.
      n6DelayInd:
        type: boolean
        description: >
          Indication of whether the N6 delay measurement is requested to be considered or not.
          The N6 delay measurement is requested to be considered if it is set to true.
          The N6 delay measurement is not requested to be considered if it is set to false.
        nullable: true
    nullable: true

  AnGwAddress:
    description: Describes the address of the access network gateway control node.
    type: object
    anyOf:
      - required: [anGwIpv4Addr]
      - required: [anGwIpv6Addr]
    properties:
      anGwIpv4Addr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
      anGwIpv6Addr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'

  Flows:
    description: Identifies the flows.
    type: object
    required:
      - medCompN
    properties:
      contVers:
        type: array
        items:
          $ref: '#/components/schemas/ContentVersion'
        minItems: 1
      fNums:
        type: array
        items:
          type: integer
        minItems: 1
      medCompN:
        type: integer

  EthFlowDescription:
    description: Identifies an Ethernet flow.
    type: object
    required:
      - ethType
    properties:
      destMacAddr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
      ethType:
        type: string
      fDesc:
        $ref: '#/components/schemas/FlowDescription'
      fDir:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/FlowDirection'
      sourceMacAddr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
      vlanTags:
        type: array
        items:
          type: string
        minItems: 1
        maxItems: 2
      srcMacAddrEnd:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
```

```
    destMacAddrEnd:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'

ResourcesAllocationInfo:
  description: Describes the status of the PCC rule(s) related to certain media components.
  type: object
  properties:
    mcResourcStatus:
      $ref: '#/components/schemas/MediaComponentResourcesStatus'
    flows:
      type: array
      items:
        $ref: '#/components/schemas/Flows'
      minItems: 1
    altSerReq:
      type: string
      description: >
        Indicates whether NG-RAN supports alternative QoS parameters. The default value false
        shall apply if the attribute is not present. It shall be set to false to indicate that
        the lowest priority alternative QoS profile could not be fulfilled.

TemporalValidity:
  description: Indicates the time interval(s) during which the AF request is to be applied.
  type: object
  properties:
    startTime:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    stopTime:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'

QosNotificationControlInfo:
  description: >
    Indicates whether the QoS targets for a GRB flow are not guaranteed or guaranteed again.
  type: object
  required:
    - notifType
  properties:
    notifType:
      $ref: '#/components/schemas/QosNotifType'
    flows:
      type: array
      items:
        $ref: '#/components/schemas/Flows'
      minItems: 1
    altSerReq:
      type: string
      description: >
        Indicates the alternative service requirement NG-RAN can guarantee. When it is omitted
        and the notifType attribute is set to NOT_GUAARANTEED it indicates that the lowest
        priority alternative alternative service requirement could not be fulfilled by NG-RAN.
    altSerReqNotSuppInd:
      type: boolean
      description: >
        When present and set to true it indicates that Alternative Service Requirements are not
        supported by NG-RAN.

AcceptableServiceInfo:
  description: Indicates the maximum bandwidth that shall be authorized by the PCF.
  type: object
  properties:
    accBwMedComps:
      type: object
      additionalProperties:
        $ref: '#/components/schemas/MediaComponent'
      description: >
        Indicates the maximum QoS parameters that shall be authorized by the PCF for
        each media component of the map. The key of the map is the media component number.
      minProperties: 1
    marBwUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    marBwDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    mirBwUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    mirBwDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
    tsnQos:
      $ref: '#/components/schemas/TsnQosContainer'
```

```
      addAccQosCombs:
        type: array
        items:
          $ref: '#/components/schemas/AcceptableServiceInfo'
        minItems: 1
        description: Contains additional combinations of acceptable service information.

    UeIdentityInfo:
      description: Represents 5GS-Level UE identities.
      type: object
      anyOf:
        - required: [gpsi]
        - required: [pei]
        - required: [supi]
      properties:
        gpsi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
        pei:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'

    AccessNetChargingIdentifier:
      description: Describes the access network charging identifier.
      type: object
      oneOf:
        - required: [accNetChaIdValue]
        - required: [accNetChargIdString]
      properties:
        accNetChaIdValue:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ChargingId'
        accNetChargIdString:
          type: string
          description: A character string containing the access network charging identifier.
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1

    OutOfCreditInformation:
      description: >
        Indicates the SDFs without available credit and the corresponding termination action.
      type: object
      required:
        - finUnitAct
      properties:
        finUnitAct:
          $ref: 'TS32291_Nchf_ConvergedCharging.yaml#/components/schemas/FinalUnitAction'
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1

    QosMonitoringInformation:
      description: >
        Indicates the QoS Monitoring information to report, i.e. UL and/or DL and or
        round trip delay.
      type: object
      properties:
        repThreshDl:
          type: integer
        repThreshUl:
          type: integer
        repThreshRp:
          type: integer
        repThreshDatRateUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        repThreshDatRateDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        conThreshDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        conThreshUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        avlBitrateUlThrs:
          type: array
          items:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
      minItems: 1
      description: Indicates a list of thresholds for uplink available bitrate reporting.
    avlBitrateDlThrs:
      type: array
      items:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
      minItems: 1
      description: Indicates a list of thresholds for downlink available bitrate reporting.

PduSessionTsnBridge:
  description: >
    Contains the new TSC user plane node information and may contain the DS-TT port and/or
    NW-TT port management information.
  type: object
  required:
    - tsnBridgeInfo
  properties:
    tsnBridgeInfo:
      $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/TsnBridgeInfo'
    tsnBridgeManCont:
      $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/BridgeManagementContainer'
    tsnPortManContDstt:
      $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
    tsnPortManContNwtts:
      type: array
      items:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/PortManagementContainer'
      minItems: 1
    ueIpv4Addr:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
    dnn:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
    snssai:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
    ipDomain:
      type: string
      description: IPv4 address domain identifier.
    ueIpv6AddrPrefix:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'

QosMonitoringInformationRm:
  description: >
    This data type is defined in the same way as the QosMonitoringInformation data type, but
    with the OpenAPI nullable property set to true.
  type: object
  properties:
    repThreshDl:
      type: integer
      nullable: true
    repThreshUl:
      type: integer
      nullable: true
    repThreshRp:
      type: integer
      nullable: true
    repThreshDatRateUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
    repThreshDatRateDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
    conThreshDl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/UintegerRm'
    conThreshUl:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/UintegerRm'
    avlBitrateUlThrs:
      type: array
      items:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
      minItems: 1
    avlBitrateDlThrs:
      type: array
      items:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRateRm'
      minItems: 1
  nullable: true

PcscfRestorationRequestData:
  description: Indicates P-CSCF restoration.
```

```
      type: object
      oneOf:
        - required: [ueIpv4]
        - required: [ueIpv6]
      properties:
        dnn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        ipDomain:
          type: string
        sliceInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        ueIpv4:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
        ueIpv6:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'

    QosMonitoringReport:
      description: QoS Monitoring reporting information.
      type: object
      properties:
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1
        ulDelays:
          type: array
          items:
            type: integer
          minItems: 1
        dlDelays:
          type: array
          items:
            type: integer
          minItems: 1
        rtDelays:
          type: array
          items:
            type: integer
          minItems: 1
        pdmf:
          type: boolean
          description: Represents the packet delay measurement failure indicator.
        ulConInfo:
          type: array
          items:
            type: integer
          minItems: 1
        dlConInfo:
          type: array
          items:
            type: integer
          minItems: 1
        ulDataRate:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        dlDataRate:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'

    TsnQosContainer:
      description: Indicates TSC Traffic QoS.
      type: object
      properties:
        maxTscBurstSize:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ExtMaxDataBurstVol'
        maxTscBurstSizeSmall:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MaxDataBurstVol'
        tscPackDelay:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketDelBudget'
        maxPer:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketErrRate'
        tscPrioLevel:
          $ref: '#/components/schemas/TscPriorityLevel'
      not:
        required: [maxTscBurstSize, maxTscBurstSizeSmall]

    TsnQosContainerRm:
```

```
      description: Indicates removable TSC Traffic QoS.
      type: object
      properties:
        maxTscBurstSize:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ExtMaxDataBurstVolRm'
        maxTscBurstSizeSmall:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MaxDataBurstVolRm'
        tscPackDelay:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketDelBudgetRm'
        maxPer:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketErrRateRm'
        tscPrioLevel:
          $ref: '#/components/schemas/TscPriorityLevelRm'
      nullable: true
      not:
        required: [maxTscBurstSize, maxTscBurstSizeSmall]

    TscaiInputContainer:
      description: Indicates TSC Traffic pattern.
      type: object
      properties:
        periodicity:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        burstArrivalTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        surTimeInNumMsg:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        surTimeInTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        burstArrivalTimeWnd:
          $ref: 'TS29122_CommonData.yaml#/components/schemas/TimeWindow'
        periodicityRange:
          $ref: '#/components/schemas/PeriodicityRange'
      nullable: true

    AppDetectionReport:
      description: >
        Indicates the start or stop of the detected application traffic and the application
        identifier of the detected application traffic.
      type: object
      required:
        - adNotifType
        - afAppId
      properties:
        adNotifType:
          $ref: '#/components/schemas/AppDetectionNotifType'
        afAppId:
          $ref: '#/components/schemas/AfAppId'

    PduSessionEventNotification:
      description: >
        Indicates PDU session related events information.
      type: object
      required:
        - evNotif
      properties:
        evNotif:
          $ref: '#/components/schemas/AfEventNotification'
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        ueIpv4:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
        ueIpv6:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'
        ueMac:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
        status:
          $ref: '#/components/schemas/PduSessionStatus'
        pcfInfo:
          $ref: '#/components/schemas/PcfAddressingInfo'
        dnn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        snssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        gpsi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'

    PcfAddressingInfo:
```

```
      description: Contains PCF address information.
      type: object
      properties:
        pcfFqdn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Fqdn'
        pcfIpEndPoints:
          type: array
          items:
            $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/IpEndPoint'
          minItems: 1
          description: IP end points of the PCF hosting the Npcf_PolicyAuthorization service.
        bindingInfo:
          type: string
          description: contains the binding indications of the PCF.

    AlternativeServiceRequirementsData:
      description: Contains an alternative QoS related parameter set.
      type: object
      required:
        - altQosParamSetRef
      properties:
        altQosParamSetRef:
          type: string
          description: Reference to this alternative QoS related parameter set.
        gbrUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        gbrDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        pdb:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketDelBudget'
        per:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PacketErrRate'
        averWindow:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AverWindow'
        maxDataBurstVol:
          $ref: '#/components/schemas/MaxDataBurstVol'
        pduSetQosDl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSetQosParaRm'
        pduSetQosUl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSetQosParaRm'

    EventsSubscPutData:
      description: >
        Identifies the events the application subscribes to within an Events Subscription
        sub-resource data. It may contain the notification of the already met events.
      anyOf:
        - $ref: '#/components/schemas/EventsSubscReqData'
        - $ref: '#/components/schemas/EventsNotification'

    PeriodicityRange:
      description: >
        Contains the acceptable range (which is formulated as lower bound and upper bound of
        the periodicity of the start twobursts in reference to the external GM) or
        acceptable periodicity value(s) (which is formulated as a list of values for
        the periodicity).
      type: object
      oneOf:
        - required: [lowerBound, upperBound]
        - required: [periodicVals]
      properties:
        lowerBound:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        upperBound:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        periodicVals:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
          minItems: 1
        addPeriodicVals:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
          minItems: 1

    BatOffsetInfo:
      description: >
        Indicates the offset of the BAT and the optionally adjusted periodicity.
```

```
      type: object
      required:
        - ranBatOffsetNotif
      properties:
        ranBatOffsetNotif:
          type: integer
          description: >
            Indicates the BAT offset of the arrival time of the data burst in units
            of milliseconds.
        adjPeriod:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1
          description: >
            Identification of the flows. If no flows are provided, the BAT offset applies
            for all flows of the AF session.

    PdvMonitoringReport:
      description: Packet Delay Variation reporting information.
      type: object
      properties:
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1
          description: Identification of the flows.
        ulPdv:
          type: integer
          description: Uplink packet delay variation in units of milliseconds.
        dlPdv:
          type: integer
          description: Downlink packet delay variation in units of milliseconds.
        rtPdv:
          type: integer
          description: Round trip packet delay variation in units of milliseconds.

    AddFlowDescriptionInfo:
      description: Contains additional flow description information.
      type: object
      properties:
        spi:
          type: string
          description: >
            4-octet string representing the security parameter index of the IPSec packet
            in hexadecimal representation.
        flowLabel:
          type: string
          description: >
            3-octet string representing the IPv6 flow label header field in hexadecimal
            representation.
        flowDir:
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/FlowDirection'

    L4sSupport:
      description: >
        Indicates whether the ECN marking for L4S support is not available or available
        again in 5GS.
      type: object
      required:
        - notifType
      properties:
        notifType:
          $ref: '#/components/schemas/L4sNotifType'
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1

    DirectNotificationReport:
      description: >
        Represents the QoS monitoring parameters that cannot be directly notified for
        the indicated flows.
      type: object
```

```
      required:
        - qosMonParamType
      properties:
        qosMonParamType:
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/QosMonitoringParamType'
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1

    RttFlowReference:
      description: >
        Contains the shared key with the media subcomponent that shares the subscription to
        round trip time measurements in the complementary direction.
      type: object
      required:
        - sharedKey
      properties:
        flowDir:
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/FlowDirection'
        sharedKey:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint32'

    RttFlowReferenceRm:
      description: >
        It is defined as the RttFlowRerence data type but with the OpenAPI nullable true property.
      type: object
      required:
        - sharedKey
      properties:
        flowDir:
          $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/FlowDirection'
        sharedKey:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint32'
      nullable: true

    CapabilityReport:
      description: Contains capability support information.
      type: object
      properties:
        capReport:
          $ref: '#/components/schemas/NotifCap'
        capType:
          $ref: '#/components/schemas/NotifCapType'
      required:
        - capReport
        - capType

    AfHeaderHandlingControlInfo:
      description: Describes AF requirements on handling of payload headers.
      type: object
      required:
        - hDetectionReference
      properties:
        hDetectionReference:
          description: Indication of header detection reference.
          type: string
        hDetectionSuppInfo:
          description: Indication of transparent dynamic information.
          type: string
          nullable: true
        notifUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UriRm'
        notifId:
          type: string
          nullable: true
        spVal:
          $ref: '#/components/schemas/SpatialValidityRm'
        tempVals:
          type: array
          items:
            $ref: '#/components/schemas/TemporalValidity'
          minItems: 1
          nullable: true
        hHndlgUl:
          type: array
          items:
```

```
          $ref: '#/components/schemas/HeaderHandlingActionRequest'
        minItems: 1
        nullable: true
        description: >
          Contains the list of header handling action request parameters in the uplink
          direction.
      hHndlgDl:
        type: array
        items:
          $ref: '#/components/schemas/HeaderHandlingActionRequest'
        minItems: 1
        nullable: true
        description: >
          Contains the list of header handling action request parameters in the downlink
          direction.
    nullable: true

  HeaderHandlingActionRequest:
    description: >
      Represents the header handling action request.
    properties:
      hHndlgCtrlRef:
          type: string
      hHndlgAction:
        type: array
        items:
          $ref: '#/components/schemas/HeaderHandlingAction'
        minItems: 1
      hInfo:
        type: string
      hVal:
        type: string
      hHndlgCond:
          $ref: '#/components/schemas/HeaderHandlingCond'
      hHndlgRep:
          $ref: '#/components/schemas/HeaderHandlingReporting'

  OnPathN6SigInfo:
    description: >
      Represents the on path N6 signaling information.
    nullable: true
    properties:
      onPathN6Method:
        $ref: '#/components/schemas/OnPathN6Method'
      asProxyAddr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/IpAddr'
    required:
      - onPathN6Method

  HeaderHandlingReporting:
    description: >
      Indicates that reporting is requested for the performed Header Handling Action.
    properties:
      notifFlag:
        type: boolean
        description: >
          Indicates whether reporting is requested for the performed Header Handling Action.
          True indicates a reporting is requested.
          False indicates a reporting is not requested.
      repSuggInfo:
        $ref: 'TS29564_Nupf_EventExposure.yaml#/components/schemas/ReportingSuggestionInformation'
      oneTimeInd:
        type: boolean
        description: >
          Indicates whether the reporting of a first occurrence of the action per packet flow
          is enough.
          True indicates that the reporting applies to the first occurrence.
          False indicates that the reporting applies to all occurrences.

  RateLimitRepo:
    description: >
      Contains the rate limit information for the non-GRB flows.
    properties:
      rateLimitRepoUl:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        minItems: 1
```

```
          description: >
            Indicates the maximum uplink data rate authorized for the non-GBR service data flow(s)
            as indicated in the attribute "flows". If no flows are provided, the maximum data rate
            applies for all the flows.
        rateLimitRepoDl:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
          minItems: 1
          description: >
            Indicates the maximum downlink data rate authorized for the non-GBR service data flow(s)
            as indicated in the attribute "flows". If no flows are provided, the maximum data rate
            applies for all the flows.
        flows:
          type: array
          items:
            $ref: '#/components/schemas/Flows'
          minItems: 1
          description: Identifications of the non-GBR service data flows.

    MpxMediaInfo:
      description: Contains the (S)RTP Multiplexed Media Identification Information.
      type: object
      properties:
        ssrcId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint32'
        payloadType:
          type: integer
          minimum: 0
          maximum: 127
        identificationTag:
          type: string
        rtpSdesHdrExtId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        rtcpPacketType:
          type: integer
          minimum: 0
          maximum: 255
      anyOf:
        - required: [ ssrcId ]
        - required: [ payloadType ]
        - required: [ identificationTag ]
        - required: [ rtcpPacketType ]
        - required: [ identificationTag, rtpSdesHdrExtId ]


#
# EXTENDED PROBLEMDETAILS
#
    ExtendedProblemDetails:
      description: Extends ProblemDetails to also include the acceptable service info.
      allOf:
        - $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        - type: object
          properties:
            acceptableServInfo:
              $ref: '#/components/schemas/AcceptableServiceInfo'

#
# SIMPLE DATA TYPES
#
    AfAppId:
      description: Contains an AF application identifier.
      type: string
    AspId:
      description: Contains an identity of an application service provider.
      type: string
    CodecData:
      description: Contains codec related information.
      type: string
    ContentVersion:
      description: Represents the content version of some content.
      type: integer
    FlowDescription:
      description: Defines a packet filter of an IP flow.
      type: string
    SponId:
      description: Contains an identity of a sponsor.
```

```
      type: string
    ServiceUrn:
      description: Contains values of the service URN and may include subservices.
      type: string
    TosTrafficClass:
      description: >
        2-octet string, where each octet is encoded in hexadecimal representation. The first octet
        contains the IPv4 Type-of-Service or the IPv6 Traffic-Class field and the second octet
        contains the ToS/Traffic Class mask field.
      type: string
    TosTrafficClassRm:
      description: >
        This data type is defined in the same way as the TosTrafficClass data type, but with the
        OpenAPI nullable property set to true.
      type: string
      nullable: true
    MultiModalId:
      description: >
        This data type contains a multi-modal service identifier.
      type: string
    TscPriorityLevel:
      description: Represents the priority level of TSC Flows.
      type: integer
      minimum: 1
      maximum: 8
    TscPriorityLevelRm:
      description: >
        This data type is defined in the same way as the TscPriorityLevel data type, but with the
        OpenAPI nullable property set to true.
      type: integer
      minimum: 1
      maximum: 8
      nullable: true

    DurationMilliSec:
      description: Indicates the time interval in units of milliseconds.
      type: integer

    DurationMilliSecRm:
      description: >
        This data type is defined in the same way as the "DurationMillisec" data type, but with the
        OpenAPI nullable property set to true.
      type: integer

    MaxDataBurstVol:
      type: integer
      minimum: 1
      maximum: 2000000
      nullable: true
      description: Unsigned integer indicating Maximum Data Burst Volume value.
#
# ENUMERATIONS DATA TYPES
#
    MediaType:
      description: Indicates the media type of a media component.
      anyOf:
        - type: string
          enum:
            - AUDIO
            - VIDEO
            - DATA
            - APPLICATION
            - CONTROL
            - TEXT
            - MESSAGE
            - OTHER
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    MpsAction:
      description: >
        Indicates whether it is an invocation, a revocation or an invocation with authorization of
        the MPS for DTS or Messaging service.
      anyOf:
        - type: string
          enum:
```

```
          - DISABLE_MPS_FOR_DTS
          - ENABLE_MPS_FOR_DTS
          - AUTHORIZE_AND_ENABLE_MPS_FOR_DTS
          - AUTHORIZE_AND_ENABLE_MPS_FOR_AF_SIGNALLING
          - DISABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING
          - ENABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING
      - type: string
        description: >
          This string provides forward-compatibility with future extensions to the enumeration
          and is not used to encode content defined in the present version of this API.

  ReservPriority:
    description: Indicates the reservation priority.
    anyOf:
      - type: string
        enum:
          - PRIO_1
          - PRIO_2
          - PRIO_3
          - PRIO_4
          - PRIO_5
          - PRIO_6
          - PRIO_7
          - PRIO_8
          - PRIO_9
          - PRIO_10
          - PRIO_11
          - PRIO_12
          - PRIO_13
          - PRIO_14
          - PRIO_15
          - PRIO_16
      - type: string
        description: >
          This string provides forward-compatibility with future extensions to the enumeration
          and is not used to encode content defined in the present version of this API.

  ServAuthInfo:
    description: Indicates the result of the Policy Authorization service request from the AF.
    anyOf:
    - type: string
      enum:
        - TP_NOT_KNOWN
        - TP_EXPIRED
        - TP_NOT_YET_OCURRED
        - ROUT_REQ_NOT_AUTHORIZED
        - DIRECT_NOTIF_NOT_POSSIBLE
        - MPX_MEDIA_NOT_SUPPORTED_IN_UE
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  SponsoringStatus:
    description: Indicates whether sponsored data connectivity is enabled or disabled/not enabled.
    anyOf:
    - type: string
      enum:
        - SPONSOR_DISABLED
        - SPONSOR_ENABLED
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  AfEvent:
    description: Represents an event to notify to the AF.
    anyOf:
    - type: string
      enum:
        - ACCESS_TYPE_CHANGE
        - ANI_REPORT
        - APP_DETECTION
        - CHARGING_CORRELATION
        - EPS_FALLBACK
        - EXTRA_UE_ADDR
        - FAILED_QOS_UPDATE
        - FAILED_RESOURCES_ALLOCATION
```

```
        - OUT_OF_CREDIT
        - PDU_SESSION_STATUS
        - PLMN_CHG
        - QOS_MONITORING
        - QOS_MON_CAP_REPO
        - QOS_NOTIF
        - RAN_NAS_CAUSE
        - REALLOCATION_OF_CREDIT
        - SAT_CATEGORY_CHG
        - SUCCESSFUL_QOS_UPDATE
        - SUCCESSFUL_RESOURCES_ALLOCATION
        - TSN_BRIDGE_INFO
        - UP_PATH_CHG_FAILURE
        - USAGE_REPORT
        - UE_REACH_STATUS_CH
        - BAT_OFFSET_INFO
        - URSP_ENF_INFO
        - PACK_DEL_VAR
        - L4S_SUPP
        - RT_DELAY_TWO_QOS_FLOWS
        - RATE_LIMIT_INFO_REPO
        - UP_PATH_CHANGE
        - CN_HEALTH_MONITOR
        - UE_TEMPORARILY_UNAVAILABLE
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  AfNotifMethod:
    description: Represents the notification methods that can be subscribed for an event.
    anyOf:
    - type: string
      enum:
        - EVENT_DETECTION
        - ONE_TIME
        - PERIODIC
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  QosNotifType:
    anyOf:
    - type: string
      enum:
        - GUARANTEED
        - NOT_GUARANTEED
        - NOT_GUARANTEED_DL
        - NOT_GUARANTEED_UL
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.
    description: |
      Indicates the notification type for QoS Notification Control.
      Possible values are:
      - GUARANTEED: The QoS targets of one or more SDFs are guaranteed. When ExtQoSR19 feature
        is enabled, The QoS targets of one or more SDFs are guaranteed again in both DL and UL
        directions.
      - NOT_GUARANTEED: The QoS targets of one or more SDFs are not being guaranteed. When
        ExtQoSR19 feature is enabled, The QoS targets of one or more SDFs are not being
        guaranteed in both DL and UL directions.
      - NOT_GUARANTEED_DL: The QoS targets of one or more SDFs are not being guaranteed in DL
        direction.
      - NOT_GUARANTEED_UL: The QoS targets of one or more SDFs are not being guaranteed in DL
        direction.

  TerminationCause:
    description: >
      Indicates the cause behind requesting the deletion of the Individual Application Session
      Context resource.
    anyOf:
    - type: string
      enum:
        - ALL_SDF_DEACTIVATION
        - PDU_SESSION_TERMINATION
        - PS_TO_CS_HO
```

```
        - INSUFFICIENT_SERVER_RESOURCES
        - INSUFFICIENT_QOS_FLOW_RESOURCES
        - SPONSORED_DATA_CONNECTIVITY_DISALLOWED
        - REQUEST_QOS_NOT_SUPPORTED_IN_PLMN
        - UE_ADDR_RELEASE
        - SMF_FAILURE
        - REFLECTIVE_QOS_NOT_SUPPORTED_IN_UE
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  MediaComponentResourcesStatus:
    description: Indicates whether the media component is active or inactive.
    anyOf:
    - type: string
      enum:
        - ACTIVE
        - INACTIVE
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  FlowUsage:
    description: Describes the flow usage of the flows described by a media subcomponent.
    anyOf:
    - type: string
      enum:
        - NO_INFO
        - RTCP
        - AF_SIGNALLING
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  FlowStatus:
    description: Describes whether the IP flow(s) are enabled or disabled.
    anyOf:
    - type: string
      enum:
        - ENABLED-UPLINK
        - ENABLED-DOWNLINK
        - ENABLED
        - DISABLED
        - REMOVED
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  RequiredAccessInfo:
    description: Indicates the access network information required for an AF session.
    anyOf:
    - type: string
      enum:
        - USER_LOCATION
        - UE_TIME_ZONE
        - UE_SAT_INFO
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

  SipForkingIndication:
    description: >
      Indicates whether several SIP dialogues are related to an "Individual Application Session
      Context" resource.
    anyOf:
      - type: string
        enum:
          - SINGLE_DIALOGUE
          - SEVERAL_DIALOGUES
      - type: string
        description: >
          This string provides forward-compatibility with future extensions to the enumeration
          and is not used to encode content defined in the present version of this API.
```

```
    AfRequestedData:
      description: Represents the information that the AF requested to be exposed.
      anyOf:
        - type: string
          enum:
            - UE_IDENTITY
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    ServiceInfoStatus:
      description: Represents the preliminary or final service information status.
      anyOf:
        - type: string
          enum:
            - FINAL
            - PRELIMINARY
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    PreemptionControlInformation:
      description: Represents Pre-emption control information.
      anyOf:
        - type: string
          enum:
            - MOST_RECENT
            - LEAST_RECENT
            - HIGHEST_BW
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    PrioritySharingIndicator:
      description: Represents the Priority sharing indicator.
      anyOf:
        - type: string
          enum:
            - ENABLED
            - DISABLED
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    PreemptionControlInformationRm:
      description: >
        This data type is defined in the same way as the PreemptionControlInformation data type, but
        with the OpenAPI nullable property set to true.
      anyOf:
        - $ref: '#/components/schemas/PreemptionControlInformation'
        - $ref: 'TS29571_CommonData.yaml#/components/schemas/NullValue'

    AppDetectionNotifType:
      description: Indicates the notification type for Application Detection Control.
      anyOf:
      - type: string
        enum:
          - APP_START
          - APP_STOP
      - type: string
        description: >
          This string provides forward-compatibility with future extensions to the enumeration
          and is not used to encode content defined in the present version of this API.

    PduSessionStatus:
      description: Indicates whether the PDU session is established or terminated.
      anyOf:
      - type: string
        enum:
          - ESTABLISHED
          - TERMINATED
      - type: string
        description: >
```

```
      This string provides forward-compatibility with future extensions to the enumeration
      and is not used to encode content defined in the present version of this API.

UplinkDownlinkSupport:
  description: >
    Represents whether an indication or capability is supported for the UL, the DL or both,
    UL and DL.
  anyOf:
    - type: string
      enum:
        - UL
        - DL
        - UL_DL
    - type: string
      description: >
        This string provides forward-compatibility with future extensions to the enumeration
        and is not used to encode content defined in the present version of this API.

L4sNotifType:
  description: Indicates the notification type for ECN marking for L4S support in 5GS.
  anyOf:
  - type: string
    enum:
      - AVAILABLE
      - NOT_AVAILABLE
  - type: string
    description: >
      This string provides forward-compatibility with future extensions to the enumeration
      and is not used to encode content defined in the present version of this API.

NotifCap:
  description: Indicates whether the notified capability is supported or not supported.
  anyOf:
  - type: string
    enum:
      - SUPPORTED
      - NOT_SUPPORTED
  - type: string
    description: >
      This string provides forward-compatibility with future extensions to the enumeration
      and is not used to encode content defined in the present version of this API.

HeaderHandlingAction:
  anyOf:
  - type: string
    enum:
      - DETECT
      - REMOVE
      - REPLACE
      - INSERT
  - type: string
    description: >
      This string provides forward-compatibility with future extensions to the enumeration but
      is not used to encode content defined in the present version of this API.
  description: |
    Represents the type of header handling actions.
    Possible values are:
    - DETECT: Indicates that the request is for the detection of a header field.
    - REMOVE: Indicates that the request is for the removal of a header field.
    - REPLACE: Indicates that the request is for the replacement of information in a header
      field.
    - INSERT: Indicates that the request is for the addition of a header field.

HeaderHandlingCond:
  anyOf:
  - type: string
    enum:
      - EVERY_MATCH
      - FIRST_MATCH
  - type: string
    description: >
      This string provides forward-compatibility with future extensions to the enumeration but
      is not used to encode content defined in the present version of this API.
  description: |
    Represents the condition to apply header handling actions.
    Possible values are:
    - EVERY_MATCH: Indicates that the header handling action is applied to every match.
    - FIRST_MATCH: Indicates that the header handling action is applied only to the first
```

```
      match.

OnPathN6Method:
  anyOf:
  - type: string
    enum:
      - CONNECT_UDP
  - type: string
    description: >
      This string provides forward-compatibility with future extensions to the enumeration but
      is not used to encode content defined in the present version of this API.
  description: |
    Represents the method of on-path N6 signaling.
    Possible values are:
    - CONNECT_UDP: Indicates that the method connect UDP is supported for on-path N6 signaling

NotifCapType:
  anyOf:
  - type: string
    enum:
      - PACKET_DELAY
      - CONGESTION
      - AVAILABLE_BITRATE
  - type: string
    description: >
      This string provides forward-compatibility with future extensions to the enumeration
      and is not used to encode content defined in the present version of this API.
  description: |
    Indicates which type of QoS Monitoring capability report is applied.
    Possible values are:
    - PACKET_DELAY: Indication the Packet delay monitoring capability is monitored.
    - CONGESTION: Indication the Congestion information monitoring capability is monitored.
    - AVAILABLE_BITRATE: Indication the Available bitrate monitoring capability is monitored.

PcSessionRecoveryStatus:
  anyOf:
  - type: string
    enum:
      - REST_REQUEST
      - REST_TRIGGERED
      - REST_NOT_TRIGGERED
      - NOT_FOUND
  - type: string
    description: >
      This string provides forward-compatibility with future extensions to the enumeration
      and is not used to encode content defined in the present version of this API.
  description: |
    Describes whether the session is being restored or the session is not found.
    Possible values are:
    - REST_REQUEST: Information about whether PDU session is being restored is requested.
    - REST_TRIGGERED: The PDU session is being restored.
    - REST_NOT_TRIGGERED: The PDU session exists and it is not being restored.
    - NOT_FOUND: The PDU session was not found.
```

Annex B (normative): IMS Related P-CSCF Procedures over N5

B.1 Provision of Service Information at P-CSCF

When the "IMS_SBI" feature is supported, the P-CSCF shall send service information to the PCF upon every SIP message that includes an SDP answer payload for the purpose of authorizing the IP flows and the QoS resources required for a negotiated IMS session, unless the SDP payload only relates to a circuit-switched bearer (i.e. "c=" line set to "PSTN" and an "m=" line set to "PSTN", refer to 3GPP TS 24.292 [29]). The service information shall be derived both from the SDP offer and the SDP answer. This ensures that the PCF receives proper information to perform media authorization for all possible IMS session set-up scenarios, and that the PCF is also capable of handling session modifications. The P-CSCF may include "servInfStatus" attribute set to "FINAL".

Additionally, the P-CSCF may send service information to the PCF when receiving a SIP message that includes an SDP offer payload for the purpose of performing an early bandwidth authorization check, or for enabling pre-authorization for a UE terminated IMS session establishment or modification with UE initiated resource reservation, or for the retrieval of network provided access network information (see clause B.8.2).

If the PCF supports "SmfFailureDetection" feature and the PCF due to operator configuration detects that the SMF has failed upon receiving service information from P-CSCF, the PCF may notify the P-CSCF as defined in clause 4.2.2.2, clause 4.2.3.2 and clause 4.2.5.3.

The P-CSCF shall send service information to the PCF when receiving a SIP message that includes an SDP offer payload when the IMS session is an MPS session that requires priority treatment. For a UE terminated session the P-CSCF may send the service information derived from the SDP offer when the SDP offer either does not include any preconditions information or includes preconditions information indicating that the local preconditions (i.e. the preconditions related to the remote peer) are already met. In this case, the P-CSCF shall derive the service information only from the SDP offer and shall include "servInfStatus" attribute set to "PRELIMINARY".

NOTE 1: For a UE terminated session setup, when the SDP offer either does not include any preconditions information or includes preconditions information indicating that the local preconditions (i.e. the preconditions related to the remote peer) are already met, the terminating UE can request a resource modification prior to sending the SDP answer. Even if the IP address and port information in the session information derived from the SDP offer can be insufficient for PCC rule authorization, the policy to handle such UE initiated requests at the PCF can take into account the fact that an IMS session establishment is ongoing, for instance in deciding whether to authorize the request and in selecting an appropriate charging key and a gating policy.

The P-CSCF shall derive the value of the "fDescs" attribute within the service information from the SDP as follows:

- An uplink entry in the "fDescs" attribute shall be formed as follows: The destination address shall be taken from the SDP information received by the P-CSCF in downlink direction, while the source IP address may be formed from the address present in the SDP received by the P-CSCF in uplink direction (taking into account only the 64 bit prefix of the Ipv6 address) Source and destination ports shall be derived according to rules provided in 3GPP TS 29.513 [7] clause 7.2.

EXAMPLE 1: Assuming UE A sends an SDP to UE B, the PCF of UE B uses the address present in this SDP for the destination address of UE B's uplink entry in the "fDescs" attribute, while the PCF of the UE A uses the 64 bit prefix of the same address for the source address of UE A's uplink entry in the "fDescs" attribute. If the source address is not formed from the 64 bit prefix, the source address shall be wildcarded.

- A downlink entry in the "fDescs" attribute shall be formed as follows: The destination address shall be taken from the SDP information received by the P-CSCF in uplink direction, while the source IP address may be formed (in order to reduce the possibilities of QoS flow misuse) from the destination address in the SDP received by the P-CSCF in downlink direction (taking into account only the 64 bit prefix of the Ipv6 address) Source and destination ports shall be derived according to rules provided in 3GPP TS 29.513 [7] clause 7.2.

EXAMPLE 2: Assuming UE A sends an SDP to UE B, the PCF of UE A uses the address present in this SDP for the destination address of UE A's downlink entry in the "fDescs" attribute, while the PCF of UE B uses the 64 bit prefix of the same address for the source address of UE B's downlink entry in the "fDescs" attribute. If the source address is not formed from the 64 bit prefix, the source address shall be wildcarded.

The P-CSCF shall derive the bandwidth information within the service information, from the "b=AS" SDP parameter and "a=bw-info" SDP parameter, if available. If "a=bw-info" is used for bandwidth derivation, the P-CSCF shall use the SDP attribute line that contains the bandwidth properties for the IP version used by the UE, as detailed in 3GPP TS 29.513 [7] clause 7.2. If the received "a=bw-info" SDP attribute line(s) contain only bandwidth properties for an IP version that is not used by the UE, the P-CSCF shall re-compute the bandwidth properties for the used IP version and use that value for the bandwidth derivation as defined in 3GPP TS 26.114 [30].

NOTE 2: If no IP version is included for any of the "a=bw-info" SDP attribute lines related to a certain payload type and direction then IPv6 is assumed for all bandwidth properties related to the same direction and payload type, on all of the related "a=bw-info" SDP attribute lines, see clause 19 of 3GPP TS 26.114 [30].

If "a=bw-info" is used for bandwidth derivation and it includes both known and unknown bandwidth properties, the P-CSCF shall only consider the known bandwidth properties to derive the bandwidth information and ignore the unknown ones. If the" a=bw-info" line is received with an unknown directionality, then the entire "a=bw-info" line shall be ignored.

For the possibly associated RTCP IP flows, the P-CSCF shall use the SDP "b=RR" and "b=RS" parameters, if present, as specified in 3GPP TS 29.513 [7] clause 7.2. The "b=AS", "b=RR" and "b=RS" parameters in the SDP contain all the overhead coming from the IP-layer and the layers above, e.g. IP, UDP, RTP and RTCP payload, or IP, UDP and RTCP.

For multiplexed RTP/RTCP flows (as negotiated using the "a=rtcp-mux" SDP attribute defined in IETF RFC 5761 [31], a P-CSCF supporting RTP/RTCP transport multiplexing shall derive the bandwidth information within the service information as specified in 3GPP TS 29.513 [7] clause 7.2.

However, if service information is received containing the "b=TIAS" SDP parameter that corresponds to an SDP answer payload, and if the P-CSCF supports this parameter, the P-CSCF may derive the bandwidth from this parameter rather than from the "b=AS" SDP parameter, as detailed in 3GPP TS 29.513 [7] clause 7.2.

When available, the P-CSCF shall also indicate to PCF, as a complement to the Service Information, the IMS Communication Service Identifier within the "afAppId" attribute. The originating P-CSCF shall take the IMS Communication Service Identifier value from the SIP response. The terminating P-CSCF shall take the IMS Communication Service Identifier value from the SIP request. Otherwise, the P-CSCF may not be able to provide an IMS Communication Service Identifier value to the PCF. The format and specific headers where IMS communication service identifiers are transported within SIP are defined in 3GPP TS 24.229 [32].

NOTE 3: In order to indicate the IMS Communication Service Identifier to the PCF, the originating P-CSCF sets the "afAppId" attribute to the ICSI contained in the topmost occurrence of the "+g.3gpp.icsi-ref" header field parameter of the Feature-Caps header field(s) of 18x or 2xx SIP response (Feature-Caps: *;+g.3gpp.icsi-ref="urn%Aurn-7%A3gpp-service.ims.icsi.mmtel") and the terminating P-CSCF sets the "afAppId" attribute to the ICSI of the P-Asserted-Service header information received in the SIP request (e.g. P-Asserted-Service: urn:urn-7:3gpp-service.ims.icsi.mmtel). Since the headers and the format of the ICSI can vary depending on the case, the PCF has to be prepared to accept the complete ICSI information received in different formats, as described in clause 7.2A.8.2 in 3GPP TS 24.229 [32].

Additionally, if "ResourceSharing" feature is supported, the P-CSCF may include the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute within a media component of the "medComponents" attribute in order to indicate the PCF that resource sharing should apply for the media components in the related direction with the same value for the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute.

Additionally, if "PrioritySharing" feature is supported, the P-CSCF may provide the "prioSharingInd" attribute within a media component of the "medComponents" attribute as described in clause 4.2.2.21 and 4.2.3.21.

NOTE 4: The P-CSCF obtains this information from the Application Server as described in 3GPP TS 23.228 [33], clause 5.4.7.9.

NOTE 5: RTCP flows are not subject to resource sharing. This requirement cannot be met for multiplexed RTP/RTCP flows as in this case there is no mechanism in the current release to distinguish between RTP and RTCP flows.

For IMS emergency services provided by a PLMN or an SNPN if the "servUrn" attribute does not include an emergency service URN, i.e. a top-level service type of "sos" as specified in IETF RFC 5031 [34] and possibly additional sub-service information on the type of the emergency service and the PCF binds the IMS service session to a PDU session established to an Emergency DNN, the PCF shall return the application error UNAUTHORIZED_NON_EMERGENCY_SESSION to the P-CSCF. Upon receiving an application error UNAUTHORIZED_NON_EMERGENCY_SESSION the P-CSCF shall apply the procedures defined in 3GPP TS 24.229 [32].

NOTE 6: The PCF determines whether a PDU session is established to an Emergency DNN based on the information received over N7 and operator configuration.

If the "afReqData" attribute is provided in the "ascReqData" attribute indicating "5GS-level UE Identities required", the PCF shall provide the available user information for the PDU session in the serving network (either a PLMN or an SNPN) within the "ueIds" attribute included in the "ascRespData" attribute, where each entry shall contain the IMSI (for PLMN access) or either IMSI or NAI (for SNPN access) within the "supi", and/or the MSISDN within the "gpsi" and/or the IMEI(SV) within the "pei" attributes.

The PCF may decide not to authorize requested service information. The PCF will indicate it to the P-CSCF by rejecting the HTTP request with an HTTP "403 Forbidden" response message including the "cause" attribute set to "REQUESTED_SERVICE_NOT_AUTHORIZED". Upon receiving an HTTP "403 Forbidden" response message including the "cause" attribute set to the value "REQUESTED_SERVICE_NOT_AUTHORIZED" the P-CSCF shall apply the procedures defined in 3GPP TS 24.229 [32].

B.2 Enabling of IP Flows

B.2.1 General

Prior to the completion of the SIP session set-up, i.e. until the 2xx response to the INVITE request is received, the P-CSCF may enable or disable media IP flows depending on operator policy, thus allowing or forbidding early media in forward and/or backward direction. The P-CSCF may set the values of the "fStatus" attribute derived from the SDP direction attributes as defined in 3GPP TS 29.513 [7] clause 7.3.3 or set the values of the "fStatus" attributes considering the em-param of the P-Early-Media header field according to clause B.2.2 or downgrade the values of the "fStatus" attribute derived from the SDP direction attribute based on the configuration in the P-CSCF according to clause B.2.3. However for multiplexed RTP/RTCP flows (as negotiated using the "a=rtcp-mux" SDP attribute defined in IETF RFC 5761 [31]), a P-CSCF supporting RTP/RTCP transport multiplexing shall set the "fStatus" attribute to "ENABLED" to prevent that RTCP is blocked. If the P-CSCF chooses to modify the values of the "fStatus" as received from the SDP direction attribute, the P-CSCF shall store the last received SDP.

When a 2xx response is received, the P-CSCF shall enable all media IP flows according to the direction attribute within the last received SDP, as specified in 3GPP TS 29.513 [7] clause 7.3.3. When a 2xx response is received and the P-CSCF previously provided the values of the "fStatus" attributes different from the value derived from the SDP direction attribute in the session information, the P-CSCF shall provide service information with values of the "fStatus" attributes corresponding to the last received SDP.

NOTE: In most cases a 2xx response is a 200 (OK) response.

If the P-CSCF receives SDP answers after the completion of the SIP session set-up, i.e. after the 2xx response to the INVITE request is received, the P-CSCF shall provide the "fStatus" attribute, based on the last received SDP answer. The "fStatus" attribute shall be derived from the SDP according to 3GPP TS 29.513 [7] clause 7.3.3.

B.2.2 Gate control procedures considering the P-Early-Media header field

Prior to the completion of the SIP session set-up, the P-CSCF may use the em-param of the P-Early-Media header field defined in IETF RFC 5009 [35] in order to enable or disable early media in forward and/or backward direction. If the P-CSCF uses the em-param of the P-Early-Media header field for the gate control of early media, the P-CSCF shall perform the following procedures.

In the terminating P-CSCF, when a SIP message with the P-Early-Media header field is received from the UE and the policies configured in the P-CSCF indicate that the UE is authorized to send early media, then:

1) the P-CSCF shall set the "fStatus" attribute to "ENABLED" if:

   - the received em-param(s) in the P-Early-Media header field includes "sendrecv" and the last received SDP direction attribute from the UE is "sendrecv" or no SDP direction attribute has been received; or

2) the P-CSCF shall set the "fStatus" attribute to "ENABLED-UPLINK" if:

   - the received em-param(s) in the P-Early-Media header field includes "sendrecv" and the last received SDP direction attribute from the UE is "sendonly"; or

   - the received em-param(s) in the P-Early-Media header field includes "sendonly" and the last received SDP direction attribute from the UE is "sendrecv" or "sendonly" or no SDP direction attribute has been received; or

3) the P-CSCF shall set the "fStatus" attribute to "ENABLED-DOWNLINK" if:

   - the received em-param(s) in the P-Early-Media header field includes "sendrecv" and the last received SDP direction attribute from the UE is "recvonly"; or

   - the received em-param(s) in the P-Early-Media header field includes "recvonly" and the last received SDP direction attribute from the UE is "sendrecv" or "recvonly" or no SDP direction attribute has been received; or

4) the P-CSCF shall set the "fStatus" attribute to "DISABLED" if either the received em-param(s) in the P-Early-Media header field or the last received SDP direction attribute from the UE includes "inactive"; or

5) the P-CSCF may set the "fStatus" attribute to "DISABLED" or apply the rules defined in clause B.2.2 if the received em-param(s) in the P-Early-Media header field includes "sendonly" or "recvonly" and the last received SDP direction attribute from the UE is "recvonly" or "sendonly" respectively.

NOTE 1: If the UE is authorized to send early media, the P-CSCF will not remove or modify the P-Early-Media header field according to 3GPP TS 24.229 [32].

When a SIP message with the P-Early-Media header field is received from the functional entity within the trust domain, and if:

- the P-Early-Media header field includes the "gated" parameter, then the P-CSCF may decide not to perform the gate control of early media; or

- the P-Early-Media header field does not include the "gated" parameter, then the P-CSCF shall perform the following procedures:

  1) the P-CSCF shall set the "fStatus" attribute to "ENABLED" if:

     - the received em-param(s) in the P-Early-Media header field includes "sendrecv" and the last received SDP direction attribute from the functional entity is "sendrecv" or no SDP direction attribute has been received; or

  2) the P-CSCF shall set the "fStatus" attribute to "ENABLED-DOWNLINK" if:

     - the received em-param(s) in the P-Early-Media header field includes "sendrecv" and the last received SDP direction attribute from the functional entity is "sendonly"; or

     - the received em-param(s) in the P-Early-Media header field includes "sendonly" and the last received SDP direction attribute from the functional entity is "sendrecv" or "sendonly" or no SDP direction parameter has been received; or

  3) the P-CSCF shall set the "fStatus" attribute to "ENABLED-UPLINK" if:

     - the received em-param(s) in the P-Early-Media header field includes "sendrecv" and the last received SDP direction attribute from the functional entity is "recvonly"; or

- the received em-param(s) in the P-Early-Media header field includes "recvonly" and the last received SDP direction attribute from the functional entity is "sendrecv" or "recvonly" or no SDP direction parameter has been received; or

4) the P-CSCF shall set the "fStatus" attribute to "DISABLED" if either the received em-param(s) in the P-Early-Media header field or the last received SDP direction attribute from the functional entity includes "inactive"; or

5) the P-CSCF may set the "fStatus" attribute to "DISABLED" or apply the rules defined in clause A.2.2 if the received em-param(s) in the P-Early-Media header field includes "sendonly" or "recvonly" and the last received SDP direction attribute from the functional entity is "recvonly" or "sendonly" respectively.

NOTE 2: According to IETF RFC 5009 [35], the non-direction parameter "gated" can be included after the direction parameter (e.g. "sendrecv") in the parameter list. The proxy performing gating of early media can add the parameter before forwarding the SIP message.

When a SIP message without the P-Early-Media header field is received from either the functional entity within the trust domain or the UE that is authorized to send early media, then the P-CSCF may set the "fStatus" attribute to "DISABLED" or apply the rules defined in clause B.2.3 or apply the rules defined in 3GPP TS 29.513 [7] clause 7.3.3.

NOTE 3: As indicated in IETF RFC 5009 [35] the applicable preconditions need to be met in order to allow early media in a particular direction.

When a SIP message is received from the functional entity other than the functional entity within the trust domain or the UE that is authorized to send early media, then the P-CSCF shall not use the received em-param(s) in the P-Early-Media header field and may apply the rules defined in clause B.2.2 or apply the rules defined in 3GPP TS 29.513 [7] clause 7.3.3.

NOTE 4: The P-CSCF will remove or modify the P-Early-Media header field in the above case.

B.2.3 Gate control procedures based on the configuration in the P-CSCF

Prior to the completion of the SIP session set-up, the P-CSCF may downgrade the values of the "fStatus" attributes derived from the SDP direction attributes based on the configuration in the P-CSCF. If the P-CSCF has the configuration for the gate control of early media, the P-CSCF shall perform the following procedures.

NOTE: The gate control of early media can be configured in the P-CSCF per UE basis.

When the "fStatus" attribute derived from the SDP direction attribute is "ENABLED", then the P-CSCF may downgrade the value of the "fStatus" attribute to the value "DISABLED", "ENABLED_UPLINK", or "ENABLED_DOWNLINK" based on the configuration in the P-CSCF.

When the "fStatus" attribute derived from the SDP direction attribute is "ENABLED_UPLINK" or "ENABLED_DOWNLINK", then the P-CSCF may downgrade the value of the "fStatus" attribute to the value "DISABLED" based on the configuration in the P-CSCF.

B.3 Support for SIP forking

B.3.0 General

The P-CSCF shall be able to handle forking when PCC is applied and the "IMS_SBI" feature is supported. Forking can occur as specified in 3GPP TS 23.228 [33]. The related UE procedures are described in 3GPP TS 24.229 [32].

B.3.1 PCC rule provisioning for early media for forked responses

When a SIP session has been originated by a connected UE, the P-CSCF may receive multiple provisional responses due to forking before the first final answer is received. Multiple early media session may be established during this process.

The UE and the P-CSCF become aware of the forking only when a subsequent provisional response arrives for a new early dialogue. After the first early media session is established, for each subsequent provisional response establishing an additional early media session, the P-CSCF shall use an Npcf_PolicyAuthorization_Update service operation containing the "sipForkInd" attribute with value "SEVERAL_DIALOGUES" and include the service information derived from the latest provisional response.

The P-CSCF shall also provision the service information derived from any subsequent SDP offer-answer exchange within an early dialogue (e.g. in PRACK and OK(PRACK), or UPDATE and OK(UPDATE)) using an Npcf_PolicyAuthorization_Update service operation containing the "sipForkInd" attribute with value "SEVERAL_DIALOGUES" and the derived service information.

When receiving an Npcf_PolicyAuthorization_Update service operation containing the "sipForkInd" attribute with value "SEVERAL_DIALOGUES", the PCF shall identify the existing "Individual Application Session Context" resource with existing authorization information.

The PCF shall send additional PCC Rules or individual service data flow filters to already provided PCC rules as required by the "fDescs" attribute within the AF session context information to the SMF. The PCF shall authorize any additional media components and any increased QoS requirements for the previously authorized media components, as requested within the service information.

The PCF shall authorize the maximum bandwidth required by any of the dialogues, but not the sum of the bandwidths required by all dialogues. Thus, the QoS authorized for a media component is equal to the highest QoS requested for that media component by any of the forked responses.

The PCF shall open or close the gates for service flows depending on the flow status that is being provisioned. However, if a flow ID has been enabled in uplink or downlink direction or both way within previous service information, it shall remain enabled even if the PCF receives service information that disable this flow ID within an Npcf_PolicyAuthorization_Update service operation containing the "sipForkInd" attribute with value "SEVERAL_DIALOGUES".

If the P-CSCF provides one or more media components within the "medComponents" attribute with the "fStatus" attribute set to "REMOVED" for previously authorized media component(s) the media component shall remain as authorized and the PCF shall not take any action on that media component(s).

NOTE: There can be cases where a forked response could not support some of the media components included in the SDP Offer (e.g. when early session disposition SDP as described in Annex B.6 applies, the forked response related to the early session could include the port set to zero for those media components not related to the early session or when a subsequent SDP Offer-Answer to indicate that some media is disabled). For those cases the P-CSCF will indicate the PCF about the removal of the corresponding media component. However this media component is already supported by other UEs and the PCF needs to maintain the corresponding PCC rules until the final SDP answer is received in the P-CSCF in order to avoid the release of resources in the network.

B.3.2 Updating the provisioned PCC rules at the final answer

The P-CSCF shall store the SDP information for each early dialogue separately till the first final SIP answer is received. Then the related early dialogue is progressed to an established dialogue to establish the final SIP session. All the other early dialogues are terminated. The service information for the SIP session is updated to match the requirements of the remaining early dialogue only.

When receiving the first final SIP response, the P-CSCF shall send an Npcf_PolicyAuthorization_Update service operation setting to null the "sipForkInd" attribute and shall include the service information derived from the SDP corresponding to the dialogue of the final response. The P-CSCF shall provision the full service information including the applicable "fDescs" attribute and "fStatus" attribute.

When receiving an Npcf_PolicyAuthorization_Update service operation with a "sipForkInd" attribute with value "SINGLE_DIALOGUE", the PCF shall update installed PCC Rules information and Authorized-QoS information to match only the requirements of the service information within this Npcf_PolicyAuthorization_Update service operation. The PCF should immediately remove PCC Rule(s) or individual service data flow filters not matching IP flow(s) in the updated Service Information, to reduce the risk for initial clipping of the media stream, and to minimize possible misuse of resources. The PCF shall also open or close the gates for service flows according to the flow status in the received service information.

B.4 Notification of AF Signalling Transmission Path Status

When the P-CSCF receives an initial REGISTER SIP message from an attached UE, the P-CSCF may subscribe to notifications of the status of the AF signalling transmission path using the procedures specified in clause 4.2.6.7. Once the P-CSCF has subscribed, the P-CSCF may receive notifications from the PCF according to clause 4.2.5.10.

NOTE: This procedure is not applicable for IMS registrations for Emergency sessions.

The P-CSCF shall cancel the subscription to notification of the status of the AF signalling transmission path when the AF signalling to that particular user is terminated (i.e. when the user is de-REGISTERED from the IM CN subsystem).

When the P-CSCF receives a notification of loss of signalling connectivity from the PCF, the P-CSCF shall behave as defined in 3GPP TS 24.229 [32].

B.5 Indication of Emergency Registration and Session Establishment

When the P-CSCF receives an initial REGISTER SIP message for an IMS emergency registration or an INVITE SIP message for an emergency session and the P-CSCF determines that there are no IMS-level roaming interfaces, and the "IMS_SBI" feature is supported the P-CSCF may request the PCF to provide the 5GS-Level UE identities (GPSI, SUPI, PEI) available for that PDU session in the serving network (either a PLMN or an SNPN) using the procedure as specified in this clause (for an IMS emergency registration) or B.1 (for an IMS emergency session establishment).

A P-CSCF may request the PCF to provide the 5GS-level identities (GPSI, SUPI, PEI) available for that PDU session when no service information is available in the P-CSCF. To do so, the P-CSCF shall create an "Individual Application Session Context" resource in the PCF for the AF signalling using an Npcf_PolicyAuthorization_Create service operation. The P-CSCF shall provide the UE's IP address (using either the "ueIpv4" attribute or the "ueIpv6" attribute) and the "afReqData" attribute set to "UE_IDENTITY". The P-CSCF shall include the "servUrn" attribute set to the value "sos", in order to indicate that the new AF session context relates to emergency traffic that is not related to a specific emergency service.

If the P-CSCF supports the SBI Message Priority mechanism for an emergency session, it shall include the "3gpp-Sbi-Message-Priority" custom HTTP header towards the PCF as described in clause 6.8.2 of 3GPP TS 29.500 [5].

NOTE 1: If the P-CSCF supports the SBI Message Priority mechanism for an emergency session, the P-CSCF includes the "3gpp-Sbi-Message-Priority" custom HTTP header based on P-CSCF policies in relation to valid values of the "servUrn" attribute. The highest user priority value is mapped to the corresponding lowest value of the "3gpp-Sbi-Message-Priority" custom HTTP header.

When the PCF receives an Npcf_PolicyAuthorization_Create service operation as described in the preceding paragraphs from the P-CSCF, the PCF shall perform session binding as described in 3GPP TS 29.513 [7]. When the PCF receives the "servUrn" attribute indicating an emergency session, the PCF may apply special policies, for instance prioritising service flows relating to the AF session context or allowing these service flows free of charge.

When the "servUrn" attribute indicates that the AF session context relates to emergency traffic and the "afReqData" attribute is received indicating "UE_IDENTITY", the PCF shall provide the requested available user information (MSISDN, IMSI (for PLMN access) or either IMSI or NAI (for SNPN access), IMEI(SV)) for the PDU session within "ueIds" attribute within the "ascRespData" in the HTTP "201 Created" response.

When the P-CSCF receives the HTTP "201 Created" response with the 5GS-level UE identities from the PCF, the P-CSCF stores the "ueIds" received within "Individual Application Session Context" resource returned in the HTTP "201 Created" response and behaves as defined in 3GPP TS 24.229 [32].

NOTE 2: The user information received within the "ueIds" attribute can be used to support PSAP callback functionality for anonymous IMS emergency sessions. See 3GPP TS 23.167 [40] for further information.

The P-CSCF may decide to delete the "Individual Application Session Context" resource at any time. In that case, the Npcf_PolicyAuthorization_Delete service operation, as described in clause 4.2.4.2.

A SIP INVITE request can contain a service URN as defined in IETF RFC 5031 [34] within the request URI. If the service within this URN is "sos", possibly with additional sub-service information, the P-CSCF shall provision this service and sub-service information within the "servUrn" attribute towards the PCF. The P-CSCF may also provision possible information about other services received within the service URN.

B.6 Support for Early Session disposition SDP

B.6.1 General

As a network option, when the "IMS_SBI" feature is supported, the P-CSCF may support the PCC procedures in the present clause to handle "early session" disposition type SDP, as standardised in IETF RFC 3959 [38].

B.6.2 Service Information Provisioning for Early Media

The P-CSCF can receive "early session" disposition SDP in addition to "session" disposition SDP in SIP early dialogues.

The P-CSCF shall then provision service information derived both from the "early session" disposition SDP and "session" disposition SDP applying the procedures in clauses B.1, B.2, and B.3, and in the present clause.

The P-CSCF shall apply the mapping rules in Annex C to derive the flow identifiers from "early session" disposition SDP.

If a single media line with one media type (e.g. "audio" or "video") is contained in "early session" disposition SDP and a single media line with the same media type is contained in the "session" disposition SDP of the same SIP dialogue, and both media lines describe service flows of the same directionality (uplink, downlink, or bidirectional), the P-CSCF should describe those SDP media lines in the same session information media component (with the same flow ID).

The "early session" disposition SDP can also contain media lines of a type not included in the "session" disposition SDP, or several media lines of the same type. Such media components shall be described in own media components in the service information.

If the P-CSCF desires to invoke special policies or separate event notifications for an "early session" disposition media line, it may choose to provision a separate session information media component even if a media line with the same media type and directionality is contained in "session" disposition SDP.

NOTE 1: A PCF is then likely to supply separate PCC rules for early media and the corresponding final media. This may lead to an over provisioning of resources during call establishment and a subsequent reconfiguration of the radio bearer, or even to a call failure if the extra resources are not authorized or available.

If the P-CSCF receives "early session" disposition SDP before any "session" disposition SDP and supplies service information derived from the "early session" disposition SDP at this point of time, it shall use dedicated media components relating only to the "early session" disposition SDP in the service information.

NOTE 2: The "session" disposition SDP offer will frequently occur before the "early session" disposition SDP offer, but can also occur in parallel or in exceptional cases afterwards. The "session" disposition SDP answer can be contained in the same SIP message as the "early session" disposition SDP offer, or can be sent in a 200 OK (INVITE), i.e. after the "early session" disposition SDP answer.

If the P-CSCF includes any media component relating both to "early session" disposition SDP and "session" disposition SDP in the service information, the P-CSCF shall:

- provision the service information derived from "early session" disposition SDP and the service information derived from "session" disposition SDP in separate Npcf_PolicyAuthorization_Update requests (to the same "Individual Application Session Context" resource), and shall send a new Npcf_PolicyAuthorization_Update request only after any previous Npcf_PolicyAuthorization_Update request has been acknowledged; and

- provision the first service information (either derived from "early session" disposition SDP or "session" disposition SDP) without the "sipForkInd" attribute, or with "sipForkInd" attribute with value "SINGLE_DIALOGUE"; and

- provision all subsequent service information during ongoing call establishment with the "sipForkInd" attribute with value SEVERAL_DIALOGUES; and

- if an SDP answer has been received and codecs are provisioned within the "codecs" attribute included in a media component of the "medComponents" attribute, provision within a "codecs" attribute the codec derived from the corresponding offer together with a codec derived from the SDP answer.

NOTE 3: The P-CSCF needs to provision the service information derived from "early session" disposition SDP and the service information derived from "session" disposition SDP in separate Npcf_PolicyAuthorization_Update requests because the encoding of the media component does not allow for the simultaneous provisioning of two corresponding filters.

NOTE 4: The PCF will treat service information containing the "sipForkInd" attribute as described in clause B.3.

B.6.3 Updating the Provisioned Service Information when Dialogue is established

The P-CSCF shall store the SDP information for the "session" disposition type until the first final SIP answer is received. Then the early media described in the "early session" disposition type SDP are terminated.

The P-CSCF shall then update the service information to match the requirements of the media described in the "session" disposition type SDP only:

- If the P-CSCF included any media component relating both to "early session" disposition SDP and "session" disposition SDP in the service information, the P-CSCF shall send an Npcf_PolicyAuthorization_Update request without the "sipForkInd" attribute or with a "sipForkInd" attribute with value SINGLE_DIALOGUE and shall include the service information derived from the "session" disposition SDP. The P-CSCF shall provision the full service information including the applicable "fDescs" attribute and "fStatus" attribute.

- The P-CSCF shall disable any media component(s) in the service information that relate to early media only by setting their flow status to "REMOVED".

B.7 Provision of Signalling Flow Information at P-CSCF

When the P-CSCF has successfully concluded the initial registration of an attached UE, i.e., when the P-CSCF has sent to the UE a SIP 200 (OK) response to the SIP REGISTER request, the P-CSCF may provision information about the SIP signalling flows between the UE and itself using the procedure specified in clause 4.2.2.16. If the P-CSCF already has created an "Individual Application Session Context" resource with the PCF related to the signalling with the UE, e.g. one that has been opened according to the procedure described in clause B.4, the P-CSCF shall reuse the already open session to provision the SIP Signalling IP Flow information using the procedure specified in clause 4.2.3.17.

NOTE: This procedure is not applicable for IMS registrations for Emergency sessions.

If the P-CSCF provisions information about SIP signalling flows, the P-CSCF shall ensure that for each signalling IP flow information it provides, the flow descriptions within the "fDescs" attribute shall accurately reflect the IP flow information as seen in the IP header 'on the wire'. The P-CSCF shall set the value of the "afSigProtocol" attribute to "SIP".

When the P-CSCF de-registers the UE and terminates SIP Signalling to the UE, the P-CSCF shall de-provision the SIP Signalling IP flow information from the PCRF as described in clauses 4.2.2.16 and 4.2.3.17.

B.8 Retrieval of network provided location information

B.8.1 General

According to clause E.7 of 3GPP TS 23.228 [33], the P-CSCF can use PCC to retrieve network provided location information. Information flows related to the distribution of network provided location information within the IMS are provided in Annex R of 3GPP TS 23.228 [33].

The following clauses provide optional PCC procedures to support the retrieval of network provided location information.

The originating P-CSCF can, depending on operator policy, retrieve the user location and/or UE Time Zone information either before sending the INVITE or MESSAGE towards the terminating side or upon reception of the SDP answer from the terminating side.

The terminating P-CSCF can, depending on operator policy, retrieve the user location and/or UE Time Zone information either upon reception of a SIP INVITE or upon reception of a SIP response.

The originating and terminating P-CSCF can, depending on operator policy, retrieve the user location and/or UE Time Zone information at mid call. e.g., when the P-CSCF learns about the access type change.

NOTE: The P-CSCF uses PCC to retrieve the serving satellite information is defined in clause B.20.

B.8.2 Retrieval of network provided location information at originating P-CSCF for inclusion in SIP Request

If the originating P-CSCF is required by operator policy to retrieve network provided location information before forwarding a SIP INVITE request, upon reception of the SIP INVITE/UPDATE request, the P-CSCF shall invoke:

- the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14 (SIP INVITE request); or
- the Npcf_PolicyAuthorization_Update service operation according to clause 4.2.3.14 or the Npcf_PolicyAuthorization_Subscribe service operation according to clause 4.2.6.6 (SIP INVITE/UPDATE request);

including in the corresponding HTTP request:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
  a) the "event" attribute set to the value "ANI_REPORT"; and
  b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information).

If the SIP INVITE request is an initial SIP INVITE request, the P-CSCF shall create a new "Individual Application Session Context" for the new SIP session with the Npcf_PolicyAuthorization_Subscribe service operation according to clause 4.2.6.6 (if no session information is included) or with the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14 (if preliminary session information is included).

The P-CSCF will receive the access network information from the PCF within the Npcf_PolicyAuthorization_Notify service operation as described in clause 4.2.5.11 and should include this access network information in the SIP INVITE/UPDATE requests that it forwards. When the retrieved access network information corresponds to the "tnapId" or "twapId" attribute, the P-CSCF may also map the retrieved access network information to a Geographical Identifier for routing, as specified in clause E.8 of 3GPP TS 23.228 [33].

If the originating P-CSCF is required by operator policy to retrieve network provided location information before forwarding a SIP MESSAGE request, upon reception of a MESSAGE request, the P-CSCF shall invoke the Npcf_PolicyAuthorization_Subscribe service operation including in the corresponding HTTP request:

- the IP address (IPv4 or IPv6) of the UE in the "ueIpv4" or "ueIpv6" attribute;
- a media component within the "medComponents" attribute including:
  a) the "medCompN" attribute set to "0"; and
  b) a single media subcomponent within the "medSubComps" attribute with:
    i. the "flowUsage" attribute set to the value "AF_SIGNALLING"; and
    ii. if the procedures for AF provisioning of AF signalling flow information do not apply, the "fNum" attribute set to "0".
- an entry of the "AfEventSubscription" data type in the "events" attribute with:
  a) the "event" attribute set to the value "ANI_REPORT"; and
  b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information).

The P-CSCF will receive the access network information from the PCF within the Npcf_PolicyAuthorization_Notify service operation as described in clause 4.2.5.11 and should include this access network information in the SIP MESSAGE requests that it forwards. When the retrieved access network information corresponds to the "tnapId" or "twapId" attribute, the P-CSCF may also map the retrieved access network information to a Geographical Identifier for routing, as specified in clause E.8 of 3GPP TS 23.228 [33].

If the AF application session context is only used for retrieval of network provided location information, at reception of this information, the AF may delete the context of application session information using the Npcf_PolicyAuthorization_ Delete service operation.

B.8.3 Retrieval of network provided location information at originating P-CSCF for inclusion in SIP response confirmation

If an originating P-CSCF is required by operator policy to retrieve network provided location information before forwarding an SDP answer, the P-CSCF shall apply the following procedures.

Upon reception of an SDP offer, the P-CSCF may invoke the Npcf_PolicyAuthorization_Create service operation to the PCF according to clause B.1 and may include in the corresponding HTTP POST request:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
  a) the "event" attribute set to the value "ANI_REPORT"; and
  b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information).

Upon reception of an SDP answer, the P-CSCF will invoke the Npcf_PolicyAuthorization_Update service operation to the PCF according to clause B.1. If the P-CSCF has not requested access network information upon reception of the SDP offer, the P-CSCF shall include in the corresponding HTTP PATCH request:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
  a) the "event" attribute set to the value "ANI_REPORT"; and
  b) the "notifMethod" attribute set to the value "ONE_TIME"; and

- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information).

The P-CSCF will receive the access network information from the PCF in the Npcf_PolicyAuthorization_Notify service operation and should include this access network information in the SIP message with the response confirmation before forwarding it. When the retrieved access network information corresponds to the "tnapId" or "twapId" attribute, the P-CSCF may also map the retrieved access network information to a Geographical Identifier for routing, as specified in clause E.8 of 3GPP TS 23.228 [33].

B.8.4 Retrieval of network provided location information at terminating P-CSCF

If a terminating P-CSCF is required by operator policy to retrieve network provided location information at session establishment and/or modification, the P-CSCF shall apply the following procedures.

The terminating P-CSCF may request network provided location information upon reception of a SIP INVITE request in the following manner:

- if the SIP INVITE request is an initial SIP INVITE request, the P-CSCF shall create a new "Individual Application Session Context" for the new SIP session with the Npcf_PolicyAuthorization_Subscribe service operation according to clause 4.2.6.6 (if no session information is included) or with the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14 (if preliminary session information is included);

- if the SIP INVITE contains an SDP offer, the P-CSCF shall include in the corresponding HTTP request:

  a) an entry of the "AfEventSubscription" data type in the "events" attribute with:

     (i) the "event" attribute set to the value "ANI_REPORT"; and

     (ii) the "notifMethod" attribute set to the value "ONE_TIME";

  b) the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information);

  c) service information derived from the SDP offer; and

  d) the "servInfStatus" attribute with the value set to "PRELIMINARY"; and

- if the SIP INVITE does not contain an SDP offer, the P-CSCF shall include in the corresponding HTTP request:

  a) an entry of the "AfEventSubscription" data type in the "events" attribute with:

     (i) the "event" attribute set to the value "ANI_REPORT"; and

     (ii) the "notifMethod" attribute set to the value "ONE_TIME"; and

  b) the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information).

Upon reception of a SIP response that requires the inclusion of access network information, if the P-CSCF has not already requested network provided location information upon reception of the corresponding SIP INVITE request, the P-CSCF shall request network provided location information in the following manner:

- if an "Individual Application Session Context" related to service data has not yet been created, the P-CSCF shall create an "Individual Application Session Context" for the new SIP session with the Npcf_PolicyAuthorization_Subscribe service operation according to clause 4.2.6.6 (if no session information is included) or with the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14 (if session information is included);

- if the SIP response includes an SDP answer, the P-CSCF shall send an HTTP request to the PCF according to clause B.1; the P-CSCF shall include in this HTTP request:

  a) an entry of the "AfEventSubscription" data type in the "events" attribute with:

(i) the "event" attribute set to the value "ANI_REPORT"; and

(ii) the "notifMethod" attribute set to the value "ONE_TIME"; and

b) the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information);

- if the SIP response does not contain an SDP body, the P-CSCF shall include in the corresponding HTTP request:

a) an entry of the "AfEventSubscription" data type in the "events" attribute with:

(i) the "event" attribute set to the value "ANI_REPORT"; and

(ii) the "notifMethod" attribute set to the value "ONE_TIME"; and

b) the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information); and

- if the SIP response includes an SDP offer, the P-CSCF shall include in the corresponding HTTP request:

- a) an entry of the "AfEventSubscription" data type in the "events" attribute with:

(i) the "event" attribute set to the value "ANI_REPORT"; and

(ii) the "notifMethod" attribute set to the value "ONE_TIME";

b) the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information);

c) service information derived from the SDP offer; and

d) the "servInfStatus" attribute with the value set to "PRELIMINARY".

The P-CSCF will receive the access network information from the PCF in the Npcf_PolicyAuthorization_Notify service operation and should include this access network information in the appropriate SIP response before forwarding it. When the retrieved access network information corresponds to the "tnapId" or "twapId" attribute, the P-CSCF may also map the retrieved access network information to a Geographical Identifier for routing, as specified in clause E.8 of 3GPP TS 23.228 [33].

If the terminating P-CSCF is required by operator policy to retrieve network provided location information upon reception of a SIP MESSAGE response, the P-CSCF shall behave according to B.8.2.

B.8.5 Provisioning of network provided location information at SIP session release

If a P-CSCF is required by operator policy to include network provided location information in SIP session release signalling, the P-CSCF shall apply the following procedures:

Upon reception of a SIP session release request that requires the inclusion of network provided location information, the P-CSCF will invoke the Npcf_PolicyAuthorization_Delete service operation to the PCF according to clause 4.2.4.6 and shall include in the HTTP request:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:

a) the "event" attribute set to the value "ANI_REPORT"; and

b) the "notifMethod" attribute set to the value "ONE_TIME"; and

- the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information).

The P-CSCF will receive the access network information from the PCF in the Npcf_PolicyAuthorization_Delete service operation according to clause 4.2.4.6 and shall include this access network information in the appropriate SIP message before forwarding it. When the retrieved access network information corresponds to the "tnapId" or "twapId" attribute,

the P-CSCF may also map the retrieved access network information to a Geographical Identifier for routing, as specified in clause E.8 of 3GPP TS 23.228 [33].

B.8.6 Provisioning of network provided location information at mid call

If a P-CSCF is required by operator policy to include network provided location information at mid call, the P-CSCF shall apply the following procedures:

Upon reception of a trigger (e.g., when the P-CSCF learns about the access change) that requires the inclusion of network provided location information in a SIP message, the P-CSCF will invoke the Npcf_PolicyAuthorization_Update and shall include in the corresponding HTTP request:

a) an entry of the "AfEventSubscription" data type in the "events" attribute with:

   (i) the "event" attribute set to the value "ANI_REPORT"; and

   (ii) the "notifMethod" attribute set to the value "ONE_TIME";

b) the "reqAnis" attribute, with the required access network information, i.e. user location and/or user time zone information);

c) service information related to the service according to clause 4.2.3.2.

The P-CSCF will receive the access network information from the PCF in the Npcf_PolicyAuthorization_Notify service operation, and should include this access network information in the appropriate SIP message before forwarding it. When the retrieved access network information corresponds to the "tnapId" or "twapId" attribute, the P-CSCF may also map the retrieved access network information to a Geographical Identifier for routing, as specified in clause E.8 of 3GPP TS 23.228 [33].

B.9 Resource Sharing

The P-CSCF may indicate to the PCF that media of an "Individual Application Session Context" resource may share resources with media belonging to other "Individual Application Session Context" resources according to 3GPP TS 23.228 [33].

If the P-CSCF determines that resource sharing is possible, it may at creation of a new "Individual Application Session Context" resource, include the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute indicating that media resources may be shared in the related direction. The P-CSCF shall assign a distinct value for the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute for each media component within the "medComponents" attribute.

NOTE 1: When resource sharing applies to both directions for a certain media component, the P-CSCF can assign the same value for "sharingKeyUl" attribute and "sharingKeyDl" attribute within the same media component.

The P-CSCF shall not include the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute within the media components in the "medComponents" attribute when the "Individual Application Session Context" resource relates to an Emergency Session.

The PCF shall not include the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute for those PCC/QoS Rules related to the RTCP traffic.

Trigger conditions that require applying or stopping resource sharing are described in 3GPP TS 24.229 [32].

NOTE 2: When P-CSCF needs to stop sharing according to the procedures described in 3GPP TS 24.229 [32], the P-CSCF will provide "null" value for the "sharingKeyUl" attribute and/or "sharingKeyDl" attribute within the media component in the "medComponents" attribute.

B.10 Handling of MCPTT priority call

B.10.1 General

Within the framework of MCPTT, when the SIP Core (3GPP TS 24.379 [41]) is implemented by an IMS core network, if the P-CSCF receives a SIP request message including a Resource-Priority header field with a namespace field and priority value defined for MCPTT for adjusting the priority of an MCPTT session, the P-CSCF shall provide the "resPrio" attribute and the "mcpttId" in the Npcf_PolicyAuthorization_Create request as defined in clause B.13.2 to allow the PCF to set the corresponding PCC rule(s) according to the prioritized MCPTT service. Additionally, if "PrioritySharing" feature is supported, the P-CSCF may provide the "prioSharingInd" attribute within the media component included in the "medComponents" attribute as described in clause B.1. For MCPTT the service priority and the priority sharing indicator are defined in 3GPP TS 24.379 [41].

NOTE 1: The process of adjusting priority may occur several times during the course of one session, e.g. a normal MCPTT group call elevated to an MCPTT emergency group call, returned to a normal priority MCPTT group call, elevated to an MCPTT imminent peril group call and returned to a normal priority MCPTT group call.

NOTE 2: Upon reception of a request that requires the adjustment of the MCPTT priority, the PCF is expected to derive the PCC Rules corresponding to the this MCPTT session, as appropriate according to operator policies.

NOTE 3: The PCF can identify an MCPTT call using the IMS Communication Service Identifier specific to MCPTT, which is provided by the P-CSCF in the "afAppId" attribute in the Npcf_PolicyAuthorization_Create request sent to PCF.

B.10.2 Determination of MCPTT priority parameter values

When the P-CSCF receives an authorized Resource-Priority header field containing an appropriate namespace and priority value used for MCPTT in SIP signalling, the P-CSCF shall include the "mcpttId" attribute and the "resPrio" attribute in the corresponding Npcf_PolicyAuthorization service operation towards the PCF.

The "mcpttId" attribute shall include the namespace defined for MCPTT as received within the Resource-Priority header field.

NOTE: Two different values are defined for the MCPTT-Identifier AVP, one for each namespace value defined for MCPTT (see IETF RFC 8101 [42]).

The "resPrio" attribute shall contain the priority value of the Resource-Priority header; the lowest priority shall be mapped to PRIO_1 (Resource-Priority header value 0), the next after the lowest to PRIO_2 (Resource-Priority header value 1), and so on up to the highest priority which shall be mapped to PRIO_16 (Resource-Priority header value 15).

Additionally, when the P-CSCF receives information about priority sharing from an MCPTT server that supports simultaneous sessions and that needs to share a common priority for several MCPTT sessions and if "PrioritySharing" feature is supported, the P-CSCF may include the "prioSharingInd" attribute within the media component received in the "medComponents" attribute in the corresponding Npcf_PolicyAuthorization service operation. See 3GPP TS 24.379 [41] for further information.

B.11 Handling of MCVideo priority call

B.11.1 General

Within the framework of MCVideo, when the SIP Core (3GPP TS 23.281 [34]) is implemented by an IMS core network, if the P-CSCF receives a SIP request message including a Resource-Priority header field with a namespace field and priority value defined for MCVideo for adjusting the priority of an MCVideo session, the P-CSCF shall provide the "resPrio" attribute and the "mcVideoId" in the Npcf_PolicyAuthorization_Create request as defined in clause B.15.2 to allow the PCF to set the corresponding PCC rule(s) according to the prioritized MCVideo service.

NOTE 1: The process of adjusting priority may occur several times during the course of one session, e.g. a normal MCVideo group call elevated to an MCVideo emergency group call and returned to a normal priority MCVideo group call, elevated to an MCVideo imminent peril group call and returned to a normal priority MCVideo group call.

NOTE 2: Upon reception of a request that requires the adjustment of the MCVideo priority, the PCF is expected to derive the PCC Rules corresponding to the this MCVideo session, as appropriate according to operator policies.

NOTE 3: The PCF can identify an MCVideo call using the IMS Communication Service Identifier specific to MCVideo, which is provided by the P-CSCF in the "afAppId" attribute in the Npcf_PolicyAuthorization_Create request sent to PCF.

B.11.2 Determination of MCVideo priority parameter values

When the P-CSCF receives an authorized Resource-Priority header field containing an appropriate namespace and priority value used for MCVideo in SIP signalling, the P-CSCF shall include the "mcVideoId" attribute and the "resPrio" attribute in the corresponding Npcf_PolicyAuthorization service operation towards the PCF.

The "mcVideoId" attribute shall include the namespace defined for MCVideo as received within the Resource-Priority header field.

The "resPrio" attribute shall contain the priority value of the Resource-Priority header; the lowest priority shall be mapped to PRIO_1 (Resource-Priority header value 0), the next after the lowest to PRIO_2 (Resource-Priority header value 1), and so on up to the highest priority which shall be mapped to PRIO_16 (Resource-Priority header value 15).

B.12 Notification Access Type Change

When the P-CSCF receives an initial SIP REGISTER message or a SIP INVITE message from an attached UE, the P-CSCF may request from the PCF the information about the access type the UE is attached to using the procedure specified in clauses 4.2.2.2, 4.2.3.2 and 4.2.6.2.

NOTE 1: This procedure is not applicable for IMS registrations for Emergency sessions.

NOTE 2: The P-CSCF can request information about the access type as part of the SIP session setup when it is only interested in the related information when the IMS session is ongoing.

If the P-CSCF requests information about the access type, the P-CSCF shall also subscribe within the same Npcf_PolicyAuthorization service operation to notifications for changes of the access type used by the UE. The P-CSCF shall include an entry of the "AfEventSubscription" data type in the "events" attribute with the "event" attribute set to the value "ACCESS_TYPE_CHANGE".

When the P-CSCF receives from the PCF the access type:

- in the subscription request response within the HTTP response; or
- in the notification of access type change in an HTTP POST request from the PCF,

the P-CSCF shall store the access type information received within the "accessType" attribute and the RAT type information received within "ratType" attribute and use the received information as per P-CSCF procedures in 3GPP TS 24.229 [32].

The P-CSCF may receive subsequent notifications for changes of the access type from the PCF according to clause 4.2.5.2. When the P-CSCF receives a notification of the change of the access type used by the UE, the P-CSCF shall store the new access type information and RAT type information and use the received information as per P-CSCF procedures in 3GPP TS 24.229 [32].

NOTE 3: The subscription to receive information about the access type will be cancelled when the corresponding Individual Application Session Context resource is removed by the P-CSCF (i.e. when the UE is de-REGISTERED or the related SIP call is released).

B.13 Notification of PLMN Change

When the P-CSCF receives an initial SIP REGISTER message from an attached UE, the P-CSCF may subscribe to notifications of PLMN changes corresponding to the identity of the network (either a PLMN or an SNPN) where the UE is located using the procedure specified in clauses 4.2.2.2, 4.2.3.2 and 4.2.6.2.

NOTE: For a UE located in an SNPN the SNPN Identifier consisting of the PLMN Identifier and the NID is provided.

When the P-CSCF receives the subscription request response in an HTTP response or the notification of PLMN change in an HTTP POST request from the PCF, the P-CSCF shall store the PLMN Identifier and, if available, the NID received within the "plmnId" attribute and use the received information as per P-CSCF procedures in 3GPP TS 24.229 [32].

The P-CSCF shall cancel the subscription to notification for changes of the PLMN used by the UE when the user is de-registered from the IM CN subsystem.

B.14 Coverage and Handoff Enhancements using Multimedia error robustness feature (CHEM)

As a network option, the P-CSCF may support the PCC procedures in the present clause to handle the Coverage and Handoff Enhancements using Multimedia error robustness feature (CHEM).

NOTE: When the CHEM feature is supported, improved error robustness might be enabled by packet-loss handling procedures of the codec, codec mode, or codec configuration to avoid, delay, or reduce the need to handoff a terminal due to degradation in the media quality. Communicating the level of robustness of the media to the network enables the eNB to use this information to determine a threshold for when the terminal should be handed off to another cell, domain (circuit-switched vs. packet-switched), or radio access technology.

When a session is initiated or modified the P-CSCF supporting the CHEM feature shall derive the "maxPacketLossRateDl" attribute and "maxPacketLossRateUl" attribute based on the PLR_adapt and maxe2e-PLR attribute values in both the SDP offer and/or SDP answer to determine the maximum tolerable end-to-end PLR budget distributed across the uplink and downlink in a media transport path as described in 3GPP TS 29.513 [7] clause 7.2.3.

Upon reception of SDP offer and answer, the P-CSCF should check whether "a= PLR_adapt" line is present in both SDP offer and answer to derive "maxPacketLossRateDl" attribute and "maxPacketLossRateUl" attribute in "medComponents" attribute else "maxPacketLossRateDl" and "maxPacketLossRateUl" attributes are not included by the P-CSCF.

The originating P-CSCF should derive "maxPacketLossRateDl" attribute to the maximum value of MaxPacketLossRateDl among all the RTP payload types. For each RTP payload type MaxPacketLossRateDl is computed as described in 3GPP TS 29.513 [7] clause 7.2.3.

- If maxe2e-PLR is included in the SDP offer then the MaxPacketLossRateDl for a payload type is derived as value of maxe2e-PLR in the SDP offer minus maxUL-PLR in the SDP answer if present else the MaxPacketLossRateDlis ½ maxe2e-PLR value present in the SDP offer.

- If maxe2e-PLR is not included in the SDP offer then the MaxPacketLossRateDl for a payload type is derived from the default value in end-to-end Maximum End-to-End Packet Loss Rate for the decoder of the RTP payload type as recommended in 3GPP TS 26.114 [30]clause X.1.2 for application layer redundancy or X.1.1 for partial redundancy minus maxUL-PLR in the SDP answer if present else the MaxPacketLossRateDl ½ default value in end-to-end Maximum End-to-End Packet Loss Rate for the decoder of the RTP payload type as recommended in 3GPP TS 26.114 [30] clause X.1.2 for application layer redundancy or X.1.1 for partial redundancy.

The originating P-CSCF should derive "maxPacketLossRateUl" attribute to the maximum value of MaxPacketLossRateUl among all the RTP payload types. For each RTP payload type MaxPacketLossRateUl is computed as described in 3GPP TS 29.513 [7] clause 7.2.3.

- If maxe2e-PLR is included in the SDP answer then the MaxPacketLossRateUl for a payload type is derived as value of maxe2e-PLR in the SDP answer minus maxDL-PLR in the SDP answer if present else the MaxPacketLossRateUl is ½ maxe2e-PLR value present in the SDP answer.

- If maxe2e-PLR is not included in the SDP answer then the MaxPacketLossRateUl for a payload type is derived as the ½ default value in end-to-end Maximum End-to-End Packet Loss Rate for the decoder of the RTP payload type as recommended in 3GPP TS 26.114 [30] clause X.1.2 for application layer redundancy or X.1.1 for partial redundancy.

The terminating P-CSCF should derive "maxPacketLossRateDl" attribute to the maximum value of MaxPacketLossRateDl among all the RTP payload types. For each RTP payload type MaxPacketLossRateDl is computed as described in 3GPP TS 29.513 [7] clause 7.2.3.

- If maxe2e-PLR is included in the SDP answer then the MaxPacketLossRateDl for a payload type is derived as value of maxDL-PLR in the SDP answer if present else the MaxPacketLossRateDl is ½ maxe2e-PLR value present in the SDP answer.

- If maxe2e-PLR is not included in the SDP answer then the MaxPacketLossRateDl for a payload type is derived as the ½ default value in end-to-end Maximum End-to-End Packet Loss Rate for the decoder of the RTP payload type as recommended in 3GPP TS 26.114 [30] clause X.1.2 for application layer redundancy or X.1.1 for partial redundancy.

The terminating P-CSCF should derive "maxPacketLossRateUl" attribute to the maximum value of MaxPacketLossRateUl among all the RTP payload types. For each RTP payload type MaxPacketLossRateUl is computed as described in 3GPP TS 29.513 [7] clause 7.2.3.

- If maxe2e-PLR is included in the SDP offer then the MaxPacketLossRateUl for a payload type is derived as value of maxUL-PLR in the SDP answer if present else the MaxPacketLossRateUl is ½ maxe2e-PLR value present in the SDP offer.

- If maxe2e-PLR is not included in the SDP offer then the MaxPacketLossRateUl for a payload type is derived as the ½ default value in end-to-end Maximum End-to-End Packet Loss Rate for the decoder of the RTP payload type as recommended in 3GPP TS 26.114 [30] clause X.1.2 for application layer redundancy or X.1.1 for partial redundancy.

B.15 Handling of a FLUS session

If the P-CSCF receives a SIP request that requires provisioning of a service information to the PCF, the "FLUS" feature is supported and an SDP attribute "a=label:flus…" is included in one or more of the received SDP media descriptions, the P-CSCF shall provide the string after "a=label:" starting with "flus" within the "flusId" attribute for each affected media components within the "medComponents" attribute in the corresponding Npcf_PolicyAuthorization service operation towards the PCF.

NOTE: During the first interaction with the PCF, the P-CSCF does not know if the "FLUS" feature is supported by the PCF. In this case the P-CSCF will include the information as if the feature is supported.

If additionally the P-CSCF receives the "a=3gpp-qos-hint" media-level SDP attribute in the SIP request, the P-CSCF shall provide the PCF with the "desMaxLatency" attribute and/or "desMaxLoss" attribute as described in 3GPP TS 29.513 [7], clause 7.2.3.

Upon receiving the information from the P-CSCF and if the "FLUS" feature is supported, the PCF shall derive the QoS information as described in 3GPP TS 29.513 [7], clause 7.3.3.

B.16 QoS hint support for data channel media

If the P-CSCF receives a SIP request or SIP response that requires provisioning of a service information to the PCF, the QoSHint feature is supported and an SDP attribute "a=3gpp-qos-hint" is included in one or more of the received data channel media descriptions, the P-CSCF may provide the "desMaxLatency" attribute and/or "desMaxLoss" attribute for each affected application media component within the "medComponents" attribute in the corresponding Npcf_PolicyAuthorization service operation towards the PCF.

NOTE: During the first interaction with the PCF, the P-CSCF does not know if the QoSHint feature is supported by the PCF. In this case the P-CSCF will include the information as if the feature is supported.

Upon receiving the information from the P-CSCF and if the QoSHint feature is supported, the PCF shall derive the QoS information as described in 3GPP TS 29.513 [7], clause 7.3.3.

B.17 Handling of MPS Session

When the P-CSCF receives an authorised Resource-Priority header field or when the P-CSCF adds a temporarily authorised Resource-Priority header field containing an appropriate namespace and priority value in SIP signaling, and recognizes the need for priority treatment as specified in 3GPP TS 24.229 [32], and the "IMS_SBI" feature is supported, the P-CSCF shall include the "mpsId" attribute and the "resPrio" attribute in the corresponding Npcf_PolicyAuthorization service operation towards the PCF. The "mpsId" attribute shall contain the national variant for MPS service name indicating an MPS session. The "resPrio" attribute shall be determined based on the resource value received in the "wps" namespace of the SIP Resource-Priority header field, and shall be included at "AppSessionContextReqData" data type level as well as the "MediaComponent" data type level. The "resPrio" attribute shall be populated with a default value if the priority value is unknown.

NOTE 1: Various mechanisms can be applied to recognize the need for priority treatment in the P-CSCF (e.g., based on the dialled digits), according to national regulation and network configuration, as stated in 3GPP TS 24.229 [32].

NOTE 2: Highest user priority level (lowest numerical resource value of the SIP Resource-Priority header field) is mapped to the highest enumerated value of the "resPrio" attribute.

If the P-CSCF supports the SBI Message Priority mechanism for an MPS session, the P-CSCF shall include the "3gpp-Sbi-Message-Priority" custom HTTP header with a priority value based on the value of the "resPrio" attribute. The highest "resPrio" value is mapped to the corresponding lowest value of the "3gpp-Sbi-Message-Priority" custom HTTP header.

Upon reception of a request that requires MPS treatment, the PCF shall derive the PCC rules corresponding to the MPS session, as appropriate. The PCF shall take specific actions on the corresponding PDU session to ensure that the MPS session is prioritized, as described in 3GPP TS 29.512 [8], clause 4.2.6.2.12.3.

When the P-CSCF detects that the MPS session has ended, the P-CSCF deletes in the PCF the "Individual Application Session Context" resource corresponding to the MPS session. The PCF shall delete the PCC rules corresponding to the MPS session and shall revoke the actions related to the prioritization of the MPS session in the corresponding PDU session, as described in 3GPP TS 29.512 [8], clause 4.2.6.2.12.3.

B.18 Handling of RAN/NAS release cause values

If the P-CSCF is required by operator policy to provide the RAN/NAS release cause information, it includes this information in the corresponding SIP message as specified in 3GPP TS 24.229 [32] when received from the PCF (see clause 4.4.4, 4.4.6.1, 4.4.6.2, 4.4.6.3 and 4.4.6.7).

B.19 Priority for AF Signalling flow for MPS for Messaging

If the feature MPSforMessaging is supported, when the P-CSCF obtains MPS for Messaging indication set to enabled as specified in 3GPP TS 23.228 [33], the P-CSCF shall include the "mpsAction" attribute set to "ENABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING" towards the PCF. The P-CSCF shall provision the AF Signalling IP flow information as described in clauses 4.2.2.12.4 and 4.2.3.12.

Upon reception of a request that requires MPS treatment for Messaging for IMS Signalling IP flow, the PCF shall derive the PCC Rules corresponding to the IMS Signalling IP flow if not done before. The PCF shall assign 5QI and/or ARP values on the corresponding PCC Rules to ensure that the IMS signalling IP flow is prioritized and provision them to the NF service consumer if not done before.

When the P-CSCF obtains MPS for Messaging indication set to disabled, the P-CSCF shall include the "mpsAction" attribute set to "DISABLE_MPS_FOR_MESSAGING_FOR_AF_SIGNALLING" towards the PCF. The PCF shall check if the IMS signalling IP flow still needs priority treatment related to other services and shall modify the PCC Rules if needed to ensure that the IMS signalling IP flow gets the proper values according to PCF decision.

B.20 Retrieval of serving satellite information

B.20.1 General

The following clauses provide optional PCC procedures to support the retrieval of serving satellite information.

The originating P-CSCF can retrieve serving satellite identifier upon reception of a SIP INVITE.

The terminating P-CSCF can retrieve serving satellite identifier upon reception of a SIP INVITE.

B.20.2 Retrieval of serving satellite information at originating P-CSCF

If the originating P-CSCF receives the SIP INVITE request indicates that the UE is accessing from NR satellites access in non roaming scenario, the P-CSCF shall invoke:

- the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14 (SIP INVITE request);

including in the corresponding HTTP request:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:

  a) the "event" attribute set to the value "ANI_REPORT"; and

  b) the "notifMethod" attribute set to the value "ONE_TIME"; and

- the "reqAnis" attribute, with the required access network information, i.e. if the feature "UeSatUeComm" feature is supported, identifier of the serving satellite of the UE.

If the SIP INVITE request is an initial SIP INVITE request, the P-CSCF shall create a new "Individual Application Session Context" for the new SIP session with the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14 (if preliminary session information is included).

The P-CSCF will receive the serving satellite identifier from the PCF within the Npcf_PolicyAuthorization_Notify service operation as described in clause 4.2.5.11 and should include this serving satellite identifier in the SIP INVITE requests that it forwards.

If the P-CSCF application session context is only used for retrieval of the required access network information, at reception of this information, the P-CSCF may delete the context of application session information using the Npcf_PolicyAuthorization_Delete service operation.

B.20.3 Retrieval of serving satellite information at terminating P-CSCF

If the terminating P-CSCF based on e.g. access information in the SIP REGISTER request (or other SIP messages) which indicates that the terminating UE is accessing from NR satellites access in a non-roaming scenario, the P-CSCF upon reception of the first SIP 18X response from the terminating UE shall invoke:

- the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14;

including in the corresponding HTTP request:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:

a) the "event" attribute set to the value "ANI_REPORT"; and

b) the "notifMethod" attribute set to the value "ONE_TIME"; and

- the "reqAnis" attribute, with the required access network information, i.e. if the feature "UeSatUeComm" feature is supported, identifier of the serving satellite of the UE.

If the SIP 18X response is for an initial SIP INVITE request, the P-CSCF shall create a new "Individual Application Session Context" for the new SIP session with the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14.

The terminating P-CSCF will receive the serving satellite identifier from the PCF within the Npcf_PolicyAuthorization_Notify service operation as described in clause 4.2.5.11 and compare with the serving satellite identifier received from the originating P-CSCF. If the two serving statellite identifiers are the same or connected with ISL(s), the terminating P-CSCF determines to active the UE-Satellite-UE communication in IMS.

If the P-CSCF application session context is only used for retrieval of network provided location information, at reception of this information, the P-CSCF may delete the context of application session information using the Npcf_PolicyAuthorization_Delete service operation.

B.21 Subscription to user plane management events at change of satellite

When the feature "UeSatUeComm" is supported and if UE-Satellite-UE communications in IMS is activated, the P-CSCF, in order to be notified of satellite change and perform IMS-AGW relocation at change of satellite, subscribes to the notification of user plane management events by invoking the Npcf_PolicyAuthorization_Update service operation according to clause 4.2.3.8, and the P-CSCF shall include in the HTTP PATCH request message in the "ascReqData" attribute, the updated values of the "upPathChgSub" attribute with:

- the "dnaiChgType" attribute in the "UpPathChgEvent" data type is set to value "EARLY_LATE";

- the notification URI where the P-CSCF is receiving the Npcf_PolicyAuthorization_Notify service operation in the "notificationUri" attribute; and

- the notification correlation identifier assigned by the P-CSCF in the "notifCorreId" attribute.

The P-CSCF may include an indication of P-CSCF acknowledgement to be expected as an "afAckInd" attribute within the "upPathChgSub" attribute.The PCF shall reply to the P-CSCF as described in clause 4.2.3.2.

B.22 Request of 5GC network functions health monitoring information at terminating P-CSCF

According to clause 7.2.2 of 3GPP TS 23.380 [39], the P-CSCF can use PCC to retrieve 5GC network functions health monitoring information when P-CSCF does not receive a SIP response after some time out.

P-CSCF shall request the 5GC network functions health monitoring information in the following manner:

- The P-CSCF shall create an "Individual Application Session Context" for the new SIP session with the Npcf_PolicyAuthorization_Create service operation according to clause 4.2.2.14 (if session information is included);

- The P-CSCF shall send an HTTP POST request to the PCF including an entry of the "AfEventSubscription" data type in the "events" attribute with:

  (a) the "event" attribute set to the value "CN_HEALTH_MONITOR"; and

  (b) the "notifMethod" attribute set to the value "ONE_TIME".

If PCF rejects the Npcf_PolicyAuthorization_Create service operation with an HTTP "500 Internal Server Error" status code with the response body including the ProblemDetails data structure with the "cause" attribute set to

"PDU_SESSION_NOT_AVAILABLE"; the P-CSCF should send a SIP "503 Service Unavailable" error message towards S-CSCF to trigger HSS/UDM based P-CSCF restoration procedures.

If the P-CSCF receives the Npcf_PolicyAuthorization_Notify service operation from PCF:

- the P-CSCF should send a SIP "500 Internal Server Error" message towards S-CSCF, if "sessionRecovStatus" attribute is set to "REST_TRIGGERED" value within the "EventsNotification" data type included in the body of the HTTP POST request message;
- the P-CSCF should continue with the current procedure (e.g. not answering to S-CSCF), if "sessionRecovStatus" attribute is set to "REST_NOT_TRIGGERED" value within the "EventsNotification" data type included in the body of the HTTP POST request message; or
- the P-CSCF should send a SIP "503 Service Unavailable" error message towards S-CSCF, to trigger HSS/UDM based P-CSCF restoration procedures in the following cases:

  a) if the Npcf_PolicyAuthorization_Notify service operation is received with "sessionRecovStatus" attribute set to "NOT_FOUND" value inside the "EventsNotification" data type included in the body of the HTTP POST request message; or

  b) if "SmfFailureDetection" feature is supported and the Npcf_PolicyAuthorization_Notify service operation is received with "TerminationInfo" data type is set to "SMF_FAILURE" within the body of the HTTP POST request.

Annex C (normative): Flow identifiers: Format definition and examples

C.1 Format of a flow identifier

C.1.1 General

A flow identifier is expressed as a 2-tuple as follows:

> <The ordinal number of the position of the media component description in the SDI. The ordinal number of the IP flow(s) within the media component description assigned in the order of increasing downlink port numbers as detailed below.>

where both are numbered starting from 1. The encoding of the flow identifier is as indicated in 3GPP TS 24.008 [36].

The rules for the allocation of flow identifiers to IP flows are defined in 3GPP TS 29.214 [20], Annex B.1.1. Derivation of flow identifiers from SDP are described in 3GPP TS 29.214 [20], Annex B.1.2, and examples are covered in 3GPP TS 29.214 [20], Annex B2, B3, B4 and B5.

Annex D (normative): Wireless and wireline convergence access support

D.1 Scope

This annex provides the stage 3 definition of the Policy Authorization Service for wireless and wireline convergence access support for 5GS.

The stage 2 definition and procedures of the Policy Authorization Service for wireless and wireline convergence access support for 5GS are contained in 3GPP TS 23.316 [44].

D.2 Npcf_PolicyAuthorization Service

D.2.1 Service Description

D.2.1.1 Overview

The overview defined in clause 4.1.1 applies with the exception that the UE is replaced by the 5G-RG and the W-AGF, which is acting as a UE towards the 5GC on behalf of the FN-RG.

D.2.1.2 Service Architecture

The service architecture defined in clause 4.1.2 applies.

D.2.1.3 Network Functions

D.2.1.3.1 Policy Control Function (PCF)

The PCF functionality defined in clause 4.1.3.1 shall apply with the following modifications for W-5GAN and for the Npcf_PolicyAuthorization service:

- The 5G-RG and the W-AGF, acting as a UE towards the 5GC on behalf of the FN-RG, replace the UE.
- The PCF provides Policy Authorization as described in this Annex.

D.2.1.3.2 NF Service Consumers

The NF service consumer functionality defined in clause 4.1.3.2 shall apply with the following exceptions for the traffic of a PDU session over wireline access:

- Indication that the QoS targets can no longer (or can again) be guaranteed does not apply.
- Invocation of Multimedia Priority Services does not apply in this release of the specification.
- Indication of PLMN change does not apply.
- Indication of TSN 5GS Bridge Information does not apply.
- Reporting RAN/NAS Release Cause over wireline does not apply.
- The Maximum Packet Loss Rate for UL and DL is not forwarded to the wireline access. CHEM feature does not apply.

D.3 Service Operations

D.3.1 Introduction

Service procedures covered in clause 4.2.1 shall apply.

D.3.2 Npcf_PolicyAuthorization_Create Service Operation

D.3.2.1 General

The procedures specified in clause 4.2.2 shall apply with the following differences:

- Subscriptions to notifications of Service Data Flow QoS targets are not supported. Clause 4.2.2.6 does not apply for the traffic of a PDU session over wireline access.
- Invocation of Multimedia Priority Services is not supported. Clause 4.2.2.12 does not apply for the traffic of a PDU session over wireline access.
- The PEI that may be returned as available user information within the "ueIds" attribute described in clause 4.2.2.18 shall have one of the following representations:
  i. When the UE supports only wireline access, the PEI shall be a MAC address.
  ii. When the UE supports at least one 3GPP access technology, the PEI shall be the allocated IMEI or IMEISV.
- Subscription and notification of PLMN change does not apply for the traffic of a PDU session over wireline access.
- Indication of TSN 5GS Bridge Information does not apply. Clauses 4.2.2.24, 4.2.2.25 and 4.2.2.31 do not apply.
- The Maximum Packet Loss Rate for UL and DL is not forwarded to the wireline access. Clause 4.2.2.28, Support of CHEM feature, does not apply for the traffic of a PDU session over wireline access.
- When the NF service consumer subscribes to the Access Type Change event, the event is met, and the 5G-RG or FN-RG is connected to the 5GC via wireline access, the reported wireline transmission technology is encoded in the "ratType" attribute, within either the EventsNotification data type or the AdditionalAccessInfo data type, as applicable.

D.3.3 Npcf_PolicyAuthorization_Update Service Operation

D.3.3.1 General

The procedures specified in clause 4.2.3 shall apply with the following differences:

- Subscriptions to notifications of Service Data Flow QoS targets are not supported. Clause 4.2.3.6 does not apply for the traffic of a PDU session over wireline access.
- Invocation of Multimedia Priority Services is not supported. Clause 4.2.3.12 does not apply for the traffic of a PDU session over wireline access.
- Subscription and notification of PLMN change does not apply for the traffic of a PDU session over wireline access.
- Indication of TSN 5GS Bridge Information does not apply. Clauses 4.2.3.24, and 4.2.3.25 do not apply.
- The Maximum Packet Loss Rate for UL and DL is not forwarded to the wireline access. Clause 4.2.3.27, Support of CHEM feature, does not apply for the traffic of a PDU session over wireline access.

- When the NF service consumer subscribes to the Access Type Change event, the event is met, and the 5G-RG or FN-RG is connected to the 5GC via wireline access, the reported wireline transmission technology is encoded in the "ratType" attribute, within either the EventsNotification data type or the AdditionalAccessInfo data type, as applicable.

D.3.4 Npcf_PolicyAuthorization_Delete Service Operation

D.3.4.1 General

The procedures specified in clause 4.2.4 shall apply with the following differences:

- When the report of access network information described in clause 4.2.4.6 includes the user location information, the "n3gaLocation" attribute shall be included in the "ueLoc" attribute and shall encode:

  a) if the UE connects via W-5GBAN access:

     - shall encode the Global Line Identifier in the "gli" attribute; and

     - may include the "w5gbanLineType" attribute to indicate whether the W-5GBAN access is DSL or PON; or

  b) if the UE connects via W-5GCAN access, the HFC Node Identifier in the "hfcNodeId" attribute.

- Reporting RAN/NAS Release Cause over wireline does not apply. Clause 4.2.4.10 does not apply.

D.3.5 Npcf_PolicyAuthorization_Notify Service Operation

D.3.5.1 General

The procedures specified in clause 4.2.5 shall apply with the following differences:

- Subscriptions to notifications of Service Data Flow QoS targets are not supported. Clause 4.2.5.4 does not apply for the traffic of a PDU session over wireline access.

- Invocation of Multimedia Priority Services is not supported. Clause 4.2.4.5 does not apply for the traffic of a PDU session over wireline access.

- When the report of access network information described in clause 4.2.5.11 includes the user location information, the "n3gaLocation" attribute shall be included in the "ueLoc" attribute and shall encode:

  a) if the UE connects via W-5GBAN access:

     - shall encode the Global Line Identifier in the "gli" attribute; and

     - may include the "w5gbanLineType" attribute to indicate whether the W-5GBAN access is DSL or PON; or

  b) if the UE connects via W-5GCAN access, the HFC Node Identifier in the "hfcNodeId" attribute.

- Notification of PLMN changes does not apply for the traffic of a PDU session over wireline access.

- Indication of TSN 5GS Bridge Information does not apply. Clauses 4.2.5.13 and 4.2.5.16 do not apply.

- Reporting RAN/NAS Release Cause over wireline does not apply. Clauses 4.2.5.5 and 4.2.5.10 do not apply.

- When the 5G-RG or FN-RG connects to the 5GC via W-5GAN, and the Access Type Change event is met, the reported wireline transmission technology is encoded in the "ratType" attribute, within either the EventsNotification data type or the AdditionalAccessInfo data type, as applicable.

D.3.6 Npcf_PolicyAuthorization_Subscribe Service Operation

D.3.6.1 General

The procedures specified in clause 4.2.6 shall apply with the following differences:

- When the NF service consumer subscribes to the Access Type Change event, the event is met, and the 5G-RG or FN-RG is connected to the 5GC via wireline access, the reported wireline transmission technology is encoded in the "ratType" attribute, within either the EventsNotification data type or the AdditionalAccessInfo data type, as applicable.
- Subscription to PLMN change does not apply for the traffic of a PDU session over wireline access.

D.3.7 Npcf_PolicyAuthorization_Unsubscribe Service Operation

D.3.7.1 General

The procedures specified in clause 4.2.7 shall apply.

Annex E (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2017-10						TS skeleton of Policy Authorization Service specification	0.0.0
2017-03	CT3#92					Inclusion of pCRs agreed during CT3#92	0.1.0
2018-01	CT3#94					Inclusion of documents agreed in CT3#94: C3-180036, C3-180038, C3-180212, C3-180213, C3-180214, C3-180217, C3-180218, C3-180243, C3-180313, C3-180314, C3-180315, C3-180316.	0.2.0
2018-03	CT3#95					Inclusion of documents agreed in CT3#95: C3-181229, C3-181338, C3-181231, C3-181232, C3-181339, C3-181323	0.3.0
2018-04	CT3#96					Inclusion of documents agreed in CT3#96: C3-182057, C3-182333, C3-182235, C3-182334, C3-182474, C3-182336, C3-182337, C3-182338, C3-182339, C3-182245, C3-182475, C3-182247, C3-182248, C3-182249, C3-182250, C3-182251	0.4.0
2018-06	CT3#97					Inclusion of documents agreed in CT3#97: C3-183220, C3-183222, C3-183230, C3-183233, C3-183234, C3-183239, C3-183281, C3-183300, C3-183301, C3-183517, C3-183518, C3-183520, C3-183521, C3-183522, C3-183523, C3-183524, C3-183525, C3-183526, C3-183577, C3-183579, C3-183580, C3-183581, C3-183582, C3-183583, C3-183584, C3-183585, C3-183586, C3-183587, C3-183588, C3-183589, C3-183590, C3-183591, C3-183592, C3-183820, C3-183821, C3-183822, C3-183879, C3-183882.	0.5.0
2018-06	CT#80					TS sent to plenary for approval	1.0.0
2018-06	CT#80					TS approved by plenary	15.0.0
2018-09	CT#81	CP-182015	0001	2	F	DNAI change notification type	15.1.0
2018-09	CT#81	CP-182015	0002	1	F	Definition of FlowStatus data type	15.1.0
2018-09	CT#81	CP-182015	0003	2	F	Temporal validity update	15.1.0
2018-09	CT#81	CP-182015	0004		F	Modification of Traffic Routing Information provided at AF session level	15.1.0
2018-09	CT#81	CP-182015	0005	1	F	Missing AF Transaction Identifier	15.1.0
2018-09	CT#81	CP-182015	0006	2	B	Solution to IPv4 overlapping	15.1.0
2018-09	CT#81	CP-182015	0007	2	B	Subscription and notification of resources allocation outcome, data model	15.1.0
2018-09	CT#81	CP-182015	0008	1	B	Subscription to resources allocation outcome, service procedures	15.1.0
2018-09	CT#81	CP-182101	0009	3	B	Notification of resource allocation outcome, service procedures	15.1.0
2018-09	CT#81	CP-182015	0010	2	B	Subscription and notification of out of credit events, data model	15.1.0
2018-09	CT#81	CP-182015	0011	1	B	Subscription to out of credit notification, service procedures	15.1.0
2018-09	CT#81	CP-182015	0012	3	B	Out of credit notification, service procedures	15.1.0
2018-09	CT#81	CP-182015	0013	1	F	References to Data Types defined in 5G Technical Specifications	15.1.0
2018-09	CT#81	CP-182015	0014	1	F	Removal of error UNAUTHORIZED_TRAFFIC_ROUTING_REQUEST	15.1.0
2018-09	CT#81	CP-182015	0015	3	F	OpenAPI corrections	15.1.0
2018-09	CT#81	CP-182015	0016	1	F	Description of Structured data types	15.1.0
2018-08	CT#81	CP-182015	0017		F	Correction on TemporalValidity	15.1.0
2018-08	CT#81	CP-182015	0018	2	F	Resource structure presentation	15.1.0
2018-08	CT#81	CP-182015	0019		F	Corrections related to Feature negotiation	15.1.0
2018-08	CT#81	CP-182040	0020	1	F	Cardinality of optional arrays and maps	15.1.0
2018-08	CT#81	CP-182015	0021		F	Application Error: SUBSCRIPTION_NOT_FOUND	15.1.0
2018-08	CT#81	CP-182015	0022	2	F	Completion and clarification of non-3GPP access location information	15.1.0
2018-08	CT#81	CP-182015	0023	1	B	Support of Priority Services	15.1.0
2018-08	CT#81	CP-182015	0024	3	F	Correction of PRA information	15.1.0
2018-08	CT#81	CP-182015	0025	1	F	Updates in clause 4.2.6.3 to detail session binding	15.1.0
2018-08	CT#81	CP-182100	0026	2	B	Support of content versioning for a media component, service procedures	15.1.0
2018-08	CT#81	CP-182015	0027		B	Support of content versioning for a media component, data model	15.1.0
2018-08	CT#81	CP-182103	0028	2	B	Updates of QoS Notification Control description and data model	15.1.0
2018-08	CT#81	CP-182015	0029	2	B	Requested Service Temporarily not authorized	15.1.0
2018-08	CT#81	CP-182102	0030	2	B	Support of notification of content version during service data flow deactivation	15.1.0
2018-08	CT#81	CP-182015	0031	1	F	Transfer of RouteToLocation Data Type to TS 29.571	15.1.0
2018-08	CT#81	CP-182015	0032	2	F	Addition of FlowUsage Information	15.1.0

2018-08	CT#81	CP-182015	0033		F	Correction of evsNotif attribute	15.1.0
2018-08	CT#81	CP-182015	0034	1	F	Completing definition of re-used data types	15.1.0
2018-08	CT#81	CP-182015	0035		F	Correction of AppSessionContextReqData	15.1.0
2018-08	CT#81	CP-182015	0036		F	Correction of evNotif array attribute	15.1.0
2018-08	CT#81	CP-182015	0037		F	Removal of Editor's note in subclause 5.6.2.6	15.1.0
2018-08	CT#81	CP-182015	0038		F	Corrections on TosTrafficClass data type	15.1.0
2018-12	CT#82	CP-183205	0043		F	Usage of EventsSubscReqData data type	15.2.0
2018-12	CT#82	CP-183205	0044		F	Reference update: RFC 7396	15.2.0
2018-12	CT#82	CP-183205	0045		F	Supported content types	15.2.0
2018-12	CT#82	CP-183205	0046		F	Update of sponsored data connectivity indication	15.2.0
2018-12	CT#82	CP-183205	0047	3	F	Npcf_PolicyAuthorization API Authorization based on OAuth2	15.2.0
2018-12	CT#82	CP-183205	0050	1	F	Removal of references to 3GPP TS 29.508	15.2.0
2018-12	CT#82	CP-183205	0051	1	F	Correction of 404 error	15.2.0
2018-12	CT#82	CP-183205	0052		F	Corrections on Spatial Validity in OpenAPI	15.2.0
2018-12	CT#82	CP-183125	0053	2	F	Corrections on Data Types	15.2.0
2018-12	CT#82	CP-183205	0054	5	F	Adding "nullable" property to OpenAPI definitions of data types	15.2.0
2018-12	CT#82	CP-183205	0055		F	Correction of figure 4.2.4.2-1 to include 204 status code	15.2.0
2018-12	CT#82	CP-183125	0056	1	F	Corrections on OpenAPI file	15.2.0
2018-12	CT#82	CP-183205	0058	1	F	Adding the externalDocs field in the OpenAPI	15.2.0
2018-12	CT#82	CP-183205	0059		F	Default value for apiRoot	15.2.0
2018-12	CT#82	CP-183205	0060	1	F	Incorrect references	15.2.0
2018-12	CT#82	CP-183205	0061	1	F	OpenAPI: HTTP status codes alignment	15.2.0
2018-12	CT#82	CP-183205	0062		F	OpenAPI: usage of the "tags" keyword	15.2.0
2018-12	CT#82	CP-183205	0063		F	Presence conditions in OpenAPI file	15.2.0
2018-12	CT#82	CP-183205	0064		F	Location header field in OpenAPI	15.2.0
2018-12	CT#82	CP-183205	0065		F	Correction of resource URIs	15.2.0
2018-12	CT#82	CP-183205	0066	1	F	New data type for subscriptions to UP Path management events	15.2.0
2018-12	CT#82	CP-183205	0067	2	F	Mandatory traffic routing information for AF influence on traffic routing	15.2.0
2018-12	CT#82	CP-183205	0068		F	Incorrect use of Link data type	15.2.0
2018-12	CT#82	CP-183125	0069	1	F	Corrections on QNC trigger name	15.2.0
2018-12	CT#82	CP-183205	0070	1	F	Miscellaneous Corrections	15.2.0
2018-12	CT#82	CP-183205	0071		F	Removal of SUBSCRIPTION_NOT_FOUND error from service procedures	15.2.0
2018-12	CT#82	CP-183125	0072		F	Update of supported AF events	15.2.0
2019-03	CT#83	CP-190112	0074		F	Add GPSI in N5	15.3.0
2019-03	CT#83	CP-190112	0077	1	F	Miscellaneous corrections	15.3.0
2019-03	CT#83	CP-190112	0078	2	F	Retry-After header definition in OpenAPI	15.3.0
2019-03	CT#83	CP-190112	0079	1	F	OpenAPI Version number update	15.3.0
2019-03	CT#83	CP-190070	0076	1	F	Indication of acceptable service information	16.0.0
2019-03	CT#83	CP-190069	0080	3	F	OpenAPI version update	16.0.0
2019-06	CT#84	CP-191076	0082	1	A	Correction to the encoding of the initial POST request callback URI	16.1.0
2019-06	CT#84	CP-191076	0084		A	Storage of OpenAPI specification file	16.1.0
2019-06	CT#84	CP-191076	0088	2	A	Correction to EthFlowDescripiont data type	16.1.0
2019-06	CT#84	CP-191076	0093	1	A	Precedence of OpenAPI file	16.1.0
2019-06	CT#84	CP-191071	0094	2	B	AF acknowledgement to be expected	16.1.0
2019-06	CT#84	CP-191071	0095	2	B	UE IP address preservation Indication	16.1.0
2019-06	CT#84	CP-191076	0097		A	Missing resPrio attribute	16.1.0
2019-06	CT#84	CP-191076	0101	1	A	Copyright Note in YAML file	16.1.0
2019-06	CT#84	CP-191101	0105	2	F	OpenAPI Version number update	16.1.0
2019-09	CT#85	CP-192155	0109	1	B	Support of "Access Network Charging Correlation Information" notification	16.2.0
2019-09	CT#85	CP-192155	0110	1	B	Support of "Out of credit" notification	16.2.0
2019-09	CT#85	CP-192155	0111		B	Support of the AF charging identifier	16.2.0
2019-09	CT#85	CP-192155	0112		B	Support of "Access Network Information Notification"	16.2.0
2019-09	CT#85	CP-192202	0114	1	B	Support a set of MAC addresses in traffic filter	16.2.0
2019-09	CT#85	CP-192144	0116	1	A	Support of Ethernet scenarios	16.2.0
2019-09	CT#85	CP-192155	0117	1	B	IMS related P-CSCF procedures and Service Information Status	16.2.0
2019-09	CT#85	CP-192155	0118		B	IMS related P-CSCF procedures, setting flow status and flow number	16.2.0
2019-09	CT#85	CP-192155	0119		B	IMS related P-CSCF procedures, Support of SIP Forking	16.2.0
2019-09	CT#85	CP-192155	0120	1	B	IMS related P-CSCF procedures, support of RTCP flows	16.2.0
2019-09	CT#85	CP-192155	0121	2	B	Subscription to notification of Signalling Path Status	16.2.0
2019-09	CT#85	CP-192222	0122		B	Provisioning of Signalling Flow Information	16.2.0
2019-09	CT#85	CP-192155	0123	1	B	Resource Sharing Support	16.2.0
2019-09	CT#85	CP-192155	0124		B	Support of Mission Critical Push To Talk	16.2.0
2019-09	CT#85	CP-192155	0125	1	B	Support of Mission Critical Video	16.2.0

2019-09	CT#85	CP-192155	0126	1	B	Priority Sharing Indication	16.2.0
2019-09	CT#85	CP-192155	0127		B	IMS emergency services	16.2.0
2019-09	CT#85	CP-192144	0129		A	Correction to Policy Authorization Update	16.2.0
2019-09	CT#85	CP-192152	0130	1	B	Support of wireline and wireless access convergence, Annex Skeleton	16.2.0
2019-09	CT#85	CP-192223	0131	1	B	Support of wireline and wireless access convergence, NFs	16.2.0
2019-09	CT#85	CP-192173	0133		F	OpenAPI version update	16.2.0
2019-12	CT#86	CP-193181	0135	1	F	Open issue for AddrReservation feature	16.3.0
2019-12	CT#86	CP-193186	0137		F	Correction to appReloc attribute	16.3.0
2019-12	CT#86	CP-193196	0138	1	B	P-CSCF procedures to support Access Type Change notification	16.3.0
2019-12	CT#86	CP-193196	0139	1	B	P-CSCF procedures to subscribe to PLMN Change notification	16.3.0
2019-12	CT#86	CP-193196	0140	2	B	Support of the RAN-NAS Release Cause	16.3.0
2019-12	CT#86	CP-193235	0141	6	B	QoS Handling for V2X Communication	16.3.0
2019-12	CT#86	CP-193181	0142	3	B	QoS Monitoring for Service Data Flows	16.3.0
2019-12	CT#86	CP-193196	0144		B	NetLoc Correction	16.3.0
2019-12	CT#86	CP-193222	0145	2	B	Transport of TSN information and containers between PCF and AF	16.3.0
2019-12	CT#86	CP-193222	0146	1	B	Transport of TSC assistance information between PCF and AF	16.3.0
2019-12	CT#86	CP-193260	0147	5	B	Indication of PS to CS Handover for 5G SRVCC	16.3.0
2019-12	CT#86	CP-193215	0148	4	B	Coverage and Handover Enhancements for Media (CHEM)	16.3.0
2019-12	CT#86	CP-193197	0149	2	B	Update of API version and TS version in OpenAPI file	16.3.0
2019-12	CT#86	CP-193186	0151		A	Correct VLAN tag description	16.3.0
2019-12	CT#86	CP-193186	0153		A	Corrections to several mistakes	16.3.0
2019-12	CT#86	CP-193228	0154	2	B	Report of Wireline Location Information	16.3.0
2019-12	CT#86	CP-193191	0155	1	B	Support of 5WWC, supported PEI format	16.3.0
2019-12	CT#86	CP-193229	0156	2	B	Support of Trusted non-3GPP accesses	16.3.0
2019-12	CT#86	CP-193196	0157		F	Correction of AF Charging Identifier data type	16.3.0
2019-12	CT#86	CP-193196	0158	2	B	P-CSCF restoration	16.3.0
2019-12	CT#86	CP-193196	0159		B	Support of Maximum Supported Bandwidth and Minimum Desired Bandwidth	16.3.0
2019-12	CT#86	CP-193212	0161	1	F	Update of API version and TS version in OpenAPI file	16.3.0
2020-03	CT#87e	CP-200215	0162		B	Support of Framework for Live Uplink Streaming (FLUS) in Npcf_PolicyAuthorization service	16.4.0
2020-03	CT#87e	CP-200207	0174		B	DNN Clarification	16.4.0
2020-03	CT#87e	CP-200265	0176	3	B	Complete the QoS Monitoring	16.4.0
2020-03	CT#87e	CP-200206	0177		B	Network provided location information at SIP session release	16.4.0
2020-03	CT#87e	CP-200231	0180	1	B	Report of EPS Fallback	16.4.0
2020-03	CT#87e	CP-200201	0181	1	B	Update of the indication of PS to CS Handover	16.4.0
2020-03	CT#87e	CP-200254	0182	3	B	Configuration of one or more NW-TT port management information containers	16.4.0
2020-03	CT#87e	CP-200218	0183		B	DS-TT port MAC address as UE MAC address	16.4.0
2020-03	CT#87e	CP-200218	0184	2	B	TSCAI input container and TSN QoS container	16.4.0
2020-03	CT#87e	CP-200256	0185	2	B	Notification about TSN port detection and/or port management information, AF session exists	16.4.0
2020-03	CT#87e	CP-200255	0186	1	B	Notification about TSN port detection and/or port management information, no AF session exists	16.4.0
2020-03	CT#87e	CP-200212	0187		F	Modification of Alternative Service Requirements	16.4.0
2020-03	CT#87e	CP-200212	0188		F	Service Procedures for AF session with required QoS functionality	16.4.0
2020-03	CT#87e	CP-200207	0189		B	Adding "ProblemDetails" data type in table 5.6.1-2	16.4.0
2020-03	CT#87e	CP-200214	0190		F	Enumeration PreemptionControlInformationRm and "nullable" keyword	16.4.0
2020-03	CT#87e	CP-200202	0191		F	Correcting 5G_URLLC errors in clause 5.6	16.4.0
2020-03	CT#87e	CP-200206	0192		F	OpenAPI: property containing the pre-emption control information	16.4.0
2020-03	CT#87e	CP-200206	0193		F	Correcting eIMS5G_SBA errors in clause 5.6	16.4.0
2020-03	CT#87e	CP-200261	0194		F	Adding info about removable attributes "maxPacketLossRateDl" and "maxPacketLossRateUl"	16.4.0
2020-03	CT#87e	CP-200212	0195	1	F	Correction to QoS notification Control	16.4.0
2020-03	CT#87e	CP-200216	0197		F	Update of OpenAPI version and TS version in externalDocs field	16.4.0
2020-06	CT#88e	CP-201219	0199	1	A	Correction to response for PUT request for Events Subscription	16.5.0
2020-06	CT#88e	CP-201252	0200	1	F	Correction to bridge information report and port management information container provisioning	16.5.0
2020-06	CT#88e	CP-201252	0201	1	B	Correction to TSCAI provisioning	16.5.0
2020-06	CT#88e	CP-201228	0203		F	Removal of MAC address	16.5.0
2020-06	CT#88e	CP-201228	0204		F	Solving ENs related to a global line identity	16.5.0
2020-06	CT#88e	CP-201228	0205		F	Solving ENs related to NetLoc support for wireline access	16.5.0

2020-06	CT#88e	CP-201213	0206		F	Adding QosMonitoringInformationRm in table 5.6.1-1	16.5.0
2020-06	CT#88e	CP-201232	0207		F	Miscellaneous corrections	16.5.0
2020-06	CT#88e	CP-201246	0208		F	Support of FLUS feature	16.5.0
2020-06	CT#88e	CP-201246	0209		F	Names of "maxPacketLossRateDl" and "maxPacketLossRateUl" attributes	16.5.0
2020-06	CT#88e	CP-201252	0210		B	Adding support of NID	16.5.0
2020-06	CT#88e	CP-201228	0211		F	Correction to Access Network Information for Trusted non-3GPP access	16.5.0
2020-06	CT#88e	CP-201228	0212		B	Solving Editor's notes on report of location for Trusted non-3GPP access	16.5.0
2020-06	CT#88e	CP-201229	0213	3	B	Access Type Report for a MA PDU session	16.5.0
2020-06	CT#88e	CP-201232	0214	3	F	Correction to NetLoc feature	16.5.0
2020-06	CT#88e	CP-201252	0215	1	B	Correction to TSCAI UL and DL description	16.5.0
2020-06	CT#88e	CP-201252	0216	3	B	Update of TSN related events	16.5.0
2020-06	CT#88e	CP-201244	0217	1	F	Storage of YAML files in ETSI Forge	16.5.0
2020-06	CT#88e	CP-201228	0218	3	B	Access Type report for WWC	16.5.0
2020-06	CT#88e	CP-201246	0219	1	B	Support of applications with specific QoS hints	16.5.0
2020-06	CT#88e	CP-201272	0221	1	B	Introduction of Bridge management information	16.5.0
2020-06	CT#88e	CP-201219	0222	1	A	Correction of Policy Authorization Delete API 200 OK response body content	16.5.0
2020-06	CT#88e	CP-201252	0224	1	B	DS-TT MAC address derivation	16.5.0
2020-06	CT#88e	CP-201273	0225	1	B	Max bitrate of TSN QoS information	16.5.0
2020-06	CT#88e	CP-201252	0226	1	B	Port management on TSN AF	16.5.0
2020-06	CT#88e	CP-201252	0227	1	F	Service information provisioning for TSN	16.5.0
2020-06	CT#88e	CP-201337	0228	1	B	TSN QoS Information derivation on the TSN AF	16.5.0
2020-06	CT#88e	CP-201256	0230	1	F	URI of the Npcf_PolicyAuthorization service	16.5.0
2020-06	CT#88e	CP-201219	0232		A	OpenAPI: adding Location header field in 303 response	16.5.0
2020-06	CT#88e	CP-201228	0233	1	B	Events not supported in wireline access	16.5.0
2020-06	CT#88e	CP-201270	0234	1	B	Reallocation of credit	16.5.0
2020-06	CT#88e	CP-201252	0235	1	B	Indication of Application Sessions resource	16.5.0
2020-06	CT#88e	CP-201252	0236	1	B	TSN AF selection by PCF	16.5.0
2020-06	CT#88e	CP-201219	0238	1	A	Correction to Subscription operation	16.5.0
2020-06	CT#88e	CP-201244	0241		F	Optionality of ProblemDetails	16.5.0
2020-06	CT#88e	CP-201252	0242		B	Providing NID to the P-CSCF	16.5.0
2020-06	CT#88e	CP-201232	0243	1	F	"PCSCF-Restoration-Enhancement" feature corrections	16.5.0
2020-06	CT#88e	CP-201244	0244		F	Required field in OpenAPI file	16.5.0
2020-06	CT#88e	CP-201244	0245	1	F	Supported headers, Resource Data type, Operation Name	16.5.0
2020-06	CT#88e	CP-201233	0247	1	B	Description of enhanced PCC features in NF description clauses	16.5.0
2020-06	CT#88e	CP-201252	0248	1	B	Description of TSN features in NF description clauses	16.5.0
2020-06	CT#88e	CP-201213	0249		B	Description of URLLC features in NF description clauses	16.5.0
2020-06	CT#88e	CP-201238	0250		B	Description of V2X features in NF description clauses	16.5.0
2020-06	CT#88e	CP-201255	0253		F	Update of OpenAPI version and TS version in externalDocs field	16.5.0
2020-09	CT#89e	CP-202065	0256	1	F	Data type correction of the reqAni	16.6.0
2020-09	CT#89e	CP-202062	0257	1	F	Removal on Editor's notes on traffic forwarding for a MA PDU session	16.6.0
2020-09	CT#89e	CP-202065	0258		F	Correction to Trusted Non-3GPP location information	16.6.0
2020-09	CT#89e	CP-202065	0259	1	F	Correction of handling of non-3GPP location information by the P-CSCF	16.6.0
2020-09	CT#89e	CP-202065	0260	1	F	Handling of MPS Session by the P-CSCF	16.6.0
2020-09	CT#89e	CP-202084	0261		F	Update of OpenAPI version and TS version in externalDocs field	16.6.0
2020-12	CT#90e	CP-203139	0262	1	F	Essential Corrections and alignments	16.7.0
2020-12	CT#90e	CP-203127	0263	1	F	SBI Message Priority mechanism for emergency session	16.7.0
2020-12	CT#90e	CP-203116	0265		A	Correction to ACCESS_TYPE_CHANGE	16.7.0
2020-12	CT#90e	CP-203150	0266	1	F	Remove the NW-TT port from the TSN bridge info	16.7.0
2020-12	CT#90e	CP-203132	0267	1	F	Correction to Alternative QoS Parameter	16.7.0
2020-12	CT#90e	CP-203116	0269	1	A	Correction to referenced attributes	16.7.0
2020-12	CT#90e	CP-203111	0270		F	Corrections on QoS monitoring	16.7.0
2020-12	CT#90e	CP-203111	0271	1	F	QoS monitoring report at PDU session termination	16.7.0
2020-12	CT#90e	CP-203139	0272	1	F	Storage of YAML files in ETSI Forge	16.7.0
2020-12	CT#90e	CP-203110	0273	1	F	Correction to support redirection codes	16.7.0
2020-12	CT#90e	CP-203152	0274		F	Update of OpenAPI version and TS version in externalDocs field	16.7.0
2021-03	CT#91e	CP-210210	0275	4	F	Disable UE notifications at changes related to Alternative QoS Profiles	16.8.0
2021-03	CT#91e	CP-210202	0276	1	F	Correction to location information	16.8.0
2021-03	CT#91e	CP-210192	0277	1	F	mandate notifCorreId for QoS monitoring subscrtiption	16.8.0
2021-03	CT#91e	CP-210191	0278	3	F	Correction to resource identifiers descriptions used in notifications	16.8.0
2021-03	CT#91e	CP-210237	0279	1	F	Correction to TSN scenarios.	16.8.0

2021-03	CT#91e	CP-210197	0287	1	A	Correction to PATCH method	16.8.0
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2021-03	CT#91e	CP-210239	0291		F	Update of OpenAPI version and TS version in externalDocs field	16.8.0
2021-03	CT#91e	CP-210219	0280	1	F	Adding "description" field for map data types	17.0.0
2021-03	CT#91e	CP-210218	0281		F	OpenAPI reference	17.0.0
2021-03	CT#91e	CP-210221	0283	1	F	Adding some missing description fields to data type definitions in OpenAPI specification files	17.0.0
2021-03	CT#91e	CP-210219	0284		F	Support of optional HTTP custom header fields	17.0.0
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2021-06	CT#92e	CP-211226	0293	3	B	AF Session for control of MPS for DTS	17.1.0
2021-06	CT#92e	CP-211257	0294	2	B	Adding PCF as the consumer of the Npcf_PolicyAuthorization service to support DCAMP	17.1.0
2021-06	CT#92e	CP-211257	0295	4	B	Support of subscription to application detection notification for a PDU session	17.1.0
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2021-06	CT#92e	CP-211205	0302	2	A	Correction on 404 Not Found	17.1.0
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2021-06	CT#92e	CP-211262	0309	1	A	Removal of tsnBridgeInfo from EventsNotification data type	17.1.0
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2021-06	CT#92e	CP-211274	0324	1	B	Support of TSCAI time domain.	17.1.0
2021-06	CT#92e	CP-211211	0325	1	B	AF influence on traffic routing related events and errors report.	17.1.0
2021-06	CT#92e	CP-211265	0327		F	Update of OpenAPI version and TS version in externalDocs field	17.1.0
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2021-09	CT#93e	CP-212211	0329	1	B	TSCTSF support for Time Sensitive Communication	17.2.0
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2022-06	CT#96	CP-221154	0413		F	Alignment with the SBI template	17.5.0
2022-06	CT#96	CP-221158	0416	1	F	Correction to traffic routing requirements	17.5.0
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2022-12	CT#98e	CP-223200	0450	1	F	Correction to QoS monitoring	18.0.0
2022-12	CT#98e	CP-223178	0451	1	B	SNPN mobility	18.0.0
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2023-06	CT#100	CP-231143	0501	3	F	Support of BAT offset and adjusted periodicity	18.2.0
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2023-12	CT#102	CP-233228	0558	1	B	HTTP RFC uplifting	18.4.0
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2023-12	CT#102	CP-233245	0560	3	B	Event notification for AF requested QoS for a UE or group of UEs not identifed by a UE address	18.4.0
2023-12	CT#102	CP-233233	0562		B	Definition of L4S feature	18.4.0
2023-12	CT#102	CP-233253	0566	1	F	Adding reference and correcting the terms to align with stage-2	18.4.0
2023-12	CT#102	CP-233253	0567	1	F	Corrections related to EnSatBackhaulCatChg feature	18.4.0
2023-12	CT#102	CP-233228	0569		F	ProblemDetails RFC 7807 obsoleted by RFC 9457	18.4.0
2023-12	CT#102	CP-233249	0573		B	Completion of direct event notification of TSC management information support	18.4.0
2023-12	CT#102	CP-233265	0574	1	A	Wrong attribute name for Access Network Gateway Address	18.4.0
2023-12	CT#102	CP-233234	0575	1	B	Update for the PDU Set QoS related data	18.4.0
2023-12	CT#102	CP-233234	0576	1	F	Update for the Periodicity information and the terminology	18.4.0
2023-12	CT#102	CP-233237	0579		F	Update of info and externalDocs fields	18.4.0
2024-03	CT#103	CP-240159	0557	6	B	Protocol description update	18.5.0
2024-03	CT#103	CP-240159	0561	5	B	Completion of QoS monitoring data model	18.5.0
2024-03	CT#103	CP-240160	0572	3	B	Multi-modal services update	18.5.0
2024-03	CT#103	CP-240159	0577	3	B	Support of QoS monitoring requests for multiple QoS flows	18.5.0
2024-03	CT#103	CP-240158	0582	1	F	Clarification of feature dependency on QoSMonitoring for EnQoSMon	18.5.0
2024-03	CT#103	CP-240158	0583	1	B	indirect feature negotiation for EnQoSMon	18.5.0
2024-03	CT#103	CP-240158	0584	1	F	Resolve the Editor's Notes for QoS monitoring for congestion	18.5.0
2024-03	CT#103	CP-240159	0587	2	F	Resolve the Editor's Notes for QoS monitoring for PDV	18.5.0
2024-03	CT#103	CP-240160	0588	3	B	Completion of PDU Set handling functionality	18.5.0
2024-03	CT#103	CP-240158	0589		B	Interactions between ECN marking for L4S and Congestion Monitoring	18.5.0
2024-03	CT#103	CP-240158	0591	1	F	Editor's note removal and service provisioning requirements changes	18.5.0
2024-03	CT#103	CP-240160	0596	1	F	Correction of QosMonitoringReport	18.5.0
2024-03	CT#103	CP-240160	0597	1	F	Correction of EventsSubscReqDataRm and QosMonitoringInformationRm	18.5.0
2024-03	CT#103	CP-240186	0598	2	B	Redundant steering mode Resource allocation notification to AF	18.5.0
2024-03	CT#103	CP-240160	0599	2	F	XRM feature name update	18.5.0
2024-03	CT#103	CP-240159	0600		F	Data rate monitoring and QoS monitoring editors note removal	18.5.0
2024-03	CT#103	CP-240247	0602	2	F	Editor's note removal and the subscription of RTT monitoring updates	18.5.0
2024-03	CT#103	CP-240159	0603		F	Editor's note removal for multi-modal communication services	18.5.0
2024-03	CT#103	CP-240159	0606	1	F	Resolve the Editor's Notes for RT latency	18.5.0
2024-03	CT#103	CP-240160	0607	1	B	Support of the measurement of Round-Trip delay over two QoS flows	18.5.0

2024-03	CT#103	CP-240159	0608	1	B	Support the provisioning of periodicity information at service data flow level	18.5.0
2024-03	CT#103	CP-240170	0611		A	Corrections on QoS monitoring reports	18.5.0
2024-03	CT#103	CP-240176	0612	1	B	Resolution of Editor's Notes	18.5.0
2024-03	CT#103	CP-240160	0613	1	B	Solving Editor's Note on Notification Correlation Id and Notification URI on per flow subscription	18.5.0
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2024-06	CT#104	CP-241080	0615	1	F	Inapplicable measurement failure report for PDV and RTT QoS monitoring	18.6.0
2024-06	CT#104	CP-241080	0616	1	F	adding dependency feature for packet delay measurement failure report	18.6.0
2024-06	CT#104	CP-241080	0617		F	remove avrgWndw attribute from packet delay monitoring procedure	18.6.0
2024-06	CT#104	CP-241080	0618		F	Corrections on MediaSubComponent and MediaSubComponentRm	18.6.0
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2024-06	CT#104	CP-241081	0626	2	F	Clean up of Editor's Notes	18.6.0
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2024-06	CT#104	CP-241081	0630	4	B	Completion of Round Trip Time over two QoS flows	18.6.0
2024-06	CT#104	CP-241103	0631	1	F	Various GMEC related corrections	18.6.0
2024-06	CT#104	CP-241097	0632	2	B	Missing PDU Session Type in URSP rule enforcement information	18.6.0
2024-06	CT#104	CP-241097	0633	1	F	URSP rule enforcement corrections	18.6.0
2024-06	CT#104	CP-241081	0634	1	F	Completion of EnQoSMon feature description in 5.8	18.6.0
2024-06	CT#104	CP-241081	0635	1	F	Remove default boolean value for attributes in modification request	18.6.0
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2024-06	CT#104	CP-241081	0643		F	Miscellaneous corrections on QoS monitoring	18.6.0
2024-06	CT#104	CP-241082	0644	1	B	Support of UL and DL policy control based on Round-Trip latency requirements	18.6.0
2024-06	CT#104	CP-241128	0645	1	F	Corrections to TSC user plane node information reporting	18.6.0
2024-06	CT#104	CP-241085	0646		F	Update of info and externalDocs fields	18.6.0
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2024-09	CT#105	C3-244273	0654		A	Corrections on the transfer policy	18.7.0
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2024-09	CT#105	C3-244552	0660	1	A	Miscellaneous Corrections	18.7.0
2024-09	CT#105	C3-244416	0661	1	F	Correction to the report of extra UE Address	18.7.0
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2024-09	CT#105	C3-244529	0665	1	F	Correction to the PCF report of available containers	18.7.0
2024-09	CT#105	C3-244523	0666	1	F	Corrections on the feature name and terminology	18.7.0
2024-09	CT#105	C3-244453	0667	1	F	Fix the misalignment for congestion monitoring	18.7.0
2024-09	CT#105	CP-242121	0648	1	F	Completion of presence condition for qosMonReports and congestReports	19.0.0
2024-09	CT#105	CP-242118	0663	1	F	Corrections to the application detection report	19.0.0
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2024-12	CT#106	CP-243104	0684	1	B	Capability network support handling	19.1.0

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2024-12	CT#106	CP-243091	0697	1	F	Event notification corrections	19.1.0
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History

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